
Riemannian Geometry

— *EYNTKA* Series —

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Part I

Metric Riemannian Geometry

Riemannian and Pseudo-Riemannian Metrics

A smooth manifold, on its own, carries no geometry. The constructions of ‘NateDiffGeo’ produce tangent spaces, vector fields, tensor bundles, and differential forms, but nothing among them measures the length of a tangent vector, the angle between two of them, or the volume of a region. These are the first things one wants of a space called geometric, and they are supplied by a single additional datum: a smoothly varying inner product on the tangent spaces, the *Riemannian metric*.

This chapter installs that datum and develops its immediate consequences. The metric is defined in the generality of arbitrary signature, so that the Riemannian and the pseudo-Riemannian (in particular the Lorentzian) cases are treated on the same footing from the outset; positive definiteness is invoked only where a result genuinely needs it. After establishing that Riemannian metrics always exist and that pseudo-Riemannian ones need not, the chapter fixes the model spaces that recur throughout the book, the round sphere, hyperbolic space, and the flat torus, and the isometries relating them. The algebraic apparatus the metric provides, the musical isomorphisms raising and lowering indices, the gradient and divergence, and the Riemannian volume form, is assembled next, and the chapter closes with the metric induced on a submanifold, the setting of classical surface theory. Everything built in later chapters, the connection, the geodesics, the curvature, is forced by the metric introduced here.

1.1 Metrics of Arbitrary Signature

A smooth manifold, on its own, remembers almost nothing of geometry. It knows which functions are smooth, which curves are smooth, and how tangent spaces are glued together as one moves from chart to chart; it does not know how long a curve is, what angle two curves make where they cross,

or whether one region is larger than another. All of that geometric information is the content of a single extra piece of data laid over the manifold: an inner product on each tangent space, varying smoothly from point to point. This section introduces that data in the generality that the rest of the theory requires, allowing the inner product to be indefinite, and isolates the one invariant that separates the cases.

The reason to allow indefiniteness from the outset is not abstraction for its own sake. The geometry of spacetime in general relativity is governed by a symmetric bilinear form that is not positive definite: the squared length of a tangent vector can be positive, negative, or zero, and those three signs encode the physical distinction between space, time, and light. The positive-definite (Riemannian) case and the indefinite (pseudo-Riemannian) case share a large body of formalism, and much of the connection and curvature theory in later chapters is written once and applies to both. Positive-definiteness is a genuine hypothesis in exactly a few places, most notably where lengths of curves must be nonnegative so that a distance function can be defined, and those places are flagged as they arise. Here the two notions are defined side by side, and the invariant that tells them apart, the signature, is shown to be locally constant, hence constant on each connected piece of the manifold.

Throughout, M denotes a smooth manifold of dimension n , and all objects are C^∞ . The Einstein summation convention is in force: an index appearing once as a subscript and once as a superscript in a single term is summed from 1 to n . In a chart with coordinates $(x^1, \dots, x^n)$ the coordinate vector fields are written $\partial_i = \partial/\partial x^i$ and the coordinate covectors dx^i .

Definition 1.1.1: Riemannian Metric

Let M be a smooth manifold. A *Riemannian metric* on M is a smooth symmetric covariant 2-tensor field g on M that is positive definite at every point: for each $p \in M$ the bilinear form

$$g_p: T_p M \times T_p M \rightarrow \mathbb{R}$$

is symmetric, $g_p(v, w) = g_p(w, v)$ for all $v, w \in T_p M$, and positive definite, $g_p(v, v) > 0$ for every nonzero $v \in T_p M$. A *Riemannian manifold* is a pair (M, g) consisting of a smooth manifold together with a Riemannian metric. The value $g_p(v, w)$ is written $\langle v, w \rangle$ or $\langle v, w \rangle_g$, and the induced *norm* of a tangent vector is $|v|_g = \sqrt{\langle v, v \rangle_g} \geq 0$.

Unwinding the definition of a covariant 2-tensor field, a Riemannian metric assigns to each point p an inner product on the tangent space $T_p M$, and it does so smoothly. Smoothness is the requirement that in every chart the component functions be smooth. Writing $g_{ij} = g(\partial_i, \partial_j)$ for the components in a chart with coordinates $(x^1, \dots, x^n)$, one has

$$g = g_{ij} dx^i dx^j,$$

where the product of covectors is the symmetric product, and g is smooth precisely when each $g_{ij}: U \rightarrow \mathbb{R}$ is a smooth function on the chart domain U . Symmetry says the matrix (g_{ij}) is symmetric at every point; positive-definiteness says that this symmetric matrix is positive definite at every point. The tensor field machinery, the meaning of a smooth symmetric covariant 2-tensor field and its transformation law under a change of chart, is developed in [3]; the content added here is the pointwise positivity condition. Geometrically, once such a g is fixed, the length of a tangent vector, the angle $\arccos(\langle v, w \rangle / (|v| |w|))$ between two nonzero vectors, and eventually the length of a curve, all acquire meaning; without g none of them do.

The first example is the flat model against which everything else is measured.

Example 1.1.2: The Euclidean and Minkowski Metrics

On $M = \mathbb{R}^n$, using the standard global coordinates $(x^1, \dots, x^n)$, the *Euclidean metric* is

$$\bar{g} = \delta_{ij} dx^i dx^j = (dx^1)^2 + \dots + (dx^n)^2,$$

so that $\bar{g}_{ij} = \delta_{ij}$ is the identity matrix at every point. Under the canonical identification of $T_p \mathbb{R}^n$ with $\mathbb{R}^n$, the value $\bar{g}_p(v, w) = \delta_{ij} v^i w^j$ is the usual dot product $v \cdot w$, which is symmetric and positive definite, and the component functions are constants, hence smooth. Thus $(\mathbb{R}^n, \bar{g})$ is a Riemannian manifold, the *Euclidean space* of dimension n ; its norm $|v|_{\bar{g}} = \sqrt{(v^1)^2 + \dots + (v^n)^2}$ is the Euclidean length.

Now alter a single sign. On $M = \mathbb{R}^{1,n}$, meaning $\mathbb{R}^{n+1}$ with coordinates written $(x^0, x^1, \dots, x^n)$, the *Minkowski metric* is

$$\eta = -(dx^0)^2 + (dx^1)^2 + \dots + (dx^n)^2,$$

with constant component matrix $\eta = \text{diag}(-1, 1, \dots, 1)$. The form η_p is symmetric, and its matrix is invertible at every point, so it is nondegenerate; but it is not positive definite, since $\eta_p(\partial_0, \partial_0) = -1 < 0$. Hence η is not a Riemannian metric. It is instead the basic pseudo-Riemannian example, defined precisely below, and it separates tangent vectors into three classes according to the sign of $\eta_p(v, v)$: a nonzero v is *timelike* if $\eta_p(v, v) < 0$, *spacelike* if $\eta_p(v, v) > 0$, and *null* (or *lightlike*) if $\eta_p(v, v) = 0$. For $v = (v^0, v^1, \dots, v^n)$ this reads $-(v^0)^2 + \sum_{i \geq 1} (v^i)^2$, so v is timelike when its x^0 -component dominates the spatial part, spacelike when the spatial part dominates, and null on the cone $-(v^0)^2 + \sum_{i \geq 1} (v^i)^2 = 0$. Unlike the Riemannian case, nonzero null vectors exist and have vanishing η -norm, so $\sqrt{\eta_p(v, v)}$ is not defined for all v ; the classification into timelike, spacelike, and null takes the place of a norm.

The Euclidean and Minkowski forms differ in exactly one entry of their diagonal, and that single sign is what places one inside the Riemannian world and the other outside it. To capture both at once, the positive-definiteness requirement is relaxed to nondegeneracy, and the number of negative signs is recorded as a separate invariant.

Recall that a symmetric bilinear form b on a finite-dimensional real vector space V is *nondegenerate* if the only $v \in V$ with $b(v, w) = 0$ for all $w \in V$ is $v = 0$; equivalently, in any basis, the matrix of b is invertible. By Sylvester's law of inertia, for a nondegenerate symmetric b there is a basis in which the matrix of b is diagonal with entries ± 1 , and the number ν of entries equal to -1 is independent of the diagonalizing basis. This number ν is the *index* of b , and the pair (number of -1 's, number of $+1$'s) is its *signature*. The index is exactly the maximal dimension of a subspace on which b is negative definite, a description that will be used in the proof that the index cannot jump.

Definition 1.1.3: Pseudo-Riemannian Metric

Let M be a smooth manifold. A *pseudo-Riemannian metric* on M is a smooth symmetric covariant 2-tensor field g that is *nondegenerate* at every point and whose *index* ν , the number of -1 's in a Sylvester diagonalization of g_p , is the same integer at every point of M . A pair (M, g) with g a pseudo-Riemannian metric is a *pseudo-Riemannian manifold*, and its *signature* is the pair $(\nu, n - \nu)$. The metric is called

1. *Riemannian* when $\nu = 0$, that is, when g_p is positive definite for every p ; and
2. *Lorentzian* when $\nu = 1$, in which case, following the mostly-plus convention, the signature is $(-, +, \dots, +)$.

The adjective *(pseudo-)Riemannian* refers to a pseudo-Riemannian metric of unspecified index, covering both the definite and indefinite cases.

Two conditions in this definition do work that the Riemannian definition handled automatically. First, nondegeneracy replaces positive-definiteness as the pointwise linear-algebraic requirement; it is exactly what is needed for the metric to identify tangent vectors with covectors (the musical isomorphisms of section 1.4) and thereby to raise and lower indices, and a positive-definite form is in particular nondegenerate, so definition 1.1.1 is the case $\nu = 0$ of definition 1.1.3. Second, the requirement that the index be a single fixed integer over all of M is a genuine hypothesis for a possibly disconnected manifold: on a manifold with two components one could in principle place a Riemannian metric on one piece and a Lorentzian metric on the other, and the resulting field would be smooth, symmetric, and nondegenerate but of no single signature, so it is excluded by fiat. On a *connected* manifold this constancy is automatic, and proposition 1.1.6 below proves exactly that. The index is a discrete invariant, an integer between 0 and n , so once it is known to vary continuously it can only be constant on a connected space; the content of the proposition is that it does vary continuously.

The running indefinite example is $(\mathbb{R}^{1,n}, \eta)$ of example 1.1.2: the matrix $\text{diag}(-1, 1, \dots, 1)$ is already in Sylvester form with a single -1 , so η has index $\nu = 1$, signature $(-, +, \dots, +)$, and $(\mathbb{R}^{1,n}, \eta)$ is a Lorentzian manifold, called *Minkowski space*. The classification of its tangent vectors into timelike, spacelike, and null carried out in that example is the model for the causal structure of any Lorentzian manifold.

A warning about terminology is worth stating before the word “metric” accumulates a second, incompatible meaning.

1.1.4 Note: A Riemannian Metric Is Not a Distance Function The word *metric* is overloaded. A Riemannian metric g is a smoothly varying inner product on tangent spaces; it is a tensor field, evaluated on pairs of tangent vectors at a single point, and $\langle v, w \rangle_g$ is a real number attached to $v, w \in T_p M$. It is not a metric in the sense of metric-space theory, that is, it is not a function $d: M \times M \rightarrow [0, \infty)$ measuring the separation of pairs of points. A Riemannian metric does *induce* such a distance function, by taking $d_g(p, q)$ to be the infimum of the lengths of curves from p to q , and (M, d_g) is then a metric space whose topology is the manifold topology; but that construction requires the length functional and a minimization argument, and it is carried out only in section 3.4, where the phrase *distance function* is reserved for d_g . Until then, “metric” means the tensor g , and no notion of point-to-point distance is in play. The distinction matters in the indefinite case for a structural reason: a Lorentzian metric assigns negative and zero squared lengths to vectors, so the naive infimum-of-lengths construction does not produce a metric space at all, and the passage from g to a distance function is special to the Riemannian setting.

The second family of examples, the round spheres, will be a recurring test case; here it appears only in its first form, as a metric induced by an ambient Euclidean metric, with the fuller treatment postponed.

Example 1.1.5: The Round Metric on the Sphere, First Look

Let $S^n = \{x \in \mathbb{R}^{n+1} \mid |x|_{\bar{g}} = 1\}$ be the unit sphere, with inclusion $\iota: S^n \hookrightarrow \mathbb{R}^{n+1}$, and let $\bar{g}$ be the Euclidean metric on $\mathbb{R}^{n+1}$ from example 1.1.2. The inclusion is a smooth embedding, so at each $p \in S^n$ the differential $d\iota_p: T_p S^n \rightarrow T_p \mathbb{R}^{n+1}$ is an injective linear map, identifying $T_p S^n$ with the hyperplane

$$\{w \in \mathbb{R}^{n+1} \mid w \cdot p = 0\} \subset \mathbb{R}^{n+1}$$

tangent to the sphere. The *round metric* $\hat{g}$ on S^n is the pullback

$$\hat{g} = \iota^* \bar{g}, \quad \hat{g}_p(v, w) = \bar{g}_p(d\iota_p(v), d\iota_p(w)),$$

that is, the restriction of the ambient dot product to the tangent hyperplane. This is symmetric because $\bar{g}$ is, and it is smooth because ι is smooth and pullback of a smooth tensor field along a smooth map is smooth (see [3]). It is positive definite: for nonzero $v \in T_p S^n$ the vector $d\iota_p(v)$ is nonzero, since $d\iota_p$ is injective, and $\bar{g}$ is positive definite, so $\hat{g}_p(v, v) = \bar{g}_p(d\iota_p v, d\iota_p v) > 0$. Hence $\hat{g}$ is a Riemannian metric and $(S^n, \hat{g})$ is a Riemannian manifold. The sphere of radius r , $S_r^n = \{x \mid |x|_{\bar{g}} = r\}$, carries the analogous induced metric. The general statement that an immersion into a Riemannian manifold pulls back to a Riemannian metric, together with an explicit coordinate form of $\hat{g}$, belongs to section 1.5, and the description of $(S_r^n, \hat{g})$ as a homogeneous model space to section 1.3; here it suffices that the sphere carries a natural Riemannian metric, obtained by restricting the flat metric of the surrounding space.

It remains to justify the assertion, built into definition 1.1.3 as a hypothesis and claimed there to be automatic on connected manifolds, that the index of a nondegenerate metric cannot vary from point to point on a connected manifold. The precise statement is that the index is a locally constant function; connectedness then forces it to be globally constant.

Proposition 1.1.6: Constancy of the Signature

Let M be a connected smooth manifold and let g be a smooth symmetric covariant 2-tensor field on M that is nondegenerate at every point. Then the index $\nu(p)$ of g_p is the same for all $p \in M$; consequently the signature of a nondegenerate g is well defined, and the constancy required in definition 1.1.3 is automatic on a connected manifold.

Proof:

The strategy is to show that the index function $\nu: M \rightarrow \{0, 1, \dots, n\}$ is locally constant. A locally constant function on a connected space is constant, since for any value k the sets $\nu^{-1}(k)$ and $\nu^{-1}(\{0, \dots, n\} \setminus \{k\})$ are both open, partition M , and connectedness forbids a nontrivial partition into two open sets. It therefore suffices to prove that every point of M has a neighborhood on which ν is constant.

Step 1: reduction to a symmetric matrix of functions. Fix $p_0 \in M$ and choose a chart (U, φ) with coordinates $(x^1, \dots, x^n)$ about p_0 , with U connected (shrink it to a coordinate ball if necessary). On U the metric has smooth components $g_{ij} = g(\partial_i, \partial_j): U \rightarrow \mathbb{R}$, and at each $p \in U$ the index $\nu(p)$ is the index of the symmetric matrix

$$G(p) = (g_{ij}(p))_{1 \leq i, j \leq n},$$

because $(\partial_1|_p, \dots, \partial_n|_p)$ is a basis of $T_p M$ and $G(p)$ is the matrix of the bilinear form g_p in that basis; the index of a symmetric form is a property of the form, computed in any basis. The entries $p \mapsto g_{ij}(p)$ are smooth, in particular continuous, so $p \mapsto G(p)$ is a continuous map from U into the space of symmetric $n \times n$ real matrices. By hypothesis $G(p)$ is invertible for every $p \in U$, so no eigenvalue of $G(p)$ is zero at any point. It suffices to show ν is constant on U ; since every point of M lies in such a connected coordinate neighborhood, this establishes local constancy on M .

Step 2: the index counts negative eigenvalues, and none of them can cross zero. The index of a symmetric matrix equals the number of its negative eigenvalues, counted with multiplicity; this is one form of Sylvester's law of inertia, obtained by orthogonally diagonalizing the symmetric matrix and normalizing the basis. Fix a point $q \in U$ and let $\lambda_1(q) \leq \lambda_2(q) \leq \dots \leq \lambda_n(q)$ be the eigenvalues of $G(q)$, listed in increasing order with multiplicity; all are real because $G(q)$ is symmetric. The number of these that are negative is $\nu(q)$. Because $G(q)$ is invertible, 0 is never an eigenvalue, so each eigenvalue is either strictly positive or strictly negative at every point of U .

It remains to show that the count of negative eigenvalues cannot change as q moves within U . This follows from continuity of the eigenvalues together with the fact that none may pass through 0. Concretely, for a fixed $q \in U$ let

$$m = \min_{1 \leq k \leq n} |\lambda_k(q)| > 0,$$

the smallest absolute value of an eigenvalue of $G(q)$, which is positive since $G(q)$ is invertible. The characteristic polynomial $\det(G(p) - tI)$ has coefficients that are polynomials in the entries $g_{ij}(p)$, hence continuous functions of p ; and its roots depend continuously on those coefficients, so the ordered eigenvalue functions $p \mapsto \lambda_k(p)$ are continuous on U (this continuity of ordered eigenvalues is a standard consequence of continuous dependence of roots on the coefficients of a

real-rooted polynomial). Choose an open neighborhood $V \subseteq U$ of q small enough that

$$|\lambda_k(p) - \lambda_k(q)| < m \quad \text{for all } p \in V \text{ and all } k = 1, \dots, n,$$

which is possible by continuity of each λ_k at q (take the intersection of the finitely many neighborhoods produced for the individual k). For $p \in V$ and each k : if $\lambda_k(q) > 0$ then $\lambda_k(q) \geq m$, so $\lambda_k(p) > \lambda_k(q) - m \geq 0$, whence $\lambda_k(p) > 0$; and if $\lambda_k(q) < 0$ then $\lambda_k(q) \leq -m$, so $\lambda_k(p) < \lambda_k(q) + m \leq 0$, whence $\lambda_k(p) < 0$. Thus each eigenvalue keeps its sign throughout V , and the number of negative eigenvalues is the same at p as at q . Therefore $\nu(p) = \nu(q)$ for all $p \in V$, so ν is constant on a neighborhood of q .

Step 3: conclusion. Since $q \in U$ was arbitrary, ν is locally constant on U , and U is connected, so ν is constant on U . As $p_0 \in M$ was arbitrary and each point admits such a connected coordinate neighborhood, ν is locally constant on all of M . By the opening remark, a locally constant integer-valued function on the connected manifold M is constant. Hence the index of g is a single integer, the signature is well defined, and no separate constancy hypothesis is needed on a connected manifold, as we sought to show.

The proof isolates why the signature is a robust invariant rather than an accident of a chosen basis or a chosen point: the index is an integer, and integers cannot vary continuously except by staying fixed, so the only way for the signature to change would be for an eigenvalue of the component matrix to pass through zero. Nondegeneracy forbids exactly that, at every point simultaneously, and connectedness ties the local values together. This is also the precise sense in which a Lorentzian metric can never degenerate into a Riemannian one across a connected region: the transition would require a null eigenvalue somewhere, contradicting nondegeneracy. The signature, once fixed, is a permanent feature of a connected (pseudo-)Riemannian manifold, and it is the single discrete datum distinguishing the Riemannian theory of the following sections from its Lorentzian counterpart.

1.2 Existence of Metrics

A Riemannian metric is a piece of geometric data laid on top of a smooth manifold, and nothing in the definition of a smooth manifold forces such data to exist. It is reasonable to ask whether the definition in definition 1.1.1 is ever vacuous: is there a smooth manifold so twisted that no smooth, symmetric, positive-definite 2-tensor field can be placed on it? The answer is no, and the reason is the mildest possible one. The obstruction that might have appeared, a failure to fit local choices of inner product together into one globally smooth field, is dissolved by the partition of unity, which is available on every smooth manifold that is Hausdorff and second countable. Every smooth manifold carries a Riemannian metric.

This should be compared against the situation for pseudo-Riemannian metrics, where existence can fail. The difference is the arithmetic of positive-definiteness: a positive combination of positive-definite forms is again positive-definite, so the gluing that averages local metrics never destroys the property that makes the result a metric. Nondegeneracy of a fixed index enjoys no such stability, and the failure is not an accident of the construction but a topological obstruction. This section proves the existence theorem in the Riemannian case, isolates the algebraic fact that powers it, and then records, with its topological reason, why the Lorentzian case is different.

Throughout, M is a smooth manifold of dimension n , meaning Hausdorff, second countable, and C^∞ . The Einstein summation convention (sum over a repeated upper and lower index) is in force.

The engine of the existence proof is a single algebraic observation, promoted to a statement about tensor fields. Two symmetric 2-tensors at a point can be added, and a symmetric 2-tensor can be scaled by a nonnegative number; the question is whether positive-definiteness survives these operations. It does, and with room to spare: as long as at least one summand contributes at a given point, the sum is already positive-definite there.

Proposition 1.2.1: Convex Combinations of Metrics

Let M be a smooth manifold and let $\{\rho_\alpha\}_{\alpha \in A}$ be a smooth partition of unity on M , so that each $\rho_\alpha: M \rightarrow \mathbb{R}$ is smooth with $\rho_\alpha \geq 0$, the supports $\{\text{supp } \rho_\alpha\}$ form a locally finite family, and $\sum_{\alpha \in A} \rho_\alpha \equiv 1$. For each α let $V_\alpha \subseteq M$ be an open set with $\text{supp } \rho_\alpha \subseteq V_\alpha$, and let g_α be a Riemannian metric on V_α . Suppose that for each α the product $\rho_\alpha g_\alpha$, extended by 0 outside $\text{supp } \rho_\alpha$, is a smooth symmetric 2-tensor field on M . Then

$$g = \sum_{\alpha \in A} \rho_\alpha g_\alpha$$

is a Riemannian metric on M .

Proof:

Step 1: the sum is a well-defined smooth symmetric 2-tensor field. Fix $p \in M$. Local finiteness of $\{\text{supp } \rho_\alpha\}$ gives an open neighborhood U of p meeting only finitely many of the supports, say those indexed by $\alpha_1, \dots, \alpha_m$. On U every other term $\rho_\alpha g_\alpha$ vanishes identically, so on U the sum defining g is the finite sum $\rho_{\alpha_1} g_{\alpha_1} + \dots + \rho_{\alpha_m} g_{\alpha_m}$ of smooth symmetric 2-tensor fields. A finite sum of smooth symmetric 2-tensor fields is a smooth symmetric 2-tensor field, so g restricts to one on U . Since such neighborhoods U cover M and the local expressions agree on overlaps (they are all the same locally finite sum), g is a globally defined smooth symmetric 2-tensor field. By construction g is symmetric at each point, because each g_α is.

Step 2: positive-definiteness. Fix $p \in M$ and a nonzero vector $v \in T_p M$. Here the value of the term $\rho_\alpha g_\alpha$ at p means its value as the globally defined tensor field of the hypothesis, namely $\rho_\alpha(p) g_{\alpha,p}$ when $p \in V_\alpha$ and 0 otherwise. Evaluate:

$$g_p(v, v) = \sum_{\alpha \in A} (\rho_\alpha g_\alpha)_p(v, v),$$

a finite sum by local finiteness. Consider a single index α . If $\rho_\alpha(p) = 0$ the term $(\rho_\alpha g_\alpha)_p(v, v)$ vanishes. If $\rho_\alpha(p) > 0$ then $p \in \text{supp } \rho_\alpha \subseteq V_\alpha$, so $g_{\alpha,p}$ is defined; it is positive-definite, whence $g_{\alpha,p}(v, v) > 0$, and the term equals $\rho_\alpha(p) g_{\alpha,p}(v, v) > 0$. In either case the term is nonnegative, so $g_p(v, v) \geq 0$. It remains to exclude equality. Because $\sum_{\alpha} \rho_\alpha(p) = 1$, at least one index β has $\rho_\beta(p) > 0$. For that index $p \in V_\beta$ and $\rho_\beta(p) g_{\beta,p}(v, v) > 0$, being a product of two strictly positive numbers, so the whole sum is strictly positive:

$$g_p(v, v) \geq \rho_\beta(p) g_{\beta,p}(v, v) > 0$$

Thus g_p is positive-definite for every p , and g is a Riemannian metric, as we sought to show.

The mechanism is worth stating in words, since it is exactly what fails for indefinite metrics. Positive-definiteness is a *closed cone* condition that is preserved under nonnegative combination:

the positive-definite symmetric bilinear forms on a fixed vector space form a convex cone, and the weights $\rho_\alpha(p)$ pick out a convex combination of points of that cone (after normalizing by the value $\sum \rho_\alpha(p) = 1$). A convex combination of points of a convex set lies in that set, and the strict conclusion above records that the combination cannot slip to the boundary as long as some weight is positive. No hypothesis links the g_α to one another; they may be wildly different metrics on the overlaps of their supports, and the sum is still a metric. This freedom is what lets the metrics on separate charts be chosen with no coordination at all.

A concrete instance makes the averaging visible. Take $M = \mathbb{R}$, covered by two overlapping intervals, and let $g_1 = dx^2$ and $g_2 = 4dx^2$ be two metrics on $\mathbb{R}$ (each a positive multiple of the standard one). Choose a partition of unity ρ_1, ρ_2 with $\rho_1 + \rho_2 \equiv 1$ and $0 \leq \rho_1 \leq 1$; for instance ρ_1 a smooth function increasing from 0 to 1 and $\rho_2 = 1 - \rho_1$. Then

$$g = \rho_1 dx^2 + \rho_2 (4dx^2) = (\rho_1 + 4\rho_2) dx^2 = (4 - 3\rho_1) dx^2$$

The coefficient $4 - 3\rho_1$ ranges over $[1, 4]$, so it is bounded below by 1 and never vanishes; g is a metric interpolating smoothly between the two scales, agreeing with dx^2 where $\rho_1 = 1$ and with $4dx^2$ where $\rho_1 = 0$. At every point at least one weight is positive, and the coefficient stays positive, which is proposition 1.2.1 in miniature.

With the algebra in hand, the existence theorem is a matter of choosing local metrics on chart domains and letting the partition of unity assemble them. The only inputs are the flat metric read off in each chart and the gluing guaranteed on any smooth manifold.

Theorem 1.2.2: Existence of Riemannian Metrics

Every smooth manifold M admits a Riemannian metric.

Proof:

Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ be a smooth atlas for M , with each $\varphi_\alpha: U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$ a diffeomorphism onto an open subset of $\mathbb{R}^n$. The family $\{U_\alpha\}$ is an open cover of M .

Step 1: a metric on each chart domain. Let $\bar{g} = \delta_{ij} dx^i dx^j$ denote the Euclidean metric on $\mathbb{R}^n$, the standard inner product $\bar{g}_x(v, w) = \sum_{i=1}^n v^i w^i$ at each point x . On the chart domain U_α define

$$g_\alpha = \varphi_\alpha^* \bar{g},$$

the pullback of $\bar{g}$ under φ_α (pullback of a covariant tensor field along a smooth map is treated in [3]). This g_α is a smooth symmetric 2-tensor field on U_α , being a pullback of one. It is positive-definite: for $p \in U_\alpha$ and nonzero $v \in T_p M$,

$$g_{\alpha,p}(v, v) = \bar{g}_{\varphi_\alpha(p)}(d(\varphi_\alpha)_p v, d(\varphi_\alpha)_p v) = |d(\varphi_\alpha)_p v|^2,$$

the squared Euclidean length of $d(\varphi_\alpha)_p v$. Since φ_α is a diffeomorphism its differential $d(\varphi_\alpha)_p$ is a linear isomorphism, so $d(\varphi_\alpha)_p v \neq 0$ whenever $v \neq 0$, and the squared length is strictly positive. Thus g_α is a Riemannian metric on the open submanifold U_α .

Step 2: gluing by a partition of unity. Since M is a smooth manifold (Hausdorff and second countable), there exists a smooth partition of unity $\{\rho_\alpha\}_{\alpha \in A}$ subordinate to the cover $\{U_\alpha\}$: each $\rho_\alpha: M \rightarrow \mathbb{R}$ is smooth with $\rho_\alpha \geq 0$ and $\text{supp } \rho_\alpha \subseteq U_\alpha$, the family of supports is locally finite, and $\sum_{\alpha \in A} \rho_\alpha \equiv 1$ (see [3] for the construction of a partition of unity subordinate to an open cover). For each α the product $\rho_\alpha g_\alpha$ is defined and smooth on U_α , and it extends by 0 to a

smooth symmetric 2-tensor field on all of M : on the open set $M \setminus \text{supp } \rho_\alpha$ the factor ρ_α vanishes together with all its derivatives, so the extension is smooth across the boundary of $\text{supp } \rho_\alpha$, and the two definitions agree on the overlap $U_\alpha \setminus \text{supp } \rho_\alpha$, where both are 0.

Step 3: assembly. Take $V_\alpha = U_\alpha$ in proposition 1.2.1: each g_α is a Riemannian metric on the open set $U_\alpha \supseteq \text{supp } \rho_\alpha$, the family $\{\rho_\alpha\}$ is a partition of unity, and the tensor fields $\rho_\alpha g_\alpha$, extended by 0 from Step 2, are globally defined, smooth, and symmetric. The hypotheses of that proposition are met, and it gives that

$$g = \sum_{\alpha \in A} \rho_\alpha g_\alpha$$

is a Riemannian metric on M . Hence M admits a Riemannian metric, completing the proof.

The proof extracts geometry from the flat model and spreads it over M with no compatibility conditions to check, because the convex-cone argument absorbs whatever discrepancies the charts introduce. What it does *not* produce is any particular metric: the output depends on the atlas and on the choice of partition of unity, and two such choices generally give metrics that are not isometric. The theorem asserts the existence of geometric structure, not its uniqueness or canonicity. A single manifold typically carries an infinite-dimensional space of Riemannian metrics, and much of the subject is the study of how the geometry, curvature, geodesics, distance, varies as the metric varies within that space.

When a manifold arises as a quotient of one that already carries a metric, the existence theorem can be bypassed by a more economical construction that descends the metric through the quotient, and the result is canonical rather than arbitrary. The next example carries this out for real projective space, which inherits a metric from the round sphere and thereby acquires a specific Riemannian structure without any recourse to a partition of unity.

Example 1.2.3: The Round Metric Descends to $\mathbb{RP}^n$

Real projective space $\mathbb{RP}^n$ is the quotient of the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ by the antipodal identification $x \sim -x$; write $\pi: S^n \rightarrow \mathbb{RP}^n$ for the quotient map, which sends x to the line $[x] = \{x, -x\}$. The existence theorem theorem 1.2.2 already guarantees that $\mathbb{RP}^n$, being a smooth manifold, admits some Riemannian metric. A specific and canonical one is obtained instead by pushing the round metric g_{S^n} of example 1.3.4 down through π , using that the antipodal map is a symmetry of the round sphere.

Let $a: S^n \rightarrow S^n$, $a(x) = -x$, be the antipodal map, the restriction to S^n of $-\text{id}$ on $\mathbb{R}^{n+1}$. Its differential at x is again $-\text{id}$, now as a map $T_x S^n \rightarrow T_{-x} S^n$ (both tangent spaces are the same hyperplane $x^\perp = (-x)^\perp \subseteq \mathbb{R}^{n+1}$). The round metric is the restriction of the Euclidean inner product to each tangent space, so for $u, v \in T_x S^n$,

$$(a^* g_{S^n})_x(u, v) = (g_{S^n})_{-x}(da_x u, da_x v) = \langle -u, -v \rangle = \langle u, v \rangle = (g_{S^n})_x(u, v)$$

Thus $a^* g_{S^n} = g_{S^n}$, so a is an isometry of (S^n, g_{S^n}) in the sense of definition 1.3.1.

This is exactly what is needed for the metric to descend. The quotient map π is a smooth covering with $\pi \circ a = \pi$, and $da_x: T_x S^n \rightarrow T_{-x} S^n$ is precisely the identification of the two tangent spaces lying over the point $[x] \in \mathbb{RP}^n$ under $d\pi$. To define a metric $\bar{g}$ on $\mathbb{RP}^n$, set, for $\bar{u}, \bar{v} \in T_{[x]} \mathbb{RP}^n$,

$$\bar{g}_{[x]}(\bar{u}, \bar{v}) = (g_{S^n})_x(u, v), \quad u = (d\pi_x)^{-1} \bar{u}, \quad v = (d\pi_x)^{-1} \bar{v},$$

where $d\pi_x: T_x S^n \rightarrow T_{[x]} \mathbb{RP}^n$ is a linear isomorphism because π is a local diffeomorphism. The

formula uses a choice of preimage x over $[x]$; the two preimages are x and $-x$, and independence of the choice is exactly the isometry property just verified. Working over $-x$ replaces u, v by $da_x u = -u$ and $da_x v = -v$ and evaluates g_{S^n} at $-x$, giving $(g_{S^n})_{-x}(-u, -v) = (g_{S^n})_x(u, v)$, the same number. Hence $\bar{g}_{[x]}$ is well defined independently of the representative.

It remains to see that $\bar{g}$ is a smooth Riemannian metric. Smoothness is local, and near any $[x]$ the covering π restricts to a diffeomorphism from a neighborhood $\tilde{U}$ of x onto a neighborhood of $[x]$, with local inverse $s = \pi|_{\tilde{U}}^{-1}$. On that neighborhood the definition reads $\bar{g} = s^* g_{S^n}$, a pullback of a smooth 2-tensor field along a smooth map, hence smooth. Symmetry of $\bar{g}_{[x]}$ is inherited from symmetry of $(g_{S^n})_x$. For positive definiteness, if $\bar{u} \neq 0$ then $u = (d\pi_x)^{-1}\bar{u} \neq 0$ since $d\pi_x$ is an isomorphism, so $\bar{g}_{[x]}(\bar{u}, \bar{u}) = (g_{S^n})_x(u, u) > 0$. Therefore $\bar{g}$ is a Riemannian metric on $\mathbb{RP}^n$, and $\pi: (S^n, g_{S^n}) \rightarrow (\mathbb{RP}^n, \bar{g})$ is by construction a local isometry.

The output is not the arbitrary metric the partition-of-unity proof would have produced. It is the unique metric making π a local isometry, and it records all the local geometry of the round sphere while forgetting only the global antipodal identification. The same argument descends a metric through any quotient by a group of isometries acting freely and properly, a theme developed when quotient and covering metrics are treated in their own right.

The role of second countability deserves a word, since it is the one global hypothesis doing work. It is what guarantees a partition of unity, and hence what makes the gluing legal. On a manifold that is smooth and Hausdorff but not second countable, such as the long line, the argument breaks at Step 2, and the existence of a Riemannian metric is no longer automatic. For the manifolds of this book, all second countable by convention, no such pathology arises.

Example 1.2.4: A Metric on the Sphere by Charts

The construction can be run explicitly on S^2 , where its output can be compared against a metric obtained by other means. Cover S^2 by the two stereographic charts, φ_N from the north pole and φ_S from the south pole, each a diffeomorphism onto $\mathbb{R}^2$. Pulling back the Euclidean metric $\bar{g} = du^2 + dv^2$ on $\mathbb{R}^2$ gives two metrics $g_N = \varphi_N^* \bar{g}$ and $g_S = \varphi_S^* \bar{g}$ on the respective chart domains $S^2 \setminus \{\text{north pole}\}$ and $S^2 \setminus \{\text{south pole}\}$. Choosing a subordinate partition of unity ρ_N, ρ_S (with ρ_N supported away from the north pole and ρ_S away from the south pole), the theorem produces

$$g = \rho_N g_N + \rho_S g_S,$$

a Riemannian metric on all of S^2 . This metric is positive-definite at every point by the argument above: near the north pole only g_S contributes, near the south pole only g_N , and in the equatorial band both are present with positive weight. There is no claim that g is the round metric of example 1.1.5; it is one metric among many, built to demonstrate existence, and its curvature and symmetry depend entirely on the arbitrary weights ρ_N, ρ_S . The round metric is singled out from this crowd only later, by its symmetry properties.

The existence theorem should not be misread as saying that *any* symmetric 2-tensor structure one might want is available. The positive-definite case is the fortunate one. Allowing the signature to be indefinite changes the answer, and the change is topological rather than a defect of the gluing method.

1.2.5 Remark: Pseudo-Riemannian metrics need not exist The existence theorem is special to the Riemannian case. A smooth manifold need not admit a pseudo-Riemannian metric of a prescribed index (definition 1.1.3); for Lorentzian metrics, signature $(-, +, \dots, +)$, the obstruction is topological.

The partition-of-unity argument fails at exactly the step where proposition 1.2.1 was used. Pulling back a fixed Minkowski form (as in example 1.1.2) in each chart produces local Lorentzian metrics, but a convex combination $\sum \rho_\alpha g_\alpha$ of Lorentzian forms need not be Lorentzian: the nondegenerate symmetric forms of index 1 do not form a convex set. Averaging two Lorentzian inner products whose negative directions disagree can yield a degenerate or positive-definite form, so the sum can fail to be a metric of any signature at some points. There is nothing to glue the local timelike directions into a coherent global choice.

A coherent global choice of timelike direction is precisely a nowhere- vanishing line field on M , a smooth assignment $p \mapsto L_p$ of a 1-dimensional subspace $L_p \subseteq T_p M$, and this is where topology enters. A Lorentzian metric g on M determines such a line field, but not by the naive route of looking at unit timelike vectors: the set $\{v \in T_p M : g_p(v, v) = -1\}$ is a two-sheeted hyperboloid whose vectors point in a continuum of directions, so it singles out no line. The correct construction compares g against an auxiliary Riemannian metric h , which exists by theorem 1.2.2. At each p define the endomorphism $A_p : T_p M \rightarrow T_p M$ by

$$g_p(u, v) = h_p(A_p u, v) \quad \text{for all } u, v \in T_p M$$

Since g_p and h_p are symmetric and h_p is positive-definite, A_p is self-adjoint with respect to h_p , hence diagonalizable with real eigenvalues in some h_p -orthonormal eigenbasis $e_1, \dots, e_n$. In that basis $g_p(e_i, e_j) = h_p(A_p e_i, e_j) = \lambda_i \delta_{ij}$, where λ_i is the eigenvalue of e_i , so the signs of the eigenvalues of A_p are exactly the diagonal signs of g_p , and hence give its signature. As g has signature $(-, +, \dots, +)$, the operator A_p has exactly one negative eigenvalue, of multiplicity 1, and $n - 1$ positive ones. Let $L_p \subseteq T_p M$ be the eigenline for the negative eigenvalue. This line is well defined and varies smoothly with p (the negative eigenspace is cut out by the continuously varying spectral projection of the smoothly varying operator A), and every nonzero $v \in L_p$ satisfies $g_p(v, v) < 0$, so L is a smooth timelike line field. Conversely a smooth line field $L \subseteq TM$ together with any Riemannian metric h produces a Lorentzian metric by reversing the sign of h along L , that is, $g(v, v) = h(v, v)$ for $v \perp_h L$ and $g(v, v) = -h(v, v)$ for $v \in L$. So a Lorentzian metric exists on M if and only if M carries a smooth line field. On a closed manifold the existence of a nowhere-vanishing line field is governed by the Euler characteristic: a closed surface admits a line field, and therefore a Lorentzian metric, if and only if its Euler characteristic vanishes.

The sphere S^2 is the clean counterexample. Its Euler characteristic is $\chi(S^2) = 2 \neq 0$, so by the Poincaré–Hopf theorem every vector field on S^2 has a zero, and in fact S^2 carries no nowhere-vanishing line field either. Hence S^2 admits no Lorentzian metric, even though, by theorem 1.2.2, it admits a Riemannian one (for instance the round metric, or the chart-glued metric of example 1.2.4). The torus T^2 , with $\chi(T^2) = 0$, does admit a Lorentzian metric. The vanishing of the Euler characteristic and the Poincaré–Hopf count of zeros of a vector field belong to [3]; the point here is only that Riemannian existence is universal while Lorentzian existence is obstructed, and the obstruction has a name and a computation.

1.3 Isometries and Model Spaces

A metric turns each tangent space into an inner-product space, and a smooth map between manifolds carries tangent vectors forward. The question that organizes this section is when such a map respects the two metrics: when the length of a vector, measured before applying the map, equals the length of its image, measured afterward. A map with this property is an isometry, and it is the correct notion of sameness for Riemannian manifolds. Two manifolds that are isometric are, from the standpoint of every metric quantity built in the rest of this book (length, angle, geodesic, curvature), indistinguishable.

Isometries also let us single out the manifolds that will serve as reference points for everything that follows. Just as linear algebra is anchored by $\mathbb{R}^n$ with its standard inner product, Riemannian geometry is anchored by a short list of highly symmetric spaces, the *model spaces*: Euclidean space, the round sphere, hyperbolic space, and the flat quotients of Euclidean space. What distinguishes these is the size of their isometry groups. Each is homogeneous (its isometry group acts transitively, so the geometry looks the same at every point) and isotropic (the isometry group can rotate any direction at a point into any other, so the geometry looks the same in every direction). The bulk of this section constructs these spaces and records the symmetries that make them models.

Definition 1.3.1: Isometry

Let (M, g) and $(\tilde{M}, \tilde{g})$ be (pseudo-)Riemannian manifolds. A diffeomorphism $\varphi: M \rightarrow \tilde{M}$ is an *isometry* if $\varphi^*\tilde{g} = g$; that is, for every $p \in M$ and all $v, w \in T_pM$,

$$\tilde{g}_{\varphi(p)}(d\varphi_p(v), d\varphi_p(w)) = g_p(v, w)$$

When such a φ exists, (M, g) and $(\tilde{M}, \tilde{g})$ are *isometric*. A smooth map $\varphi: M \rightarrow \tilde{M}$ is a *local isometry* if every point of M has a neighborhood U on which φ restricts to a diffeomorphism onto its image with $(\varphi|_U)^*\tilde{g} = g|_U$; equivalently, each $d\varphi_p$ is a linear isometry of the inner-product spaces (T_pM, g_p) and $(T_{\varphi(p)}\tilde{M}, \tilde{g}_{\varphi(p)})$.

The condition $\varphi^*\tilde{g} = g$ is pointwise and linear-algebraic: at each p it asks that the differential $d\varphi_p$, a linear map between two inner-product spaces, preserve inner products. A linear map that preserves inner products is injective, so a local isometry is automatically an immersion; when M and $\tilde{M}$ have the same dimension it is a local diffeomorphism by the inverse function theorem. The distinction between a local isometry and a (global) isometry is therefore exactly the distinction between a local diffeomorphism and a diffeomorphism: a local isometry matches the metrics on small pieces but need not be injective or surjective. The flat torus below is the standard illustration that the two notions differ. The pullback $\varphi^*\tilde{g}$ is the pullback of a covariant 2-tensor field in the sense of [3]; that $\varphi^*\tilde{g}$ is again a smooth symmetric covariant 2-tensor field is part of that pullback machinery, and $\varphi^*\tilde{g}$ is positive definite whenever $\tilde{g}$ is and φ is an immersion.

Being isometric is an equivalence relation on (pseudo-)Riemannian manifolds. The identity map is an isometry; if $\varphi: M \rightarrow \tilde{M}$ is an isometry then so is φ^{-1} , because $(\varphi^{-1})^*g = (\varphi^{-1})^*\varphi^*\tilde{g} = (\varphi \circ \varphi^{-1})^*\tilde{g} = \tilde{g}$; and a composite of isometries is an isometry, since $(\psi \circ \varphi)^*\hat{g} = \varphi^*\psi^*\hat{g} = \varphi^*\tilde{g} = g$. The same computation with $\tilde{M} = M$ and $\tilde{g} = g$ shows that the isometries of a single manifold to itself are closed under composition and inversion.

Example 1.3.2: The Euclidean group as isometries

Let $(\mathbb{R}^n, \bar{g})$ be Euclidean space with $\bar{g} = \delta_{ij} dx^i dx^j$, the flat model of example 1.1.2 (the Einstein summation convention, summing over each repeated upper/lower index pair, is in force throughout this section). Under the canonical identification $T_x \mathbb{R}^n \cong \mathbb{R}^n$ the metric $\bar{g}_x$ is the standard dot product at every point. A linear map $A \in \text{GL}(n, \mathbb{R})$, viewed as the diffeomorphism $x \mapsto Ax$, has constant differential $dA_x = A$, so $A^* \bar{g} = \bar{g}$ holds iff A preserves the dot product, that is iff $A \in O(n)$. More generally, for a fixed $b \in \mathbb{R}^n$ the rigid motion $x \mapsto Ax + b$ has differential A at every point, so it is an isometry of $(\mathbb{R}^n, \bar{g})$ precisely when $A \in O(n)$. Thus every element of the Euclidean group $\mathbb{R}^n \rtimes O(n)$ is an isometry of Euclidean space. That these are *all* of the isometries requires knowing that an isometry is determined by its value and differential at one point, which follows from the geodesic machinery of Chapter 3; the containment $\mathbb{R}^n \rtimes O(n) \subseteq \text{Isom}(\mathbb{R}^n, \bar{g})$ is what the direct computation gives here.

The isometries of a fixed manifold form a group, and this group is the primary invariant that separates the model spaces from generic Riemannian manifolds.

Definition 1.3.3: Isometry group

Let (M, g) be a (pseudo-)Riemannian manifold. The set of all isometries $\varphi: M \rightarrow M$, under composition, is a group, the *isometry group* $\text{Isom}(M, g)$. Its identity is id_M and the inverse of an isometry is its inverse map, both of which are isometries by the computation above.

The group axioms hold because composition of maps is associative and, as recorded above, isometries are closed under composition and inversion and include the identity. The isometry group is a subgroup of the diffeomorphism group $\text{Diff}(M)$. For a Riemannian manifold it carries more structure than that of an abstract group: a theorem of Myers and Steenrod, taken up in Chapter 9, gives $\text{Isom}(M, g)$ the structure of a Lie group acting smoothly on M , with dimension bounded by $\frac{1}{2}n(n+1)$. That bound is attained by the constant-curvature model spaces built next, Euclidean space, the round sphere, and hyperbolic space, which is one precise sense in which they are maximally symmetric; the flat torus (example 1.3.6), homogeneous but not isotropic, does not attain it. Until Chapter 9, $\text{Isom}(M, g)$ is used only as a group; the size of its orbits already records homogeneity and isotropy.

A manifold (M, g) is *homogeneous* if $\text{Isom}(M, g)$ acts transitively on M , and *isotropic at p* if the isotropy subgroup $\{\varphi \in \text{Isom}(M, g) : \varphi(p) = p\}$ acts transitively on the set of unit vectors in $T_p M$; it is *isotropic* if it is isotropic at every point. Euclidean space is homogeneous by the translations $x \mapsto x + b$ and isotropic at the origin by the rotations $A \in O(n)$, which act transitively on unit vectors; together with homogeneity this makes it isotropic at every point. The next three examples build the remaining model spaces and verify the same two properties for each.

Example 1.3.4: The round sphere as a model space

Fix $r > 0$ and let

$$S_r^n = \{x \in \mathbb{R}^{n+1} \mid |x| = r\}$$

be the sphere of radius r , carrying the metric g_r induced by the inclusion $\iota: S_r^n \hookrightarrow (\mathbb{R}^{n+1}, \bar{g})$; that is, $g_r = \iota^* \bar{g}$, the first look at which was example 1.1.5. Concretely, for $p \in S_r^n$ the tangent space is $T_p S_r^n = \{v \in \mathbb{R}^{n+1} \mid \langle v, p \rangle = 0\}$, the orthogonal complement of the position vector, and g_r is the restriction of the Euclidean dot product to this hyperplane. This metric is positive definite because it is the restriction of a positive-definite form to a subspace, so (S_r^n, g_r) is a Riemannian manifold in the sense of definition 1.1.1.

Symmetries. The orthogonal group $O(n+1)$ acts on $\mathbb{R}^{n+1}$ by linear isometries of $\bar{g}$ and preserves $|x|$, hence maps S_r^n to itself. For $A \in O(n+1)$ the restriction $A|_{S_r^n}: S_r^n \rightarrow S_r^n$ is a diffeomorphism, and since $A^*\bar{g} = \bar{g}$ its restriction pulls $g_r = \iota^*\bar{g}$ back to itself: writing $\iota \circ (A|_{S_r^n}) = A \circ \iota$ gives $(A|_{S_r^n})^*g_r = (A|_{S_r^n})^*\iota^*\bar{g} = (A \circ \iota)^*\bar{g} = \iota^*A^*\bar{g} = \iota^*\bar{g} = g_r$. Thus $O(n+1) \subseteq \text{Isom}(S_r^n, g_r)$.

Homogeneity. Given $p, q \in S_r^n$, both have norm r , so p/r and q/r are unit vectors; extend each to an orthonormal basis and let $A \in O(n+1)$ be the orthogonal map sending the first basis to the second. Then $A(p) = q$, so $O(n+1)$ acts transitively and S_r^n is homogeneous.

Isotropy. Fix $p \in S_r^n$ and let $v, w \in T_p S_r^n$ be unit vectors, so $\{p/r, v\}$ and $\{p/r, w\}$ are each orthonormal. Extend to orthonormal bases of $\mathbb{R}^{n+1}$ agreeing in the first slot p/r ; the orthogonal map A carrying one basis to the other fixes p and sends v to w . Its differential at p is A itself (a linear map is its own differential), so A lies in the isotropy subgroup at p and carries v to w . Hence S_r^n is isotropic.

Homogeneity and isotropy together force the sectional curvature to be constant, a statement made precise in Chapter 4; there the value for (S_r^n, g_r) is computed to be $\text{sec} \equiv 1/r^2$, positive and shrinking as the radius grows. The round metric, its geodesics (great circles), and this curvature value recur as the constant-positive-curvature benchmark throughout the book.

The sphere is the model of constant positive curvature. Its constant-negative-curvature counterpart is hyperbolic space, which does not sit inside Euclidean space as a metric submanifold but does sit inside Minkowski space, where the indefinite ambient form restricts to a positive-definite metric on the appropriate hypersurface.

Example 1.3.5: Hyperbolic space and its models

Work in Minkowski space $\mathbb{R}^{1,n}$ of example 1.1.2, with coordinates $x = (x^0, x^1, \dots, x^n)$ and the Lorentzian form

$$m(x, y) = -x^0 y^0 + \sum_{i=1}^n x^i y^i,$$

a pseudo-Riemannian metric of signature $(-, +, \dots, +)$ in the sense of definition 1.1.3. Define the *hyperboloid model*

$$H^n = \{x \in \mathbb{R}^{1,n} \mid m(x, x) = -1, x^0 > 0\},$$

the upper sheet of the two-sheeted hyperboloid $-(x^0)^2 + \sum (x^i)^2 = -1$. This is a smooth hypersurface: -1 is a regular value of $f(x) = m(x, x)$ since $df_x(v) = 2m(x, v)$ is nonzero for $x \neq 0$. Its tangent space at p is $T_p H^n = \{v \in \mathbb{R}^{1,n} \mid m(p, v) = 0\} = p^\perp$, the m -orthogonal complement of p .

The induced metric is Riemannian. Fix $p \in H^n$. Because $m(p, p) = -1 < 0$, the vector p spans a timelike line, and Minkowski space splits m -orthogonally as $\mathbb{R}p \oplus p^\perp$. On the timelike line $\mathbb{R}p$ the form m is negative definite; since m has index 1, its restriction to the complement $p^\perp$ must be positive definite. Thus $g_H := m|_{p^\perp}$ is a positive-definite inner product on each $T_p H^n$, and (H^n, g_H) is a Riemannian manifold. This is the mechanism by which an indefinite ambient form produces a genuine Riemannian metric on a spacelike hypersurface.

Symmetries. Let $O(1, n)$ be the group of linear maps preserving m , and let $O^+(1, n)$ be the subgroup preserving the condition $x^0 > 0$ on the set $\{m(x, x) = -1\}$ (the time-orientation-preserving Lorentz transformations). Each $A \in O^+(1, n)$ maps H^n to itself, and exactly as in the spherical case $(A|_{H^n})^*g_H = g_H$ because $A^*m = m$. Transitivity of $O^+(1, n)$ on H^n and on unit tangent vectors is proved by the same orthonormalization argument as for the sphere, now using

m -orthonormal frames (a timelike unit vector along p together with a spacelike orthonormal basis of $p^\perp$); the analog of extending to matching frames goes through because any two such frames are related by an element of $O^+(1, n)$. Hence H^n is homogeneous and isotropic. As with the sphere, this forces constant sectional curvature, computed in Chapter 4 to be $\text{sec} \equiv -1$, negative.

The ball model. Stereographic projection of the hyperboloid from the point $(-1, 0, \dots, 0)$ onto the hyperplane $\{x^0 = 0\}$ carries H^n diffeomorphically onto the open unit ball $B^n = \{u \in \mathbb{R}^n \mid |u| < 1\}$. Its inverse is parametrized by

$$x^0 = \frac{1 + |u|^2}{1 - |u|^2}, \quad x^i = \frac{2u^i}{1 - |u|^2} \quad (1 \leq i \leq n),$$

which one checks satisfies $m(x, x) = -1$ and $x^0 > 0$. Pulling $g_H = m$ back along this parametrization gives the *Poincaré ball metric*

$$g_B = \frac{4}{(1 - |u|^2)^2} \bar{g}, \quad (1.1)$$

where $\bar{g} = \delta_{ij} du^i du^j$ is the Euclidean metric on the ball. Direct computation confirms (1.1): differentiating the parametrization and forming $m(\partial_a x, \partial_b x)$ yields $\frac{4}{(1 - |u|^2)^2} \delta_{ab}$, with the cross terms vanishing. Thus (B^n, g_B) , a conformal rescaling of the Euclidean metric that blows up at the boundary sphere, is isometric to the hyperboloid model.

The half-space model. The *upper half-space* is $U^n = \{y \in \mathbb{R}^n \mid y^n > 0\}$ with the metric

$$g_U = \frac{1}{(y^n)^2} \bar{g} = \frac{1}{(y^n)^2} ((dy^1)^2 + \dots + (dy^n)^2)$$

For $n = 2$ the map $z \mapsto i \frac{1+z}{1-z}$ carries the unit disc B^2 (coordinate $z = u^1 + iu^2$) biholomorphically onto the upper half-plane $U^2 = \{w \in \mathbb{C} \mid \text{Im} w > 0\}$; a conformal map $w = F(z)$ pulls $g_U = (\text{Im} w)^{-2} |dw|^2$ back to $|F'(z)|^2 (\text{Im} F(z))^{-2} |dz|^2$, and this equals the ball metric $\frac{4}{(1 - |z|^2)^2} |dz|^2$ of (1.1) by direct substitution. So in dimension 2 the ball and half-plane models are isometric; the upper half-plane as a complex-analytic object is treated in [2]. In every dimension the analogous inversion in a boundary sphere provides an isometry $B^n \rightarrow U^n$, so all three models, hyperboloid, ball, and half-space, present the same Riemannian manifold. This is the origin of the phrase *hyperbolic space*: it names one manifold, presented by whichever model is convenient.

Homogeneity is compatible with a small isometry group in one important way: a manifold can be homogeneous under a discrete group of translations rather than a continuous one, and then it can carry curvature-zero geometry without being Euclidean space. The flat torus is the basic example, and it is also the example that separates local isometry from global isometry.

Example 1.3.6: The flat torus

Let $\mathbb{Z}^n \subset \mathbb{R}^n$ be the integer lattice, acting on $\mathbb{R}^n$ by translations $x \mapsto x + k$ for $k \in \mathbb{Z}^n$. The action is free and properly discontinuous, so the quotient $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ is a smooth compact n -manifold and the projection $\pi: \mathbb{R}^n \rightarrow \mathbb{T}^n$ is a smooth covering map. Each translation $\tau_k(x) = x + k$ is an isometry of $(\mathbb{R}^n, \bar{g})$ (it is the rigid motion with $A = \text{id}$ in example 1.3.2), so the Euclidean metric is invariant under the group action and therefore descends to a metric on the quotient. Precisely: near any point of $\mathbb{T}^n$ the covering π restricts to a diffeomorphism from an open set

$\tilde{U} \subset \mathbb{R}^n$, and the local formula $g_{\text{flat}} := (\pi|_{\tilde{U}})_*(\bar{g}|_{\tilde{U}})$ is independent of the chosen sheet $\tilde{U}$ because any two sheets differ by a translation τ_k , which preserves $\bar{g}$. This defines the *flat torus* $(\mathbb{T}^n, g_{\text{flat}})$, and by construction $\pi: (\mathbb{R}^n, \bar{g}) \rightarrow (\mathbb{T}^n, g_{\text{flat}})$ satisfies $\pi^*g_{\text{flat}} = \bar{g}$.

π is a local isometry. The identity $\pi^*g_{\text{flat}} = \bar{g}$ says exactly that each $d\pi_x$ is a linear isometry $(T_x\mathbb{R}^n, \bar{g}_x) \rightarrow (T_{\pi(x)}\mathbb{T}^n, g_{\text{flat}})$, and π is a local diffeomorphism because it is a covering map. So π is a local isometry in the sense of definition 1.3.1, and the flat torus is *locally isometric* to Euclidean space: every point of $\mathbb{T}^n$ has a neighborhood carrying the same metric as an open set of $\mathbb{R}^n$. In particular the flat torus is flat, with the same constant zero curvature as $\mathbb{R}^n$ once curvature is defined in Chapter 4.

$\mathbb{T}^n$ is not globally isometric to $\mathbb{R}^n$. There is no isometry $\mathbb{T}^n \rightarrow \mathbb{R}^n$, and indeed no diffeomorphism, because $\mathbb{T}^n$ is compact while $\mathbb{R}^n$ is not. A local isometry between manifolds of equal dimension therefore need not be a global isometry: π matches the metrics on every small piece yet is far from injective, since $\pi(x) = \pi(x')$ whenever $x - x' \in \mathbb{Z}^n$. This is the promised separation of the two notions from definition 1.3.1.

Homogeneity, and non-uniqueness of the flat structure. The torus is homogeneous: the translations of $\mathbb{R}^n$ descend to a transitive action of $\mathbb{T}^n$ on itself by isometries (the group $\mathbb{T}^n$ acting on the manifold $\mathbb{T}^n$), since τ_b commutes with the lattice action and so passes to the quotient. It is not isotropic, however: the isometries just described are the identity on each tangent space up to lattice symmetries, and the geometry does record a preferred set of directions, those of the lattice. Replacing $\mathbb{Z}^n$ by a different lattice $\Lambda \subset \mathbb{R}^n$ produces another flat torus $\mathbb{R}^n/\Lambda$, and two such tori are isometric iff the lattices are related by an element of $O(n)$; distinct lattice shapes give non-isometric flat tori, all of them flat and all locally isometric to $\mathbb{R}^n$. The flat model space is thus not a single manifold but a family, distinguished by the metric data of the lattice rather than by curvature.

1.4 Musical Isomorphisms, Gradient, Divergence, and Volume

A metric does more than measure lengths and angles of tangent vectors. Because g_p is a nondegenerate bilinear form on each tangent space T_pM , it identifies that space with its dual T_p^*M , and this identification is the mechanism by which the metric reaches into every tensor bundle over M . On a bare smooth manifold there is no canonical way to turn a covector into a vector: the pairing $T_pM \times T_p^*M \rightarrow \mathbb{R}$ is available, but converting a functional ω into a vector requires a choice, and the tangent and cotangent bundles, though isomorphic, are not canonically so. A metric supplies exactly that choice, uniformly and smoothly across the manifold. Once vectors and covectors are interchangeable, quantities defined most directly in one bundle can be transported to the other: the differential df of a function is a covector field, but the metric lets one read it as a vector field, the gradient, whose direction is that of steepest ascent measured by g .

The same nondegeneracy that identifies TM with T^*M also singles out, on an oriented manifold, a preferred top-degree form. In Euclidean space one integrates functions against $dx^1 \wedge \cdots \wedge dx^n$ without comment; on a Riemannian manifold the analogous object is the volume form dV_g , and its coordinate expression $\sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^n$ carries the metric explicitly. With a volume form in hand one can integrate functions, and the failure of a flow to preserve dV_g becomes a scalar, the divergence of the generating vector field. This section assembles these four constructions, all of them consequences

of nondegeneracy, and records the one computation, invariance of dV_g under change of chart, that makes the volume form well defined. Throughout, (M, g) is a (pseudo-)Riemannian manifold of dimension n , the Einstein summation convention is in force (a repeated index appearing once up and once down is summed from 1 to n), and the tensor-bundle and differential-form machinery is taken from [3].

Definition 1.4.1: Musical isomorphisms

Let (M, g) be a (pseudo-)Riemannian manifold. The *flat* operator is the bundle map

$$\flat: TM \rightarrow T^*M, \quad v^\flat := g(v, -) = \langle v, - \rangle,$$

sending a tangent vector $v \in T_pM$ to the covector $w \mapsto \langle v, w \rangle$ on T_pM . Because g_p is nondegenerate, $\flat$ is a fiberwise linear isomorphism; its inverse is the *sharp* operator

$$\sharp: T^*M \rightarrow TM, \quad \omega^\sharp := (\flat)^{-1}(\omega),$$

characterized by $\langle \omega^\sharp, w \rangle = \omega(w)$ for all $w \in T_pM$. In a chart $(U, (x^i))$, writing $g = g_{ij} dx^i dx^j$ and (g^{ij}) for the inverse of the matrix (g_{ij}) , a vector $v = v^i \partial_i$ has

$$v^\flat = v_j dx^j, \quad v_j = g_{ij} v^i,$$

and a covector $\omega = \omega_j dx^j$ has

$$\omega^\sharp = \omega^i \partial_i, \quad \omega^i = g^{ij} \omega_j$$

The passages $v^i \mapsto v_j = g_{ij} v^i$ and $\omega_j \mapsto \omega^i = g^{ij} \omega_j$ are *lowering* and *raising* an index. More generally, for a tensor of type (k, l) the same contractions with g_{ij} or g^{ij} lower or raise any chosen slot, and (g^{ij}) is the *inverse metric*, the induced (pseudo-)Riemannian inner product g^{-1} on T^*M determined by $\langle \alpha, \beta \rangle_{g^{-1}} = g^{ij} \alpha_i \beta_j$.

The names are a mnemonic borrowed from musical notation: lowering an index (flat) pushes a symbol down, raising it (sharp) pushes it up. The content behind the notation is a single fact, nondegeneracy of g_p , which is exactly the statement that the linear map $v \mapsto g_p(v, -)$ from T_pM to T_p^*M has trivial kernel and hence, by equality of dimensions, is an isomorphism. In the Riemannian case positive-definiteness gives nondegeneracy for free; in the pseudo-Riemannian case nondegeneracy is imposed by definition precisely so that the musical isomorphisms exist. The matrix identity behind the coordinate formulas is $g^{ik} g_{kj} = \delta_j^i$, so raising an index one has just lowered returns the original object: $\sharp$ and $\flat$ are mutually inverse, as the notation $\sharp = \flat^{-1}$ demands. That g^{ij} defines an inner product on covectors, and not just a matrix of functions, is the assertion that these operations are geometric rather than coordinate artifacts; the pairing $\langle \alpha, \beta \rangle_{g^{-1}}$ is exactly the one for which $\flat$ and $\sharp$ are linear isometries between (T_pM, g_p) and (T_p^*M, g_p^{-1}) .

Example 1.4.2: Musical isomorphisms in Euclidean and Minkowski space

On Euclidean space $(\mathbb{R}^n, \bar{g})$ with $\bar{g} = \delta_{ij} dx^i dx^j$ (see example 1.1.2), the matrix (g_{ij}) is the identity, so $g_{ij} = g^{ij} = \delta_{ij}$ and raising or lowering an index changes nothing numerically: for $v = v^i \partial_i$ one has $v^\flat = v^i dx^i$ and, for $\omega = \omega_i dx^i$, $\omega^\sharp = \omega_i \partial_i$. This is why in elementary vector calculus the distinction between a vector field and a 1-form is invisible; the Euclidean metric silently identifies them.

On Minkowski space $\mathbb{R}^{1,n}$ with $g = -(dx^0)^2 + \delta_{ab} dx^a dx^b$ (indices a, b running over $1, \dots, n$) the identification is no longer inert. Here $(g_{ij}) = \text{diag}(-1, 1, \dots, 1)$ equals its own inverse, so for $v = v^0 \partial_0 + v^a \partial_a$ one computes

$$v^\flat = -v^0 dx^0 + v^a dx^a,$$

the timelike component acquiring a sign under lowering. Thus a timelike vector and the covector it corresponds to point in opposite temporal directions, a feature with no Euclidean analogue and the source of many sign conventions in relativity.

Among all covector fields on M , one arises on every smooth manifold without reference to a metric: the differential df of a smooth function f , defined by $df(X) = Xf$. It is precisely the object waiting to be sharpened.

Definition 1.4.3: Gradient

Let (M, g) be a (pseudo-)Riemannian manifold and $f \in C^\infty(M)$. The *gradient* of f is the vector field

$$\text{grad } f := (df)^\sharp,$$

the sharp of the differential df . Equivalently, $\text{grad } f$ is the unique vector field characterized by

$$\langle \text{grad } f, X \rangle = df(X) = Xf \quad \text{for every vector field } X$$

In a chart $(U, (x^i))$,

$$\text{grad } f = g^{ij} \partial_i f \partial_j$$

The characterizing identity $\langle \text{grad } f, X \rangle = Xf$ says that pairing the gradient against any direction X recovers the rate of change of f in that direction. The direction of $\text{grad } f$ is therefore the one in which this pairing is largest for unit vectors, that is, the direction of steepest ascent as measured by the metric, and its magnitude $|\text{grad } f|_g$ is that maximal rate. Because the identification uses g , the gradient is a metric-dependent object even though df is not: changing the metric rotates and rescales $\text{grad } f$ while leaving df untouched. The coordinate formula makes this dependence explicit through the factor g^{ij} , and reduces, on Euclidean space where $g^{ij} = \delta^{ij}$, to the familiar $\text{grad } f = \sum_i (\partial_i f) \partial_i$ of vector calculus.

Example 1.4.4: Gradient on the round sphere in stereographic coordinates

Take the round sphere S^2 and a stereographic chart in which the induced metric (whose full derivation belongs to example 1.1.5) has the conformal form $g = \lambda(x, y)^2 (dx^2 + dy^2)$ for a smooth positive function λ ; concretely, stereographic projection of the unit sphere from the north pole gives $\lambda(x, y) = 2/(1 + x^2 + y^2)$. Then $(g_{ij}) = \lambda^2 I$, so $(g^{ij}) = \lambda^{-2} I$, and for $f \in C^\infty(S^2)$

$$\text{grad } f = \lambda^{-2} (\partial_x f \partial_x + \partial_y f \partial_y)$$

The gradient points in the same coordinate direction as the Euclidean gradient of f , but its length is scaled by λ^{-1} : near the north pole, where λ is small, a fixed coordinate slope of f corresponds to a large metric gradient, reflecting that stereographic coordinates compress large regions of the sphere into a small coordinate patch. On the flat model $(\mathbb{R}^2, \bar{g})$, where $\lambda \equiv 1$, the formula collapses to the ordinary gradient.

Turning from tensors to integration, the metric selects a preferred top-degree form. The construction

requires an orientation to produce a genuine n -form; on a nonorientable manifold the same local data assembles instead into a density, against which functions can still be integrated. The distinction is recorded below and the orientable case is treated in full.

Definition 1.4.5: Riemannian volume form

Let (M, g) be an oriented (pseudo-)Riemannian manifold of dimension n . The *Riemannian volume form* dV_g is the unique smooth n -form on M that, in every oriented local chart $(U, (x^i))$ (one for which $dx^1 \wedge \cdots \wedge dx^n$ is positively oriented), is given by

$$dV_g = \sqrt{|\det(g_{ij})|} dx^1 \wedge \cdots \wedge dx^n$$

In the Riemannian case $\det(g_{ij}) > 0$ and the absolute value is unnecessary; in the pseudo-Riemannian case the determinant may be negative and $|\det(g_{ij})|$ is intended. Equivalently, dV_g is the unique positively oriented n -form assigning the value 1 to every positively oriented g -orthonormal frame. If M is not orientable, the same local expression $\sqrt{|\det(g_{ij})|} |dx^1 \wedge \cdots \wedge dx^n|$ defines instead the *Riemannian density*, against which functions are integrated in the sense of [3]; the local computations below apply verbatim.

The characterization by orthonormal frames explains why the awkward-looking factor $\sqrt{|\det(g_{ij})|}$ is the right one. An oriented g -orthonormal coframe $(e^1, \dots, e^n)$ has $e^1 \wedge \cdots \wedge e^n$ deserving the name “unit volume”, so dV_g is decreed to send the dual frame to 1; expressing this dual frame in terms of $\partial_1, \dots, \partial_n$ produces exactly the Jacobian-type factor $\sqrt{|\det(g_{ij})|}$, since the change-of-basis matrix from an orthonormal frame to the coordinate frame has determinant of absolute value $\sqrt{|\det(g_{ij})|}$. The form is what one integrates against: for $f \in C^\infty(M)$ with suitable support, $\int_M f dV_g$ generalizes the Riemann integral, and $\int_M dV_g$ is the total volume of M when M is compact. That the local formula does not depend on the chart, which is what entitles these local expressions to define a single global form, is the content of proposition 1.4.9 below.

Example 1.4.6: Volume forms of Euclidean space and the round sphere

On $(\mathbb{R}^n, \bar{g})$ the matrix (g_{ij}) is the identity, so $\det(g_{ij}) = 1$ and

$$dV_{\bar{g}} = dx^1 \wedge \cdots \wedge dx^n,$$

the standard volume form, against which integration reproduces ordinary Lebesgue integration on $\mathbb{R}^n$ (see [4]).

For the round sphere S^2 in a conformal chart $g = \lambda^2(dx^2 + dy^2)$ as in example 1.4.4, one has $(g_{ij}) = \lambda^2 I$, hence $\det(g_{ij}) = \lambda^4$ and

$$dV_g = \sqrt{\lambda^4} dx \wedge dy = \lambda^2 dx \wedge dy$$

With $\lambda = 2/(1+x^2+y^2)$ this is $dV_g = \frac{4}{(1+x^2+y^2)^2} dx \wedge dy$, and integrating over the whole plane (the image of stereographic projection) recovers the area 4π of the unit sphere, as $\int_{\mathbb{R}^2} \frac{4}{(1+x^2+y^2)^2} dx dy = 4\pi$.

The volume form also converts a first-order differential property of a vector field into a scalar. A vector field X generates a flow, and one asks whether that flow expands or contracts volume. The infinitesimal answer is the Lie derivative of dV_g along X , which, being an n -form on an n -manifold,

is a function multiple of dV_g .

Definition 1.4.7: Divergence

Let (M, g) be an oriented (pseudo-)Riemannian manifold and X a smooth vector field on M . The *divergence* $\operatorname{div} X \in C^\infty(M)$ is the unique smooth function defined by

$$\mathcal{L}_X(dV_g) = (\operatorname{div} X) dV_g,$$

where $\mathcal{L}_X$ denotes the Lie derivative along X (see [3]). Equivalently, by Cartan's formula $\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$ and $d(dV_g) = 0$ (as dV_g is a top form),

$$d(\iota_X dV_g) = (\operatorname{div} X) dV_g$$

On a nonorientable manifold the same identity, read against the Riemannian density, defines $\operatorname{div} X$; the local formula below is unaffected.

Because $\mathcal{L}_X(dV_g)$ is an n -form on an n -dimensional manifold, and dV_g is nowhere zero, the ratio $\mathcal{L}_X(dV_g)/dV_g$ is a well-defined smooth function, and this ratio is by definition $\operatorname{div} X$. The geometric reading is exact rather than metaphorical: if Φ_t is the flow of X , then $\frac{d}{dt}\big|_{t=0} \Phi_t^*(dV_g) = \mathcal{L}_X(dV_g) = (\operatorname{div} X) dV_g$, so $\operatorname{div} X$ measures the instantaneous logarithmic rate at which the flow of X changes volume, positive where the flow spreads volume out and negative where it concentrates volume. The definition uses only the volume form, hence only the metric through dV_g , and makes no reference to a connection; the coordinate expression below and the divergence theorem both follow from this alone. The connection-based formula $\operatorname{div} X = \operatorname{tr}(\nabla X)$ and its agreement with the present definition belong to the development of the Levi-Civita connection and are not needed here.

Example 1.4.8: Divergence in coordinates and on Euclidean space

Let $X = X^i \partial_i$ in an oriented chart, write $|g| := |\det(g_{ij})|$, and abbreviate $dx := dx^1 \wedge \cdots \wedge dx^n$, so that $dV_g = \sqrt{|g|} dx$. Contracting,

$$\iota_X dV_g = \sqrt{|g|} \iota_X dx = \sqrt{|g|} \sum_{i=1}^n (-1)^{i-1} X^i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n,$$

where the hat denotes omission. Applying the exterior derivative, only the ∂_i term of each summand survives the wedge, and with the sign $(-1)^{i-1}$ it lands back on dx :

$$d(\iota_X dV_g) = \sum_{i=1}^n \partial_i (\sqrt{|g|} X^i) dx = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} X^i) dV_g$$

Comparing with $d(\iota_X dV_g) = (\operatorname{div} X) dV_g$ gives the coordinate formula

$$\operatorname{div} X = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} X^i)$$

On $(\mathbb{R}^n, \bar{g})$, where $|g| \equiv 1$, this reduces to $\operatorname{div} X = \partial_i X^i = \sum_i \partial X^i / \partial x^i$, the classical divergence. Combining the gradient formula of definition 1.4.3 with this expression yields the *Laplace–Beltrami operator* $\Delta f := \operatorname{div}(\operatorname{grad} f) = \frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} g^{ij} \partial_j f)$, the metric generalization of the Laplacian.

It remains to justify the definition of dV_g : the local formula must be independent of the oriented chart used to write it, so that the local n -forms agree on overlaps and glue to a single global form. This is the one computation the section owes.

Proposition 1.4.9: Coordinate-independence of the volume form

Let (M, g) be an oriented (pseudo-)Riemannian manifold of dimension n . On the overlap of two oriented charts $(U, (x^i))$ and $(V, (y^\alpha))$, the local n -forms

$$\sqrt{|\det(g_{ij})|} dx^1 \wedge \cdots \wedge dx^n \quad \text{and} \quad \sqrt{|\det(g_{\alpha\beta})|} dy^1 \wedge \cdots \wedge dy^n$$

coincide, where (g_{ij}) and $(g_{\alpha\beta})$ are the components of g in the two charts. Consequently the local expression of definition 1.4.5 defines a single smooth n -form dV_g on M .

Proof:

Let $J = (\partial y^\alpha / \partial x^i)$ denote the Jacobian matrix of the transition map $y = y(x)$ on $U \cap V$, with entries $J^\alpha_i := \partial y^\alpha / \partial x^i$. Since both charts are positively oriented for the given orientation of M , the transition map is orientation-preserving, so

$$\det J > 0$$

Step 1: transformation of the metric components. The metric is a covariant 2-tensor, so its components transform by

$$g_{ij} = \frac{\partial y^\alpha}{\partial x^i} \frac{\partial y^\beta}{\partial x^j} g_{\alpha\beta} = J^\alpha_i J^\beta_j g_{\alpha\beta}$$

In matrix form, writing $G_x := (g_{ij})$ for the component matrix in the x -chart and $G_y := (g_{\alpha\beta})$ for that in the y -chart,

$$G_x = J^\top G_y J$$

Taking determinants and using multiplicativity together with $\det(J^\top) = \det J$,

$$\det(g_{ij}) = (\det J)^2 \det(g_{\alpha\beta})$$

Because $(\det J)^2 > 0$, the sign of the determinant is the same in both charts, consistent with the constancy of the signature of g on the connected overlap, and taking absolute values and square roots,

$$\sqrt{|\det(g_{ij})|} = |\det J| \sqrt{|\det(g_{\alpha\beta})|} = (\det J) \sqrt{|\det(g_{\alpha\beta})|},$$

the last equality using $\det J > 0$.

Step 2: transformation of the coordinate n -form. The top-degree coordinate forms are related by the same Jacobian determinant. From $dy^\alpha = J^\alpha_i dx^i$ and the antisymmetry of the wedge product,

$$dy^1 \wedge \cdots \wedge dy^n = \det\left(\frac{\partial y^\alpha}{\partial x^i}\right) dx^1 \wedge \cdots \wedge dx^n = (\det J) dx^1 \wedge \cdots \wedge dx^n$$

This is the standard identity that the pullback of the volume element under a coordinate change multiplies by the Jacobian determinant; it holds for any smooth change of coordinates.

Step 3: assembling the two transformations. Multiply the two displayed relations. Using Step 2 followed by Step 1,

$$\sqrt{|\det(g_{\alpha\beta})|} dy^1 \wedge \cdots \wedge dy^n = \sqrt{|\det(g_{\alpha\beta})|} (\det J) dx^1 \wedge \cdots \wedge dx^n$$

By Step 1 the coefficient $(\det J) \sqrt{|\det(g_{\alpha\beta})|}$ equals $\sqrt{|\det(g_{ij})|}$, so the right-hand side is

$$\sqrt{|\det(g_{ij})|} dx^1 \wedge \cdots \wedge dx^n$$

Thus the two local n -forms agree on $U \cap V$.

Since the local expressions coincide on every overlap of oriented charts, they patch together to a single well-defined n -form on M ; it is smooth because on each chart it equals $\sqrt{|\det(g_{ij})|} dx^1 \wedge \cdots \wedge dx^n$ with $\sqrt{|\det(g_{ij})|}$ a smooth nowhere-vanishing function (the determinant of the invertible matrix (g_{ij}) is nowhere zero and depends smoothly on the point). This is the asserted dV_g , completing the proof.

The orientation hypothesis entered at exactly one place: the sign $\det J > 0$, which let $|\det J|$ be dropped in favor of $\det J$ so that the two positively signed local forms could be matched. Without an orientation the coordinate n -forms $dx^1 \wedge \cdots \wedge dx^n$ transform with $\det J$, which can be negative, and no consistent choice of sign is available globally; this is precisely why the nonorientable case yields a density rather than a form, the density absorbing the sign into $|dx^1 \wedge \cdots \wedge dx^n|$ and transforming by $|\det J|$ instead. In either case the coefficient $\sqrt{|\det(g_{ij})|}$ is intrinsic, and the object it defines, form or density, is the geometric measure attached to g against which all integration over (M, g) is carried out.

1.5 Induced Metrics and Submanifolds

So far a Riemannian metric has been a structure placed on a manifold by hand: chosen, or shown to exist, or transported by an isometry. But the oldest source of metrics is more direct. A surface sitting in $\mathbb{R}^3$ inherits a way of measuring the length of tangent vectors from the ambient dot product, and this inherited measurement is what one actually computes with when doing geometry on the surface. The length of a curve on the sphere, the angle at which two great circles cross, the area of a spherical triangle: all of these are read off from the restriction of the Euclidean inner product to the tangent planes of the sphere. This is the mechanism by which classical differential geometry, the geometry of curves and surfaces in space, produced its metrics, and it is the mechanism by which most concrete Riemannian manifolds are still built today.

The construction generalizes with no change to any immersion into any (pseudo-)Riemannian manifold. Given a map $\iota: M \rightarrow \tilde{M}$ whose differential is injective at every point, and a metric $\tilde{g}$ on the target, one restricts $\tilde{g}$ to the image of each tangent space $d\iota_p(T_p M)$ and pulls the result back to $T_p M$. The outcome is a covariant 2-tensor on M , the pullback $\iota^* \tilde{g}$, and the point of this section is that when $\tilde{g}$ is Riemannian the injectivity of $d\iota$ is exactly what forces $\iota^* \tilde{g}$ to be positive definite, hence a Riemannian metric in its own right. This turns every immersed submanifold of a Riemannian manifold into a Riemannian manifold. The two running examples, the round sphere and the surfaces of revolution, are computed from this definition here; the sphere computation confirms that the metric obtained by restriction agrees with the round metric of example 1.3.4, and the surfaces of revolution supply

the warped form $dr^2 + f(r)^2 d\theta^2$ that later chapters use as a laboratory for geodesics and curvature. The Einstein summation convention, summation over each index that appears once raised and once lowered, is in force throughout.

Recall from [3] the pullback of a covariant tensor field. If $F: M \rightarrow \tilde{M}$ is smooth and A is a covariant k -tensor field on $\tilde{M}$, the pullback F^*A is the covariant k -tensor field on M defined pointwise by

$$(F^*A)_p(v_1, \dots, v_k) = A_{F(p)}(dF_p(v_1), \dots, dF_p(v_k)), \quad v_1, \dots, v_k \in T_pM$$

For a metric $\tilde{g}$, which is a covariant 2-tensor, this reads $(F^*\tilde{g})_p(v, w) = \tilde{g}_{F(p)}(dF_p v, dF_p w)$. Pullback is smooth (it takes smooth tensor fields to smooth tensor fields), is functorial $((F \circ G)^* = G^* \circ F^*)$, and commutes with the symmetrization that makes a 2-tensor symmetric. These facts are established in [3] and are used freely below.

Definition 1.5.1: Induced metric and Riemannian submanifold

Let $(\tilde{M}, \tilde{g})$ be a (pseudo-)Riemannian manifold and let $\iota: M \rightarrow \tilde{M}$ be an immersion, that is, a smooth map whose differential $d\iota_p: T_pM \rightarrow T_{\iota(p)}\tilde{M}$ is injective for every $p \in M$. The *induced metric* (or *pullback metric*) is the covariant symmetric 2-tensor field

$$\iota^*\tilde{g}$$

on M , given at each point by $(\iota^*\tilde{g})_p(v, w) = \tilde{g}_{\iota(p)}(d\iota_p v, d\iota_p w)$ for $v, w \in T_pM$. When $\iota^*\tilde{g}$ is a Riemannian metric, the pair $(M, \iota^*\tilde{g})$ is a *Riemannian submanifold* of $(\tilde{M}, \tilde{g})$, and ι is an *isometric immersion*. If M is an embedded submanifold of $\tilde{M}$ and ι is the inclusion, $\iota^*\tilde{g}$ is the restriction of $\tilde{g}$ to tangent vectors of M , and one writes $\tilde{g}|_M$.

In the classical setting of a surface in $\mathbb{R}^3$, the induced metric is called the *first fundamental form* of the surface.

The name “first fundamental form” predates the language of tensors, and it fixes the perspective from which the object is read: it is the datum that records the intrinsic geometry of the surface, everything about the surface that a two-dimensional inhabitant could measure without reference to the surrounding space. Two surfaces with the same first fundamental form (in suitable coordinates) are indistinguishable to such an inhabitant, even if they sit in space quite differently; the plane and the cylinder are the standard illustration, and the isometry between them is worked out in example 1.5.4 below. The qualifier “first” anticipates a second fundamental form, which records instead how the surface curves inside the ambient space and is therefore extrinsic; that object belongs to the theory of submanifold curvature and is not needed here.

Two points of the definition deserve a word before any example. First, the immersion hypothesis is on the differential, not on the map ι itself: ι need not be injective, so its image may cross itself, yet the induced metric is still perfectly well defined because it is computed tangent space by tangent space. A figure-eight traced out by an immersion of the circle carries an induced metric on the circle even though the image has a self-crossing. Second, positive-definiteness of the induced metric is not automatic; it is a consequence of the immersion condition together with positive-definiteness of $\tilde{g}$, and this is precisely what the next proposition proves. Dropping the immersion hypothesis breaks it: the constant map has $d\iota_p = 0$, so $\iota^*\tilde{g} = 0$, which is not a metric.

Proposition 1.5.2: Immersions induce metrics

Let $(\tilde{M}, \tilde{g})$ be a Riemannian manifold and $\iota: M \rightarrow \tilde{M}$ an immersion. Then the induced metric $\iota^*\tilde{g}$ is a Riemannian metric on M . More generally, if $\tilde{g}$ is pseudo-Riemannian then $\iota^*\tilde{g}$ is a smooth symmetric 2-tensor field on M , and it is nondegenerate at p if and only if the subspace $d\iota_p(T_p M)$ is nondegenerate for $\tilde{g}$; in the Riemannian case this last condition holds at every point.

Proof:

Step 1: smoothness and symmetry. The pullback of a smooth covariant 2-tensor field under a smooth map is again a smooth covariant 2-tensor field, by [3], so $\iota^*\tilde{g}$ is a smooth 2-tensor field on M . Symmetry is inherited pointwise: for $v, w \in T_p M$,

$$(\iota^*\tilde{g})_p(v, w) = \tilde{g}_{\iota(p)}(d\iota_p v, d\iota_p w) = \tilde{g}_{\iota(p)}(d\iota_p w, d\iota_p v) = (\iota^*\tilde{g})_p(w, v),$$

the middle equality being the symmetry of $\tilde{g}$. Thus $\iota^*\tilde{g}$ is a smooth symmetric covariant 2-tensor field, which is the type required of a (pseudo-)Riemannian metric by definition 1.1.1.

Step 2: the Riemannian case is positive definite. Fix $p \in M$ and a nonzero $v \in T_p M$. Because ι is an immersion, $d\iota_p$ is injective, so $d\iota_p v \neq 0$. Since $\tilde{g}$ is a Riemannian metric, $\tilde{g}_{\iota(p)}$ is positive definite on $T_{\iota(p)}\tilde{M}$, whence

$$(\iota^*\tilde{g})_p(v, v) = \tilde{g}_{\iota(p)}(d\iota_p v, d\iota_p v) > 0$$

For $v = 0$ both sides vanish. Therefore $(\iota^*\tilde{g})_p$ is positive definite on $T_p M$ for every p , and $\iota^*\tilde{g}$ is a Riemannian metric.

Step 3: the pseudo-Riemannian case. Now let $\tilde{g}$ be pseudo-Riemannian. Step 1 shows $\iota^*\tilde{g}$ is a smooth symmetric 2-tensor field. Fix p and set $W = d\iota_p(T_p M)$, a linear subspace of $T_{\iota(p)}\tilde{M}$ of dimension $\dim M$, the equality of dimensions holding because $d\iota_p$ is injective. The map $d\iota_p: T_p M \rightarrow W$ is a linear isomorphism, and for $v, w \in T_p M$

$$(\iota^*\tilde{g})_p(v, w) = \tilde{g}_{\iota(p)}(d\iota_p v, d\iota_p w),$$

so $(\iota^*\tilde{g})_p$ is the pullback under this isomorphism of the bilinear form $\tilde{g}_{\iota(p)}|_{W \times W}$. A pullback of a bilinear form under a linear isomorphism is nondegenerate if and only if the form itself is nondegenerate: if $\varphi: V \rightarrow W$ is a linear isomorphism and b is a bilinear form on W , then $\varphi^*b(v, \cdot)$ is the zero functional on V exactly when $b(\varphi v, \cdot)$ is the zero functional on W , since φ is surjective, so the radical of φ^*b is φ^{-1} of the radical of b . Hence $(\iota^*\tilde{g})_p$ is nondegenerate if and only if $\tilde{g}_{\iota(p)}|_{W \times W}$ is, that is, if and only if W is a nondegenerate subspace of $(T_{\iota(p)}\tilde{M}, \tilde{g}_{\iota(p)})$. In the Riemannian case every subspace is nondegenerate, because the restriction of a positive-definite form to any subspace is again positive definite, hence nondegenerate; this recovers Step 2 and completes the proof.

The pseudo-Riemannian caveat in Step 3 is not a technicality to be skipped, and one concrete case makes the danger visible. In Minkowski space $\mathbb{R}^{1,3}$ with $\tilde{g} = -dt^2 + dx^2 + dy^2 + dz^2$, the plane spanned by ∂_t and ∂_x at a point is nondegenerate and inherits a Lorentzian metric, whereas the line spanned by the null vector $\partial_t + \partial_x$ inherits the zero form, since $\tilde{g}(\partial_t + \partial_x, \partial_t + \partial_x) = -1 + 1 = 0$. An immersion whose image is tangent to such a null direction fails to induce a pseudo-Riemannian

metric there, even though it is a perfectly good immersion. Positive-definiteness of $\tilde{g}$ removes this possibility outright, which is why the Riemannian statement is clean and the pseudo-Riemannian statement carries a hypothesis. For the remainder of the section $\tilde{g}$ is Riemannian.

The computational content of an induced metric lives in coordinates. Suppose $\iota: M \rightarrow \tilde{M}$ is an immersion, $(x^1, \dots, x^n)$ are local coordinates on M , and $(y^1, \dots, y^{\tilde{n}})$ are local coordinates on $\tilde{M}$ near $\iota(p)$, in which $\tilde{g} = \tilde{g}_{ab} dy^a dy^b$. Writing the map in coordinates as $y^a = \iota^a(x^1, \dots, x^n)$, the differential sends $\partial/\partial x^i$ to $(\partial \iota^a / \partial x^i) \partial / \partial y^a$, so the components of the induced metric are

$$(\iota^* \tilde{g})_{ij} = \tilde{g}_{ab} \frac{\partial \iota^a}{\partial x^i} \frac{\partial \iota^b}{\partial x^j} \quad (1.2)$$

When $\tilde{M} = \mathbb{R}^{\tilde{n}}$ with the Euclidean metric, $\tilde{g}_{ab} = \delta_{ab}$, and (1.2) collapses to the dot product of the coordinate partials of ι , viewed as vectors in $\mathbb{R}^{\tilde{n}}$. This is the formula behind every classical first fundamental form, and it is the one applied in the two examples that follow.

Example 1.5.3: The induced metric on the sphere

Let $S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}$ with the inclusion $\iota: S^n \hookrightarrow \mathbb{R}^{n+1}$, and give $\mathbb{R}^{n+1}$ the Euclidean metric $\bar{g} = \delta_{ab} dy^a dy^b$. The inclusion is an embedding, in particular an immersion, so proposition 1.5.2 applies and $\iota^* \bar{g}$ is a Riemannian metric on S^n . The claim is that it equals the round metric described in example 1.3.4.

Work on the upper hemisphere $\{y^{n+1} > 0\}$ with the graph coordinates $(x^1, \dots, x^n)$, $|x| < 1$, under which

$$\iota(x) = (x^1, \dots, x^n, \sqrt{1 - |x|^2}), \quad |x|^2 = (x^1)^2 + \dots + (x^n)^2$$

The other coordinate patches are treated identically. Then $\iota^a = x^a$ for $a \leq n$ and $\iota^{n+1} = \sqrt{1 - |x|^2}$, so

$$\frac{\partial \iota^a}{\partial x^i} = \delta_i^a \quad (a \leq n), \quad \frac{\partial \iota^{n+1}}{\partial x^i} = \frac{-x^i}{\sqrt{1 - |x|^2}}$$

Substituting into (1.2) with $\tilde{g}_{ab} = \delta_{ab}$ gives

$$(\iota^* \bar{g})_{ij} = \sum_{a=1}^n \delta_i^a \delta_j^a + \frac{\partial \iota^{n+1}}{\partial x^i} \frac{\partial \iota^{n+1}}{\partial x^j} = \delta_{ij} + \frac{x^i x^j}{1 - |x|^2},$$

so the induced metric in these coordinates is

$$\iota^* \bar{g} = \delta_{ij} dx^i dx^j + \frac{(x^i dx^i)(x^j dx^j)}{1 - |x|^2} = |dx|^2 + \frac{(x \cdot dx)^2}{1 - |x|^2}, \quad (1.3)$$

where $|dx|^2 = \sum_i (dx^i)^2$ and $x \cdot dx = \sum_i x^i dx^i$.

To see that (1.3) is the round metric of example 1.3.4, evaluate the length it assigns to a curve and match it against the length measured in the ambient space. Let $c(t)$ be a curve on S^n and write it in the ambient coordinates as $\gamma(t) = \iota(c(t)) \in \mathbb{R}^{n+1}$. By the definition of the pullback, the $\iota^* \bar{g}$ -length of the velocity of c equals the Euclidean length of the velocity of γ :

$$|\dot{c}(t)|_{\iota^* \bar{g}}^2 = (\iota^* \bar{g})_{c(t)}(\dot{c}(t), \dot{c}(t)) = \bar{g}_{\gamma(t)}(d\iota \dot{c}(t), d\iota \dot{c}(t)) = |\dot{\gamma}(t)|^2$$

Thus the intrinsic length of any curve on S^n , measured with $\iota^* \bar{g}$, coincides with its length as a curve in $\mathbb{R}^{n+1}$, which is exactly the prescription defining the round metric as the restriction

of $\bar{g}$ to tangent vectors of the sphere in example 1.3.4. A symmetric 2-tensor is determined by the quadratic form $v \mapsto g(v, v)$ it induces on tangent vectors, by polarization, and the two tensors induce the same quadratic form on every $T_c S^n$; hence $\iota^* \bar{g}$ equals the round metric, and the inclusion of the round sphere into Euclidean space is an isometric immersion.

The sphere computation is the prototype: a submanifold is described by a parametrization or a graph, the ambient inner product is restricted, and the resulting form is the metric one uses for all intrinsic measurements. The identity $|\dot{c}|_{\iota^* \bar{g}} = |\dot{\gamma}|$ derived above holds for every isometric immersion, not only for the sphere, and it is the reason the induced metric is the correct object: it is the unique metric on M for which ι preserves the lengths of tangent vectors, so that arc length computed upstairs on M agrees with arc length of the image curve downstairs. In the language of definition 1.3.1, an isometric immersion is a map that is a local isometry onto its image equipped with the induced metric, and the sphere inclusion realizes the abstract round metric of example 1.3.4 concretely as such an image.

The second running example produces the warped normal form that later chapters return to, and it introduces the class of surfaces of revolution.

Example 1.5.4: Surfaces of revolution

Let a *profile curve* in the xz -plane be given by $r \mapsto (\varphi(r), 0, \psi(r))$ for r in an interval I , with $\varphi(r) > 0$ and $(\varphi'(r), \psi'(r)) \neq (0, 0)$ for all r ; the condition $\varphi > 0$ keeps the curve off the axis of rotation. The associated *surface of revolution* $\Sigma \subset \mathbb{R}^3$ is swept out by rotating this curve about the z -axis, and is parametrized by

$$\iota(r, \theta) = (\varphi(r) \cos \theta, \varphi(r) \sin \theta, \psi(r)), \quad r \in I, \theta \in \mathbb{R},$$

where θ is taken modulo 2π . Here r measures position along the profile and θ measures the angle of rotation.

This parametrization is an immersion. Its coordinate partials are

$$\frac{\partial \iota}{\partial r} = (\varphi' \cos \theta, \varphi' \sin \theta, \psi'), \quad \frac{\partial \iota}{\partial \theta} = (-\varphi \sin \theta, \varphi \cos \theta, 0),$$

which are linearly independent at every point. The second is nonzero because $\varphi > 0$. The first is nonzero because $(\varphi', \psi') \neq (0, 0)$, and it is not a scalar multiple of the second: the second vector is purely azimuthal, orthogonal to the radial direction $(\cos \theta, \sin \theta, 0)$ and to the z -axis, whereas the first has radial component φ' and z -component ψ' , at least one of which is nonzero. Hence $d\iota$ has rank 2 everywhere and proposition 1.5.2 applies.

Compute the induced metric from (1.2) with $\tilde{g}_{ab} = \delta_{ab}$, so that each component is the Euclidean dot product of two coordinate partials. Direct calculation gives

$$\begin{aligned} (\iota^* \bar{g})_{rr} &= \frac{\partial \iota}{\partial r} \cdot \frac{\partial \iota}{\partial r} = \varphi'^2 \cos^2 \theta + \varphi'^2 \sin^2 \theta + \psi'^2 = \varphi'^2 + \psi'^2, \\ (\iota^* \bar{g})_{r\theta} &= \frac{\partial \iota}{\partial r} \cdot \frac{\partial \iota}{\partial \theta} = -\varphi' \varphi \cos \theta \sin \theta + \varphi' \varphi \sin \theta \cos \theta + 0 = 0, \\ (\iota^* \bar{g})_{\theta\theta} &= \frac{\partial \iota}{\partial \theta} \cdot \frac{\partial \iota}{\partial \theta} = \varphi^2 \sin^2 \theta + \varphi^2 \cos^2 \theta = \varphi^2 \end{aligned}$$

Therefore

$$\iota^* \bar{g} = (\varphi'^2 + \psi'^2) dr^2 + \varphi^2 d\theta^2 \tag{1.4}$$

The vanishing of the cross term $(\iota^* \bar{g})_{r\theta}$ records that the meridians (θ constant) meet the parallels (r constant) orthogonally, a reflection of the rotational symmetry of the surface.

Two normalizations simplify (1.4). If the profile curve is parametrized by arc length, so that $\varphi'^2 + \psi'^2 \equiv 1$, the metric takes the *warped form*

$$\iota^* \bar{g} = dr^2 + f(r)^2 d\theta^2, \quad f(r) := \varphi(r),$$

in which the coefficient of dr^2 is constant and all the geometry is carried by the single warping function $f = \varphi$, the distance from a point of the surface to the axis. This is the form that reappears in the study of geodesics and curvature on rotationally symmetric surfaces.

The warped form specializes to two familiar surfaces at once. Taking $\varphi(r) = 1$ and $\psi(r) = r$ gives the right circular cylinder of radius 1, with induced metric $dr^2 + d\theta^2$; this is the flat Euclidean metric on the strip $\mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z})$, so the cylinder is locally isometric to the plane, illustrating the remark after definition 1.5.1 that the first fundamental form is an intrinsic datum blind to how the surface bends in space. Taking instead $\varphi(r) = \sin r$ and $\psi(r) = \cos r$ for $r \in (0, \pi)$, so that the profile is a unit semicircle parametrized by arc length, gives the unit sphere minus its poles, with induced metric

$$\iota^* \bar{g} = dr^2 + \sin^2 r d\theta^2,$$

the round metric in geodesic polar form, where r is the angle from the north pole. This is (1.3) rewritten in latitude and longitude, and it agrees with the round metric of example 1.3.4.

1.6 Warped Products

The metrics built so far have been either primitive, chosen or shown to exist on a single manifold, or inherited, pulled back along an immersion. Warped products are the first constructive recipe: they take two Riemannian manifolds and a positive function on the first, and assemble from these data a metric on the product of the two. The construction is simple to state and yet it accounts for a large fraction of the concrete metrics in the subject. Euclidean space written in polar coordinates, the round sphere in latitude and longitude, hyperbolic space in its several models, every surface of revolution, and the spatial slices of the standard cosmological spacetimes are all warped products, and recognizing a metric as a warped product is often the first step toward computing anything about it.

The idea begins with the ordinary product. If (M, g_M) and (N, g_N) are Riemannian manifolds, their product $M \times N$ carries a metric that measures the base direction with g_M and the fiber direction with g_N , with the two directions declared orthogonal. This product metric is the geometry of a manifold that is, at every point, a rigid orthogonal join of its two factors. A warped product loosens this join in one controlled way: the fiber metric g_N is rescaled by the square of a positive function f on the base, so that the size of the fiber over a point of M depends on where that point sits. Nothing else changes, the base metric and the orthogonality are kept, but allowing the fiber to shrink and grow is already enough to bend a flat product into a sphere or a hyperbolic space. The present section constructs both metrics at the level of the metric tensor and verifies that the warped-product tensor is a Riemannian metric; the Levi-Civita connection and the curvature of a warped product, which are where the bending becomes quantitative, are developed once the connection (Chapter 2) and the curvature tensor (Chapter 4) are available.

Throughout, $\pi_M: M \times N \rightarrow M$ and $\pi_N: M \times N \rightarrow N$ denote the two projections of the product manifold, both smooth submersions. The Einstein summation convention, summation over each index appearing once raised and once lowered, is in force.

Definition 1.6.1: Product metric

Let (M, g_M) and (N, g_N) be (pseudo-)Riemannian manifolds with projections $\pi_M: M \times N \rightarrow M$ and $\pi_N: M \times N \rightarrow N$. The *product metric* on $M \times N$ is the covariant 2-tensor field

$$g = \pi_M^* g_M + \pi_N^* g_N$$

Explicitly, at a point $(p, q) \in M \times N$ and for tangent vectors $u, v \in T_{(p,q)}(M \times N)$,

$$g_{(p,q)}(u, v) = (g_M)_p(d\pi_M u, d\pi_M v) + (g_N)_q(d\pi_N u, d\pi_N v)$$

The (pseudo-)Riemannian manifold $(M \times N, g)$ is the *Riemannian product* of (M, g_M) and (N, g_N) , written $M \times N$ or $(M \times N, g_M \oplus g_N)$.

The canonical isomorphism $T_{(p,q)}(M \times N) \cong T_p M \oplus T_q N$ turns the two projection differentials into the two coordinate projections of this direct sum, and under this identification the product metric is the orthogonal direct sum of the two inner products. A vector tangent to the base slice $M \times \{q\}$ has $d\pi_N u = 0$ and is measured purely by g_M ; a vector tangent to the fiber $\{p\} \times N$ has $d\pi_M u = 0$ and is measured purely by g_N ; and the mixed term $g(u, v)$ between a base vector and a fiber vector vanishes, because one of the two differentials annihilates each of them. Thus the tangent space splits g -orthogonally into a horizontal part tangent to M and a vertical part tangent to N , and the product metric is exactly the metric for which this splitting is orthogonal and each summand keeps its own geometry. That $\pi_M^* g_M + \pi_N^* g_N$ is a Riemannian metric when both factors are is the special case $f \equiv 1$ of proposition 1.6.3 below, so its verification is folded into that proof.

The product metric records a manifold whose two factors do not interact: the geometry along M is the same over every point of N and conversely. A cylinder $S^1 \times \mathbb{R}$ with the product of the standard circle and line metrics is the picture to keep in mind, a fiber circle of the same radius sitting over every point of the line. The warped product breaks this uniformity in precisely one place, by letting the radius of the fiber vary over the base.

Definition 1.6.2: Warped product

Let (M, g_M) and (N, g_N) be (pseudo-)Riemannian manifolds and let $f: M \rightarrow (0, \infty)$ be a smooth positive function, the *warping function*. The *warped product* $M \times_f N$ is the product manifold $M \times N$ equipped with the covariant 2-tensor field

$$g = \pi_M^* g_M + (f \circ \pi_M)^2 \pi_N^* g_N$$

Explicitly, at $(p, q) \in M \times N$ and for $u, v \in T_{(p,q)}(M \times N)$,

$$g_{(p,q)}(u, v) = (g_M)_p(d\pi_M u, d\pi_M v) + f(p)^2 (g_N)_q(d\pi_N u, d\pi_N v)$$

The manifold (M, g_M) is the *base*, (N, g_N) the *fiber*, and g is written $g_M \oplus f^2 g_N$. The product metric of definition 1.6.1 is the case $f \equiv 1$.

Under the splitting $T_{(p,q)}(M \times N) \cong T_p M \oplus T_q N$ the warped metric is $(g_M)_p \oplus f(p)^2 (g_N)_q$: the base part is untouched, the base and fiber remain orthogonal, and the fiber inner product is scaled by the positive factor $f(p)^2$. Scaling an inner product by a positive constant scales every length it measures by the square root of that constant, so over the point $p \in M$ the fiber $\{p\} \times N$ carries

the metric $f(p)^2 g_N$, a copy of (N, g_N) magnified by the factor $f(p)$. The warping function therefore controls the size of the fiber as it slides over the base, and the whole content of the construction is that this one positive function is allowed to vary. Where f is small the fiber is a small copy of N ; where f is large the fiber is a large copy; and where f is constant the warped product is locally a Riemannian product. The square on f is a convention that makes the length scale, rather than the squared length, equal to f : a fiber curve of g_N -length ℓ over p has warped length $f(p)\ell$.

The metric g is manifestly a smooth symmetric covariant 2-tensor field, being a sum of pullbacks of such fields scaled by a smooth function; the point requiring proof is positive-definiteness, and this is where the two hypotheses, positivity of f and positive-definiteness of g_M and g_N , enter.

Proposition 1.6.3: The warped product is a Riemannian metric

Let (M, g_M) and (N, g_N) be Riemannian manifolds and $f: M \rightarrow (0, \infty)$ smooth. Then the warped-product tensor $g = \pi_M^* g_M + (f \circ \pi_M)^2 \pi_N^* g_N$ of definition 1.6.2 is a Riemannian metric on $M \times N$. In particular, taking $f \equiv 1$, the product metric of definition 1.6.1 is a Riemannian metric. If instead g_M and g_N are pseudo-Riemannian of indices μ and ν , then g is pseudo-Riemannian of index $\mu + \nu$; positivity of f is all that is used of the warping function in either case.

Proof:

Step 1: smoothness and symmetry. The projections π_M and π_N are smooth, and the pullback of a smooth symmetric covariant 2-tensor field under a smooth map is again a smooth symmetric covariant 2-tensor field, by [3]. Hence $\pi_M^* g_M$ and $\pi_N^* g_N$ are smooth symmetric covariant 2-tensor fields on $M \times N$. The function $f \circ \pi_M$ is smooth and strictly positive, so $(f \circ \pi_M)^2$ is a smooth positive function, and scaling a smooth symmetric 2-tensor field by a smooth function preserves smoothness and symmetry. Therefore $g = \pi_M^* g_M + (f \circ \pi_M)^2 \pi_N^* g_N$ is a smooth symmetric covariant 2-tensor field on $M \times N$, which is the type required of a (pseudo-)Riemannian metric by definition 1.1.1.

Step 2: the orthogonal splitting of the tangent space. Fix a point $(p, q) \in M \times N$. The differentials of the projections furnish the canonical linear isomorphism

$$\Phi: T_{(p,q)}(M \times N) \longrightarrow T_p M \oplus T_q N, \quad \Phi(u) = (d\pi_M u, d\pi_N u)$$

Its inverse identifies $T_p M$ with the subspace tangent to the base slice $M \times \{q\}$ and $T_q N$ with the subspace tangent to the fiber $\{p\} \times N$. Write a tangent vector as $u = (a, b)$ under Φ , with $a = d\pi_M u \in T_p M$ the *horizontal part* and $b = d\pi_N u \in T_q N$ the *vertical part*. In these terms the warped metric reads

$$g_{(p,q)}((a, b), (a', b')) = (g_M)_p(a, a') + f(p)^2 (g_N)_q(b, b') \quad (1.5)$$

directly from the definition, since $d\pi_M u = a$ and $d\pi_N u = b$ and likewise for $u' = (a', b')$. Taking one vector horizontal, $b = 0$, and the other vertical, $a' = 0$, makes both terms of (1.5) vanish, so the horizontal subspace $T_p M \oplus 0$ and the vertical subspace $0 \oplus T_q N$ are g -orthogonal.

Step 3: positive-definiteness in the Riemannian case. Let $u = (a, b)$ be a nonzero tangent vector at (p, q) . Then

$$g_{(p,q)}(u, u) = (g_M)_p(a, a) + f(p)^2 (g_N)_q(b, b)$$

by (1.5). Both summands are nonnegative: $(g_M)_p$ and $(g_N)_q$ are positive definite because g_M and g_N are Riemannian, and the coefficient $f(p)^2$ is strictly positive because f takes values

in $(0, \infty)$. Since $u \neq 0$, at least one of a, b is nonzero. If $a \neq 0$ then $(g_M)_p(a, a) > 0$; if $b \neq 0$ then $f(p)^2 > 0$ and $(g_N)_q(b, b) > 0$, so $f(p)^2(g_N)_q(b, b) > 0$. In either case the sum of the two nonnegative terms is strictly positive, so $g_{(p,q)}(u, u) > 0$. Thus $g_{(p,q)}$ is positive definite at every point, and g is a Riemannian metric. Setting $f \equiv 1$ throughout gives the same conclusion for the product metric.

Step 4: the pseudo-Riemannian case. Now suppose g_M and g_N are pseudo-Riemannian of indices μ and ν . Step 1 already shows g is a smooth symmetric 2-tensor field, and Step 2 gives the g -orthogonal splitting $T_{(p,q)}(M \times N) = (T_p M \oplus 0) \oplus_\perp (0 \oplus T_q N)$ with g restricting on the first summand to $(g_M)_p$ and on the second to $f(p)^2(g_N)_q$, by (1.5). Because $f(p)^2 > 0$, the form $f(p)^2(g_N)_q$ has the same signature as $(g_N)_q$: scaling a symmetric bilinear form by a positive constant preserves the sign of $b \mapsto (g_N)_q(b, b)$ on every vector, hence preserves the number of positive, negative, and zero eigenvalues. The index (number of negative eigenvalues) of a g -orthogonal direct sum of two nondegenerate forms is the sum of their indices, and the form is nondegenerate exactly when each summand is. Here $(g_M)_p$ has index μ and $f(p)^2(g_N)_q$ has index ν , both nondegenerate, so $g_{(p,q)}$ is nondegenerate of index $\mu + \nu$. As $\mu + \nu$ is independent of the point, g is a pseudo-Riemannian metric of constant index $\mu + \nu$ in the sense of definition 1.1.3, completing the proof.

The verification isolates exactly why the two hypotheses cannot be dropped. Positive-definiteness of the fiber metric g_N is needed so that the vertical term $f(p)^2(g_N)_q(b, b)$ is nonnegative; positivity of f is needed so that this term does not collapse. Were f allowed to vanish at a point p_0 , the metric would degenerate along the entire fiber $\{p_0\} \times N$, where every vertical vector would have zero length, and g would fail to be a metric there. This is not an abstract worry: the polar and spherical presentations below have warping functions ($f(r) = r$, $f(r) = \sin r$) that do vanish at the endpoints of their domains, and it is precisely at those parameter values that the warped product breaks down as a manifold and a single point (the origin, or a pole) must be glued in by hand. The warped-product formalism describes the smooth geometry away from such collapse loci; the points where $f = 0$ are the seams where a warped product is capped off.

Two structural features of a warped product are worth naming before the examples, both concerning how the two factors sit inside $M \times_f N$. First, the base copy $M \times \{q\}$, for any fixed $q \in N$, is a Riemannian submanifold isometric to (M, g_M) : the inclusion $p \mapsto (p, q)$ has image the base slice, and the induced metric of definition 1.5.1 is (g_M) unchanged, since the warping factor multiplies only the vertical term, which is absent for vectors tangent to the base. The base is embedded without distortion. Second, the fiber copy $\{p\} \times N$, for fixed $p \in M$, is a Riemannian submanifold isometric to $(N, f(p)^2 g_N)$, that is, to (N, g_N) rescaled by the constant $f(p)$; different fibers are rescaled copies of the same manifold, with scale governed by f . These two observations already fix the qualitative shape of a warped product, and the examples that follow make them quantitative.

The asymmetry between base and fiber, that only g_N is warped, and only by a function on M , is the source of the construction's utility. It is also what makes the Levi-Civita connection and curvature of a warped product computable in closed form in terms of the two factor geometries and the derivatives of f : the mixed connection terms involve $\text{grad } f$ on the base, and the curvature of $M \times_f N$ splits into a base contribution, a fiber contribution, and a mixed contribution controlled by the Hessian of f . Those formulas, due in this generality to O'Neill, require the connection of Chapter 2 and the curvature tensor of Chapter 4 and are taken up there; the metric-level construction is complete as it stands, and the examples below are the payoff.

The first example is the one that motivates the whole formalism: Euclidean space, away from the origin, is a warped product over a half-line with spherical fibers, and the warping function is simply

the radial distance.

Example 1.6.4: Euclidean space as a warped product

Let $(S^{n-1}, g_{S^{n-1}})$ be the unit sphere with its round metric, the case $r = 1$ of example 1.3.4, and take the base to be the half-line $(0, \infty)$ with the standard metric dr^2 . Form the warped product

$$(0, \infty) \times_r S^{n-1}, \quad g = dr^2 + r^2 g_{S^{n-1}},$$

with warping function $f(r) = r$; here r denotes both the coordinate on the base and, as a function on the product, the warping function $f \circ \pi_M$. The claim is that this warped product is isometric to Euclidean space $\mathbb{R}^n$ with the origin removed, carrying the flat metric $\bar{g}$ of example 1.1.2.

Consider the map

$$\Psi: (0, \infty) \times S^{n-1} \longrightarrow \mathbb{R}^n \setminus \{0\}, \quad \Psi(r, \omega) = r\omega,$$

which sends a radius and a unit direction to the point at that distance in that direction. This is a diffeomorphism, with inverse $x \mapsto (|x|, x/|x|)$, the passage to polar coordinates. To compute the pullback $\Psi^*\bar{g}$, write a tangent vector to the product at (r, ω) as (a, b) with $a \in T_r(0, \infty) = \mathbb{R}$ the radial part and $b \in T_\omega S^{n-1}$ the spherical part, and note $\langle \omega, \omega \rangle = 1$ and $\langle \omega, b \rangle = 0$, the latter because b is tangent to the unit sphere at ω , hence orthogonal to the position vector ω . The differential of Ψ carries (a, b) to

$$d\Psi_{(r, \omega)}(a, b) = a\omega + r b \in \mathbb{R}^n,$$

the first term from varying the radius, the second from varying the direction. Since $\bar{g}$ is the Euclidean dot product on $\mathbb{R}^n$ under the identification $T_x \mathbb{R}^n \cong \mathbb{R}^n$,

$$(\Psi^*\bar{g})((a, b), (a, b)) = \langle a\omega + r b, a\omega + r b \rangle = a^2 \langle \omega, \omega \rangle + 2ar \langle \omega, b \rangle + r^2 \langle b, b \rangle = a^2 + r^2 |b|^2,$$

using $\langle \omega, \omega \rangle = 1$ and $\langle \omega, b \rangle = 0$. The cross term drops out precisely because the spherical directions are orthogonal to the radial direction. Now $|b|^2 = \langle b, b \rangle$ is the Euclidean square-length of b as a vector tangent to the unit sphere in $\mathbb{R}^n$, which by example 1.3.4 equals $g_{S^{n-1}}(b, b)$, the round metric being the restriction of the ambient dot product. Hence

$$(\Psi^*\bar{g})((a, b), (a, b)) = a^2 + r^2 g_{S^{n-1}}(b, b) = (dr^2 + r^2 g_{S^{n-1}})((a, b), (a, b)),$$

and by polarization $\Psi^*\bar{g} = dr^2 + r^2 g_{S^{n-1}}$. Thus Ψ is an isometry from the warped product $(0, \infty) \times_r S^{n-1}$ onto $\mathbb{R}^n \setminus \{0\}$, so the flat metric in polar form is

$$\bar{g} = dr^2 + r^2 g_{S^{n-1}}$$

This is the source of the warping function $f(r) = r$: the fiber over radius r is a sphere of that radius, a copy of the unit sphere magnified by r , exactly the geometry of the sphere $\{|x| = r\}$ inside $\mathbb{R}^n$. The warping degenerates as $r \rightarrow 0$, where $f(r) = r \rightarrow 0$ and the spheres collapse to the single missing point, the origin, which the warped product does not see. For $n = 2$ this reads $\bar{g} = dr^2 + r^2 d\theta^2$, the elementary polar-coordinate expression for the plane, with $g_{S^1} = d\theta^2$.

The polar decomposition of $\mathbb{R}^n$ is the template, and the same shape, a half-line base with spherical fibers, produces the round sphere and hyperbolic space by changing only the warping function. Replacing r by $\sin r$ gives the sphere, as already seen in the geodesic polar form of example 1.5.4; replacing it by $\sinh r$ gives hyperbolic space, which the next example verifies.

Example 1.6.5: Hyperbolic space as a warped product

Hyperbolic space H^n of example 1.3.5 admits two natural warped-product presentations, one over a half-line with spherical fibers and one over a line with Euclidean fibers. The second is the quicker to verify from the models already in hand, so it is treated in full; the first is then stated with its warping function.

Half-space form: a line warped over Euclidean fibers. Recall the upper half-space model $U^n = \{y \in \mathbb{R}^n \mid y^n > 0\}$ from example 1.3.5, with metric

$$g_U = \frac{1}{(y^n)^2} ((dy^1)^2 + \cdots + (dy^{n-1})^2 + (dy^n)^2)$$

Take the base to be $\mathbb{R}$ with coordinate t and metric dt^2 , the fiber to be $\mathbb{R}^{n-1}$ with the Euclidean metric $\bar{g}_{n-1} = (dz^1)^2 + \cdots + (dz^{n-1})^2$, and the warping function $f(t) = e^t$. The resulting warped product is

$$\mathbb{R} \times_{e^t} \mathbb{R}^{n-1}, \quad g = dt^2 + e^{2t} \bar{g}_{n-1}$$

Introduce the change of coordinates

$$\Xi: \mathbb{R} \times \mathbb{R}^{n-1} \longrightarrow U^n, \quad \Xi(t, z) = (z^1, \dots, z^{n-1}, e^{-t}),$$

a diffeomorphism onto U^n with inverse $y \mapsto (-\log y^n, y^1, \dots, y^{n-1})$. Along Ξ the horizontal coordinate is $y^i = z^i$ for $i < n$ and the vertical coordinate is $y^n = e^{-t}$, so $dy^i = dz^i$ for $i < n$ and $dy^n = -e^{-t} dt$. Substituting into g_U ,

$$\Xi^* g_U = \frac{1}{(e^{-t})^2} \left(\sum_{i=1}^{n-1} (dz^i)^2 + e^{-2t} dt^2 \right) = e^{2t} \sum_{i=1}^{n-1} (dz^i)^2 + e^{2t} e^{-2t} dt^2 = dt^2 + e^{2t} \bar{g}_{n-1}$$

Hence $\Xi^* g_U = g$, and the warped product $\mathbb{R} \times_{e^t} \mathbb{R}^{n-1}$ is isometric to the upper half-space model, so it presents hyperbolic space. In this form the fibers are flat Euclidean spaces $\mathbb{R}^{n-1}$ whose scale grows exponentially in the base coordinate t , matching the intuition, made precise once the curvature is available, that hyperbolic space spreads apart exponentially: the horospheres $t = \text{const}$ are flat, and moving in the t direction rescales them by e^t .

Geodesic polar form: a half-line warped over spherical fibers. The presentation parallel to the Euclidean polar form of example 1.6.4 is

$$(0, \infty) \times_{\sinh r} S^{n-1}, \quad g = dr^2 + \sinh^2 r g_{S^{n-1}},$$

with warping function $f(r) = \sinh r$, where r is hyperbolic distance from a fixed center point. The metric $dr^2 + \sinh^2 r g_{S^{n-1}}$ is the geodesic polar form of H^n about that center: the fiber over radius r is a sphere whose size is governed by $\sinh r$ in place of the Euclidean r , and since $\sinh r > r$ for $r > 0$ the hyperbolic spheres are larger than Euclidean spheres of the same radius, the metric statement of negative curvature. The three warping functions $f(r) = r$ (Euclidean, example 1.6.4), $f(r) = \sin r$ (spherical, example 1.5.4), and $f(r) = \sinh r$ (hyperbolic) are exactly the solutions of $f'' + \kappa f = 0$ with $f(0) = 0$, $f'(0) = 1$, for $\kappa = 0$, $\kappa = 1$, and $\kappa = -1$ respectively; this differential equation is the one that later emerges, from the curvature computation of Chapter 4, as the condition for the warped product to have constant sectional curvature κ . Its identification is deferred to that chapter, but the three model spaces already exhibit the three cases side by side. As $r \rightarrow 0$ the warping $\sinh r \rightarrow 0$, and as in the Euclidean case the collapsing fibers are capped by the single center point.

The three constant-curvature model spaces are thus all warped products of the same base-fiber shape, distinguished by their warping functions. The final example turns from these to the class that gave the construction its geometric name, and reconciles the notation of this section with the surface computation of Chapter 1.

Example 1.6.6: Surfaces of revolution as warped products

The surfaces of revolution of example 1.5.4, with the profile curve parametrized by arc length, carry the induced metric

$$dr^2 + f(r)^2 d\theta^2, \quad f(r) = \varphi(r) > 0,$$

computed there, where r runs over an interval I , θ over $\mathbb{R}/2\pi\mathbb{Z}$, and $f = \varphi$ is the distance from a point of the surface to the axis of rotation. This is exactly a warped product. Take the base to be the interval (I, dr^2) , the fiber to be the circle $(S^1, d\theta^2) = (\mathbb{R}/2\pi\mathbb{Z}, d\theta^2)$, and the warping function to be f ; then

$$I \times_f S^1, \quad g = dr^2 + f(r)^2 d\theta^2,$$

and the induced metric of example 1.5.4 is by inspection this warped-product metric. The identification is immediate because the cross term $(\iota^* \bar{g})_{r\theta}$ was shown to vanish in example 1.5.4, which is the orthogonality of base and fiber demanded by definition 1.6.2, and the two diagonal terms are dr^2 and $f(r)^2 d\theta^2$, the base and warped-fiber terms.

The symbol f carries the same meaning in both places, the warping function, so no reconciliation of notation is needed beyond noting that the fiber circle here is the unit circle $S^1 = (\mathbb{R}/2\pi\mathbb{Z}, d\theta^2)$, whose round metric is $d\theta^2$, the case $n = 1$ of the round sphere $g_{S^1} = d\theta^2$ used in example 1.6.4. Concretely, the fiber over the parameter value r is the circle of radius $f(r)$ traced out by rotating the corresponding point of the profile curve about the axis, a copy of the unit circle magnified by the distance $f(r)$ to the axis, precisely the geometry of a parallel of the surface. This is the origin of the term *warped product*: the fiber, here a circle, is warped, that is rescaled, by a function of position along the base.

The two specializations recorded in example 1.5.4 now read off as warped products of familiar type. The cylinder, $f(r) = 1$, is the unwarped Riemannian product $I \times S^1$ with metric $dr^2 + d\theta^2$, flat because it is a product of flat factors with constant warping. The unit sphere minus its poles, $f(r) = \sin r$ for $r \in (0, \pi)$, is the warped product $(0, \pi) \times_{\sin r} S^1$ with metric $dr^2 + \sin^2 r d\theta^2$, the two-dimensional case of the spherical polar form and the counterpart to the Euclidean $dr^2 + r^2 d\theta^2$ and hyperbolic $dr^2 + \sinh^2 r d\theta^2$ forms of the preceding examples. The warping function $\sin r$ vanishes at $r = 0$ and $r = \pi$, where the parallels shrink to the north and south poles, the two points at which the warped product is capped off to recover the full sphere.

The Levi-Civita Connection

The metric of chapter 1 measures vectors at a single point, but it does not by itself say how to compare vectors at different points, nor how to differentiate a vector field along a curve. On a general manifold there is no canonical way to do either: the partial derivatives of the components of a vector field do not transform tensorially, so differentiation of tangent vectors requires an extra structure, a *connection*. This chapter shows that a metric selects one connection above all others.

The development proceeds from the general notion of an affine connection and its Christoffel symbols to the two properties that single out the geometrically correct one: freedom from torsion, which makes second covariant derivatives symmetric, and compatibility with the metric, which makes parallel transport preserve lengths and angles. The fundamental theorem of Riemannian geometry asserts that these two conditions are met by exactly one connection, the Levi-Civita connection, and the Koszul formula produces it explicitly. The chapter ends with parallel transport, the concrete transport of tangent vectors along curves that the connection defines, and the induced differentiation of tensor fields of every type. With the Levi-Civita connection in hand, the notion of a straight line on a manifold becomes available, and that is the subject of the next chapter.

2.1 Affine Connections and Christoffel Symbols

On a general smooth manifold there is no canonical way to compare tangent vectors at different points. A tangent vector at p lives in T_pM and a tangent vector at a nearby point q lives in T_qM , and these are different vector spaces with no preferred identification between them. This is not a defect of the setup; it reflects the fact that a bare smooth manifold carries no notion of “the same direction” from one point to the next. The consequence is that the derivative of a vector field is not well defined. If Y is a vector field and X_p a direction at p , one would like to differentiate Y along X_p by forming the difference quotient $\frac{1}{t}(Y_{\gamma(t)} - Y_p)$ along a curve γ with $\dot{\gamma}(0) = X_p$. The obstruction is immediate: $Y_{\gamma(t)}$ and Y_p sit in different tangent spaces, so their difference has no meaning.

An affine connection is precisely the extra structure that repairs this. It prescribes, as a primitive datum, how to differentiate one vector field along another, subject to the algebraic rules that a directional derivative ought to satisfy. On $\mathbb{R}^n$ there is an obvious choice: differentiate each component function separately, which works because the coordinate frame $\partial_1, \dots, \partial_n$ furnishes a global identification of every tangent space with $\mathbb{R}^n$. On a manifold no such global frame need exist, and in general there is no distinguished connection at all; a connection is a choice. The purpose of this section is to axiomatize that choice, to record its coordinate description through the Christoffel symbols, to show that connections always exist and to measure exactly how much freedom the choice involves, and finally to transport the operation from vector fields on M to vector fields along a curve, where it becomes the covariant derivative D_t that governs geodesics and parallel transport in the sections to come. Throughout, M is a smooth n -manifold and $C^\infty(M)$ is its algebra of smooth real functions; $\mathfrak{X}(M)$ denotes the $C^\infty(M)$ -module of smooth vector fields, following [3].

Definition 2.1.1: Affine Connection

Let M be a smooth manifold. An *affine connection* on TM is a map

$$\nabla: \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M), \quad (X, Y) \mapsto \nabla_X Y,$$

called *covariant differentiation*, satisfying the following three properties for all $X, X_1, X_2, Y, Y_1, Y_2 \in \mathfrak{X}(M)$, all $a, b \in \mathbb{R}$, and all $f \in C^\infty(M)$.

1. $C^\infty(M)$ -linearity in the differentiation slot:

$$\nabla_{fX_1 + X_2} Y = f \nabla_{X_1} Y + \nabla_{X_2} Y$$

2. $\mathbb{R}$ -linearity in the differentiated slot:

$$\nabla_X (aY_1 + bY_2) = a \nabla_X Y_1 + b \nabla_X Y_2$$

3. Leibniz rule in the differentiated slot:

$$\nabla_X (fY) = (Xf)Y + f \nabla_X Y$$

The vector field $\nabla_X Y$ is the *covariant derivative* of Y in the direction X .

The asymmetry between the two slots is the heart of the definition and deserves comment. In the differentiation slot X , the connection is $C^\infty(M)$ -linear: scaling the direction by a function scales the answer by the same function, pointwise. This is what makes $\nabla_X Y$ tensorial in X , and it has a concrete consequence recorded below in proposition 2.1.5, namely that $(\nabla_X Y)_p$ depends on X only through its value $X_p \in T_p M$, not on how X extends off p . In the differentiated slot Y , by contrast, the connection is only $\mathbb{R}$ -linear and obeys the Leibniz rule; the extra term $(Xf)Y$ is exactly the derivative of the coefficient function, and it is what forces $\nabla_X Y$ to depend on how Y varies rather than only on its value at a point. A directional derivative must behave this way: differentiating f times a fixed field Y should produce a rate of change of f , which is the content of $(Xf)Y$. The failure of $C^\infty(M)$ -linearity in Y is not a blemish but the defining feature that distinguishes a derivative from a tensor.

The first worked example is the one that motivates the whole apparatus.

Example 2.1.2: The Euclidean Connection

On $\mathbb{R}^n$ with global coordinates $(x^1, \dots, x^n)$, write a vector field as $Y = Y^i \partial_i$ with $Y^i \in C^\infty(\mathbb{R}^n)$; the Einstein summation convention, summing over any index appearing once up and once down, is in force here and throughout the section. Define

$$\bar{\nabla}_X Y := (XY^i) \partial_i,$$

that is, differentiate each component function of Y in the direction X and reassemble. This is the *Euclidean* (or *flat*, or *standard*) connection.

Each axiom of definition 2.1.1 is immediate from the corresponding property of the directional derivative of functions. For property (1), $C^\infty(\mathbb{R}^n)$ -linearity of $X \mapsto XY^i$ in X gives $\bar{\nabla}_{fX_1 + X_2} Y = ((fX_1 + X_2)Y^i) \partial_i = f(X_1 Y^i) \partial_i + (X_2 Y^i) \partial_i = f \bar{\nabla}_{X_1} Y + \bar{\nabla}_{X_2} Y$. Property (2) follows because $X(aY_1^i + bY_2^i) \partial_i = aXY_1^i \partial_i + bXY_2^i \partial_i = a \bar{\nabla}_X Y_1 + b \bar{\nabla}_X Y_2$. For property (3), writing $fY = (fY^i) \partial_i$ and using the ordinary product rule $X(fY^i) = (Xf)Y^i + f(XY^i)$ gives

$$\bar{\nabla}_X (fY) = (X(fY^i)) \partial_i = (Xf)Y^i \partial_i + f(XY^i) \partial_i = (Xf)Y + f \bar{\nabla}_X Y,$$

which is the Leibniz rule. Thus $\bar{\nabla}$ is an affine connection on $\mathbb{R}^n$. The computation goes through because the coordinate frame (∂_i) is globally defined and its members are, so to speak, held fixed while the coefficients are differentiated. On a general manifold no global coordinate frame exists, so there is no canonical connection to write down; the definition above is the abstraction of what remains once the crutch of global coordinates is removed.

Away from $\mathbb{R}^n$ a connection cannot be described by “differentiate the components,” because the coordinate frames on overlapping charts differ and the frame vectors themselves must be differentiated. The bookkeeping is carried by a family of functions on each chart.

Definition 2.1.3: Christoffel Symbols

Let ∇ be an affine connection on TM and let $(U, (x^1, \dots, x^n))$ be a smooth chart with coordinate frame $(\partial_1, \dots, \partial_n)$. Each $\nabla_{\partial_i} \partial_j$ is a smooth vector field on U , hence expands in the coordinate frame. The *Christoffel symbols* of ∇ in the chart are the n^3 functions $\Gamma_{ij}^k \in C^\infty(U)$ defined by

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$$

They record the action of ∇ on the frame; the lower indices (i, j) label the direction and the field, the upper index k the output component.

Once the Christoffel symbols are known, the connection is completely determined on the chart. Writing $X = X^i \partial_i$ and $Y = Y^j \partial_j$ and applying the three axioms in turn, first C^∞ -linearity in X , then $\mathbb{R}$ -linearity in Y , then the Leibniz rule on each $Y^j \partial_j$, gives

$$\nabla_X Y = X^i \nabla_{\partial_i} (Y^j \partial_j) = X^i ((\partial_i Y^j) \partial_j + Y^j \nabla_{\partial_i} \partial_j) = X^i (\partial_i Y^j) \partial_j + X^i Y^j \Gamma_{ij}^k \partial_k$$

Relabelling the dummy index $j \rightarrow k$ in the first term collects the ∂_k -component:

$$\nabla_X Y = \left(X^i \partial_i Y^k + \Gamma_{ij}^k X^i Y^j \right) \partial_k \quad (2.1)$$

The first summand is the Euclidean answer of example 2.1.2: the plain directional derivative of the components. The second summand, quadratic in the frame data through Γ_{ij}^k , is the correction that

accounts for the variation of the frame itself. On $\mathbb{R}^n$ in Cartesian coordinates every Γ_{ij}^k vanishes, since $\bar{\nabla}_{\partial_i} \partial_j = (\partial_i \delta_j^k) \partial_k = 0$, and (2.1) reduces to $\bar{\nabla}_X Y = X^i (\partial_i Y^k) \partial_k = (XY^k) \partial_k$, recovering the flat connection. It should be stressed that the Γ_{ij}^k are not the components of a tensor: under a change of coordinates they acquire an inhomogeneous term (computed in proposition 2.1.5 below), so “the Christoffel symbols vanish” is a statement about a chart, not about a point.

Example 2.1.4: Christoffel Symbols in Polar Coordinates

Take $\mathbb{R}^2$ minus the origin with polar coordinates (r, θ) , related to Cartesian coordinates by $x = r \cos \theta$, $y = r \sin \theta$, and equip it with the Euclidean connection $\bar{\nabla}$ of example 2.1.2. The coordinate frames are related by

$$\partial_r = \cos \theta \partial_x + \sin \theta \partial_y, \quad \partial_\theta = -r \sin \theta \partial_x + r \cos \theta \partial_y$$

Since $\bar{\nabla}_{\partial_x} \partial_x = \bar{\nabla}_{\partial_x} \partial_y = \bar{\nabla}_{\partial_y} \partial_y = 0$, the connection acts on the polar frame by differentiating the coefficients above. Direct computation gives

$$\bar{\nabla}_{\partial_r} \partial_r = \bar{\nabla}_{\partial_r} (\cos \theta \partial_x + \sin \theta \partial_y) = (\partial_r \cos \theta) \partial_x + (\partial_r \sin \theta) \partial_y = 0,$$

because $\cos \theta$ and $\sin \theta$ are independent of r ; hence $\Gamma_{rr}^r = \Gamma_{rr}^\theta = 0$. Next,

$$\bar{\nabla}_{\partial_r} \partial_\theta = \bar{\nabla}_{\partial_r} (-r \sin \theta \partial_x + r \cos \theta \partial_y) = -\sin \theta \partial_x + \cos \theta \partial_y = \frac{1}{r} \partial_\theta,$$

so $\Gamma_{r\theta}^\theta = \frac{1}{r}$ and $\Gamma_{r\theta}^r = 0$. The same value arises in the direction ∂_θ on ∂_r :

$$\bar{\nabla}_{\partial_\theta} \partial_r = \bar{\nabla}_{\partial_\theta} (\cos \theta \partial_x + \sin \theta \partial_y) = -\sin \theta \partial_x + \cos \theta \partial_y = \frac{1}{r} \partial_\theta,$$

giving $\Gamma_{\theta r}^\theta = \frac{1}{r}$, $\Gamma_{\theta r}^r = 0$. Finally,

$$\bar{\nabla}_{\partial_\theta} \partial_\theta = \bar{\nabla}_{\partial_\theta} (-r \sin \theta \partial_x + r \cos \theta \partial_y) = -r \cos \theta \partial_x - r \sin \theta \partial_y = -r \partial_r,$$

so $\Gamma_{\theta\theta}^r = -r$ and $\Gamma_{\theta\theta}^\theta = 0$. The single flat connection $\bar{\nabla}$ thus has all Christoffel symbols zero in Cartesian coordinates and the nonzero symbols $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$, $\Gamma_{\theta\theta}^r = -r$ in polar coordinates. This concretely exhibits the coordinate-dependence flagged above: the Christoffel symbols are not tensorial, and their vanishing in one chart says nothing about their vanishing in another.

Two questions are now pressing. Do connections exist on every manifold, given that no global-frame recipe is available in general? And how much does a connection depend on choices, once one exists? The next result answers both. Its proof uses a partition of unity, exactly as in the construction of Riemannian metrics; the analytic input is the standard fact that a smooth manifold admits a partition of unity subordinate to any open cover [3].

Proposition 2.1.5: Existence and Local Character of Connections

Let M be a smooth manifold. Then:

1. (*Locality.*) If ∇ is an affine connection and $X, Y \in \mathfrak{X}(M)$ agree with X', Y' on an open set U , then $\nabla_X Y = \nabla_{X'} Y'$ on U . Moreover $(\nabla_X Y)_p$ depends on X only through the value $X_p \in T_p M$. Consequently ∇ restricts to a well-defined connection on every open subset, and is determined on each chart by its Christoffel symbols through (2.1).
2. (*Difference tensor.*) If ∇ and $\tilde{\nabla}$ are two affine connections on TM , then

$$A(X, Y) := \tilde{\nabla}_X Y - \nabla_X Y$$

is $C^\infty(M)$ -bilinear, hence defines a $(1, 2)$ -tensor field on M . Conversely, for any $(1, 2)$ -tensor field A and any affine connection ∇ , the map $\tilde{\nabla}_X Y := \nabla_X Y + A(X, Y)$ is an affine connection. The Christoffel symbols transform under a change of chart $(x^i) \rightarrow (\tilde{x}^a)$ by

$$\tilde{\Gamma}_{ab}^c = \frac{\partial \tilde{x}^c}{\partial x^k} \frac{\partial x^i}{\partial \tilde{x}^a} \frac{\partial x^j}{\partial \tilde{x}^b} \Gamma_{ij}^k + \frac{\partial \tilde{x}^c}{\partial x^k} \frac{\partial^2 x^k}{\partial \tilde{x}^a \partial \tilde{x}^b},$$

the second, inhomogeneous term being what disqualifies (Γ_{ij}^k) from being a tensor.

3. (*Existence.*) M admits an affine connection on TM .

Proof:

Step 1: locality of the differentiation slot. Fix an affine connection ∇ . First $C^\infty(M)$ -linearity in X (axiom (1)) is exactly the statement that $X \mapsto (\nabla_X Y)_p$ is $C^\infty(M)$ -linear; by the tensor characterization lemma for $C^\infty(M)$ -multilinear maps [3], such a map is pointwise, so $(\nabla_X Y)_p$ depends on X only through X_p . Explicitly: if $X_p = 0$ choose smooth functions f^i and vector fields E_i with $X = f^i E_i$ near p and $f^i(p) = 0$; then $(\nabla_X Y)_p = f^i(p)(\nabla_{E_i} Y)_p = 0$. Applied to $X - X'$ this shows $(\nabla_X Y)_p = (\nabla_{X'} Y)_p$ whenever $X_p = X'_p$.

Step 2: locality of the differentiated slot. Suppose $Y = Y'$ on an open set U and fix $p \in U$. Choose a bump function $\varphi \in C^\infty(M)$ with $\varphi \equiv 1$ on a neighborhood of p and $\text{supp } \varphi \subset U$, which exists by [3]. Then $\varphi Y = \varphi Y'$ on all of M , so $\nabla_X(\varphi Y) = \nabla_X(\varphi Y')$. Expanding the left side with the Leibniz rule (axiom (3)) and evaluating at p , where $\varphi(p) = 1$ and $(X\varphi)(p) = 0$ because φ is constant near p , gives

$$(\nabla_X(\varphi Y))_p = (X\varphi)(p) Y_p + \varphi(p) (\nabla_X Y)_p = (\nabla_X Y)_p,$$

and likewise $(\nabla_X(\varphi Y'))_p = (\nabla_X Y')_p$; hence $(\nabla_X Y)_p = (\nabla_X Y')_p$. Combining Steps 1 and 2, if $(X, Y) = (X', Y')$ on U then $\nabla_X Y = \nabla_{X'} Y'$ on U . In particular ∇ descends to a connection on the open submanifold U , and on a chart the value $\nabla_X Y$ at any point is computed from the chart data alone; formula (2.1), derived from the axioms, then shows the Christoffel symbols determine ∇ on the chart. This proves (1).

Step 3: the difference is a tensor. Let $\nabla, \tilde{\nabla}$ be connections and $A(X, Y) = \tilde{\nabla}_X Y - \nabla_X Y$. Additivity of A in each slot is clear. For $f \in C^\infty(M)$, the differentiation slot gives $A(fX, Y) = \tilde{\nabla}_{fX} Y - \nabla_{fX} Y = f\tilde{\nabla}_X Y - f\nabla_X Y = fA(X, Y)$ by axiom (1) for both connections. For the

differentiated slot, the Leibniz terms cancel:

$$A(X, fY) = ((Xf)Y + f\tilde{\nabla}_X Y) - ((Xf)Y + f\nabla_X Y) = f A(X, Y)$$

Thus A is $C^\infty(M)$ -bilinear, so by the tensor characterization [3] it is a $(1, 2)$ -tensor field: a section of $\text{Hom}(TM \otimes TM, TM)$. Conversely, if A is any such tensor and ∇ any connection, then $\tilde{\nabla}_X Y := \nabla_X Y + A(X, Y)$ is $C^\infty(M)$ -linear in X (sum of two such), $\mathbb{R}$ -linear in Y , and satisfies the Leibniz rule because A contributes no derivative term:

$$\tilde{\nabla}_X(fY) = \nabla_X(fY) + A(X, fY) = (Xf)Y + f\nabla_X Y + f A(X, Y) = (Xf)Y + f\tilde{\nabla}_X Y$$

Hence $\tilde{\nabla}$ is an affine connection. For the transformation law one fixes a single connection ∇ and compares its Christoffel symbols in the two charts, applying (2.1) to $\nabla_{\partial/\partial\tilde{x}^a}(\partial/\partial\tilde{x}^b)$. Using $\partial/\partial\tilde{x}^a = (\partial x^i/\partial\tilde{x}^a)\partial_i$ together with C^∞ -linearity in the direction and the Leibniz rule,

$$\nabla_{\partial/\partial\tilde{x}^a} \frac{\partial}{\partial\tilde{x}^b} = \frac{\partial x^i}{\partial\tilde{x}^a} \nabla_{\partial_i} \left(\frac{\partial x^j}{\partial\tilde{x}^b} \partial_j \right) = \frac{\partial x^i}{\partial\tilde{x}^a} \left(\partial_i \frac{\partial x^j}{\partial\tilde{x}^b} \right) \partial_j + \frac{\partial x^i}{\partial\tilde{x}^a} \frac{\partial x^j}{\partial\tilde{x}^b} \Gamma_{ij}^k \partial_k$$

The first term simplifies because $\frac{\partial x^i}{\partial\tilde{x}^a} \partial_i = \partial/\partial\tilde{x}^a$, so $\frac{\partial x^i}{\partial\tilde{x}^a} (\partial_i \frac{\partial x^j}{\partial\tilde{x}^b}) = \frac{\partial^2 x^j}{\partial\tilde{x}^a \partial\tilde{x}^b}$. Expanding $\partial_j = (\partial\tilde{x}^c/\partial x^j) \partial/\partial\tilde{x}^c$ and $\partial_k = (\partial\tilde{x}^c/\partial x^k) \partial/\partial\tilde{x}^c$ and reading off the $\partial/\partial\tilde{x}^c$ -coefficient (renaming the summed index $j \rightarrow k$ in the first term) yields

$$\tilde{\Gamma}_{ab}^c = \frac{\partial\tilde{x}^c}{\partial x^k} \frac{\partial x^i}{\partial\tilde{x}^a} \frac{\partial x^j}{\partial\tilde{x}^b} \Gamma_{ij}^k + \frac{\partial\tilde{x}^c}{\partial x^k} \frac{\partial^2 x^k}{\partial\tilde{x}^a \partial\tilde{x}^b},$$

the second term arising from differentiating the transition Jacobian. This establishes (2).

Step 4: existence. Cover M by charts $\{U_\alpha\}$ and let $\{\rho_\alpha\}$ be a smooth partition of unity subordinate to this cover [3], so $\text{supp } \rho_\alpha \subset U_\alpha$, the family $\{\text{supp } \rho_\alpha\}$ is locally finite, and $\sum_\alpha \rho_\alpha \equiv 1$. On each U_α the coordinate frame is a global frame, so the “differentiate the components” recipe of example 2.1.2 defines an affine connection ∇^α on U_α (all its Christoffel symbols vanish in that chart). Define

$$\nabla_X Y := \sum_\alpha \rho_\alpha \nabla_X^\alpha Y,$$

where $\nabla_X^\alpha Y$ is interpreted as extended by zero outside $\text{supp } \rho_\alpha$; local finiteness makes the sum a finite sum near each point, hence a smooth vector field. Properties (1) and (2) of definition 2.1.1 are inherited termwise from the ∇^α , since $C^\infty(M)$ -linearity in X and $\mathbb{R}$ -linearity in Y are preserved by multiplication by the fixed functions ρ_α and by summation. For the Leibniz rule, compute using $\nabla_X^\alpha(fY) = (Xf)Y + f\nabla_X^\alpha Y$ and $\sum_\alpha \rho_\alpha = 1$:

$$\nabla_X(fY) = \sum_\alpha \rho_\alpha ((Xf)Y + f\nabla_X^\alpha Y) = (Xf)Y \sum_\alpha \rho_\alpha + f \sum_\alpha \rho_\alpha \nabla_X^\alpha Y = (Xf)Y + f\nabla_X Y$$

The crucial point is that the convex-combination coefficients ρ_α sum to 1, so the derivative term $(Xf)Y$ appears with total weight 1 rather than being scaled; had the weights summed to anything else, the Leibniz rule would fail. Thus ∇ is an affine connection on TM , proving (3), completing the proof.

The proposition draws a clean picture of the space of connections. Part (2) says that connections

form an affine space modeled on the vector space of $(1, 2)$ -tensor fields: fix one connection ∇ , and every other is $\nabla + A$ for a unique such tensor A , while $\nabla - \nabla'$ is always a tensor even though neither connection is. This is the precise sense in which a connection is “a choice of order A ”: the ambiguity is a tensor field and can therefore be prescribed pointwise and patched. Part (1) is what makes the Christoffel symbols a legitimate local description rather than an artifact, and it licenses all coordinate computations to follow, including the polar-coordinate calculation of example 2.1.4. Existence in part (3) removes the last worry: although no connection is canonical on a bare manifold, at least one always exists, and once a Riemannian metric is present the Fundamental Theorem of the next chapter will single out a distinguished one.

With the connection understood as an operation on vector fields defined on all of M , the remaining task in this section is to differentiate a vector field that is defined only along a curve. A geodesic is a curve whose velocity does not turn, and the velocity of a curve is a vector field defined along that curve, not on any open set of M ; the flat connection cannot be applied to it directly. The locality established above is exactly what allows the connection to be pulled back to curves.

Definition 2.1.6: Covariant Derivative Along a Curve

Let ∇ be an affine connection on TM and let $\gamma: I \rightarrow M$ be a smooth curve on an interval $I \subseteq \mathbb{R}$. A *vector field along γ* is a smooth map $V: I \rightarrow TM$ with $V(t) \in T_{\gamma(t)}M$ for every $t \in I$; the set of such fields is denoted $\mathfrak{X}(\gamma)$. The *covariant derivative along γ* is the unique map

$$D_t: \mathfrak{X}(\gamma) \rightarrow \mathfrak{X}(\gamma), \quad V \mapsto D_t V,$$

written also $\frac{D}{dt}$, satisfying:

1. *$\mathbb{R}$ -linearity*: $D_t(aV_1 + bV_2) = a D_t V_1 + b D_t V_2$ for $a, b \in \mathbb{R}$;
2. *Leibniz rule*: $D_t(fV) = \dot{f}V + f D_t V$ for $f \in C^\infty(I)$;
3. *compatibility with ∇* : if V is the restriction of a vector field $\tilde{V} \in \mathfrak{X}(M)$, meaning $V(t) = \tilde{V}_{\gamma(t)}$, then $D_t V(t) = (\nabla_{\dot{\gamma}(t)} \tilde{V})$.

In a chart $(U, (x^i))$ with $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$ and $V(t) = V^k(t) \partial_k|_{\gamma(t)}$, the covariant derivative is

$$D_t V = \left(\dot{V}^k + \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i V^j \right) \partial_k|_{\gamma(t)} \quad (2.2)$$

Proof:

Existence, uniqueness, and the formula are established together. On the chart domain, if D_t exists then property (2) applied to $V = V^k \partial_k|_\gamma$, viewing each $\partial_k|_\gamma$ as the restriction of the coordinate field $\partial_k \in \mathfrak{X}(U)$, gives by $\mathbb{R}$ -linearity and the Leibniz rule

$$D_t V = \dot{V}^k \partial_k|_\gamma + V^k D_t(\partial_k|_\gamma)$$

By property (3), $D_t(\partial_k|_\gamma) = \nabla_{\dot{\gamma}} \partial_k = \dot{\gamma}^i \nabla_{\partial_i} \partial_k = \dot{\gamma}^i \Gamma_{ik}^m \partial_m|_\gamma$, using C^∞ -linearity of ∇ in the direction slot (proposition 2.1.5(1)) to pass $\dot{\gamma}^i$ outside. Substituting and relabelling the summed indices $(m, k) \rightarrow (k, j)$ yields (2.2). This proves that any D_t satisfying (1)–(3) is given by the stated formula on each chart, hence is unique. Conversely, formula (2.2) defines a smooth vector field along γ on each chart; on an overlap the two coordinate expressions agree, because both compute the same operation applied to the same V , so they patch to a globally defined

$D_t V \in \mathfrak{X}(\gamma)$. Properties (1) and (2) are read off directly from (2.2), since $V \mapsto \dot{V}^k$ is $\mathbb{R}$ -linear and satisfies the product rule $\frac{d}{dt}(fV^k) = \dot{f}V^k + f\dot{V}^k$ while the Γ -term is $\mathbb{R}$ -linear and $C^\infty(I)$ -linear in V . Property (3) holds because if $V(t) = \tilde{V}_{\gamma(t)}$ with $\tilde{V} = \tilde{V}^k \partial_k$, then $V^k(t) = \tilde{V}^k(\gamma(t))$ and the chain rule gives $\dot{V}^k = \dot{\gamma}^i (\partial_i \tilde{V}^k)$, so (2.2) becomes $(\dot{\gamma}^i \partial_i \tilde{V}^k + \Gamma_{ij}^k \dot{\gamma}^i \tilde{V}^j) \partial_k = \nabla_{\dot{\gamma}} \tilde{V}$ by (2.1), as required.

The content of this definition is that the connection on M induces, on each curve, an intrinsic notion of the rate of change of a vector field along the curve, and that this induced operation is pinned down by three requirements: it is linear, it differentiates coefficients by the product rule, and it agrees with ∇ whenever the field along the curve happens to extend to a field on M . The subtlety it resolves is real. A vector field along γ need not extend to any vector field on M , for instance when γ has a self-intersection and V takes different values at the two times the curve passes through the same point; there is then no $\tilde{V}$ to which property (3) could be applied. Formula (2.2) shows the operation is nonetheless well defined, because it refers only to the values of V and of the Christoffel symbols along γ , never to an extension. The two summands mirror those in (2.1): $\dot{V}^k$ is the componentwise rate of change of V , and $\Gamma_{ij}^k \dot{\gamma}^i V^j$ corrects for the turning of the coordinate frame as the base point $\gamma(t)$ moves.

Example 2.1.7: Covariant Derivative of the Velocity on a Circle

Return to $\mathbb{R}^2$ minus the origin with the Euclidean connection and the polar Christoffel symbols of example 2.1.4. Let $\gamma(t) = (r(t), \theta(t)) = (1, t)$ describe the unit circle traversed once as t runs over $[0, 2\pi]$, so $\dot{\gamma} = (\dot{r}, \dot{\theta}) = (0, 1)$, that is, $\dot{\gamma} = \partial_\theta$ along γ . Its velocity is the vector field $V = \dot{\gamma}$ along γ , with components $V^r = 0$, $V^\theta = 1$ (constant). Formula (2.2) gives, for the r -component,

$$(D_t \dot{\gamma})^r = \dot{V}^r + \Gamma_{ij}^r \dot{\gamma}^i V^j = 0 + \Gamma_{\theta\theta}^r \dot{\theta} V^\theta = (-r)(1)(1) = -1,$$

using $r = 1$ along γ and that the only Γ^r with both lower indices θ is $\Gamma_{\theta\theta}^r = -r$; for the θ -component,

$$(D_t \dot{\gamma})^\theta = \dot{V}^\theta + \Gamma_{ij}^\theta \dot{\gamma}^i V^j = 0 + \Gamma_{\theta\theta}^\theta \dot{\theta} V^\theta = 0,$$

since $\Gamma_{\theta\theta}^\theta = 0$. Hence $D_t \dot{\gamma} = -\partial_r$ along the circle. The velocity has constant components in polar coordinates, yet its covariant derivative is nonzero: it points radially inward, reflecting that the unit circle is not a straight line and its velocity is continually turning toward the center. As a check, in Cartesian coordinates $\gamma(t) = (\cos t, \sin t)$ has velocity $(-\sin t, \cos t)$ and $D_t \dot{\gamma} = \overline{\nabla}_{\dot{\gamma}} \dot{\gamma} = (-\cos t, -\sin t)$, the inward unit radial vector, in agreement with $-\partial_r$. A curve whose velocity has vanishing covariant derivative, $D_t \dot{\gamma} = 0$, is a geodesic; the circle is not one, and its failure is measured precisely by this radial term.

2.2 Torsion and Metric Compatibility

An affine connection, as introduced in definition 2.1.1, is an extra piece of data laid on top of a smooth manifold: a rule for differentiating one vector field along another. On a bare manifold there is no preferred such rule, and proposition 2.1.5 shows that connections come in abundance, the difference of any two being a tensor. The geometry of a Riemannian manifold, by contrast, singles out one connection above all others. The point of this section and the next is to isolate the two properties that pin it down. Neither property refers to the connection's abundance; each is a clean

structural constraint, and each has a transparent geometric reading.

The first property, *torsion-freeness*, measures the failure of the second-order behavior of ∇ to be symmetric in its two directional arguments. Torsion is the antisymmetric part of the connection that is not already accounted for by the Lie bracket of the two fields, and a connection is torsion-free precisely when its Christoffel symbols are symmetric in their lower indices. The second property, *metric compatibility*, ties the connection to the metric g of definition 1.1.1: it asks that differentiating an inner product of two fields obey the Leibniz rule with respect to ∇ , so that the metric is constant from the connection's point of view. Torsion-freeness is a property a connection has entirely on its own, before any metric is chosen; metric compatibility is a property a connection has relative to a fixed g . Chapter 2's central theorem, proved in section 2.3, is that on a (pseudo-)Riemannian manifold exactly one connection has both. This section develops the two properties separately, so that the uniqueness argument there can proceed unencumbered.

Throughout, M is a smooth n -manifold and ∇ an affine connection on TM ; when a metric is present it is denoted g , with the associated inner product $\langle v, w \rangle$ on each tangent space. The Einstein summation convention (sum over any index appearing once up and once down) is in force. Christoffel symbols Γ_{ij}^k are defined relative to a coordinate frame by $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k$, as in definition 2.1.3.

Definition 2.2.1: Torsion Tensor

Let ∇ be an affine connection on a smooth manifold M . The *torsion* of ∇ is the map $T: \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ defined by

$$T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

The connection ∇ is called *torsion-free* (or *symmetric*) if $T \equiv 0$, that is, if

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

for all $X, Y \in \mathfrak{X}(M)$.

Two things must be checked before the definition earns its name. First, T is antisymmetric, $T(X, Y) = -T(Y, X)$, which is immediate from the antisymmetry of the Lie bracket. Second, and less transparently, T is a *tensor*: although each of the three terms $\nabla_X Y$, $\nabla_Y X$, and $[X, Y]$ fails on its own to be $C^\infty(M)$ -linear in X or Y , the non-tensorial pieces cancel in the combination. The following lemma records this, and shows at the same time that torsion-freeness is equivalent to the coordinate condition $\Gamma_{ij}^k = \Gamma_{ji}^k$ promised in the contract for this section.

Lemma 2.2.2: Torsion is a tensor; symmetry of the Christoffel symbols

Let ∇ be an affine connection on M with torsion T . Then:

1. T is $C^\infty(M)$ -bilinear, hence defines a $(1, 2)$ -tensor field on M ; its value $T(X, Y)_p$ depends only on X_p and Y_p .
2. In any coordinate chart, with Γ_{ij}^k the Christoffel symbols of ∇ ,

$$T(\partial_i, \partial_j) = (\Gamma_{ij}^k - \Gamma_{ji}^k) \partial_k$$

Consequently ∇ is torsion-free if and only if $\Gamma_{ij}^k = \Gamma_{ji}^k$ for all i, j, k in every chart.

Proof:

Step 1: bilinearity. By antisymmetry $T(X, Y) = -T(Y, X)$, so it suffices to establish $C^\infty(M)$ -linearity in the second argument; linearity in the first then follows. Additivity in Y is clear from the additivity of $\nabla_X(\cdot)$, of $\nabla_{(\cdot)}X$ read off from the fixed first slot, and of the bracket, together with $\mathbb{R}$ -linearity. For the function-linearity, fix $f \in C^\infty(M)$ and compute each term of $T(X, fY)$ using the Leibniz rule for ∇ (definition 2.1.1) and the corresponding identity for the Lie bracket. The Leibniz rule gives

$$\nabla_X(fY) = (Xf)Y + f\nabla_XY,$$

while the definition of the bracket on a product gives $[X, fY] = (Xf)Y + f[X, Y]$. The term $\nabla_{fY}X = f\nabla_YX$ is function-linear directly, since ∇ is $C^\infty(M)$ -linear in its subscript. Subtracting,

$$T(X, fY) = (Xf)Y + f\nabla_XY - f\nabla_YX - (Xf)Y - f[X, Y] = fT(X, Y)$$

The two occurrences of $(Xf)Y$ cancel, leaving $T(X, fY) = fT(X, Y)$. Thus T is $C^\infty(M)$ -bilinear. A $C^\infty(M)$ -bilinear map $\mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ is a $(1, 2)$ -tensor field, and the value at p of such a map depends only on the values X_p, Y_p of the arguments at p .

Step 2: coordinate expression. In a chart, coordinate vector fields commute, $[\partial_i, \partial_j] = 0$. Hence

$$T(\partial_i, \partial_j) = \nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i - [\partial_i, \partial_j] = \Gamma_{ij}^k\partial_k - \Gamma_{ji}^k\partial_k = (\Gamma_{ij}^k - \Gamma_{ji}^k)\partial_k,$$

using $\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k\partial_k$ from definition 2.1.3. Since T is a tensor by Step 1, it vanishes identically if and only if it vanishes on every pair of coordinate frame fields in every chart, and by the display just above this occurs if and only if $\Gamma_{ij}^k = \Gamma_{ji}^k$ for all indices. This is the asserted equivalence, completing the proof.

The name *symmetric* is now transparent: a connection is torsion-free exactly when its Christoffel symbols are symmetric in the two lower indices. Geometrically, torsion detects a certain twisting. Given two commuting directions ∂_i and ∂_j , one may transport ∂_j infinitesimally in the ∂_i direction and, separately, ∂_i in the ∂_j direction; torsion is the leading-order discrepancy between the two resulting infinitesimal displacements, over and above the discrepancy already carried by the Lie bracket. When it vanishes, the connection introduces no twist of its own beyond what the bracket of the fields already prescribes. The bracket term is exactly the correction that makes T tensorial, and hence a pointwise object, rather than a differential operator on fields.

Example 2.2.3: The Euclidean connection is torsion-free

On $\mathbb{R}^n$ with the standard flat connection $\bar{\nabla}$ of example 2.1.2, all Christoffel symbols vanish in the standard coordinates: $\bar{\nabla}_{\partial_i} \partial_j = 0$, so $\Gamma_{ij}^k = 0$. Trivially $\Gamma_{ij}^k = \Gamma_{ji}^k$, and by lemma 2.2.2 the Euclidean connection is torsion-free. This can also be seen without coordinates: for $X = X^i \partial_i$ and $Y = Y^j \partial_j$ the flat connection reads off $\bar{\nabla}_X Y = X(Y^j) \partial_j$, so

$$\bar{\nabla}_X Y - \bar{\nabla}_Y X = X(Y^j) \partial_j - Y(X^j) \partial_j = [X, Y],$$

the last equality being the standard coordinate formula for the bracket on $\mathbb{R}^n$. Thus $T \equiv 0$.

For a connection with nonzero torsion, keep the same manifold $\mathbb{R}^2$ but define ∇ by prescribing $\Gamma_{12}^1 = 1$ and all other symbols zero. Then $\Gamma_{12}^1 = 1 \neq 0 = \Gamma_{21}^1$, so $T(\partial_1, \partial_2) = (\Gamma_{12}^1 - \Gamma_{21}^1) \partial_1 = \partial_1 \neq 0$. This ∇ is a perfectly good affine connection, differing from $\bar{\nabla}$ by a $(1, 2)$ -tensor as in proposition 2.1.5; it simply carries torsion. The example shows that torsion-freeness is a genuine restriction, not an automatic feature.

Torsion-freeness constrains a connection using no metric at all. The second property of interest is of the opposite character: it makes sense only once a metric g has been fixed, and it asks the connection to respect that metric. The requirement is that the product rule hold when a directional derivative meets an inner product.

Definition 2.2.4: Metric-Compatible Connection

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM . Then ∇ is *compatible with g* (or *metric-compatible*) if

$$X \langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle$$

for all vector fields $X, Y, Z \in \mathfrak{X}(M)$, where $\langle -, - \rangle = g(-, -)$ and $X \langle Y, Z \rangle$ denotes the derivative of the function $p \mapsto \langle Y_p, Z_p \rangle$ along X .

The left-hand side, $X \langle Y, Z \rangle$, is the ordinary directional derivative of a real-valued function; the right-hand side distributes that derivative across the two slots of the inner product by way of ∇ . The identity therefore says that ∇ obeys the Leibniz rule for the bilinear form g exactly as it does for the product of a function with a field. A connection satisfying it treats g as an invariant: lengths and angles measured by g are compatible with the notion of differentiation supplied by ∇ . The compact restatement is that the covariant derivative of the metric tensor vanishes, $\nabla g = 0$, and the equivalence of the two formulations is the content of proposition 2.2.6 below. Before that, the direct verification on the flat model.

Example 2.2.5: The Euclidean connection is metric-compatible

Let $\bar{g} = \delta_{ij} dx^i dx^j$ be the Euclidean metric on $\mathbb{R}^n$ (definition 1.1.1) and $\bar{\nabla}$ the flat connection of example 2.1.2. For $Y = Y^i \partial_i$ and $Z = Z^j \partial_j$ the inner product is the function $\langle Y, Z \rangle = \delta_{ij} Y^i Z^j = \sum_i Y^i Z^i$, whose derivative along X is, by the ordinary product rule,

$$X \langle Y, Z \rangle = \sum_i (X(Y^i) Z^i + Y^i X(Z^i))$$

On the other hand $\bar{\nabla}_X Y = X(Y^i) \partial_i$ and $\bar{\nabla}_X Z = X(Z^i) \partial_i$, so

$$\langle \bar{\nabla}_X Y, Z \rangle + \langle Y, \bar{\nabla}_X Z \rangle = \sum_i X(Y^i) Z^i + \sum_i Y^i X(Z^i),$$

which is the same expression. Hence $\bar{\nabla}$ is compatible with $\bar{g}$. The computation used only that the components $\bar{g}_{ij} = \delta_{ij}$ are constant, so that no derivative of the metric appears; the coordinate criterion of proposition 2.2.6 makes precise the sense in which constancy of the g_{ij} relative to a ∇ -parallel frame is what compatibility demands.

Metric compatibility admits three faces, and it is convenient to have all of them available before the fundamental theorem. The first is the vanishing of ∇g ; the second is the Leibniz identity of definition 2.2.4; the third is a statement about parallel transport, the operation of section 2.4, namely that transport along any curve is a linear isometry. The equivalence of the first two is a self-contained tensor computation and is proved in full here. The third is stated as part of the same package and proved in section 2.4, once parallel transport is available; the forward reference is precise, and no circularity results, since the argument there uses only the Leibniz identity established here.

To state the first formulation, recall from definition 1.4.1 that g extends canonically to all tensor bundles, so that ∇g is the covariant derivative of g regarded as a covariant 2-tensor field: for vector fields X, Y, Z ,

$$(\nabla_X g)(Y, Z) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z)$$

This is the standard Leibniz definition of the induced connection on the tensor bundle acting on a $(0, 2)$ -tensor, and the reader may take it as the working definition of ∇g for the purposes of this section; it is developed for general tensor bundles in section 2.4.

Proposition 2.2.6: Characterizations of metric compatibility

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM . The following are equivalent.

1. $\nabla g = 0$, i.e. $\nabla_X g = 0$ for every $X \in \mathfrak{X}(M)$.
2. For all $X, Y, Z \in \mathfrak{X}(M)$,

$$X \langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle$$

3. Parallel transport along every smooth curve in M is a linear isometry between the tangent spaces at its endpoints.

In any coordinate chart, conditions (i) and (ii) are further equivalent to

$$\partial_k g_{ij} = \Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il} \quad \text{for all } i, j, k,$$

where $g_{ij} = g(\partial_i, \partial_j)$.

Proof:

Equivalence of (i) and (ii). Unwinding the definition of ∇g recorded above, for any X, Y, Z ,

$$(\nabla_X g)(Y, Z) = X \langle Y, Z \rangle - \langle \nabla_X Y, Z \rangle - \langle Y, \nabla_X Z \rangle$$

The right-hand side is exactly the difference of the two sides of the identity in (ii). Hence $(\nabla_X g)(Y, Z) = 0$ for all Y, Z if and only if the identity of (ii) holds for all Y, Z (with this same X); requiring this for all X gives $\nabla g = 0$ if and only if (ii) holds for all X, Y, Z . Thus (i) $\iff$ (ii).

It remains to justify that the pointwise vanishing of the tensor $\nabla_X g$ is detected by the identity in (ii), i.e. that $\nabla_X g$ really is a tensor so that “ $(\nabla_X g)(Y, Z) = 0$ for all Y, Z ” is equivalent to $\nabla_X g = 0$. This is standard for the induced connection on a tensor bundle, but the tensoriality can be seen directly from the display: the map $(Y, Z) \mapsto X \langle Y, Z \rangle - \langle \nabla_X Y, Z \rangle - \langle Y, \nabla_X Z \rangle$ is $C^\infty(M)$ -bilinear. Indeed, for $f \in C^\infty(M)$,

$$X \langle fY, Z \rangle = X(f \langle Y, Z \rangle) = (Xf) \langle Y, Z \rangle + f X \langle Y, Z \rangle,$$

while $\langle \nabla_X(fY), Z \rangle = \langle (Xf)Y + f\nabla_X Y, Z \rangle = (Xf) \langle Y, Z \rangle + f \langle \nabla_X Y, Z \rangle$ and $\langle fY, \nabla_X Z \rangle = f \langle Y, \nabla_X Z \rangle$. Subtracting, the $(Xf) \langle Y, Z \rangle$ terms cancel and the expression scales by f ; linearity in Z is identical. So $\nabla_X g$ is a well-defined $(0, 2)$ -tensor and vanishes iff it vanishes on all coordinate pairs.

Coordinate form. Apply (ii) to $X = \partial_k$, $Y = \partial_i$, $Z = \partial_j$. Then $X \langle Y, Z \rangle = \partial_k g_{ij}$, while $\nabla_{\partial_k} \partial_i = \Gamma_{ki}^l \partial_l$ and $\nabla_{\partial_k} \partial_j = \Gamma_{kj}^l \partial_l$ give

$$\langle \nabla_{\partial_k} \partial_i, \partial_j \rangle + \langle \partial_i, \nabla_{\partial_k} \partial_j \rangle = \Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il}$$

Equating the two, (ii) restricted to coordinate frames is precisely the displayed identity $\partial_k g_{ij} = \Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il}$. Conversely, since both sides of the identity in (ii) are $C^\infty(M)$ -linear in Y and Z (shown above) and $\mathbb{R}$ -linear in X together with the Leibniz rule in X , an identity that holds on coordinate frames holds for all fields: write $X = X^k \partial_k$, $Y = Y^i \partial_i$, $Z = Z^j \partial_j$ and expand. Hence the coordinate identity is equivalent to (ii), and therefore to (i).

Equivalence with (iii). Parallel transport along curves is constructed in section 2.4; there proposition 2.4.4 proves that when ∇ satisfies the Leibniz identity of (ii), the transport P_{t_0, t_1}^γ along any curve γ is a linear isometry, and, conversely, that if every parallel transport is a linear isometry then (ii) holds. That argument uses only (ii) and the defining equation $D_t V = 0$ of a parallel field, so it may be invoked here without circularity. Combining it with the equivalence (i) $\iff$ (ii) above yields the equivalence of all three, as we sought to show.

The coordinate identity in proposition 2.2.6 is the working form of metric compatibility. Read as a formula for the derivatives $\partial_k g_{ij}$ in terms of the Christoffel symbols, it says that the rate of change of the metric coefficients is entirely dictated by the connection. This is precisely the constraint that will, in the next section, be solved together with the symmetry $\Gamma_{ij}^k = \Gamma_{ji}^k$ of lemma 2.2.2 to recover the Christoffel symbols uniquely from g : two linear conditions on the Γ_{ij}^k , symmetry from torsion-freeness and the derivative identity from compatibility, together determine them. The example below illustrates the derivative identity in the one setting where it is nontrivial yet fully explicit, and previews how it forces the values of the Christoffel symbols.

Example 2.2.7: Metric compatibility on a warped metric

Consider the upper half-plane coordinates (x, y) with the metric

$$g = y^2 dx^2 + dy^2, \quad y > 0,$$

so that $g_{11} = g_{22} = y^2$ and $g_{12} = g_{21} = 0$; write $\partial_1 = \partial_x$, $\partial_2 = \partial_y$. The only nonzero coordinate derivatives are $\partial_2 g_{11} = \partial_2 g_{22} = 2y$. Metric compatibility, in the form of proposition 2.2.6, is the system

$$\partial_k g_{ij} = \Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il}$$

Take, for instance, $k = 2, i = j = 1$: the left side is $\partial_2 g_{11} = 2y$, and since g is diagonal the right side is $\Gamma_{21}^1 g_{11} + \Gamma_{21}^2 g_{11} = 2y^2 \Gamma_{21}^1$. Compatibility therefore forces $\Gamma_{21}^1 = 1/y$. Taking $k = 1, i = j = 1$ gives $\partial_1 g_{11} = 0 = 2y^2 \Gamma_{11}^1$, so $\Gamma_{11}^1 = 0$; taking $k = 2, i = j = 2$ gives $\partial_2 g_{22} = 2y = 2y^2 \Gamma_{22}^2$, so $\Gamma_{22}^2 = 1/y$. Each equation of the compatibility system pins down a combination of Christoffel symbols against a known derivative of g . On its own this system does not determine every Γ_{ij}^k , since compatibility alone leaves the antisymmetric-in- (i, j) part free; adjoining the symmetry $\Gamma_{ij}^k = \Gamma_{ji}^k$ from lemma 2.2.2 removes exactly that freedom. The next section carries out this solution in general and in closed form.

The two properties are logically independent. The flat connection on $\mathbb{R}^n$ has both, as example 2.2.3 and example 2.2.5 record. A connection can be torsion-free without being compatible with a given g : on $\mathbb{R}^n$ scale the metric to $\bar{g}$ but keep some non-flat torsion-free ∇ whose symbols are symmetric yet do not satisfy $\partial_k \bar{g}_{ij} = \Gamma_{ki}^l \bar{g}_{lj} + \Gamma_{kj}^l \bar{g}_{il}$, which for the constant Euclidean metric reads $\Gamma_{ki}^l \delta_{lj} + \Gamma_{kj}^l \delta_{il} = 0$, i.e. $\Gamma_{ki}^j + \Gamma_{kj}^i = 0$; a nonzero symmetric Γ violating this fails compatibility while retaining torsion-freeness. Conversely a connection can be compatible with g while carrying torsion, since the derivative identity of proposition 2.2.6 constrains only the symmetric-in- (i, j) combination $\Gamma_{ki}^l g_{lj} + \Gamma_{kj}^l g_{il}$ and leaves the torsion $\Gamma_{ij}^k - \Gamma_{ji}^k$ unconstrained. That the two conditions are independent, yet jointly determine a connection, is what makes the theorem of the next section possible: the symmetric part of Γ is fixed by compatibility and the antisymmetric part by torsion-freeness.

2.3 The Fundamental Theorem of Riemannian Geometry

Two structures now sit on a (pseudo-)Riemannian manifold, and so far they have led separate lives. The metric g measures lengths and angles; a connection ∇ differentiates one vector field along another. The metric knows nothing about differentiation, and a general affine connection knows nothing about the metric: a smooth manifold carries a vast supply of connections (their difference is any $(1, 2)$ -tensor, by proposition 2.1.5), and none of them is singled out by the smooth structure alone. On a Riemannian manifold, though, we have grown accustomed to differentiating vectors in a way that respects lengths, as one does for the position and velocity of a curve in $\mathbb{R}^n$. The question is whether the metric picks out a connection of its own, and if so, how many.

Two conditions on a connection have already earned their names. Metric compatibility (definition 2.2.4) is the demand that ∇ and g agree, so that differentiating an inner product obeys a product rule and, equivalently, parallel transport preserves lengths; without it, a vector could grow or shrink under transport and the geometry of ∇ would drift away from the geometry of g . Torsion-freeness (definition 2.2.1) is the demand that the connection add no antisymmetric part beyond the Lie bracket, so that second covariant derivatives are as symmetric as possible and, in coordinates, $\Gamma_{ij}^k = \Gamma_{ji}^k$. Each

condition is natural on its own; neither, on its own, determines ∇ . The content of this section is that the two together determine ∇ completely and are simultaneously satisfiable. The argument is short and self-contained, and it is worth watching how the two conditions interlock: the same three-term cyclic manipulation that proves uniqueness also produces the formula that proves existence.

Theorem 2.3.1: Fundamental Theorem of Riemannian Geometry

Let (M, g) be a (pseudo-)Riemannian manifold. There exists a unique affine connection ∇ on TM that is both torsion-free and compatible with g . It is characterized by the *Koszul formula*: for all vector fields X, Y, Z on M ,

$$2 \langle \nabla_X Y, Z \rangle = X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle X, [Y, Z] \rangle + \langle Y, [Z, X] \rangle + \langle Z, [X, Y] \rangle. \quad (2.3)$$

Proof:

The proof has two halves. First uniqueness: any torsion-free, metric-compatible connection is forced to satisfy (2.3), and the right-hand side determines $\nabla_X Y$ because g is nondegenerate. Then existence: the right-hand side of (2.3) defines a $\nabla_X Y$, and this ∇ is verified to be a connection that is torsion-free and metric-compatible.

Step 1: derivation of the Koszul formula (uniqueness). Suppose ∇ is torsion-free and compatible with g . Fix vector fields X, Y, Z . Compatibility (definition 2.2.4) applied to the three pairs, cyclically, gives

$$\begin{aligned} X \langle Y, Z \rangle &= \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle, \\ Y \langle Z, X \rangle &= \langle \nabla_Y Z, X \rangle + \langle Z, \nabla_Y X \rangle, \\ Z \langle X, Y \rangle &= \langle \nabla_Z X, Y \rangle + \langle X, \nabla_Z Y \rangle \end{aligned}$$

Add the first two and subtract the third:

$$X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle + \langle \nabla_Y Z, X \rangle + \langle Z, \nabla_Y X \rangle - \langle \nabla_Z X, Y \rangle - \langle X, \nabla_Z Y \rangle$$

Now eliminate every covariant derivative whose subscript is not X by torsion-freeness (definition 2.2.1), which supplies $\nabla_X Z = \nabla_Z X + [X, Z]$, $\nabla_Y X = \nabla_X Y + [Y, X]$, and $\nabla_Y Z = \nabla_Z Y + [Y, Z]$. Substituting these into the second, third, and fourth terms on the right,

$$\begin{aligned} X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle &= \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_Z X \rangle + \langle Y, [X, Z] \rangle + \langle \nabla_Z Y, X \rangle + \langle [Y, Z], X \rangle \\ &\quad + \langle Z, \nabla_X Y \rangle + \langle Z, [Y, X] \rangle - \langle \nabla_Z X, Y \rangle - \langle X, \nabla_Z Y \rangle. \end{aligned}$$

The pair $\langle Y, \nabla_Z X \rangle - \langle \nabla_Z X, Y \rangle$ vanishes and the pair $\langle \nabla_Z Y, X \rangle - \langle X, \nabla_Z Y \rangle$ vanishes, by symmetry of g ; the two remaining $\nabla_X Y$ terms combine as $\langle \nabla_X Y, Z \rangle + \langle Z, \nabla_X Y \rangle = 2 \langle \nabla_X Y, Z \rangle$. What survives is

$$X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle = 2 \langle \nabla_X Y, Z \rangle + \langle Y, [X, Z] \rangle + \langle X, [Y, Z] \rangle + \langle Z, [Y, X] \rangle$$

Solve for $2 \langle \nabla_X Y, Z \rangle$, then normalize the brackets using $[X, Z] = -[Z, X]$ and $[Y, X] = -[X, Y]$:

$$2 \langle \nabla_X Y, Z \rangle = X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle - \langle X, [Y, Z] \rangle + \langle Y, [Z, X] \rangle + \langle Z, [X, Y] \rangle,$$

which is (2.3). Every quantity on the right depends only on g , on the bracket, and on X, Y, Z ; no reference to ∇ survives. Hence a torsion-free metric-compatible connection must satisfy (2.3).

This already gives uniqueness. Fix X, Y and a point p . As Z ranges over all vector fields, the right-hand side of (2.3) assigns to each $Z_p \in T_p M$ a number, and by inspection this assignment is $\mathbb{R}$ -linear in Z_p once X, Y are fixed (each term is $\mathbb{R}$ -linear in Z , and the value at p depends only on Z_p , a point verified in Step 2 below). Thus the right-hand side is the pairing of a well-defined covector with Z_p . Because g_p is nondegenerate (definition 1.1.3), the equation $2\langle W, Z_p \rangle = (\text{that covector})(Z_p)$ for all Z_p has at most one solution $W = (\nabla_X Y)_p$. So any two torsion-free metric-compatible connections agree.

Step 2: the Koszul formula defines a connection (existence). Define $\nabla_X Y$ by requiring that (2.3) hold for all Z ; concretely, for fixed X, Y let $\omega(Z)$ denote the right-hand side of (2.3), and set $\nabla_X Y$ to be the unique vector field with $2\langle \nabla_X Y, Z \rangle = \omega(Z)$ for all Z . For this to make sense two things must be checked: that ω is a 1-form (that is, $C^\infty(M)$ -linear and pointwise in Z , so that the musical isomorphism $\sharp$ of definition 1.4.1 produces a vector field $\nabla_X Y := \frac{1}{2}\omega^\sharp$), and that the resulting operator is a connection.

First, ω is additive in Z termwise. For $C^\infty(M)$ -linearity, replace Z by fZ with $f \in C^\infty(M)$. Each of the six terms of ω either scales by f or produces one additional term through a Leibniz rule, using $[Y, fZ] = f[Y, Z] + (Yf)Z$ and $[fZ, X] = f[Z, X] - (Xf)Z$:

$$\begin{aligned} X\langle Y, fZ \rangle &= fX\langle Y, Z \rangle + (Xf)\langle Y, Z \rangle, & Y\langle fZ, X \rangle &= fY\langle Z, X \rangle + (Yf)\langle Z, X \rangle, \\ -fZ\langle X, Y \rangle &= f(-Z\langle X, Y \rangle), & -\langle X, [Y, fZ] \rangle &= -f\langle X, [Y, Z] \rangle - (Yf)\langle X, Z \rangle, \\ \langle Y, [fZ, X] \rangle &= f\langle Y, [Z, X] \rangle - (Xf)\langle Y, Z \rangle, & \langle fZ, [X, Y] \rangle &= f\langle Z, [X, Y] \rangle \end{aligned}$$

The four terms beyond $f\omega(Z)$ are $(Xf)\langle Y, Z \rangle$, $(Yf)\langle Z, X \rangle$, $-(Yf)\langle X, Z \rangle$, and $-(Xf)\langle Y, Z \rangle$. By symmetry of g , $\langle Y, Z \rangle = \langle Z, Y \rangle$ and $\langle Z, X \rangle = \langle X, Z \rangle$, so $(Xf)\langle Y, Z \rangle - (Xf)\langle Y, Z \rangle = 0$ and $(Yf)\langle Z, X \rangle - (Yf)\langle X, Z \rangle = 0$. All four cancel, so $\omega(fZ) = f\omega(Z)$. Hence ω is $C^\infty(M)$ -linear in Z , and by the tensor characterization of 1-forms ([3]) it is a 1-form, its value at p depends only on Z_p , and $\nabla_X Y := \frac{1}{2}\omega^\sharp$ is a well-defined smooth vector field.

Now verify the connection axioms (definition 2.1.1). Write $\Phi(X, Y, Z)$ for the right-hand side of (2.3), so $2\langle \nabla_X Y, Z \rangle = \Phi(X, Y, Z)$ for all Z .

$\mathbb{R}$ -bilinearity. Φ is $\mathbb{R}$ -linear in X and in Y termwise (each metric-derivative and bracket term is $\mathbb{R}$ -linear in each slot), so $\nabla_X Y$ is $\mathbb{R}$ -linear in X and in Y .

$C^\infty(M)$ -linearity in X . Replace X by fX . The terms of $\Phi(fX, Y, Z)$ that differ from $f\Phi(X, Y, Z)$ arise where X acts as a derivation or enters a bracket, using $[Z, fX] = f[Z, X] + (Zf)X$ and $[fX, Y] = f[X, Y] - (Yf)X$:

$$\begin{aligned} Y\langle Z, fX \rangle &= fY\langle Z, X \rangle + (Yf)\langle Z, X \rangle, & -Z\langle fX, Y \rangle &= -fZ\langle X, Y \rangle - (Zf)\langle X, Y \rangle, \\ \langle Y, [Z, fX] \rangle &= f\langle Y, [Z, X] \rangle + (Zf)\langle Y, X \rangle, & \langle Z, [fX, Y] \rangle &= f\langle Z, [X, Y] \rangle - (Yf)\langle Z, X \rangle, \end{aligned}$$

while $X\langle Y, Z \rangle \mapsto fX\langle Y, Z \rangle$ and $-\langle fX, [Y, Z] \rangle = -f\langle X, [Y, Z] \rangle$ produce no extra terms. The four extra terms are $(Yf)\langle Z, X \rangle$, $-(Zf)\langle X, Y \rangle$, $+(Zf)\langle Y, X \rangle$, and $-(Yf)\langle Z, X \rangle$. The two (Yf) terms cancel outright, and by symmetry of g the two (Zf) terms give $-(Zf)\langle X, Y \rangle + (Zf)\langle Y, X \rangle = 0$. Hence $\Phi(fX, Y, Z) = f\Phi(X, Y, Z)$ for all Z , so $\nabla_{fX} Y = f\nabla_X Y$.

Leibniz in Y . Replace Y by fY . The terms that differ from $f\Phi(X, Y, Z)$ arise where Y acts as a derivation or enters a bracket, using $[fY, Z] = f[Y, Z] - (Zf)Y$ and $[X, fY] = f[X, Y] + (Xf)Y$:

$$\begin{aligned} X\langle fY, Z \rangle &= fX\langle Y, Z \rangle + (Xf)\langle Y, Z \rangle, & -Z\langle X, fY \rangle &= -fZ\langle X, Y \rangle - (Zf)\langle X, Y \rangle, \\ -\langle X, [fY, Z] \rangle &= -f\langle X, [Y, Z] \rangle + (Zf)\langle X, Y \rangle, & \langle Z, [X, fY] \rangle &= f\langle Z, [X, Y] \rangle + (Xf)\langle Z, Y \rangle, \end{aligned}$$

while $Y \langle Z, X \rangle \mapsto fY \langle Z, X \rangle$ and $\langle fY, [Z, X] \rangle = f \langle Y, [Z, X] \rangle$ produce no extra terms. The four extra terms are $(Xf) \langle Y, Z \rangle$, $-(Zf) \langle X, Y \rangle$, $+(Zf) \langle X, Y \rangle$, and $(Xf) \langle Z, Y \rangle$. The two (Zf) terms cancel, and by symmetry of g the two (Xf) terms combine to $2(Xf) \langle Y, Z \rangle$. Therefore

$$\Phi(X, fY, Z) = f \Phi(X, Y, Z) + 2(Xf) \langle Y, Z \rangle = 2f \langle \nabla_X Y, Z \rangle + 2(Xf) \langle Y, Z \rangle,$$

so $2 \langle \nabla_X(fY), Z \rangle = 2 \langle f \nabla_X Y + (Xf)Y, Z \rangle$ for all Z , and nondegeneracy gives $\nabla_X(fY) = f \nabla_X Y + (Xf)Y$. Thus ∇ is a connection.

Step 3: torsion-freeness. By the defining relation,

$$2 \langle \nabla_X Y - \nabla_Y X, Z \rangle = \Phi(X, Y, Z) - \Phi(Y, X, Z)$$

Interchanging X and Y in (2.3), the first three (symmetric) metric-derivative terms $X \langle Y, Z \rangle + Y \langle Z, X \rangle - Z \langle X, Y \rangle$ are invariant under $X \leftrightarrow Y$ (using symmetry of g in the last), so they cancel in the difference. Among the bracket terms, $-\langle X, [Y, Z] \rangle \rightarrow -\langle Y, [X, Z] \rangle$ and $\langle Y, [Z, X] \rangle \rightarrow \langle X, [Z, Y] \rangle$ and $\langle Z, [X, Y] \rangle \rightarrow \langle Z, [Y, X] \rangle$. Hence

$$\Phi(X, Y, Z) - \Phi(Y, X, Z) = -\langle X, [Y, Z] \rangle + \langle Y, [Z, X] \rangle + \langle Z, [X, Y] \rangle + \langle Y, [X, Z] \rangle - \langle X, [Z, Y] \rangle - \langle Z, [Y, X] \rangle$$

Now $[Z, Y] = -[Y, Z]$ and $[X, Z] = -[Z, X]$ and $[Y, X] = -[X, Y]$, so $-\langle X, [Y, Z] \rangle - \langle X, [Z, Y] \rangle = 0$, $\langle Y, [Z, X] \rangle + \langle Y, [X, Z] \rangle = 0$, and $\langle Z, [X, Y] \rangle - \langle Z, [Y, X] \rangle = 2 \langle Z, [X, Y] \rangle$. Therefore

$$2 \langle \nabla_X Y - \nabla_Y X, Z \rangle = 2 \langle [X, Y], Z \rangle \quad \text{for all } Z,$$

and nondegeneracy gives $\nabla_X Y - \nabla_Y X = [X, Y]$, that is, the torsion vanishes (definition 2.2.1).

Step 4: metric compatibility. Fix X, Y, Z . From the defining relation, $2 \langle \nabla_X Y, Z \rangle = \Phi(X, Y, Z)$ and $2 \langle Y, \nabla_X Z \rangle = 2 \langle \nabla_X Z, Y \rangle = \Phi(X, Z, Y)$. Add:

$$2(\langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle) = \Phi(X, Y, Z) + \Phi(X, Z, Y)$$

Writing out both and using symmetry of g , the term $X \langle Y, Z \rangle$ appears in $\Phi(X, Y, Z)$ and $X \langle Z, Y \rangle$ appears in $\Phi(X, Z, Y)$, together giving $2X \langle Y, Z \rangle$. The remaining metric-derivative terms are $Y \langle Z, X \rangle - Z \langle X, Y \rangle$ from the first and $Z \langle Y, X \rangle - Y \langle X, Z \rangle$ from the second; since $\langle Z, X \rangle = \langle X, Z \rangle$ and $\langle Y, X \rangle = \langle X, Y \rangle$, these cancel in pairs. For the bracket terms, $\Phi(X, Y, Z)$ contributes $-\langle X, [Y, Z] \rangle + \langle Y, [Z, X] \rangle + \langle Z, [X, Y] \rangle$ and $\Phi(X, Z, Y)$ contributes $-\langle X, [Z, Y] \rangle + \langle Z, [Y, X] \rangle + \langle Y, [X, Z] \rangle$; using $[Z, Y] = -[Y, Z]$, $[Y, X] = -[X, Y]$, $[X, Z] = -[Z, X]$, each pair cancels. Hence

$$\langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle = X \langle Y, Z \rangle,$$

which is exactly the compatibility identity of definition 2.2.4. So ∇ is metric-compatible.

Existence and uniqueness are both established, completing the proof.

The proof deserves a second look for what makes it work. Uniqueness is the sturdier half: it uses nothing about M beyond nondegeneracy of g , and the same cyclic combination of the three compatibility equations that eliminates the unknown ∇ from the right-hand side is the standard move whenever a symmetric bilinear form is to be recovered from its derivatives, as in the polarization identity for an inner product. Existence is the more delicate half only because one must confirm that the formula, read as a definition, respects the tensorial and Leibniz bookkeeping of a connection; the cancellations in Steps 2 through 4 are all the same cancellation, driven by the symmetry of

g and the antisymmetry of the bracket. Notice that positive-definiteness was never invoked, only nondegeneracy: the theorem holds verbatim for Lorentzian and general pseudo-Riemannian metrics, which is why every spacetime carries a canonical connection.

Definition 2.3.2: Levi-Civita Connection

Let (M, g) be a (pseudo-)Riemannian manifold. The unique torsion-free, metric-compatible affine connection on TM provided by theorem 2.3.1 is called the *Levi-Civita connection* of (M, g) , also called the *Riemannian connection*. It is denoted ∇ (or ∇^g when the metric must be displayed).

From now on, the symbol ∇ on a (pseudo-)Riemannian manifold means the Levi-Civita connection unless another connection is named. Two consequences of definition 2.3.2 organize the rest of the theory. First, every metric invariant of ∇ , the geodesics, parallel transport, and the curvature tensor built later, is an invariant of g alone, because ∇ is manufactured from g ; an isometry φ (definition 1.3.1) therefore carries the Levi-Civita connection of the source to that of the target, since $\varphi^*\tilde{g} = g$ forces φ to intertwine the two connections that g and $\tilde{g}$ uniquely determine. Second, the connection is a local object: the Koszul formula reads off $\nabla_X Y$ from g and derivatives of g on any neighborhood, so ∇ on an open set U is the Levi-Civita connection of $(U, g|_U)$. Both statements will be used repeatedly and without further comment.

To make the connection computable one converts the Koszul formula into coordinates. In a chart the coordinate frame (∂_i) has vanishing brackets, $[\partial_i, \partial_j] = 0$, so (2.3) collapses to a formula for the Christoffel symbols in terms of the metric components and their first derivatives. Throughout what follows the Einstein summation convention is in force: an index appearing once as an upper and once as a lower index is summed over $1, \dots, n$.

Proposition 2.3.3: Christoffel Symbols of the Levi-Civita Connection

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ . In any smooth chart $(U, (x^i))$, with metric components $g_{ij} = \langle \partial_i, \partial_j \rangle$ and inverse matrix (g^{ij}) , the Christoffel symbols of ∇ (definition 2.1.3) are

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

In particular $\Gamma_{ij}^k = \Gamma_{ji}^k$, and the connection is symmetric in the lower indices.

Proof:

Apply the Koszul formula (2.3) to the coordinate frame, taking $X = \partial_i$, $Y = \partial_j$, $Z = \partial_l$. Every Lie bracket of coordinate fields vanishes, $[\partial_i, \partial_j] = 0$, so the three bracket terms drop out and the three metric-derivative terms survive:

$$2 \langle \nabla_{\partial_i} \partial_j, \partial_l \rangle = \partial_i \langle \partial_j, \partial_l \rangle + \partial_j \langle \partial_l, \partial_i \rangle - \partial_l \langle \partial_i, \partial_j \rangle = \partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}$$

By definition 2.1.3, $\nabla_{\partial_i} \partial_j = \Gamma_{ij}^m \partial_m$, so the left-hand side is

$$2 \langle \Gamma_{ij}^m \partial_m, \partial_l \rangle = 2 \Gamma_{ij}^m \langle \partial_m, \partial_l \rangle = 2 \Gamma_{ij}^m g_{ml}$$

Equating the two expressions,

$$2 \Gamma_{ij}^m g_{ml} = \partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}$$

To solve for Γ_{ij}^k , contract both sides with the inverse metric g^{kl} , summing over l . Since $g^{kl}g_{ml} = \delta_m^k$,

$$2\Gamma_{ij}^m\delta_m^k = 2\Gamma_{ij}^k = g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}),$$

which is the stated formula after dividing by 2. Symmetry in i, j is visible from the right-hand side: interchanging i and j swaps the first two terms $\partial_i g_{jl} \leftrightarrow \partial_j g_{il}$ and fixes the third, using $g_{ij} = g_{ji}$; the sum is unchanged. Equivalently, symmetry follows from torsion-freeness of ∇ together with $[\partial_i, \partial_j] = 0$, as required.

The formula for Γ_{ij}^k is the working definition of the Levi-Civita connection in practice: to differentiate covariantly, one reads off g_{ij} from the chart, inverts the matrix to get g^{kl} , and differentiates. Everything downstream, the geodesic equation, parallel transport, and curvature, is assembled from these symbols. Because the Γ_{ij}^k are built from first derivatives of the metric, they encode no invariant information at a single point (they can be made to vanish at any chosen point by passing to normal coordinates, developed in section 3.2); the invariant content of the metric first appears in the second derivatives of g , which is where curvature will come from.

Example 2.3.4: Christoffel Symbols of a Surface of Revolution

Consider a surface of revolution with the warped metric

$$g = dr^2 + f(r)^2 d\theta^2, \quad f(r) > 0,$$

in coordinates $(x^1, x^2) = (r, \theta)$; this is the form computed in example 1.5.4. The metric matrix and its inverse are

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & f(r)^2 \end{pmatrix}, \quad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & f(r)^{-2} \end{pmatrix}$$

The only nonzero partial derivatives of the components are $\partial_r g_{\theta\theta} = 2ff'$, where $f' = df/dr$. Since (g^{ij}) is diagonal, the sum over l in proposition 2.3.3 collapses to the single term $l = k$:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kk} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) \quad (\text{no sum on } k)$$

Run through the components. For $k = r$ ($g^{rr} = 1$): the only surviving term needs a nonzero ∂g , so $\Gamma_{\theta\theta}^r = \frac{1}{2} \cdot 1 \cdot (-\partial_r g_{\theta\theta}) = -ff'$, and $\Gamma_{rr}^r = \Gamma_{r\theta}^r = 0$. For $k = \theta$ ($g^{\theta\theta} = f^{-2}$): $\Gamma_{r\theta}^\theta = \frac{1}{2} f^{-2} (\partial_r g_{\theta\theta} + \partial_\theta g_{r\theta} - \partial_\theta g_{r\theta}) = \frac{1}{2} f^{-2} (2ff') = f'/f$, and by symmetry $\Gamma_{\theta r}^\theta = f'/f$, while $\Gamma_{rr}^\theta = \Gamma_{\theta\theta}^\theta = 0$. The nonzero symbols are therefore

$$\Gamma_{\theta\theta}^r = -f(r)f'(r), \quad \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{f'(r)}{f(r)}$$

Two checks confirm the answer. Symmetry in the lower indices holds, as proposition 2.3.3 guarantees. And in the flat case $f(r) = r$ (polar coordinates on $\mathbb{R}^2$, whose induced metric is exactly $dr^2 + r^2 d\theta^2$) these become $\Gamma_{\theta\theta}^r = -r$ and $\Gamma_{r\theta}^\theta = 1/r$, the familiar Christoffel symbols of the plane in polar coordinates; the metric $\bar{g} = \delta_{ij} dx^i dx^j$ of example 1.1.2 carries no curvature, yet its Christoffel symbols in polar coordinates are nonzero, a reminder that the Γ_{ij}^k measure the chart, not the geometry.

The round 2-sphere is a special case of the previous computation, and its Christoffel symbols are worth having explicitly, since they drive the geodesic and curvature calculations of later chapters.

Example 2.3.5: Christoffel Symbols of the Round Sphere

On the round unit 2-sphere S^2 , away from the poles, use the standard spherical coordinates $(x^1, x^2) = (\theta, \varphi)$ in which the induced metric is

$$g = d\theta^2 + \sin^2 \theta d\varphi^2, \quad 0 < \theta < \pi,$$

with θ the polar angle measured from the north pole and φ the azimuthal angle. The metric matrix and its inverse are

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}, \quad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^{-2} \theta \end{pmatrix}$$

Only one component depends on the coordinates, and its single nonzero partial derivative is

$$\partial_\theta g_{\varphi\varphi} = 2 \sin \theta \cos \theta,$$

while every other $\partial_k g_{ij}$ vanishes. Since (g^{ij}) is diagonal, the contraction over l in proposition 2.3.3 keeps only the term $l = k$, giving

$$\Gamma_{ij}^k = \frac{1}{2} g^{kk} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) \quad (\text{no sum on } k)$$

For $k = \theta$ ($g^{\theta\theta} = 1$) the only combination that meets a nonzero derivative is $i = j = \varphi$, where the two symmetric terms $\partial_\varphi g_{\varphi\theta}$ vanish and the subtracted term survives:

$$\Gamma_{\varphi\varphi}^\theta = \frac{1}{2} \cdot 1 \cdot (-\partial_\theta g_{\varphi\varphi}) = -\frac{1}{2} (2 \sin \theta \cos \theta) = -\sin \theta \cos \theta,$$

and $\Gamma_{\theta\theta}^\theta = \Gamma_{\theta\varphi}^\theta = 0$. For $k = \varphi$ ($g^{\varphi\varphi} = \sin^{-2} \theta$) the surviving case is $\{i, j\} = \{\theta, \varphi\}$:

$$\Gamma_{\theta\varphi}^\varphi = \frac{1}{2} \sin^{-2} \theta (\partial_\theta g_{\varphi\varphi} + \partial_\varphi g_{\theta\varphi} - \partial_\varphi g_{\theta\varphi}) = \frac{1}{2} \sin^{-2} \theta (2 \sin \theta \cos \theta) = \cot \theta,$$

with $\Gamma_{\varphi\theta}^\varphi = \cot \theta$ by symmetry, and $\Gamma_{\theta\theta}^\varphi = \Gamma_{\varphi\varphi}^\varphi = 0$. The nonzero Christoffel symbols of the round sphere are therefore

$$\Gamma_{\varphi\varphi}^\theta = -\sin \theta \cos \theta, \quad \Gamma_{\theta\varphi}^\varphi = \Gamma_{\varphi\theta}^\varphi = \cot \theta,$$

and all others vanish. This is the case $f(\theta) = \sin \theta$ of example 2.3.4, with the radial coordinate r there playing the role of θ here: substituting $f = \sin \theta$ gives $-ff' = -\sin \theta \cos \theta$ and $f'/f = \cot \theta$, matching term by term, since the round sphere is the surface of revolution generated by a unit-speed meridian. The blow-up of $\cot \theta$ as $\theta \rightarrow 0, \pi$ reflects the breakdown of the coordinates at the poles, not of the geometry, exactly as $1/r$ does at the origin in polar coordinates.

2.4 Parallel Transport

The obstruction that motivated the connection in the first place was the absence of any canonical way to compare tangent vectors at different points of a manifold: $T_p M$ and $T_q M$ are distinct vector spaces with no preferred identification. A connection repairs this infinitesimally, by prescribing how to differentiate a vector field along a curve. Parallel transport is the finite, integrated form of that repair. Given a curve γ from p to q and a tangent vector at p , one drags the vector along γ so that it never changes from the connection's point of view, that is, so that its covariant derivative along γ

stays zero. The vector that arrives at q is the parallel transport of the starting vector. This furnishes exactly the missing identification $T_p M \rightarrow T_q M$, with one caveat that will run through the rest of the theory: the identification depends on the curve. Transporting the same vector along two different paths from p to q can produce two different vectors at q , and the discrepancy is the first visible symptom of curvature.

The construction is a matter of solving a linear ordinary differential equation. In a chart the condition $D_t V = 0$ becomes a first-order linear system for the component functions of V , and the existence and uniqueness theory for such systems hands over, at once, that a parallel field is determined by its value at any single time and that parallel transport is a linear isomorphism between tangent spaces. When the connection is the Levi-Civita connection of a metric, transport preserves lengths and angles: it is a linear isometry, the promised third face of metric compatibility. This section carries out the construction, records these two properties in full, and then extends the covariant derivative from the tangent bundle to all tensor bundles, so that ∇ can differentiate metrics, forms, and endomorphisms, and does so compatibly with contraction and with the raising and lowering of indices. Throughout, M is a smooth n -manifold, ∇ an affine connection on TM (definition 2.1.1), and $\gamma: I \rightarrow M$ a smooth curve on an interval $I \subseteq \mathbb{R}$; the covariant derivative D_t along γ is the operator of definition 2.1.6. The Einstein summation convention, summing over any index appearing once up and once down, is in force.

Definition 2.4.1: Parallel Vector Field and Parallel Transport

Let ∇ be an affine connection on TM and $\gamma: I \rightarrow M$ a smooth curve. A vector field $V \in \mathfrak{X}(\gamma)$ along γ is *parallel* (along γ , with respect to ∇) if

$$D_t V \equiv 0 \quad \text{on } I$$

For $t_0, t_1 \in I$, the *parallel transport* from $\gamma(t_0)$ to $\gamma(t_1)$ along γ is the map

$$P_{t_0, t_1}^\gamma: T_{\gamma(t_0)} M \rightarrow T_{\gamma(t_1)} M, \quad P_{t_0, t_1}^\gamma(v) = V(t_1),$$

where V is the parallel field along γ with $V(t_0) = v$. This is well defined once every such v extends to a unique parallel field, which is established in theorem 2.4.2 below.

A parallel field is one that the connection reads as constant along the curve. On $\mathbb{R}^n$ with the Euclidean connection $\bar{\nabla}$ the condition $D_t V = 0$ reads $\dot{V}^k = 0$ in Cartesian coordinates (all Γ_{ij}^k vanish), so parallel fields are exactly those with constant Cartesian components, and parallel transport is ordinary translation of a vector to a new base point, keeping its Euclidean components fixed. This is the picture the word *parallel* is meant to evoke: transporting a vector while keeping it, from the flat connection's vantage, pointed the same way. On a curved manifold no global Cartesian frame is available, and the condition $D_t V = 0$ couples the components of V to the Christoffel symbols along γ ; the resulting fields are no longer given by any fixed-components recipe, and, as the sphere example below shows, transporting a vector around a closed loop can return it rotated. The map P_{t_0, t_1}^γ is the whole point of the construction: it converts the infinitesimal comparison supplied by ∇ into a bona fide linear map between the two tangent spaces $T_{\gamma(t_0)} M$ and $T_{\gamma(t_1)} M$, an identification that the bare manifold could not provide.

The existence and uniqueness of parallel fields, and the linearity and invertibility of P_{t_0, t_1}^γ , all follow from reading $D_t V = 0$ as a linear system and invoking the standard theory of linear ordinary differential equations.

Theorem 2.4.2: Existence and Uniqueness of Parallel Transport

Let ∇ be an affine connection on TM and $\gamma: I \rightarrow M$ a smooth curve. For every $t_0 \in I$ and every $v \in T_{\gamma(t_0)}M$ there is a unique parallel vector field $V \in \mathfrak{X}(\gamma)$ along γ with $V(t_0) = v$. Consequently, for all $t_0, t_1 \in I$ the parallel transport

$$P_{t_0, t_1}^\gamma: T_{\gamma(t_0)}M \rightarrow T_{\gamma(t_1)}M$$

is a well-defined linear isomorphism, with inverse P_{t_1, t_0}^γ and with the composition law $P_{t_1, t_2}^\gamma \circ P_{t_0, t_1}^\gamma = P_{t_0, t_2}^\gamma$ for all $t_0, t_1, t_2 \in I$.

Proof:

Step 1: reduction to a linear system on a chart. Suppose first that the image $\gamma(I)$ lies in the domain of a single chart $(U, (x^1, \dots, x^n))$. Writing $V(t) = V^k(t) \partial_k|_{\gamma(t)}$, the local formula (2.2) for the covariant derivative gives

$$D_t V = \left(\dot{V}^k + \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i(t) V^j(t) \right) \partial_k|_{\gamma(t)}$$

Hence V is parallel if and only if its components satisfy the system

$$\dot{V}^k(t) = -\Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i(t) V^j(t), \quad k = 1, \dots, n \quad (2.4)$$

Set $A^k_j(t) := -\Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i(t)$. Each A^k_j is a smooth (in particular continuous) real function on I , since γ , $\dot{\gamma}$, and the Christoffel symbols are smooth. In the vector notation $V(t) = (V^1(t), \dots, V^n(t)) \in \mathbb{R}^n$, the system (2.4) is the linear ordinary differential equation

$$\dot{V}(t) = A(t) V(t), \quad (2.5)$$

with $A(t) = (A^k_j(t))$ a matrix of continuous functions on I .

Step 2: existence and uniqueness on a chart. For a linear system (2.5) with continuous coefficient matrix $A(\cdot)$ on an interval I and any prescribed $t_0 \in I$, $v \in \mathbb{R}^n$, the existence-and-uniqueness theorem for linear ordinary differential equations provides a unique solution $V: I \rightarrow \mathbb{R}^n$ with $V(t_0) = v$, and this solution is defined on all of I . Linearity of the equation is what guarantees the solution does not escape to infinity in finite time, so no maximal-interval shrinking occurs: the solution exists on the full parameter interval, not on a proper subinterval only. The solution is smooth because $A(\cdot)$ is smooth. Translating back, $V(t) = V^k(t) \partial_k|_{\gamma(t)}$ is the unique parallel field along γ with $V(t_0) = v$, establishing the theorem when $\gamma(I)$ lies in one chart.

Step 3: patching over the general curve. For a general smooth curve, cover the compact image of each closed subinterval by finitely many charts and propagate the solution across the overlaps. Precisely, fix $t_0 \in I$ and $v \in T_{\gamma(t_0)}M$. Consider the set

$$J = \{ t \in I : \text{a unique parallel field along } \gamma|_{[t_0, t]} \text{ (or } \gamma|_{[t, t_0]}) \text{ with value } v \text{ at } t_0 \text{ exists} \}$$

The point t_0 lies in J . For any $s \in I$, the point $\gamma(s)$ lies in some chart domain U_s , and $\gamma^{-1}(U_s)$ is an open neighborhood of s in I containing an interval $(s - \varepsilon, s + \varepsilon) \cap I$ whose image lies in U_s ; by Step 2 a unique parallel field exists over that interval with any prescribed value at any one of its points. Thus if $s \in J$ then, using the value already determined at s as the new initial condition, the solution extends uniquely to $(s - \varepsilon, s + \varepsilon) \cap I$, so J is open in I . It is also closed:

if s is a limit point of J lying in I , choose a chart interval about s as above meeting J at some point s' ; the unique solution on the chart interval agrees with the already-constructed field at s' by uniqueness (Step 2), hence extends the field to s , so $s \in J$. Since I is an interval, hence connected, and J is nonempty, open, and closed in I , one concludes $J = I$. The resulting field V is parallel on all of I and satisfies $V(t_0) = v$, and it is unique because any two such fields agree on the connected set where both the argument applies, namely all of I .

Step 4: linearity and invertibility of P_{t_0, t_1}^γ . Fix t_0, t_1 . For $v \in T_{\gamma(t_0)}M$ let V_v denote the unique parallel field with $V_v(t_0) = v$; then $P_{t_0, t_1}^\gamma(v) = V_v(t_1)$. If $v, w \in T_{\gamma(t_0)}M$ and $a, b \in \mathbb{R}$, then $aV_v + bV_w$ is parallel by $\mathbb{R}$ -linearity of D_t (definition 2.1.6) and has value $av + bw$ at t_0 ; by uniqueness it equals V_{av+bw} . Evaluating at t_1 ,

$$P_{t_0, t_1}^\gamma(av + bw) = (aV_v + bV_w)(t_1) = aV_v(t_1) + bV_w(t_1) = aP_{t_0, t_1}^\gamma(v) + bP_{t_0, t_1}^\gamma(w),$$

so P_{t_0, t_1}^γ is linear. For the composition law, let V be parallel with $V(t_0) = v$; then $V(t_1) = P_{t_0, t_1}^\gamma(v)$, and since the *same* field V is the unique parallel field with value $V(t_1)$ at t_1 , one has $P_{t_1, t_2}^\gamma(V(t_1)) = V(t_2) = P_{t_0, t_2}^\gamma(v)$. As v was arbitrary, $P_{t_1, t_2}^\gamma \circ P_{t_0, t_1}^\gamma = P_{t_0, t_2}^\gamma$. Taking $t_2 = t_0$ gives $P_{t_1, t_0}^\gamma \circ P_{t_0, t_1}^\gamma = P_{t_0, t_0}^\gamma = \text{id}_{T_{\gamma(t_0)}M}$, and symmetrically the other composite is the identity, so P_{t_0, t_1}^γ is a linear isomorphism with inverse P_{t_1, t_0}^γ , completing the proof.

The theorem does the work the connection was designed for: it delivers, for each curve γ and each pair of parameter values, a well-defined linear identification of the two tangent spaces along the curve. The isomorphisms P_{t_0, t_1}^γ depend smoothly on the endpoints and compose as a path is traversed, so a single curve organizes its tangent spaces into a coherent one-parameter family of identifications. What the theorem does *not* say is that this identification is independent of the curve. Because (2.5) depends on γ through both the Christoffel symbols evaluated along γ and the velocity $\dot{\gamma}$, two curves with the same endpoints generally produce different transports; the failure of P_{t_0, t_1}^γ to depend only on the endpoints, quantified along an infinitesimal loop, is the curvature of ∇ , to be constructed later. The next example exhibits this path-dependence in a form one can compute by hand.

Example 2.4.3: Parallel Transport on the Round Sphere

Take the unit sphere S^2 with the round metric and its Levi-Civita connection (definition 2.3.2), and transport a tangent vector around a geodesic triangle to see the path-dependence directly. Fix the north pole N and consider a spherical triangle with vertices at N and two points A, B on the equator, whose two meridian sides from N to A and from N to B subtend an angle α at N , and whose third side is the equatorial arc from A to B . All three sides are geodesics (great-circle arcs).

Transport is governed by two features of the Levi-Civita connection established below and in section 3.1. First, along a geodesic σ the velocity $\dot{\sigma}$ is itself parallel, since a geodesic satisfies $D_t \dot{\sigma} = 0$; so a vector making a fixed angle with the geodesic and of fixed length, transported along that geodesic, keeps that angle and length, because transport preserves both length and the inner product with the parallel field $\dot{\sigma}$ (the isometry property proved in proposition 2.4.4). Begin at A with the vector v tangent to the meridian AN , pointing from A toward N . Carry v up the meridian to N : it stays tangent to the meridian, arriving at N pointing back down along NA . Now carry it down the second meridian from N to B : the two meridians meet at N at angle α , and the transported vector, which points along NA at N , therefore makes angle α with the meridian NB ; transported down NB it keeps that angle α , arriving at B as a vector making angle α with the meridian and hence with the southward direction. Finally carry it along the equator from B back to A : the equator is a geodesic and the vector keeps its angle to it, so it returns to A

still rotated by the same angle relative to where it would sit had it been carried only along the meridian.

The upshot is that transport around the closed triangular loop returns v rotated, and the rotation angle equals the angular excess of the triangle. For this triangle the three interior angles are α at N and $\pi/2$ at each of A, B , summing to $\alpha + \pi$, so the excess over π is exactly α , which equals the spherical area enclosed. Transport of v around the loop is thus rotation by $\alpha \neq 0$: the identification $T_A S^2 \rightarrow T_A S^2$ produced by going around the triangle is not the identity. On the flat plane the same loop returns every vector unchanged, since Euclidean transport keeps Cartesian components fixed. The nonzero rotation on S^2 is the concrete signature of curvature, and it shows in a single computable instance that parallel transport depends on the path and not only on the endpoints.

Path-dependence is the qualitative feature of transport. Its quantitative counterpart on a (pseudo-)Riemannian manifold is that, whatever the path, transport by the Levi-Civita connection is rigid: it preserves the metric. This is the third characterization of metric compatibility promised in proposition 2.2.6, and its proof completes the equivalence stated there.

Proposition 2.4.4: Parallel Transport is a Linear Isometry

Let (M, g) be a (pseudo-)Riemannian manifold and ∇ an affine connection on TM . Then ∇ is compatible with g (definition 2.2.4) if and only if, for every smooth curve γ and all t_0, t_1 in its parameter interval, the parallel transport P_{t_0, t_1}^γ is a linear isometry:

$$\langle P_{t_0, t_1}^\gamma(v), P_{t_0, t_1}^\gamma(w) \rangle_g = \langle v, w \rangle_g \quad \text{for all } v, w \in T_{\gamma(t_0)} M$$

In particular the Levi-Civita connection of (M, g) (definition 2.3.2), being metric-compatible, transports vectors isometrically along every curve, preserving lengths and angles.

Proof:

Compatibility implies isometry. Assume ∇ is compatible with g . Let $\gamma: I \rightarrow M$ be a smooth curve and $V, W \in \mathfrak{X}(\gamma)$ two parallel fields along it, so $D_t V = 0$ and $D_t W = 0$. Consider the real function

$$h(t) := \langle V(t), W(t) \rangle_g = g_{\gamma(t)}(V(t), W(t))$$

The compatibility identity of definition 2.2.4 has a version for fields along a curve: for any $V, W \in \mathfrak{X}(\gamma)$,

$$\frac{d}{dt} \langle V, W \rangle_g = \langle D_t V, W \rangle_g + \langle V, D_t W \rangle_g \quad (2.6)$$

To verify (2.6), work in a chart $(U, (x^i))$ where $\gamma(t) = (\gamma^i(t))$, $V = V^k \partial_k|_\gamma$, $W = W^l \partial_l|_\gamma$, and $g_{kl} = g(\partial_k, \partial_l)$. Then $\langle V, W \rangle_g = g_{kl}(\gamma(t)) V^k W^l$, and the ordinary product rule gives

$$\frac{d}{dt} \langle V, W \rangle_g = (\dot{\gamma}^i \partial_i g_{kl}) V^k W^l + g_{kl} \dot{V}^k W^l + g_{kl} V^k \dot{W}^l$$

The coordinate form of metric compatibility, $\partial_i g_{kl} = \Gamma_{ik}^m g_{ml} + \Gamma_{il}^m g_{km}$ from proposition 2.2.6, converts the first term:

$$(\dot{\gamma}^i \partial_i g_{kl}) V^k W^l = \dot{\gamma}^i (\Gamma_{ik}^m g_{ml} + \Gamma_{il}^m g_{km}) V^k W^l = g_{ml} (\Gamma_{ik}^m \dot{\gamma}^i V^k) W^l + g_{km} V^k (\Gamma_{il}^m \dot{\gamma}^i W^l)$$

Adding $g_{kl} \dot{V}^k W^l + g_{kl} V^k \dot{W}^l$ and grouping by W^l and by V^k respectively, the sum becomes

$$g_{ml} (\dot{V}^m + \Gamma_{ik}^m \dot{\gamma}^i V^k) W^l + g_{km} V^k (\dot{W}^m + \Gamma_{il}^m \dot{\gamma}^i W^l) = \langle D_t V, W \rangle_g + \langle V, D_t W \rangle_g,$$

by the local formula (2.2) for $D_t V$ and $D_t W$; this is (2.6). Now for parallel V, W both covariant derivatives vanish, so $h'(t) = \langle D_t V, W \rangle + \langle V, D_t W \rangle = 0$, and h is constant on I . Given $v, w \in T_{\gamma(t_0)} M$, let V, W be the parallel fields with $V(t_0) = v$, $W(t_0) = w$ (theorem 2.4.2); then $V(t_1) = P_{t_0, t_1}^\gamma(v)$ and $W(t_1) = P_{t_0, t_1}^\gamma(w)$, and constancy of h gives

$$\langle P_{t_0, t_1}^\gamma(v), P_{t_0, t_1}^\gamma(w) \rangle_g = h(t_1) = h(t_0) = \langle v, w \rangle_g$$

Thus P_{t_0, t_1}^γ preserves g , and being also a linear isomorphism (theorem 2.4.2) it is a linear isometry.

Isometry implies compatibility. Conversely, assume that P_{t_0, t_1}^γ is a linear isometry for every curve γ and all parameters. To recover the compatibility identity of definition 2.2.4, fix $p \in M$, a tangent vector $X_p \in T_p M$, and vector fields $Y, Z \in \mathfrak{X}(M)$; it suffices to show $X_p \langle Y, Z \rangle = \langle \nabla_{X_p} Y, Z \rangle + \langle Y, \nabla_{X_p} Z \rangle$ at p , since p and X_p are arbitrary and both sides are pointwise in the direction (proposition 2.1.5). Choose a curve γ with $\gamma(0) = p$ and $\dot{\gamma}(0) = X_p$. Let $(e_1, \dots, e_n)$ be a basis of $T_p M$ and let $E_a(t) := P_{0, t}^\gamma(e_a)$ be the parallel fields extending them, so each E_a is parallel, $D_t E_a = 0$, and $(E_1(t), \dots, E_n(t))$ is a basis of $T_{\gamma(t)} M$ for every t (each $P_{0, t}^\gamma$ is an isomorphism). By hypothesis $P_{0, t}^\gamma$ is an isometry, so the functions

$$\langle E_a(t), E_b(t) \rangle_g = \langle P_{0, t}^\gamma(e_a), P_{0, t}^\gamma(e_b) \rangle_g = \langle e_a, e_b \rangle_g$$

are constant in t . Expand Y and Z along γ in this parallel frame: $Y(\gamma(t)) = Y^a(t) E_a(t)$ and $Z(\gamma(t)) = Z^a(t) E_a(t)$ with smooth coefficient functions Y^a, Z^a . Because each E_a is parallel, the covariant derivative along γ of Y restricted to γ is

$$D_t(Y^a E_a) = \dot{Y}^a E_a + Y^a D_t E_a = \dot{Y}^a E_a,$$

and by definition 2.1.6(3) this equals $\nabla_{\dot{\gamma}} Y$; at $t = 0$, $\nabla_{X_p} Y = \dot{Y}^a(0) E_a(0) = \dot{Y}^a(0) e_a$, and similarly $\nabla_{X_p} Z = \dot{Z}^a(0) e_a$. On the other hand, writing $c_{ab} := \langle e_a, e_b \rangle_g = \langle E_a(t), E_b(t) \rangle_g$ (constant in t),

$$\langle Y, Z \rangle_g(\gamma(t)) = \langle Y^a E_a, Z^b E_b \rangle_g = Y^a(t) Z^b(t) c_{ab},$$

so differentiating this function of t and evaluating at 0, where the c_{ab} carry no t -dependence,

$$X_p \langle Y, Z \rangle = \left. \frac{d}{dt} \right|_0 (Y^a Z^b c_{ab}) = \dot{Y}^a(0) Z^b(0) c_{ab} + Y^a(0) \dot{Z}^b(0) c_{ab}$$

The first summand is $\langle \dot{Y}^a(0) e_a, Z^b(0) e_b \rangle_g = \langle \nabla_{X_p} Y, Z \rangle$ and the second is $\langle Y, \nabla_{X_p} Z \rangle$, both at p . Hence $X_p \langle Y, Z \rangle = \langle \nabla_{X_p} Y, Z \rangle + \langle Y, \nabla_{X_p} Z \rangle$, which is the compatibility identity. As p, X_p, Y, Z were arbitrary, ∇ is compatible with g .

The Levi-Civita connection is metric-compatible by definition 2.3.2, so the forward direction applies to it, completing the proof.

The proposition supplies the geometric reading of metric compatibility that its algebraic definition only hinted at. A metric-compatible connection is one whose parallel transport is a rigid motion between tangent spaces: it carries orthonormal bases to orthonormal bases, keeps the length of a transported vector fixed, and keeps the angle between two transported vectors fixed. For the Levi-Civita connection this holds along every curve, so a vector's length is a transport invariant even though the vector's direction, in any external sense, is not. With the equivalence proposition 2.2.6(iii)

now proved, all three formulations of compatibility, $\nabla g = 0$, the Leibniz identity for inner products, and isometric transport, stand on equal footing, and one may invoke whichever is most convenient. The isometry property is what makes the sphere computation of example 2.4.3 legitimate: the angle and length arguments used there rest precisely on transport preserving g .

Example 2.4.5: Isometric Transport and Constant Speed of Geodesics

One immediate consequence of proposition 2.4.4 is worth isolating, as it recurs throughout the geodesic theory. Let γ be a geodesic of a (pseudo-)Riemannian manifold, meaning $D_t \dot{\gamma} = 0$, so that the velocity $\dot{\gamma}$ is a parallel field along γ (definition 2.4.1). Then for any t ,

$$\dot{\gamma}(t) = P_{t_0, t}^\gamma(\dot{\gamma}(t_0)),$$

since both sides are the value at t of the parallel field with initial value $\dot{\gamma}(t_0)$, and these agree by uniqueness (theorem 2.4.2). Applying the isometry property to $v = w = \dot{\gamma}(t_0)$ gives

$$|\dot{\gamma}(t)|_g^2 = \langle \dot{\gamma}(t), \dot{\gamma}(t) \rangle_g = \langle P_{t_0, t}^\gamma \dot{\gamma}(t_0), P_{t_0, t}^\gamma \dot{\gamma}(t_0) \rangle_g = \langle \dot{\gamma}(t_0), \dot{\gamma}(t_0) \rangle_g = |\dot{\gamma}(t_0)|_g^2,$$

so $|\dot{\gamma}|_g$ is constant along γ : geodesics have constant speed. The same conclusion is reached by direct differentiation, since $\frac{d}{dt} \langle \dot{\gamma}, \dot{\gamma} \rangle = 2 \langle D_t \dot{\gamma}, \dot{\gamma} \rangle = 0$ by (2.6) and $D_t \dot{\gamma} = 0$; the transport formulation makes the invariance conceptual rather than computational. This constant-speed property is taken up in its own right in section 3.1, where geodesics are the subject; here it serves to show the isometry of transport at work on the one parallel field every geodesic carries, its own velocity.

The covariant derivative has so far acted only on vector fields and on vector fields along curves. Curvature, the Hessian of a function, the second fundamental form, and the very tensor g that has been differentiated informally above all live in tensor bundles built from TM and T^*M . The connection extends to all of these in exactly one way consistent with the operations that relate them, namely the Leibniz rule, contraction, and the pairing of a vector with a covector.

Definition 2.4.6: Induced Connection on Tensor Bundles

Let ∇ be an affine connection on TM of a smooth manifold M . The *induced connection*, also denoted ∇ , on each tensor bundle $T^{(r,s)}M$ of (r, s) -tensors is the unique family of operators ∇_X on tensor fields, for $X \in \mathfrak{X}(M)$, determined by the following requirements.

1. On functions $((0, 0)$ -tensors), $\nabla_X f = Xf$; on vector fields $((1, 0)$ -tensors), ∇_X is the given connection on TM .
2. $\mathbb{R}$ -linearity and the Leibniz rule over tensor products: for tensor fields S, T ,

$$\nabla_X(S \otimes T) = (\nabla_X S) \otimes T + S \otimes (\nabla_X T)$$

3. *Commutation with contractions*: for every contraction C of an upper against a lower index, $\nabla_X(CT) = C(\nabla_X T)$.

Equivalently, on a covector field ω the induced connection is fixed by requiring the pairing $\langle \omega, Y \rangle = \omega(Y)$ to obey the product rule,

$$(\nabla_X \omega)(Y) = X(\omega(Y)) - \omega(\nabla_X Y), \quad (2.7)$$

and on a general (r, s) -tensor T by the Leibniz-type formula

$$(\nabla_X T)(\omega^1, \dots, \omega^r, Y_1, \dots, Y_s) = X(T(\omega^1, \dots, \omega^r, Y_1, \dots, Y_s)) - \sum_a T(\dots, \nabla_X \omega^a, \dots) - \sum_b T(\dots, \nabla_X Y_b, \dots)$$

Proof:

The requirements determine ∇_X on all tensor bundles and are consistent. On a 1-form ω , apply requirement (3) to the contraction $C(\omega \otimes Y) = \omega(Y)$ and requirement (2) to the product $\omega \otimes Y$:

$$X(\omega(Y)) = \nabla_X(\omega(Y)) = \nabla_X C(\omega \otimes Y) = C(\nabla_X(\omega \otimes Y)) = (\nabla_X \omega)(Y) + \omega(\nabla_X Y),$$

using requirement (1) for the first equality (the contraction $\omega(Y)$ is a function). Solving for $(\nabla_X \omega)(Y)$ gives (2.7), so the action on 1-forms is forced. That (2.7) defines a genuine 1-form, that is, that $Y \mapsto (\nabla_X \omega)(Y)$ is $C^\infty(M)$ -linear, follows from a cancellation of the derivative term: for $f \in C^\infty(M)$,

$$(\nabla_X \omega)(fY) = X(f\omega(Y)) - \omega((Xf)Y + f\nabla_X Y) = (Xf)\omega(Y) + fX(\omega(Y)) - (Xf)\omega(Y) - f\omega(\nabla_X Y) = f(\nabla_X \omega)(Y),$$

the $(Xf)\omega(Y)$ terms cancelling. On a general (r, s) -tensor T , iterating the Leibniz rule (2) across the tensor product and contracting against the arguments $\omega^1, \dots, \omega^r, Y_1, \dots, Y_s$ yields the displayed Leibniz-type formula; the same cancellation as for 1-forms shows the right-hand side is $C^\infty(M)$ -linear in each argument, so $\nabla_X T$ is again an (r, s) -tensor field. Conversely, the operators so defined satisfy (1), (2), and (3) by construction: (2) because the formula was built from the product rule, and (3) because contracting the Leibniz-type formula in a pair of slots reproduces the covariant derivative of the contracted tensor, the extra terms matching. Hence the family is the unique extension of ∇ compatible with (1)–(3), completing the proof.

The induced connection is what lets ∇ differentiate any tensorial object of the theory, and the

requirements defining it are exactly the compatibilities one needs downstream. Requirement (2), the Leibniz rule over tensor products, means ∇ never needs a separate definition for each new bundle; every tensor is assembled from vectors and covectors, and its covariant derivative is assembled the same way. Requirement (3), commutation with contractions, is the statement that differentiating and then contracting gives the same result as contracting and then differentiating, so that traces and pairings are transparent to ∇ . Two consequences are used constantly. First, the covariant derivative commutes with the musical isomorphisms of definition 1.4.1 when ∇ is metric-compatible: since $\nabla g = 0$ (proposition 2.2.6), lowering an index $Y \mapsto Y^\flat = g(Y, -)$ commutes with ∇_X , because $\nabla_X(Y^\flat) = \nabla_X(C(g \otimes Y)) = C((\nabla_X g) \otimes Y + g \otimes \nabla_X Y) = (\nabla_X Y)^\flat$, the first term vanishing; raising is the inverse operation and commutes likewise. So for a metric-compatible connection one may freely raise and lower indices before or after differentiating. Second, applied to g itself, requirement (2) recovers the informal expansion of ∇g used already in proposition 2.2.6: contraction of $\nabla_X(g \otimes Y \otimes Z)$ produces $(\nabla_X g)(Y, Z) = X \langle Y, Z \rangle - \langle \nabla_X Y, Z \rangle - \langle Y, \nabla_X Z \rangle$, the definition on which the whole compatibility discussion rested.

Example 2.4.7: Covariant Derivative of a One-Form on the Plane

Take $\mathbb{R}^2$ minus the origin in polar coordinates (r, θ) with the Euclidean connection $\bar{\nabla}$, whose nonzero Christoffel symbols are $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = 1/r$ and $\Gamma_{\theta\theta}^r = -r$ (example 2.1.4). Differentiate the coordinate 1-form $d\theta$ in the radial direction $X = \partial_r$. Formula (2.7) gives, on each coordinate field ∂_j ,

$$(\bar{\nabla}_{\partial_r} d\theta)(\partial_j) = \partial_r(d\theta(\partial_j)) - d\theta(\bar{\nabla}_{\partial_r} \partial_j)$$

The pairing $d\theta(\partial_j) = \delta_j^\theta$ is constant, so its ∂_r -derivative is zero, and $d\theta$ selects the θ -component of $\bar{\nabla}_{\partial_r} \partial_j$. For $j = \theta$: $\bar{\nabla}_{\partial_r} \partial_\theta = \Gamma_{r\theta}^k \partial_k = \frac{1}{r} \partial_\theta$, whose θ -component is $1/r$, so $(\bar{\nabla}_{\partial_r} d\theta)(\partial_\theta) = -1/r$. For $j = r$: $\bar{\nabla}_{\partial_r} \partial_r = 0$, so $(\bar{\nabla}_{\partial_r} d\theta)(\partial_r) = 0$. Therefore

$$\bar{\nabla}_{\partial_r} d\theta = -\frac{1}{r} d\theta$$

The sign is opposite to that seen when the same connection acts on the vector field ∂_θ , where $\bar{\nabla}_{\partial_r} \partial_\theta = +\frac{1}{r} \partial_\theta$; the covector transforms contravariantly to its dual vector, exactly as (2.7) dictates through the minus sign. As a consistency check with commutation over contraction, the pairing $\langle d\theta, \partial_\theta \rangle = d\theta(\partial_\theta) = 1$ is constant, so its covariant derivative must vanish; indeed $\bar{\nabla}_{\partial_r}(d\theta(\partial_\theta)) = (\bar{\nabla}_{\partial_r} d\theta)(\partial_\theta) + d\theta(\bar{\nabla}_{\partial_r} \partial_\theta) = (-\frac{1}{r})(1) + (1)(\frac{1}{r}) = 0$, confirming requirement (3) of definition 2.4.6 in this instance.

Geodesics and the Exponential Map

The Levi-Civita connection of chapter 2 makes it possible to ask which curves are straight. A straight line in Euclidean space is one whose velocity does not change, and the covariant derivative supplies the correct notion of an unchanging velocity on a manifold: a curve is a *geodesic* when its acceleration, the covariant derivative of its own velocity, vanishes. This chapter develops geodesics and the map that organizes them, the exponential map, and establishes the sense in which geodesics are the shortest paths.

The geodesic equation is a second-order ordinary differential equation, so a geodesic is determined by an initial point and an initial velocity; the exponential map packages these solutions by sending a tangent vector to the point reached by travelling along the corresponding geodesic for unit time. Near a point it is a diffeomorphism, and it produces the normal coordinates in which the metric is standard to first order. The Gauss lemma, which records that the exponential map sends radii to orthogonal geodesics, then yields the local minimizing property: within a small ball the radial geodesic is the unique shortest path, and the metric induces a genuine distance function on the manifold, resolving the terminological caution raised in chapter 1. The chapter closes with the variational picture, in which geodesics reappear as the critical points of the length and energy functionals, the point of view that governs the comparison theory of later chapters.

3.1 The Geodesic Equation

A straight line in Euclidean space is the trace of a particle that feels no force: its velocity is constant, so its acceleration vanishes. On a curved manifold there is no ambient vector space in which to demand that the velocity vector literally not change, but the Levi-Civita connection of definition 2.3.2 supplies the intrinsic replacement. The velocity $\dot{\gamma}$ of a curve γ is a vector field along γ , and its intrinsic rate of change is the covariant derivative $D_t \dot{\gamma}$ of definition 2.1.6. Requiring this to vanish is the manifold analogue of “the velocity does not change,” and the curves that satisfy it are the

geodesics. They are the straightest curves the geometry allows, the paths of free particles, and, as the next chapter shows, the locally shortest ones.

This section extracts the differential equation that governs geodesics and settles the two questions any second-order equation raises: given a starting point and a starting velocity, does a solution exist, and is it unique. The answers come from the theory of ordinary differential equations, applied to the coordinate form of the equation $D_t\dot{\gamma} = 0$. A geodesic turns out to be determined by its initial position and initial velocity exactly as a Newtonian trajectory is determined by initial data, and it depends smoothly on both. Throughout, (M, g) is a (pseudo-)Riemannian manifold with its Levi-Civita connection ∇ , and the Einstein summation convention, summing over any index appearing once as an upper and once as a lower index, is in force.

Definition 3.1.1: Geodesic

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ , and let $\gamma: I \rightarrow M$ be a smooth curve on an interval $I \subseteq \mathbb{R}$. Then γ is a *geodesic* if its velocity field is parallel along γ , that is,

$$D_t\dot{\gamma} = 0 \quad \text{on } I,$$

where D_t is the covariant derivative along γ of definition 2.1.6. The vector field $D_t\dot{\gamma} \in \mathfrak{X}(\gamma)$ is the *acceleration* of γ ; a geodesic is thus a curve of zero acceleration, an *autoparallel* curve.

The word autoparallel names the geometric content: a geodesic transports its own velocity parallel to itself. The velocity vector at one instant is carried, by the connection, exactly onto the velocity vector at the next, so the curve never turns relative to the geometry it lives in. On the flat model $(\mathbb{R}^n, \bar{g})$ the Levi-Civita connection is the Euclidean connection, whose covariant derivative along a curve is componentwise ordinary differentiation, so $D_t\dot{\gamma} = \ddot{\gamma}$ and the geodesic condition reads $\ddot{\gamma} = 0$: the geodesics of Euclidean space are the affinely parametrized straight lines $t \mapsto p + tv$, recovering the classical picture. One caution is built into the definition. The condition $D_t\dot{\gamma} = 0$ constrains not only the trace of the curve but its parametrization: a straight line traversed with varying speed, such as $t \mapsto p + t^2v$, has $\ddot{\gamma} = 2v \neq 0$ and is not a geodesic in this sense, even though its image is a geodesic image. Geodesics are curves, parametrization included, not point sets.

To compute with geodesics one passes to coordinates. In a chart $(U, (x^1, \dots, x^n))$ write $\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$; the velocity has components $\dot{\gamma}^k$ and, by the coordinate formula (2.2) for the covariant derivative along a curve applied to $V = \dot{\gamma}$ (so $V^k = \dot{\gamma}^k$ and $\dot{V}^k = \ddot{\gamma}^k$),

$$D_t\dot{\gamma} = \left(\ddot{\gamma}^k + \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i \dot{\gamma}^j \right) \partial_k|_{\gamma(t)}$$

Since the coordinate frame (∂_k) is a basis of each tangent space, the vanishing of $D_t\dot{\gamma}$ is the vanishing of every component. This yields the local form of the geodesic condition.

$$\ddot{\gamma}^k + \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i \dot{\gamma}^j = 0, \quad k = 1, \dots, n \quad (3.1)$$

Equation (3.1) is the *geodesic equation*. It is a system of n coupled second-order ordinary differential equations for the coordinate functions $\gamma^k(t)$, one for each k . The leading term $\ddot{\gamma}^k$ is the naive coordinate acceleration; the quadratic term $\Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j$ is the correction that accounts for the curving of the coordinate frame, exactly as in the covariant derivative it descends from. The whole geometric content of the connection enters through the Christoffel symbols Γ_{ij}^k , which by proposition 2.3.3 are built from the metric g_{ij} and its first derivatives. Two features of the equation should be registered

before it is solved. The coefficients $\Gamma_{ij}^k(\gamma(t))$ depend on the unknown γ itself, so the system is nonlinear; and the nonlinearity is quadratic in the velocities $\dot{\gamma}^i$, never higher, which is what will make the reduction to a first-order system in the next theorem clean.

Example 3.1.2: Meridians on a Surface of Revolution

Take a surface of revolution with the warped metric $g = dr^2 + f(r)^2 d\theta^2$, $f(r) > 0$, in coordinates $(x^1, x^2) = (r, \theta)$, whose Christoffel symbols were computed in example 2.3.4: the only nonzero symbols are

$$\Gamma_{\theta\theta}^r = -f(r)f'(r), \quad \Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{f'(r)}{f(r)}$$

The geodesic equation (3.1) has one component for each of $k = r$ and $k = \theta$. Reading off the nonzero symbols, the r -component receives a contribution only from the pair $(i, j) = (\theta, \theta)$, and the θ -component only from the pairs (r, θ) and (θ, r) , which are equal and so combine with a factor 2:

$$\begin{aligned} \ddot{r} - f(r)f'(r)\dot{\theta}^2 &= 0, \\ \ddot{\theta} + \frac{2f'(r)}{f(r)}\dot{r}\dot{\theta} &= 0 \end{aligned}$$

Consider a *meridian*, a curve of constant longitude $\theta(t) \equiv \theta_0$ traced out in the r -direction; parametrize it by $\gamma(t) = (r(t), \theta_0)$. Then $\dot{\theta} \equiv 0$ and $\ddot{\theta} \equiv 0$, so the second equation is satisfied identically, and the first equation reduces to $\ddot{r} = 0$, whose solutions are the affine functions $r(t) = r_0 + ct$. Thus every meridian, traversed at constant r -speed, is a geodesic. The geometric reading is that a meridian is the profile curve of the surface, and moving along it the velocity never turns sideways, because the rotational symmetry of the surface leaves no preferred direction for it to turn toward. A parallel of constant latitude $r(t) \equiv r_0$, by contrast, has $\dot{r} \equiv 0$ and $\dot{\theta} = \text{const}$, and the first equation demands $-f(r_0)f'(r_0)\dot{\theta}^2 = 0$; since $f > 0$ this forces $f'(r_0) = 0$. A circle of latitude is therefore a geodesic exactly when it sits at a critical radius of the profile function f , for instance the equator of a sphere of revolution, where f is maximal. Specializing to the flat plane $f(r) = r$ recovers the earlier finding: the equations become $\ddot{r} - r\dot{\theta}^2 = 0$ and $\ddot{\theta} + \frac{2}{r}\dot{r}\dot{\theta} = 0$, the polar-coordinate form of $\ddot{\gamma} = 0$, whose radial-line solutions $\theta \equiv \theta_0$, $r = r_0 + ct$ are precisely the straight lines through the origin.

The geodesic equation is nonlinear and second order, so its solvability is not automatic; it is delivered by the existence and uniqueness theory for ordinary differential equations, once the second-order system is recast as a first-order one on a suitable space. The natural space is the tangent bundle TM , whose points record a position together with a velocity, precisely the data that a geodesic evolves.

Theorem 3.1.3: Existence and Uniqueness of Geodesics

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection. For each $p \in M$ and each $v \in T_p M$ there is a unique *maximal geodesic*

$$\gamma_v: I_v \rightarrow M, \quad \gamma_v(0) = p, \quad \dot{\gamma}_v(0) = v,$$

defined on an open interval $I_v \subseteq \mathbb{R}$ containing 0, in the sense that any geodesic with these initial conditions is the restriction of γ_v to a subinterval. Moreover γ_v depends smoothly on the initial data: the map $(t, p, v) \mapsto \gamma_v(t)$ is smooth on its (open) domain of definition inside $\mathbb{R} \times TM$.

Proof:

The argument reduces the second-order geodesic equation to a first-order system on the tangent bundle, invokes the standard theorem on ordinary differential equations, and then globalizes the local solutions into a single maximal one.

Step 1: reduction to a first-order system in a chart. Fix a chart $(U, (x^i))$ and identify a curve γ in U with its coordinate functions $(\gamma^1, \dots, \gamma^n)$. Introduce the velocity coordinates $w^k := \dot{\gamma}^k$, doubling the number of unknowns to $2n$, namely (γ^k, w^k) . The geodesic equation (3.1) then becomes the first-order system

$$\dot{\gamma}^k = w^k, \quad \dot{w}^k = -\Gamma_{ij}^k(\gamma) w^i w^j, \quad k = 1, \dots, n \quad (3.2)$$

A curve γ in U solves (3.1) if and only if the pair $(\gamma, \dot{\gamma}) = (\gamma^k, w^k)$ solves (3.2): the first block of equations is the definition $w^k = \dot{\gamma}^k$, and substituting it into the second reproduces $\ddot{\gamma}^k = \dot{w}^k = -\Gamma_{ij}^k(\gamma) \dot{\gamma}^i \dot{\gamma}^j$. Write $\xi = (\gamma^1, \dots, \gamma^n, w^1, \dots, w^n)$ for the unknown taking values in the open set $\pi^{-1}(U) \cong U \times \mathbb{R}^n \subseteq TM$, where $\pi: TM \rightarrow M$ is the bundle projection; the system (3.2) has the form $\dot{\xi} = F(\xi)$ with

$$F(\gamma, w) = (w^1, \dots, w^n, -\Gamma_{ij}^k(\gamma) w^i w^j)$$

The map F is smooth: its first n components are the linear projections $\xi \mapsto w^k$, and its last n components are polynomials of degree two in the w 's whose coefficients Γ_{ij}^k are smooth functions of γ on U , by proposition 2.3.3. Thus F is a smooth vector field on the open subset $\pi^{-1}(U)$ of TM .

Step 2: local existence, uniqueness, and smooth dependence. The system $\dot{\xi} = F(\xi)$ with F smooth is the setting of the fundamental theorem on ordinary differential equations, equivalently the theorem on the local flow of a smooth vector field [3]. It provides, for each initial value $\xi_0 = (p, v) \in \pi^{-1}(U)$, a number $\varepsilon > 0$ and a unique solution $\xi: (-\varepsilon, \varepsilon) \rightarrow \pi^{-1}(U)$ with $\xi(0) = \xi_0$; uniqueness means any two solutions with the same initial value agree on the intersection of their domains. The theorem further furnishes smooth dependence on the initial value: there is a neighborhood of ξ_0 and a single $\varepsilon > 0$ such that the map $(t, \xi_0) \mapsto \xi(t; \xi_0)$ is smooth in t and in ξ_0 jointly. Translating back, the first block of ξ is a curve γ in U solving (3.1) with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$, and the smooth dependence of ξ on $\xi_0 = (p, v)$ is exactly the asserted smooth dependence of $\gamma_v(t)$ on (t, p, v) within the chart. Uniqueness of the geodesic follows: two geodesics through p with velocity v lift to two solutions of (3.2) with the same initial value, hence coincide where both are defined.

Step 3: the solution is independent of the chart. The first-order system (3.2) was written in a particular chart, but the condition it encodes, $D_t \dot{\gamma} = 0$, is coordinate-free: the covariant derivative along a curve is chart-independent by definition 2.1.6. Consequently, if a geodesic passes from the domain of one chart into that of another, the two coordinate representations describe the same curve on the overlap, and by the uniqueness of Step 2 they patch. This makes the local geodesics globally well defined as curves in M and lets the maximality argument proceed on M rather than in a fixed chart.

Step 4: the maximal geodesic. Fix p and v . Consider the collection of all geodesics $\gamma: I \rightarrow M$ with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$, where I ranges over open intervals containing 0; this collection is nonempty by Step 2. Let I_v be the union of all such intervals I , again an open interval containing 0. Define $\gamma_v: I_v \rightarrow M$ by setting $\gamma_v(t) = \gamma(t)$ for any geodesic γ in the collection whose domain contains t . This is well defined: if $\gamma_1: I_1 \rightarrow M$ and $\gamma_2: I_2 \rightarrow M$ are two geodesics with the same initial data, then the set $\{t \in I_1 \cap I_2 : \gamma_1(t) = \gamma_2(t) \text{ and } \dot{\gamma}_1(t) = \dot{\gamma}_2(t)\}$ is nonempty (it contains 0), closed in the connected set $I_1 \cap I_2$ by continuity, and open by the local uniqueness of Step 2 applied at each of its points (a geodesic through $\gamma_1(t_0)$ with velocity $\dot{\gamma}_1(t_0)$ is unique near t_0 , forcing $\gamma_1 = \gamma_2$ on a neighborhood of t_0); hence it is all of $I_1 \cap I_2$, so γ_1 and γ_2 agree on the overlap. Therefore the assignment $\gamma_v(t) = \gamma(t)$ does not depend on which γ is used, and γ_v is a well-defined curve on I_v . It is a geodesic, because the geodesic equation is local in t and near every $t_0 \in I_v$ the curve γ_v coincides with some member γ of the collection, which satisfies the equation there; and it is smooth for the same reason. By construction $\gamma_v(0) = p$, $\dot{\gamma}_v(0) = v$, and every geodesic with these initial conditions has domain contained in I_v and agrees with γ_v there, so it is a restriction of γ_v . This is the asserted maximal geodesic, and it is unique because any curve with the same defining property has the same domain I_v and the same values.

Finally, the smooth dependence on (t, p, v) is inherited from Step 2. The domain

$$\mathcal{D} = \{(t, p, v) \in \mathbb{R} \times TM : t \in I_v\}$$

is open: around any $(t_0, p_0, v_0) \in \mathcal{D}$, the local solution of Step 2 exists for initial data near (p_0, v_0) and for times in a neighborhood of t_0 , obtained by chaining finitely many local flow boxes along the compact segment $\gamma_{v_0}([0, t_0])$; on that neighborhood $(t, p, v) \mapsto \gamma_v(t)$ agrees with a composition of smooth local flows, hence is smooth. As this holds near every point of $\mathcal{D}$, the map $(t, p, v) \mapsto \gamma_v(t)$ is smooth on $\mathcal{D}$, completing the proof.

The theorem is the geometric counterpart of the determinism of Newtonian mechanics: a geodesic is fixed by where it starts and how fast, in what direction, it leaves, and nothing more. The passage to TM is the reason the second-order equation on M becomes a first-order flow: a point of TM carries both the position and the velocity, so evolving it forward advances both at once, and the geodesics are exactly the projections to M of the integral curves of the vector field F built in the proof. That field, defined here only in charts, is a genuine global object on TM , the *geodesic spray*, and its flow is the *geodesic flow*; the exponential map of section 3.2 is nothing but this flow read off at time one, and the smooth dependence just proved is what makes the exponential map smooth. One point deserves emphasis. The interval I_v need not be all of $\mathbb{R}$: a geodesic can run off the manifold in finite time, as it does on $\mathbb{R}^n \setminus \{0\}$ with the Euclidean metric, where the geodesic aimed at the deleted origin reaches it at a finite parameter and cannot continue. When every maximal geodesic is defined on all of $\mathbb{R}$ the manifold is called *geodesically complete*, a condition analyzed in the Hopf–Rinow theorem of a later chapter.

Uniqueness of the geodesic through given initial data is a computational tool as much as a structural

fact: to identify all geodesics of a model space it suffices to exhibit, through each point and in each direction, one curve that solves (3.1), and theorem 3.1.3 then forces it to be the geodesic. The two constant-curvature models make this concrete.

Example 3.1.4: Great circles are the geodesics of the sphere

On the round 2-sphere in coordinates $(x^1, x^2) = (\theta, \varphi)$ with $g = d\theta^2 + \sin^2 \theta d\varphi^2$, the Christoffel symbols were computed in example 2.3.5: the only nonzero ones are

$$\Gamma_{\varphi\varphi}^{\theta} = -\sin \theta \cos \theta, \quad \Gamma_{\theta\varphi}^{\varphi} = \Gamma_{\varphi\theta}^{\varphi} = \cot \theta$$

The geodesic equation (3.1) therefore has the two components

$$\begin{aligned} \ddot{\theta} - \sin \theta \cos \theta \dot{\varphi}^2 &= 0, \\ \ddot{\varphi} + 2 \cot \theta \dot{\theta} \dot{\varphi} &= 0, \end{aligned}$$

the θ -component receiving its contribution from the pair $(i, j) = (\varphi, \varphi)$ and the φ -component from the pairs (θ, φ) and (φ, θ) , equal and combining with a factor 2.

The equator. Set $\theta(t) \equiv \pi/2$, so $\dot{\theta} \equiv 0$, $\ddot{\theta} \equiv 0$, and $\cos \theta \equiv 0$. The first equation reads $0 - 0 = 0$, satisfied identically, and the second reduces to $\ddot{\varphi} = 0$, whose solutions are $\varphi(t) = \varphi_0 + ct$. Thus the equator, traversed at constant φ -speed, is a geodesic. It is a great circle, the intersection of S^2 with the plane $\{x^3 = 0\}$ through the center.

The meridians. Set $\varphi(t) \equiv \varphi_0$, so $\dot{\varphi} \equiv 0$ and $\ddot{\varphi} \equiv 0$. The second equation holds identically, and the first reduces to $\ddot{\theta} = 0$, giving $\theta(t) = \theta_0 + ct$. Each meridian, a half great circle of constant longitude, is a geodesic; this is the specialization of example 3.1.2 to the sphere of revolution with profile $f(\theta) = \sin \theta$.

General great circles. Every remaining geodesic is obtained from these by symmetry, without solving the equations again. Fix $p \in S^2$ and a unit $v \in T_p S^2$, and let $\Pi \subset \mathbb{R}^3$ be the plane through the origin spanned by the position vector p and the ambient velocity v ; the great circle $C = S^2 \cap \Pi$ passes through p in the direction v . Let $\rho: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the reflection fixing Π pointwise and negating the orthogonal direction. Then ρ is a linear isometry of $\mathbb{R}^3$ preserving S^2 , hence an isometry of the round sphere, and its fixed-point set on S^2 is exactly C . An isometry carries geodesics to geodesics, because $\varphi^*g = g$ preserves the Levi-Civita connection and so takes solutions of (3.1) to solutions; therefore $\rho \circ \gamma_v$ is a geodesic. It has the same initial data as γ_v , namely $\rho(p) = p$ and $d\rho_p(v) = v$ (both p and v lie in the fixed plane Π), so by the uniqueness in theorem 3.1.3, $\rho \circ \gamma_v = \gamma_v$. Thus γ_v is fixed pointwise by ρ , which means its image lies in the fixed set C . Since some suitably rotated equator realizes any prescribed (p, v) (rotations act transitively on unit tangent vectors), every geodesic of S^2 is a great circle traversed at constant speed, and conversely each great circle is a geodesic. The same reflection argument identifies the geodesics of S^n for every n as its great circles.

The hyperbolic plane furnishes the negatively curved counterpart, where the geodesics are visibly not the Euclidean straight lines of the coordinate model.

Example 3.1.5: Geodesics of the hyperbolic plane

Take the upper half-plane model of example 1.3.5 in dimension two, with coordinates $(x^1, x^2) =$

(x, y) on $U^2 = \{(x, y) \mid y > 0\}$ and metric

$$g = \frac{1}{y^2}(dx^2 + dy^2)$$

Here $g_{11} = g_{22} = y^{-2}$, $g_{12} = 0$, so the inverse metric is $g^{11} = g^{22} = y^2$, and the only nonzero partial derivatives of the components are $\partial_y g_{11} = \partial_y g_{22} = -2y^{-3}$. Feeding these into proposition 2.3.3 gives the nonzero Christoffel symbols

$$\Gamma_{xy}^x = \Gamma_{yx}^x = -\frac{1}{y}, \quad \Gamma_{xx}^y = \frac{1}{y}, \quad \Gamma_{yy}^y = -\frac{1}{y}$$

For instance $\Gamma_{xx}^y = \frac{1}{2}g^{yy}(2\partial_x g_{xy} - \partial_y g_{xx}) = \frac{1}{2}y^2(0 - (-2y^{-3})) = y^{-1}$, and the others follow by the same computation. The geodesic equation (3.1) for $\gamma(t) = (x(t), y(t))$ becomes

$$\begin{aligned} \ddot{x} - \frac{2}{y}\dot{x}\dot{y} &= 0, \\ \ddot{y} + \frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 &= 0 \end{aligned}$$

The vertical rays. Set $x(t) \equiv x_0$, so $\dot{x} \equiv 0$ and $\ddot{x} \equiv 0$; the first equation holds identically. The second reduces to $\ddot{y} = \dot{y}^2/y$, solved by $y(t) = y_0 e^{ct}$, for which $\dot{y} = cy_0 e^{ct}$, $\ddot{y} = c^2 y_0 e^{ct}$, and $\dot{y}^2/y = c^2 y_0 e^{ct}$ agree. So each vertical line $x = x_0$, traversed as $y = y_0 e^{ct}$, is a geodesic. Its speed $g(\dot{\gamma}, \dot{\gamma})^{1/2} = |\dot{y}|/y = |c|$ is constant, consistent with the parametrization being affine; the exponential dependence, rather than a linear one, is what constant speed demands in this metric, since intervals near the boundary $y = 0$ are metrically long.

The semicircles on the real axis. For the non-vertical geodesics, consider the curve

$$x(t) = a + R \tanh t, \quad y(t) = \frac{R}{\cosh t} \quad (R > 0),$$

whose image is the upper semicircle $(x - a)^2 + y^2 = R^2$, $y > 0$, centered on the real axis. Direct differentiation gives $\dot{x} = R \operatorname{sech}^2 t$, $\ddot{x} = -2R \operatorname{sech}^2 t \tanh t$, $\dot{y} = -R \operatorname{sech} t \tanh t$, and $\ddot{y} = -R \operatorname{sech} t (\operatorname{sech}^2 t - \tanh^2 t)$. For the first equation,

$$\frac{2}{y}\dot{x}\dot{y} = \frac{2}{R \operatorname{sech} t}(R \operatorname{sech}^2 t)(-R \operatorname{sech} t \tanh t) = -2R \operatorname{sech}^2 t \tanh t = \ddot{x},$$

so $\ddot{x} - \frac{2}{y}\dot{x}\dot{y} = 0$. For the second, using $\operatorname{sech}^2 t + \tanh^2 t = 1$,

$$\frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 = R \operatorname{sech}^3 t - R \operatorname{sech} t \tanh^2 t = R \operatorname{sech} t (\operatorname{sech}^2 t - \tanh^2 t) = -\ddot{y},$$

so $\ddot{y} + \frac{1}{y}\dot{x}^2 - \frac{1}{y}\dot{y}^2 = 0$. Both equations hold, and the speed is computed from $y^{-2} = R^{-2} \operatorname{sech}^{-2} t$ and $\dot{x}^2 + \dot{y}^2 = R^2 \operatorname{sech}^2 t (\operatorname{sech}^2 t + \tanh^2 t)$: the R^2 and the $\operatorname{sech}^{\pm 2} t$ factors cancel, leaving $g(\dot{\gamma}, \dot{\gamma}) = y^{-2}(\dot{x}^2 + \dot{y}^2) = \operatorname{sech}^2 t + \tanh^2 t = 1$ by the identity $\operatorname{sech}^2 t + \tanh^2 t = 1$, so the parametrization is by arc length.

These exhaust the geodesics. Through a point $p = (x_0, y_0)$ with $y_0 > 0$, a unit direction v is either vertical, matched by the vertical ray through p , or non-vertical, in which case exactly one circle centered on the real axis passes through p tangent to v (its center is the intersection of the axis

with the Euclidean normal line to v at p), realized by a suitable (a, R) above. In every case a solution of the geodesic equation with initial data (p, v) has been produced, so by the uniqueness in theorem 3.1.3 it is *the* geodesic through (p, v) . The geodesics of the hyperbolic plane are therefore precisely the vertical rays and the semicircles meeting the real axis at right angles, a family that looks nothing like the straight lines of the coordinate plane it is drawn in.

The remaining structural fact about geodesics is that they move at constant speed. This is not built into the definition; it is a consequence of metric compatibility, the property of the Levi-Civita connection that its covariant derivative respects the inner product.

Proposition 3.1.6: Geodesics Have Constant Speed

Let (M, g) be a (pseudo-)Riemannian manifold and let γ be a geodesic. Then the quantity $\langle \dot{\gamma}, \dot{\gamma} \rangle$ is constant along γ . In the Riemannian case the speed $|\dot{\gamma}|_g = \langle \dot{\gamma}, \dot{\gamma} \rangle^{1/2}$ is constant, and γ is either constant or an immersion with nowhere-vanishing velocity.

Proof:

Consider the smooth real function $s(t) = \langle \dot{\gamma}(t), \dot{\gamma}(t) \rangle$ along γ , where $\langle -, - \rangle = g(-, -)$. Its derivative is computed by the along-a-curve Leibniz rule for the covariant derivative, the compatibility identity (2.6) established in section 2.4: for any two vector fields $V, W \in \mathfrak{X}(\gamma)$,

$$\frac{d}{dt} \langle V, W \rangle = \langle D_t V, W \rangle + \langle V, D_t W \rangle$$

This identity is the along-a-curve counterpart of the compatibility of definition 2.2.4, and, unlike the on- M statement, it holds for every field along γ whether or not it extends to a field on M ; that is why it applies to $\dot{\gamma}$ directly. Taking $V = W = \dot{\gamma}$ gives

$$\dot{s}(t) = \frac{d}{dt} \langle \dot{\gamma}, \dot{\gamma} \rangle = 2 \langle D_t \dot{\gamma}, \dot{\gamma} \rangle$$

Since γ is a geodesic, $D_t \dot{\gamma} = 0$ by definition 3.1.1, so $\dot{s} \equiv 0$ and s is constant on the interval of definition. Hence $\langle \dot{\gamma}, \dot{\gamma} \rangle$ is constant, as claimed.

In the Riemannian case g is positive definite, so $s(t) = |\dot{\gamma}|_g^2 \geq 0$ and $|\dot{\gamma}|_g = s^{1/2}$ is a constant $c \geq 0$. If $c = 0$ then $\dot{\gamma} \equiv 0$, so γ is constant. If $c > 0$ then $\dot{\gamma}(t) \neq 0$ for all t , so γ is an immersion, completing the proof.

Constant speed has two consequences that shape everything downstream. First, a geodesic comes with a preferred class of parametrizations. If γ is a geodesic and $t \mapsto at + b$ is an affine change of parameter, then $\sigma(t) = \gamma(at + b)$ satisfies $\dot{\sigma} = a \dot{\gamma}$ and $D_t \dot{\sigma} = a^2 (D_t \dot{\gamma}) \circ (at + b) = 0$, so σ is again a geodesic; but a nonaffine reparametrization changes the speed and destroys the geodesic property, precisely because constant speed is forced. Geodesics are therefore *affinely parametrized* curves, and among all parametrizations of a given geodesic image only the affine ones, running at constant speed, are geodesics. Second, for a nonconstant Riemannian geodesic the speed is a positive constant, so one may rescale time to arrange unit speed $|\dot{\gamma}|_g \equiv 1$; this arc-length parametrization is the standing normalization when geodesics are compared with arbitrary curves as candidate shortest paths, the theme taken up through the Gauss lemma and the distance function in the sections that follow. In the pseudo-Riemannian case the proof shows only that $\langle \dot{\gamma}, \dot{\gamma} \rangle$ is constant, not that a speed can be

normalized to one: the constant may be negative or zero, and a geodesic is called timelike, spacelike, or null according to the sign of this conserved quantity, its causal character preserved for all time.

3.2 The Exponential Map and Normal Coordinates

The geodesic equation of section 3.1 produces, for each starting point p and each initial velocity $v \in T_p M$, a unique maximal geodesic γ_v (theorem 3.1.3). The data of the problem live in the tangent space: a point p is fixed, and the free parameter is the vector v . It is natural to assemble these solutions into a single map that reads a tangent vector and returns the point the geodesic reaches. Fixing the travel time at one unit, the map sends $v \in T_p M$ to $\gamma_v(1)$, the endpoint of the geodesic that leaves p with velocity v after unit time. This is the exponential map at p , and it is the device that transports the linear structure of the tangent space onto the manifold near p .

The payoff is a coordinate system tailored to the geometry rather than to an arbitrary chart. The Christoffel symbols Γ_{ij}^k measure the chart, not the geometry (this was the closing lesson of section 2.3, where polar coordinates on the flat plane had nonzero symbols), and one would like coordinates in which the metric looks as Euclidean as a chart can be forced to look. The exponential map delivers exactly this: its inverse is a chart, the *normal coordinates*, in which the metric agrees with the Euclidean metric to first order at p , the Christoffel symbols vanish at p , and every geodesic through p is a straight ray. These coordinates are the standard tool for computing pointwise metric invariants and for reducing tensor identities at a point to their flat-space form.

Definition 3.2.1: Exponential Map

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and fix $p \in M$. For $v \in T_p M$ let γ_v denote the maximal geodesic with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$ (theorem 3.1.3). Let

$$\mathcal{E}_p = \{v \in T_p M \mid \gamma_v \text{ is defined on an interval containing } [0, 1]\}$$

be the set of vectors for which the geodesic exists up to time 1. The *exponential map* at p is

$$\exp_p: \mathcal{E}_p \rightarrow M, \quad \exp_p(v) = \gamma_v(1)$$

Two features of this definition ask to be recorded at once. The domain $\mathcal{E}_p$ is a subset of the vector space $T_p M$, and it contains the origin, since γ_0 is the constant geodesic at p , defined for all time, so $\exp_p(0) = p$. It need not be all of $T_p M$: a geodesic can run off the manifold in finite time, as on an open ball in $\mathbb{R}^n$, where a geodesic aimed at the boundary ceases to exist once it reaches the edge, so long vectors v fall outside $\mathcal{E}_p$. A manifold on which every γ_v extends to all of $\mathbb{R}$, so that $\mathcal{E}_p = T_p M$ for every p , is called geodesically complete; the equivalence of this condition with metric completeness is the content of the Hopf–Rinow theorem of a later chapter, and is not needed here. What is needed is that $\mathcal{E}_p$ is an open neighborhood of 0 containing a ball, which the smooth dependence of geodesics on their initial data guarantees and which the differential computation below makes precise.

The defining feature that makes $\exp_p$ tractable is the homogeneity of the geodesic equation under rescaling of the velocity: travelling twice as fast for half as long lands at the same point. This is the rescaling identity, and it is what lets the single travel time $t = 1$ in the definition capture geodesics of every length.

Proposition 3.2.2: Rescaling Identity for Geodesics

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, $v \in T_p M$, and $\lambda \in \mathbb{R}$. Then for every t for which either side is defined,

$$\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$$

Consequently, if $v \in \mathcal{E}_p$ then $tv \in \mathcal{E}_p$ for all $t \in [0, 1]$, and

$$\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t);$$

that is, $t \mapsto \exp_p(tv)$ is the geodesic γ_v itself, reparametrized on $[0, 1]$.

Proof:

Fix v and λ and set $\sigma(t) := \gamma_v(\lambda t)$, defined for all t with λt in the domain of γ_v . This σ is a smooth curve with $\sigma(0) = \gamma_v(0) = p$, and by the chain rule its velocity is $\dot{\sigma}(t) = \lambda \dot{\gamma}_v(\lambda t)$, so $\dot{\sigma}(0) = \lambda \dot{\gamma}_v(0) = \lambda v$. It remains to check that σ is a geodesic. The geodesic condition is $D_t \dot{\sigma} = 0$ (definition 3.1.1), where D_t is the covariant derivative along the curve. In a chart, $\sigma^k(t) = \gamma_v^k(\lambda t)$, so $\dot{\sigma}^k(t) = \lambda \dot{\gamma}_v^k(\lambda t)$ and $\ddot{\sigma}^k(t) = \lambda^2 \ddot{\gamma}_v^k(\lambda t)$, and the local form of the geodesic equation (3.1) gives

$$\ddot{\sigma}^k + \Gamma_{ij}^k(\sigma) \dot{\sigma}^i \dot{\sigma}^j = \lambda^2 \ddot{\gamma}_v^k(\lambda t) + \Gamma_{ij}^k(\gamma_v(\lambda t)) \lambda \dot{\gamma}_v^i(\lambda t) \lambda \dot{\gamma}_v^j(\lambda t) = \lambda^2 \left(\ddot{\gamma}_v^k + \Gamma_{ij}^k(\gamma_v) \dot{\gamma}_v^i \dot{\gamma}_v^j \right) \Big|_{\lambda t}$$

The bracketed quantity vanishes for all arguments because γ_v is a geodesic, so $D_t \dot{\sigma} = 0$. Thus σ is a geodesic with $\sigma(0) = p$ and $\dot{\sigma}(0) = \lambda v$. By uniqueness of geodesics (theorem 3.1.3), $\sigma = \gamma_{\lambda v}$ on their common domain, which is the claimed identity $\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$.

For the consequence, suppose $v \in \mathcal{E}_p$, so γ_v is defined on an interval containing $[0, 1]$. Fix $t \in [0, 1]$. Applying the identity with $\lambda = t$ shows $\gamma_{tv}(s) = \gamma_v(ts)$ is defined for all s with $ts \in [0, 1]$, in particular for $s \in [0, 1]$ since $t \in [0, 1]$; hence $tv \in \mathcal{E}_p$. Evaluating at $s = 1$, $\exp_p(tv) = \gamma_{tv}(1) = \gamma_v(t)$, as we sought to show.

The rescaling identity has a clean picture behind it. The image $\exp_p(\{tv : t \in [0, 1]\})$ of a radial segment in $T_p M$ is precisely the geodesic γ_v run from p to $\exp_p(v)$; the exponential map straightens each geodesic through p into a ray emanating from the origin of $T_p M$. This is why $\exp_p$ deserves to be thought of as spreading the tangent space out over the manifold along geodesics, and it is the geometric fact that the normal coordinates below make into an algebraic one.

Example 3.2.3: The exponential map of Euclidean space

On $(\mathbb{R}^n, \bar{g})$ with $\bar{g} = \delta_{ij} dx^i dx^j$ (example 1.1.2) the Christoffel symbols vanish in the standard chart, so the geodesic equation (3.1) reads $\ddot{\gamma}^k = 0$, whose solutions are the affine lines $\gamma_v(t) = p + tv$. These are defined for all $t \in \mathbb{R}$, so $\mathcal{E}_p = T_p \mathbb{R}^n$ for every p , and under the canonical identification $T_p \mathbb{R}^n \cong \mathbb{R}^n$,

$$\exp_p(v) = \gamma_v(1) = p + v$$

The exponential map at p is translation by p , a diffeomorphism of $\mathbb{R}^n$ onto itself. The rescaling identity $\gamma_{\lambda v}(t) = p + \lambda tv = \gamma_v(\lambda t)$ is visible directly, and the radial rays of $T_p \mathbb{R}^n$ map to the straight lines through p . Euclidean space is the case in which $\exp_p$ is a global diffeomorphism, not only a local one, and the normal coordinates it produces are the standard Cartesian coordinates translated to place p at the origin.

On the round sphere the picture is already less trivial, and it exhibits the finite-time behaviour that Euclidean space suppresses.

Example 3.2.4: The exponential map of the round sphere

On the unit sphere (S^n, g) of example 1.3.4 the geodesics through a point p are the great circles traced at constant speed; a unit vector $v \in T_p S^n$ generates the great circle

$$\gamma_v(t) = (\cos t)p + (\sin t)v,$$

a curve on S^n (since $|\gamma_v(t)|^2 = \cos^2 t + \sin^2 t = 1$ using $\langle p, v \rangle = 0$) with $\gamma_v(0) = p$, $\dot{\gamma}_v(0) = v$, and unit speed. For a general $v \in T_p S^n$ with $|v| = \rho > 0$, the rescaling identity gives $\gamma_v(t) = (\cos \rho t)p + (\sin \rho t)v/\rho$, so

$$\exp_p(v) = \gamma_v(1) = (\cos |v|)p + (\sin |v|)\frac{v}{|v|}, \quad \exp_p(0) = p$$

Every geodesic extends to all of $\mathbb{R}$, so $\mathcal{E}_p = T_p S^n$. Here $\exp_p$ is far from injective: the vector v with $|v| = \pi$ sends every direction to the antipode $-p$, and adding $2\pi v/|v|$ returns to the same image. The map is a diffeomorphism only on the open ball $\{v \mid |v| < \pi\}$, whose image is S^n minus the antipode. This is the generic situation the next proposition addresses: $\exp_p$ is a local diffeomorphism near 0, and how far the local statement extends is a global question governed by the injectivity radius $\text{inj}(p)$, which for the unit sphere equals π .

The differential of $\exp_p$ at the origin is the computation that turns the exponential map into a coordinate chart. The key is again the rescaling identity: it forces the derivative of $\exp_p$ along a radial direction to reproduce that same direction, so that $\exp_p$ moves the origin at unit rate in every direction at once.

Proposition 3.2.5: The Differential of the Exponential Map at the Origin

Let (M, g) be a (pseudo-)Riemannian manifold and $p \in M$. Under the canonical identification $T_0(T_p M) \cong T_p M$ of the tangent space to the vector space $T_p M$ at its origin with $T_p M$ itself, the differential of $\exp_p$ at 0 is the identity:

$$d(\exp_p)_0 = \text{id}_{T_p M}$$

Consequently there is an open neighborhood of 0 in $T_p M$ on which $\exp_p$ is a diffeomorphism onto an open neighborhood of p in M .

Proof:

The domain $\mathcal{E}_p$ is a neighborhood of 0 in $T_p M$: by smooth dependence of geodesics on their initial data (theorem 3.1.3), the set of (q, w) for which the geodesic exists up to time 1 is open in the tangent bundle, and it contains $(p, 0)$, so its slice $\mathcal{E}_p$ over p is an open neighborhood of 0. Hence $\exp_p$ is a smooth map between open subsets of manifolds of the same dimension n , and its differential $d(\exp_p)_0: T_0(T_p M) \rightarrow T_p M$ is defined.

Because $T_p M$ is a vector space, the canonical identification $T_0(T_p M) \cong T_p M$ sends a vector $v \in T_p M$ to the velocity at $s = 0$ of the straight line $s \mapsto sv$ in $T_p M$. Compute $d(\exp_p)_0$ on this vector by differentiating $\exp_p$ along that line. For $|s|$ small enough that $sv \in \mathcal{E}_p$, the rescaling

identity (proposition 3.2.2) gives

$$\exp_p(sv) = \gamma_v(s),$$

so the curve $s \mapsto \exp_p(sv)$ in M is exactly the geodesic γ_v . By the chain rule, applied to the composition of $s \mapsto sv$ with $\exp_p$,

$$d(\exp_p)_0(v) = \left. \frac{d}{ds} \right|_{s=0} \exp_p(sv) = \left. \frac{d}{ds} \right|_{s=0} \gamma_v(s) = \dot{\gamma}_v(0) = v$$

Thus $d(\exp_p)_0(v) = v$ for every $v \in T_p M$, which is the identity map under the stated identification.

The identity map is invertible, so $d(\exp_p)_0$ is a linear isomorphism. By the inverse function theorem, $\exp_p$ restricts to a diffeomorphism from some open neighborhood of 0 in $T_p M$ onto an open neighborhood of $\exp_p(0) = p$ in M , completing the proof.

The statement that $d(\exp_p)_0$ is the identity is the precise form of the intuition that $\exp_p$ agrees with the linear structure of $T_p M$ to first order at the base point. It is a first-order statement only: the second derivatives of $\exp_p$ at 0 encode the metric's departure from flatness, and this is where the curvature will eventually enter (through the Jacobi equation of Chapter 5). For the present purposes, first order is enough. The inverse function theorem now upgrades the local isomorphism of differentials to a local diffeomorphism of maps, and it is this diffeomorphism whose inverse serves as a chart.

Definition 3.2.6: Normal Neighborhoods and Normal Coordinates

Let (M, g) be a (pseudo-)Riemannian manifold and $p \in M$. An open neighborhood U of p is a *normal neighborhood* of p if there is a star-shaped open neighborhood V of 0 in $T_p M$ (meaning $v \in V$ implies $tv \in V$ for all $t \in [0, 1]$) such that $\exp_p$ restricts to a diffeomorphism $\exp_p: V \rightarrow U$. When V is an open ball $\{v \mid |v| < \varepsilon\}$, its image U is a *geodesic ball* (or *normal ball*) $B(p, \varepsilon)$ of radius ε .

Given a normal neighborhood U of p and an orthonormal basis $(e_1, \dots, e_n)$ of $(T_p M, g_p)$, let $E: \mathbb{R}^n \rightarrow T_p M$ be the linear isomorphism $E(x^1, \dots, x^n) = x^i e_i$ (Einstein summation over the repeated index is in force). The map

$$\varphi := E^{-1} \circ (\exp_p|_V)^{-1}: U \rightarrow \mathbb{R}^n$$

is a smooth chart on U , called a system of *normal coordinates* (or *geodesic normal coordinates*) centered at p . In these coordinates p has coordinates 0, and the geodesic γ_v with $v = v^i e_i$ is the straight ray $t \mapsto (tv^1, \dots, tv^n)$.

The three ingredients of the chart deserve to be separated. The exponential map converts a neighborhood of p into a star-shaped set in the vector space $T_p M$; the choice of an orthonormal basis identifies that vector space with $\mathbb{R}^n$ in a way that makes g_p the standard dot product; and the composite is the coordinate map. The requirement that V be star-shaped is exactly what makes the last sentence of the definition hold: for $v \in V$ the whole segment $\{tv : t \in [0, 1]\}$ lies in V , so by the rescaling identity (proposition 3.2.2) its image $t \mapsto \exp_p(tv) = \gamma_v(t)$ is a geodesic lying in U , and in coordinates this image is the straight line through the origin with direction $(v^1, \dots, v^n)$. Normal coordinates are the coordinates in which radial geodesics from p are literally the straight lines through the origin, parametrized proportionally to arc length. The freedom in the construction

is exactly the choice of orthonormal basis, so any two normal coordinate systems at p differ by an element of $O(n)$ (or of the appropriate pseudo-orthogonal group in the indefinite case) acting on $\mathbb{R}^n$.

Example 3.2.7: Normal coordinates on the sphere are geodesic polar coordinates

On the unit sphere (S^n, g) of example 1.3.4, fix a point p and an orthonormal basis $(e_1, \dots, e_n)$ of $T_p S^n$. From example 3.2.4, for $v = v^i e_i$ with $|v| = \rho < \pi$,

$$\exp_p(v) = (\cos \rho) p + (\sin \rho) \frac{v}{\rho},$$

and this restricts to a diffeomorphism on the ball $V = \{v \mid |v| < \pi\}$, which is star-shaped about 0. The image $U = \exp_p(V)$ is S^n minus the antipode $-p$, a normal neighborhood, and $\varphi = E^{-1} \circ (\exp_p|_V)^{-1}$ is a normal coordinate chart. A point at normal-coordinate distance $\rho = |v|$ from p lies on the great circle in direction v/ρ at arc length ρ from p , so the coordinate $|\varphi(q)|$ of a point q is exactly its distance from p along the connecting geodesic. Writing $\rho = |v|$ and letting $\theta = v/|v|$ range over the unit sphere of $T_p S^n$ recovers the classical geodesic polar coordinates (ρ, θ) on S^n , in which the metric will take the warped form $d\rho^2 + \sin^2 \rho d\theta^2$ once the Gauss lemma of section 3.3 is available. The failure of $\exp_p$ to be a diffeomorphism at $|v| = \pi$, where the entire coordinate sphere collapses to $-p$, is the reason the normal chart cannot be extended past radius π .

The value of normal coordinates is that the metric assumes its simplest possible pointwise form at the center, and its first derivatives there vanish. This is the sharp version of the claim that $\exp_p$ agrees with the flat model to first order, transferred from the map to the metric components.

Proposition 3.2.8: The Metric in Normal Coordinates

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, and let $(x^1, \dots, x^n)$ be normal coordinates centered at p arising from an orthonormal basis $(e_1, \dots, e_n)$ of $(T_p M, g_p)$ (definition 3.2.6). Write g_{ij} for the components of g and Γ_{ij}^k for the Christoffel symbols of the Levi-Civita connection in these coordinates. Then, at the center p :

1. $g_{ij}(p) = \varepsilon_i \delta_{ij}$, where $\varepsilon_i = \langle e_i, e_i \rangle_g \in \{+1, -1\}$; in the Riemannian case $g_{ij}(p) = \delta_{ij}$;
2. $\Gamma_{ij}^k(p) = 0$ for all i, j, k ;
3. $\partial_k g_{ij}(p) = 0$ for all i, j, k .

Proof:

The three statements are proved in order; each rests on the fact, from definition 3.2.6 via the rescaling identity (proposition 3.2.2), that in normal coordinates every geodesic through p is a straight ray from the origin.

Step 1: the metric at p . The coordinate vector field ∂_i at p corresponds under the chart to the basis vector e_i . Indeed, the normal coordinate map is $\varphi = E^{-1} \circ (\exp_p|_V)^{-1}$, so its inverse is $\varphi^{-1} = \exp_p \circ E$; the coordinate vector $\partial_i|_p$ is the image under $d(\varphi^{-1})_0$ of the standard basis vector $\partial/\partial x^i$ of $\mathbb{R}^n$ at 0. Now $d(\varphi^{-1})_0 = d(\exp_p)_0 \circ dE_0$; the linear map E is its own differential and sends $\partial/\partial x^i$ to e_i , and $d(\exp_p)_0 = \text{id}$ by proposition 3.2.5. Hence $\partial_i|_p = e_i$. Therefore

$$g_{ij}(p) = \langle \partial_i, \partial_j \rangle_p = \langle e_i, e_j \rangle_g = \varepsilon_i \delta_{ij},$$

since $(e_1, \dots, e_n)$ is g_p -orthonormal, so $\langle e_i, e_j \rangle_g = 0$ for $i \neq j$ and $\langle e_i, e_i \rangle_g = \pm 1$; in the Riemannian case all $\varepsilon_i = +1$ and $g_{ij}(p) = \delta_{ij}$.

Step 2: the Christoffel symbols at p . Fix any $v = v^i e_i \in T_p M$ small enough that the ray $t \mapsto tv$ lies in V . By definition 3.2.6, the geodesic γ_v has, in normal coordinates, the components

$$\gamma_v^k(t) = t v^k,$$

linear in t . Substituting into the local geodesic equation (3.1), and using $\dot{\gamma}_v^k(t) = v^k$ and $\ddot{\gamma}_v^k(t) = 0$, gives for every t (with the point $\gamma_v(t)$ in the chart)

$$0 = \ddot{\gamma}_v^k + \Gamma_{ij}^k(\gamma_v(t)) \dot{\gamma}_v^i \dot{\gamma}_v^j = \Gamma_{ij}^k(\gamma_v(t)) v^i v^j$$

Evaluate at $t = 0$, where $\gamma_v(0) = p$:

$$\Gamma_{ij}^k(p) v^i v^j = 0 \quad \text{for every } v = (v^1, \dots, v^n) \in \mathbb{R}^n \text{ near } 0$$

Both sides are homogeneous of degree 2 in v , so the identity, holding on a neighborhood of 0, holds for all $v \in \mathbb{R}^n$. For fixed k , the map $v \mapsto \Gamma_{ij}^k(p) v^i v^j$ is a quadratic form whose associated symmetric matrix is $\frac{1}{2}(\Gamma_{ij}^k(p) + \Gamma_{ji}^k(p)) = \Gamma_{ij}^k(p)$, the last equality because the Levi-Civita connection is torsion-free, so $\Gamma_{ij}^k = \Gamma_{ji}^k$ (proposition 2.3.3). A symmetric bilinear form whose quadratic form vanishes identically is zero: taking $v = e_i$ gives $\Gamma_{ii}^k(p) = 0$, and taking $v = e_i + e_j$ gives $\Gamma_{ii}^k(p) + 2\Gamma_{ij}^k(p) + \Gamma_{jj}^k(p) = 0$, whence $\Gamma_{ij}^k(p) = 0$. Thus $\Gamma_{ij}^k(p) = 0$ for all i, j, k .

Step 3: the first derivatives of the metric at p . The Christoffel symbols and the first derivatives of the metric determine one another. By proposition 2.3.3,

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

Lowering the free index by contracting with g_{km} and summing over k , using $g_{km} g^{kl} = \delta_m^l$, yields the equivalent relation

$$g_{km} \Gamma_{ij}^k = \frac{1}{2} (\partial_i g_{jm} + \partial_j g_{im} - \partial_m g_{ij})$$

Evaluate at p . The left-hand side vanishes by Step 2, so at p

$$\partial_i g_{jm}(p) + \partial_j g_{im}(p) - \partial_m g_{ij}(p) = 0 \quad \text{for all } i, j, m \quad (3.3)$$

To solve this for the individual derivatives, abbreviate $a_{ijm} := \partial_i g_{jm}(p)$. This array is symmetric in its last two indices, $a_{ijm} = a_{imj}$, because $g_{jm} = g_{mj}$. In this notation, and rewriting the third term of (3.3) as $\partial_m g_{ij}(p) = a_{mij} = a_{mji}$, the relation reads

$$a_{ijm} + a_{jim} = a_{mij} \quad \text{for all } i, j, m \quad (3.4)$$

Form the three relations obtained from (3.4) by cyclically rotating the ordered triple (i, j, m) : the relation itself, its version under $(i, j, m) \mapsto (j, m, i)$, and its version under $(i, j, m) \mapsto (m, i, j)$,

$$a_{ijm} + a_{jim} = a_{mij}, \quad a_{jmi} + a_{mji} = a_{ijm}, \quad a_{mij} + a_{imj} = a_{jmi}$$

Add all three. Using the last-two-index symmetry to identify $a_{jmi} = a_{jim}$, $a_{mji} = a_{mij}$, and $a_{imj} = a_{ijm}$, every term on the right of one equation is matched by an identical term on the left of another, and the total collapses to

$$a_{ijm} + a_{jim} + a_{mij} = 0 \quad (3.5)$$

Substitute the original relation (3.4), in the form $a_{ijm} + a_{jim} = a_{mij}$, into (3.5):

$$a_{mij} + a_{mij} = 0, \quad \text{hence} \quad a_{mij} = 0$$

The indices are arbitrary, so $a_{ijm} = 0$ for every i, j, m ; that is, $\partial_i g_{jm}(p) = 0$, completing the proof.

The three properties are the working content of normal coordinates. The first two say that at the center p the metric is the standard inner product and the connection has no first-order part: covariant differentiation at p agrees with ordinary partial differentiation, because the correction terms $\Gamma_{ij}^k(p)$ are gone. The third is the sharper statement that the metric is flat to first order at p , so the Taylor expansion of g_{ij} about p begins $g_{ij}(x) = \varepsilon_i \delta_{ij} + O(|x|^2)$ with no linear term. What survives is the quadratic term, and its coefficients are the second derivatives of the metric at p ; these cannot in general be removed by any choice of coordinates, and their obstruction to being made to vanish is precisely the curvature of Chapter 4. Normal coordinates are thus the exact expression of the principle that a (pseudo-)Riemannian metric is Euclidean to first order but not to second: the first-order deviation is coordinate artifact, the second-order deviation is geometry.

The computational use of proposition 3.2.8 is constant in what follows. Any tensor identity that is $C^\infty(M)$ -linear, hence pointwise, may be verified at an arbitrary point p by passing to normal coordinates there, where the metric is $\varepsilon_i \delta_{ij}$ and all Christoffel symbols vanish. Terms involving ∇ collapse to partial derivatives at p , and expressions that would be forbidding in an arbitrary chart become tractable. The device is the geometer's counterpart of choosing an orthonormal basis in linear algebra, and it will be used to establish the symmetries of the curvature tensor, the divergence and Laplacian formulas, and the second variation, each time by reducing an invariant statement at a point to its flat-space form.

Example 3.2.9: Normal coordinates confirm the flat model

On $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, the exponential map at p is $\exp_p(v) = p + v$ (example 3.2.3), so with the standard orthonormal basis the normal coordinate map $\varphi(q) = q - p$ is translation carrying p to the origin. In these coordinates $g_{ij} \equiv \delta_{ij}$ is constant, so every claim of proposition 3.2.8 holds not only at p but everywhere: $g_{ij}(p) = \delta_{ij}$, the Christoffel symbols $\Gamma_{ij}^k = \frac{1}{2} \delta^{kl} (\partial_i \delta_{jl} + \partial_j \delta_{il} - \partial_l \delta_{ij}) = 0$ vanish identically, and $\partial_k g_{ij} \equiv 0$. This is the degenerate case in which the second-order term also vanishes, consistent with the flatness of Euclidean space; on a curved manifold such as the sphere the analogous normal coordinates make $g_{ij}(p) = \delta_{ij}$, $\Gamma_{ij}^k(p) = 0$, $\partial_k g_{ij}(p) = 0$ hold at the center but leave a nonzero $\partial_k \partial_l g_{ij}(p)$, the seed of the curvature computation, as for the sphere of example 3.2.7.

3.3 The Gauss Lemma

The exponential map $\exp_p$ of definition 3.2.1 turns a neighborhood of the origin in the tangent space $T_p M$ into a neighborhood of p in M , and by proposition 3.2.5 it does so as a diffeomorphism, its differential at the origin being the identity. This much is a statement about the point p alone. What it does not yet reveal is how $\exp_p$ distorts the linear geometry of $T_p M$ as one moves away from the origin. In $T_p M$ there is a clean picture: the rays through the origin are orthogonal to the spheres $|v| = r$ centered at the origin, and travelling out along a ray at unit speed increases r at unit rate. The exponential map carries the rays to geodesics through p and carries the spheres to certain hypersurfaces in M ; whether it also carries the orthogonality of that flat picture to M is not obvious, since $\exp_p$ is only a diffeomorphism, not an isometry, and its differential away from the origin is unknown.

The Gauss lemma answers this. It states that $\exp_p$ preserves exactly the part of the flat orthogonality that involves the radial direction: radial geodesics from p remain orthogonal to the images of the spheres, and the differential of $\exp_p$ sends the radial direction of $T_p M$ to the unit-speed velocity of

the radial geodesic without any distortion in length. The tangential directions may be stretched or sheared, and how much they are is the business of curvature; but the radial–tangential angle is rigid. From this single orthogonality statement the metric near p acquires a normal form in polar coordinates, $g = dr^2 + h_r$, in which the radial part is exactly Euclidean and all the geometry is packed into the family h_r of metrics on the geodesic spheres. That normal form is the engine of the local minimizing property proved in the next section, and it is the starting point for the comparison theory of later chapters. Throughout, the Einstein summation convention is in force: an index that appears once raised and once lowered is summed over $1, \dots, n$.

Definition 3.3.1: Geodesic Balls and Geodesic Spheres

Let (M, g) be a Riemannian manifold, $p \in M$, and let $\mathcal{O} \subseteq T_p M$ be a star-shaped open neighborhood of the origin on which $\exp_p$ (definition 3.2.1) restricts to a diffeomorphism onto a normal neighborhood $U = \exp_p(\mathcal{O})$ (definition 3.2.6). For $r > 0$ small enough that the open ball $B_r(0) = \{v \in T_p M \mid |v|_g < r\}$ has closure contained in $\mathcal{O}$, the image

$$B_r(p) := \exp_p(B_r(0))$$

is called the *geodesic ball* (or *normal ball*) of radius r about p , and the image of the sphere,

$$S_r(p) := \exp_p(\{v \in T_p M \mid |v|_g = r\}),$$

is called the *geodesic sphere* of radius r about p . The number r is the *radius* of the geodesic ball or sphere.

The set $\{v \mid |v|_g = r\}$ is a round sphere of radius r in the inner product space $(T_p M, g_p)$, and since $\exp_p$ is a diffeomorphism on $\mathcal{O}$, its image $S_r(p)$ is an embedded hypersurface of M diffeomorphic to S^{n-1} , bounding the geodesic ball $B_r(p)$. The name is a warning as much as a description: a geodesic sphere is not in general a metric sphere for any obvious distance, nor need it look round from inside M ; it is only the exponential image of a Euclidean sphere in the tangent space. What the Gauss lemma will show is that, despite this, the radial geodesics leaving p cross every geodesic sphere at a right angle, just as radii cross spheres in the flat model.

Example 3.3.2: Geodesic Spheres in Euclidean Space and on the Round Sphere

On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2 the geodesics through a point p are the straight lines $t \mapsto p + tv$, so $\exp_p(v) = p + v$ for every $v \in T_p M \cong \mathbb{R}^n$, and $\exp_p$ is a global diffeomorphism. A geodesic ball $B_r(p)$ is then the ordinary open Euclidean ball of radius r about p , and the geodesic sphere $S_r(p)$ is the ordinary Euclidean sphere of radius r . Here the geodesic sphere really is a round sphere, and the radial lines $t \mapsto p + tv/|v|$ meet it orthogonally by the elementary geometry of $\mathbb{R}^n$; the Gauss lemma below recovers this as a special case.

On the round sphere S_r^n of example 1.3.4 the picture is already less trivial. Fix a point p ; the geodesics through p are the great circles, traversed at constant speed. A geodesic ball $B_\rho(p)$ of radius $\rho < \pi r$ (a scale on which $\exp_p$ is still a diffeomorphism) is a spherical cap, and the geodesic sphere $S_\rho(p)$ is the boundary circle of that cap, the set of points at great-circle distance ρ from p . This boundary circle is a Euclidean circle in the ambient $\mathbb{R}^{n+1}$, but its radius as measured inside S_r^n is not ρ : for $n = 2$ and unit sphere $r = 1$ the circumference of $S_\rho(p)$ is $2\pi \sin \rho$, not $2\pi\rho$. The tangential directions are compressed relative to the flat model, precisely the distortion the Gauss

lemma does not control. What it does control is that the great circles through p still meet each cap boundary at a right angle, which one sees directly from the rotational symmetry of the sphere about the axis through p .

The proof of the Gauss lemma rests on a single computation with a two-parameter family of geodesics, so it is worth isolating the mechanism before stating the result. Let $v \in T_p M$ lie in the domain $\mathcal{O}$ of definition 3.3.1, and let $w \in T_p M$ be any tangent vector, which under the canonical identification $T_v(T_p M) \cong T_p M$ is regarded as a tangent vector to $T_p M$ at the point v . The vector $d(\exp_p)_v(w) \in T_{\exp_p(v)} M$ is computed by running a curve in $T_p M$ through v with velocity w , applying $\exp_p$, and differentiating. The next lemma records the one identity about such families that the argument needs: the symmetry of the mixed covariant derivatives of a variation, a consequence of the connection being torsion-free.

Lemma 3.3.3: Symmetry of the Mixed Covariant Derivative

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and let $\Gamma: (-\varepsilon, \varepsilon) \times [a, b] \rightarrow M$, $(s, t) \mapsto \Gamma(s, t)$, be a smooth map (a *variation*). Write $\partial_s \Gamma$ and $\partial_t \Gamma$ for the coordinate velocity fields of Γ , each a smooth vector field along Γ , and let D_s, D_t denote covariant differentiation along the s - and t -parameter curves (definition 2.1.6). Then

$$D_s \partial_t \Gamma = D_t \partial_s \Gamma$$

Proof:

The identity is local, so it suffices to verify it in a chart $(U, (x^k))$ meeting the image of Γ , using the coordinate expression for D_t from definition 2.1.6. Write the components of Γ as $\Gamma^k(s, t)$. Then

$$\partial_t \Gamma = \frac{\partial \Gamma^k}{\partial t} \partial_k, \quad \partial_s \Gamma = \frac{\partial \Gamma^k}{\partial s} \partial_k,$$

where (∂_k) is the coordinate frame. By the local formula for the covariant derivative of a vector field along a curve, applied to the s -curves,

$$D_s \partial_t \Gamma = \left(\frac{\partial^2 \Gamma^k}{\partial s \partial t} + \Gamma_{ij}^k \frac{\partial \Gamma^i}{\partial s} \frac{\partial \Gamma^j}{\partial t} \right) \partial_k,$$

the second term arising because the frame vectors ∂_k themselves vary along the s -curve, contributing $\frac{\partial \Gamma^i}{\partial s} \nabla_{\partial_i} \partial_j = \frac{\partial \Gamma^i}{\partial s} \Gamma_{ij}^k \partial_k$ against the j -component $\frac{\partial \Gamma^j}{\partial t}$ of $\partial_t \Gamma$. Interchanging the roles of s and t ,

$$D_t \partial_s \Gamma = \left(\frac{\partial^2 \Gamma^k}{\partial t \partial s} + \Gamma_{ij}^k \frac{\partial \Gamma^i}{\partial t} \frac{\partial \Gamma^j}{\partial s} \right) \partial_k$$

The pure second-derivative terms agree because mixed partial derivatives of the smooth functions Γ^k commute. The connection terms agree because the Christoffel symbols are symmetric in their lower indices, $\Gamma_{ij}^k = \Gamma_{ji}^k$ (proposition 2.3.3), which lets the relabelling $i \leftrightarrow j$ turn $\Gamma_{ij}^k \frac{\partial \Gamma^i}{\partial t} \frac{\partial \Gamma^j}{\partial s}$ into $\Gamma_{ji}^k \frac{\partial \Gamma^j}{\partial s} \frac{\partial \Gamma^i}{\partial t}$. Both bracketed expressions are therefore equal, so $D_s \partial_t \Gamma = D_t \partial_s \Gamma$, as claimed.

Symmetry of the Christoffel symbols is the whole of the connection input to the lemma, and it is what makes the Gauss lemma a statement about the Levi-Civita connection in particular; a connection with torsion would introduce a discrepancy $\partial_s \Gamma$ and $\partial_t \Gamma$ equal to the torsion tensor evaluated on the

two velocity fields, and the orthogonality below would fail. With this in hand the Gauss lemma is a short computation.

Theorem 3.3.4: The Gauss Lemma

Let (M, g) be a (pseudo-)Riemannian manifold, $p \in M$, and let $v \in T_p M$ be a nondegenerate vector, meaning $\langle v, v \rangle_g \neq 0$, lying in the domain on which $\exp_p$ is defined and a local diffeomorphism (definition 3.2.1). Identify $T_v(T_p M)$ with $T_p M$ in the canonical way. Then for every $w \in T_p M$,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g = \langle v, w \rangle_g$$

In particular:

1. $d(\exp_p)_v(v) = \dot{\gamma}_v(1)$, the velocity at $t = 1$ of the radial geodesic $\gamma_v(t) = \exp_p(tv)$, and $\langle d(\exp_p)_v(v), d(\exp_p)_v(v) \rangle_g = \langle v, v \rangle_g$: the radial direction is carried to the radial geodesic without change of the conserved quantity $\langle \dot{\gamma}_v, \dot{\gamma}_v \rangle_g$; in the Riemannian case this reads $|d(\exp_p)_v(v)|_g = |v|_g$, no distortion of length; and
2. if $\langle v, w \rangle_g = 0$, then $d(\exp_p)_v(v) \perp d(\exp_p)_v(w)$: in the Riemannian case, where $\langle v, w \rangle_g = 0$ means $w \perp v$ in the inner product space $(T_p M, g_p)$, this says that radial geodesics from p meet the geodesic spheres $S_r(p)$ of definition 3.3.1 orthogonally.

Proof:

Fix v in the domain with $\langle v, v \rangle_g \neq 0$, and let $w \in T_p M$ be arbitrary. Both sides of the asserted identity are $\mathbb{R}$ -linear in w with v fixed, so it suffices to treat the two cases $w = v$ and $\langle v, w \rangle_g = 0$ separately; the general case follows in Step 3 by writing w as the sum of its component along v and its g_p -orthogonal complement and using linearity.

Step 1: the radial direction. By the rescaling identity of definition 3.2.1, $\gamma_v(t) = \exp_p(tv)$ is the geodesic with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$, defined at least for $t \in [0, 1]$. Differentiating $t \mapsto \exp_p(tv)$ at $t = 1$ using the chain rule, and noting that the curve $t \mapsto tv$ in $T_p M$ has velocity v at the point v ,

$$d(\exp_p)_v(v) = \left. \frac{d}{dt} \right|_{t=1} \exp_p(tv) = \dot{\gamma}_v(1)$$

Along a geodesic the quantity $\langle \dot{\gamma}_v, \dot{\gamma}_v \rangle_g$ is constant (proposition 3.1.6), so $\langle \dot{\gamma}_v(1), \dot{\gamma}_v(1) \rangle_g = \langle \dot{\gamma}_v(0), \dot{\gamma}_v(0) \rangle_g = \langle v, v \rangle_g$, which gives both $d(\exp_p)_v(v) = \dot{\gamma}_v(1)$ and, taking $w = v$,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(v) \rangle_g = \langle \dot{\gamma}_v(1), \dot{\gamma}_v(1) \rangle_g = \langle v, v \rangle_g$$

This proves the identity when $w = v$ and establishes claim (1).

Step 2: the orthogonal directions. Suppose now $\langle v, w \rangle_g = 0$; the goal is $\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g = 0$. Choose a curve $s \mapsto v(s)$ in $T_p M$ with $v(0) = v$, with $\langle v(s), v(s) \rangle_g = \langle v, v \rangle_g$ constant in s , and with $v'(0) = w$. Such a curve exists: the level set $\Sigma = \{u \in T_p M \mid \langle u, u \rangle_g = \langle v, v \rangle_g\}$ is, near v , a smooth hypersurface of $T_p M$, because the function $u \mapsto \langle u, u \rangle_g$ has differential $2\langle v, - \rangle_g$ at v , which is nonzero: the inner product g_p is nondegenerate, and $v \neq 0$ since $\langle v, v \rangle_g \neq 0$ by hypothesis. The tangent space to Σ at v is the kernel of this differential, namely $v^\perp = \{u \mid \langle v, u \rangle_g = 0\}$. Since $w \in v^\perp$ by hypothesis, there is a smooth curve in Σ through v with velocity w at $s = 0$,

and any such curve serves as $v(s)$. Along it $\langle v(s), v(s) \rangle_g$ is constant. Define the variation of geodesics

$$\Gamma(s, t) := \exp_p(t v(s)), \quad (s, t) \in (-\varepsilon, \varepsilon) \times [0, 1],$$

which is smooth by smoothness of $\exp_p$ and of $s \mapsto v(s)$, and is defined on this rectangle for ε small because v lies in the open domain of $\exp_p$. For each fixed s the curve $t \mapsto \Gamma(s, t) = \gamma_{v(s)}(t)$ is the geodesic with initial velocity $v(s)$, so

$$D_t \partial_t \Gamma = 0 \quad \text{and} \quad \langle \partial_t \Gamma(s, t), \partial_t \Gamma(s, t) \rangle_g = \langle v(s), v(s) \rangle_g = \langle v, v \rangle_g \quad (3.6)$$

for all (s, t) , the first because each t -curve is a geodesic (definition 3.1.1) and the second because $\langle \partial_t \Gamma, \partial_t \Gamma \rangle_g$ is constant along each geodesic t -curve (proposition 3.1.6), together with the constancy of $\langle v(s), v(s) \rangle_g$ in s .

Now consider the function

$$f(s, t) := \langle \partial_t \Gamma, \partial_s \Gamma \rangle_g(s, t),$$

the g -inner product of the two velocity fields of the variation along Γ . The claim of Step 2 is exactly that $f(0, 1) = 0$, since at $(s, t) = (0, 1)$ one has $\partial_t \Gamma(0, 1) = \dot{\gamma}_v(1) = d(\exp_p)_v(v)$ by Step 1, and

$$\partial_s \Gamma(0, 1) = \left. \frac{\partial}{\partial s} \right|_{s=0} \exp_p(v(s)) = d(\exp_p)_v(v'(0)) = d(\exp_p)_v(w),$$

using $t = 1$ so that $t v(s) = v(s)$ and $v'(0) = w$. The strategy is to differentiate f in t and integrate.

Differentiate f along the t -curve, using metric compatibility of ∇ (definition 2.2.4), which turns the derivative of an inner product into inner products of covariant derivatives:

$$\frac{\partial f}{\partial t} = \langle D_t \partial_t \Gamma, \partial_s \Gamma \rangle_g + \langle \partial_t \Gamma, D_t \partial_s \Gamma \rangle_g$$

The first term vanishes by the geodesic equation (3.6). For the second term, the symmetry lemma (lemma 3.3.3) replaces $D_t \partial_s \Gamma$ by $D_s \partial_t \Gamma$, so

$$\frac{\partial f}{\partial t} = \langle \partial_t \Gamma, D_s \partial_t \Gamma \rangle_g = \frac{1}{2} \frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle_g,$$

where the second equality is again metric compatibility, now applied to the s -derivative of $\langle \partial_t \Gamma, \partial_t \Gamma \rangle_g$. By (3.6) the quantity $\langle \partial_t \Gamma, \partial_t \Gamma \rangle_g$ equals $\langle v, v \rangle_g$, a constant independent of both s and t , so its s -derivative is zero. Therefore

$$\frac{\partial f}{\partial t}(s, t) = 0 \quad \text{for all } (s, t),$$

and $f(s, t)$ does not depend on t : for each fixed s , $f(s, \cdot)$ is constant.

It remains to evaluate this constant. Consider the value at $t = 0$. For every s , $\Gamma(s, 0) = \exp_p(0) = p$, so the s -curve $s \mapsto \Gamma(s, 0)$ is constant at p , and hence its velocity vanishes: $\partial_s \Gamma(s, 0) = 0$. Consequently

$$f(s, 0) = \langle \partial_t \Gamma(s, 0), \partial_s \Gamma(s, 0) \rangle_g = \langle \partial_t \Gamma(s, 0), 0 \rangle_g = 0$$

Since $f(s, \cdot)$ is constant in t , it follows that $f(s, t) = 0$ for all t , in particular $f(0, 1) = 0$. As computed above, $f(0, 1) = \langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g$, so

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g = 0 = \langle v, w \rangle_g,$$

which is the asserted identity when $\langle v, w \rangle_g = 0$ and establishes claim (2).

Step 3: the general case. For arbitrary $w \in T_p M$, decompose $w = \lambda v + w^\perp$ with $\lambda \in \mathbb{R}$ and $\langle v, w^\perp \rangle_g = 0$; this is possible with $\lambda = \langle v, w \rangle_g / \langle v, v \rangle_g$ because $\langle v, v \rangle_g \neq 0$ by hypothesis, taking $w^\perp = w - \lambda v$, which satisfies $\langle v, w^\perp \rangle_g = \langle v, w \rangle_g - \lambda \langle v, v \rangle_g = 0$. By linearity of $d(\exp_p)_v$ and bilinearity of g , using Step 1 for the λv part and Step 2 for the $w^\perp$ part,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g = \lambda \langle v, v \rangle_g + 0 = \langle v, w \rangle_g$$

This proves the identity for all w . The orthogonality assertion of claim (2) is the special case $\langle v, w \rangle_g = 0$, saying that the radial geodesic γ_v , whose velocity at $t = 1$ is $d(\exp_p)_v(v)$, is g -orthogonal to every vector $d(\exp_p)_v(w)$ with $\langle v, w \rangle_g = 0$. In the Riemannian case these vectors $d(\exp_p)_v(w)$, for $w \perp v$ in $(T_p M, g_p)$, are exactly the vectors tangent to the geodesic sphere $S_{|v|_g}(p)$, since they are the images under $d(\exp_p)_v$ of the vectors w tangent to the Euclidean sphere through v ; so the radial geodesic meets that geodesic sphere orthogonally. This completes the proof.

The lemma is named for Gauss because the surface case, that the radial geodesics from a point cut the geodesic circles at right angles, is one of the computations behind his *Theorema Egregium*; the argument above is the modern packaging, and it isolates precisely the two ingredients that are used, metric compatibility (to differentiate the inner product) and the torsion-free symmetry (through lemma 3.3.3). Both are the defining properties of the Levi-Civita connection, so the Gauss lemma is a property of the metric alone. Notice what is *not* asserted: nothing is claimed about $\langle d(\exp_p)_v(w), d(\exp_p)_v(w') \rangle_g$ for w, w' both tangential. Those inner products measure how the geodesic spheres are shaped, and they are governed by curvature, which is why the round sphere and Euclidean space, which share the flat radial structure, have geodesic spheres of different sizes as example 3.3.2 records. The rigidity the Gauss lemma provides is only in the radial direction, but that is enough to put the metric into a normal form.

Proposition 3.3.5: The Metric in Geodesic Polar Coordinates

Let (M, g) be a Riemannian manifold, $p \in M$, and let $B_R(p)$ be a geodesic ball about p (definition 3.3.1). On the punctured ball $B_R(p) \setminus \{p\}$ introduce *geodesic polar coordinates* (r, θ) , where $r = |\exp_p^{-1}(\cdot)|_g \in (0, R)$ is the radial distance parameter and θ ranges over the unit sphere $\{u \in T_p M \mid |u|_g = 1\} \cong S^{n-1}$, so that $q = \exp_p(r\theta)$. Then in these coordinates the metric takes the form

$$g = dr^2 + h_r,$$

where for each fixed $r \in (0, R)$ the expression h_r is a Riemannian metric on the geodesic sphere $S_r(p)$, and dr^2 and h_r contain no cross term: g has no $dr d\theta$ component. Equivalently, the radial vector field ∂_r is a g -unit vector field everywhere orthogonal to each geodesic sphere, and h_r is the metric induced on $S_r(p)$ by restriction of g .

Proof:

The map $\Psi: (0, R) \times S^{n-1} \rightarrow B_R(p) \setminus \{p\}$, $\Psi(r, \theta) = \exp_p(r\theta)$, is a diffeomorphism: it is the composition of the diffeomorphism $\exp_p$ on the punctured ball $B_R(0) \setminus \{0\} \subset T_p M$ with the polar-coordinate diffeomorphism $(0, R) \times S^{n-1} \rightarrow B_R(0) \setminus \{0\}$, $(r, \theta) \mapsto r\theta$, of the punctured Euclidean

ball in $(T_p M, g_p)$. The coordinate vector fields of Ψ are computed by pushing forward the corresponding vector fields on $(0, R) \times S^{n-1}$. At a point $q = \exp_p(r_0 \theta_0)$, write $v = r_0 \theta_0 \in T_p M$, so $|v|_g = r_0$.

The radial field. The r -coordinate curve through q is $r \mapsto \exp_p(r \theta_0) = \gamma_{\theta_0}(r)$, the unit-speed radial geodesic in direction θ_0 . Hence

$$\partial_r|_q = \dot{\gamma}_{\theta_0}(r_0) = d(\exp_p)_v(\theta_0),$$

and by claim (1) of the Gauss lemma (theorem 3.3.4), applied with the radial vector $v = r_0 \theta_0$ and using linearity of $d(\exp_p)_v$ so that $d(\exp_p)_v(\theta_0) = \frac{1}{r_0} d(\exp_p)_v(v)$,

$$g(\partial_r, \partial_r)|_q = |d(\exp_p)_v(\theta_0)|_g^2 = \frac{1}{r_0^2} |d(\exp_p)_v(v)|_g^2 = \frac{1}{r_0^2} |v|_g^2 = \frac{1}{r_0^2} r_0^2 = 1$$

So ∂_r is a g -unit vector field, which accounts for the coefficient 1 of dr^2 and shows the dr^2 term is exactly Euclidean.

No cross term. Let ∂_{θ^α} ($\alpha = 1, \dots, n-1$) be any coordinate vector field of the sphere factor S^{n-1} , tangent to the level set $\{r = r_0\}$. Its image under Ψ at q is

$$\partial_{\theta^\alpha}|_q = d(\exp_p)_v(r_0 \partial_{\theta^\alpha} \theta_0),$$

where $\partial_{\theta^\alpha} \theta_0 \in T_{\theta_0} S^{n-1}$ is tangent to the unit sphere at θ_0 and hence g_p -orthogonal to θ_0 , so that the vector $w := r_0 \partial_{\theta^\alpha} \theta_0$ satisfies $\langle v, w \rangle_g = r_0^2 \langle \theta_0, \partial_{\theta^\alpha} \theta_0 \rangle_g = 0$. By claim (2) of the Gauss lemma (theorem 3.3.4),

$$g(\partial_r, \partial_{\theta^\alpha})|_q = \frac{1}{r_0} \langle d(\exp_p)_v(v), d(\exp_p)_v(w) \rangle_g = \frac{1}{r_0} \langle v, w \rangle_g = 0$$

So g has no $dr d\theta^\alpha$ cross terms, and in the coordinate frame $(\partial_r, \partial_{\theta^1}, \dots, \partial_{\theta^{n-1}})$ the matrix of g is block diagonal, with the 1×1 radial block equal to 1 and the remaining $(n-1) \times (n-1)$ block equal to the restriction of g to the level set $\{r = r_0\} = S_{r_0}(p)$.

The tangential block is a metric on the sphere. Define h_r to be this tangential block, that is the 2-tensor on $S_r(p)$ obtained by restricting g to vectors tangent to the geodesic sphere; it is the first fundamental form of $S_r(p)$ as a submanifold of the Riemannian manifold (M, g) in the sense of definition 1.5.1. By proposition 1.5.2, the restriction of the Riemannian metric g to the tangent spaces of the embedded hypersurface $S_r(p)$ is positive definite, hence h_r is a Riemannian metric on $S_r(p)$ for each $r \in (0, R)$. Assembling the two blocks, in geodesic polar coordinates

$$g = dr^2 + h_r,$$

with h_r a Riemannian metric on the geodesic sphere of radius r and no cross term, which is the assertion, completing the proof.

The form $g = dr^2 + h_r$ separates the geometry near p into two pieces of unequal difficulty. The radial piece dr^2 is universal, the same on every Riemannian manifold, and it is the Gauss lemma that certifies this; travelling out along a radial geodesic, one gains arclength at exactly the rate dr , with no interference from the tangential directions. The tangential piece h_r carries everything that distinguishes one manifold from another near p . For Euclidean space, $\exp_p(r\theta) = p + r\theta$ gives

$h_r = r^2 g_{S^{n-1}}$, the round metric on the sphere of radius r , so $g = dr^2 + r^2 g_{S^{n-1}}$ recovers the polar form of the flat metric of example 1.1.2; the coefficient r^2 is what makes the geodesic sphere of radius r have the Euclidean size. On the round unit sphere S^n the same construction yields $h_r = \sin^2 r g_{S^{n-1}}$, and the deficit between $\sin^2 r$ and r^2 is the tangential compression seen in example 3.3.2, the first appearance of positive curvature. The rate at which h_r departs from the Euclidean $r^2 g_{S^{n-1}}$ as r grows is measured by the second derivatives of the metric, which is where the curvature tensor of the next chapter enters; the Gauss lemma has already fixed the first-order radial behaviour, so the polar form is the correct coordinate system in which to read curvature off the metric.

Example 3.3.6: The Polar Form on a Surface of Revolution

The normal form $g = dr^2 + h_r$ is not only an abstract consequence of the Gauss lemma; on a surface it is a coordinate system one can write down. Take the warped metric

$$g = d\rho^2 + f(\rho)^2 d\varphi^2, \quad f(\rho) > 0,$$

of a surface of revolution as in example 1.5.4, in coordinates (ρ, φ) , and suppose $f(0) = 0$ with $f'(0) = 1$, the condition that makes $\rho = 0$ a smooth pole rather than a cone point. The meridian curves $\varphi = \text{const}$ are the curves along which only ρ varies; their velocity has g -length $|\partial_\rho|_g = \sqrt{g_{\rho\rho}} = 1$, so each meridian is parametrized by arclength, and a direct check from the Christoffel symbols of this warped metric, whose only nonzero entries are $\Gamma_{\rho\varphi}^\rho = -f f'$ and $\Gamma_{\rho\varphi}^\varphi = \Gamma_{\varphi\rho}^\varphi = f'/f$ by proposition 2.3.3, shows the meridians satisfy the geodesic equation (3.1): with $\dot{\varphi} = 0$ one has $\ddot{\rho} = -\Gamma_{\varphi\varphi}^\rho \dot{\varphi}^2 = 0$ and $\ddot{\varphi} + 2\Gamma_{\rho\varphi}^\varphi \dot{\rho}\dot{\varphi} = 0$, both satisfied identically. So the meridians are unit-speed radial geodesics emanating from the pole $\rho = 0$, and ρ is exactly the radial distance parameter r of proposition 3.3.5. Reading off the metric, $dr^2 = d\rho^2$ and

$$h_r = f(r)^2 d\varphi^2,$$

a metric on the geodesic circle $S_r(p)$ (here a one-dimensional sphere, so h_r is a single positive coefficient $f(r)^2$). The Gauss lemma is visible in the absence of a $d\rho d\varphi$ cross term in the warped metric, which was built in from the start; the content of proposition 3.3.5 is that this absence is forced by the geometry, not a lucky feature of the coordinates. The circumference of the geodesic circle of radius r is $\int_0^{2\pi} \sqrt{h_r} d\varphi = 2\pi f(r)$, and the pole condition $f(0) = 0$, $f'(0) = 1$ makes this $2\pi r + O(r^3)$ for small r , matching the Euclidean circumference to leading order exactly as the polar form $g = dr^2 + h_r$ with $h_r \rightarrow r^2 g_{S^{n-1}}$ predicts.

3.4 Geodesics Locally Minimize and the Distance Function

A geodesic was defined in section 3.1 as a curve of vanishing acceleration, a curve that is as straight as the connection allows. Straightness and shortness are separate ideas, and the passage from one to the other is the business of this section. On $\mathbb{R}^n$ a straight segment is the shortest path between its endpoints, and every student meets the two descriptions of a line as interchangeable; on a curved manifold they come apart. A great circle on the sphere is a geodesic, yet the long arc of a great circle between two nearby points is a geodesic that is far from shortest, and no path realizes the distance between antipodal points uniquely. The correct statement is local: near any point a geodesic really is the shortest path to nearby points, and it is the only shortest path up to how it is parametrized. The tool that makes this precise is the Gauss lemma of section 3.3, which says that inside a normal ball the geodesics emanating from the center are orthogonal to the spheres they cross. That single

orthogonality forces any competing path to spend length crossing spheres it need not cross.

Once shortest paths exist locally, the length of the shortest path becomes a candidate for the distance between two points, and the infimum of lengths over all connecting curves defines a function $d_g: M \times M \rightarrow [0, \infty)$. This section constructs d_g , proves that it is a metric in the sense of metric-space theory, and proves that the topology it generates is the topology M already carried as a manifold. That last statement closes the loop opened in box 1.1.4: the Riemannian metric, a tensor, and the distance function, a metric-space datum, are different objects, but the second is built from the first, and the resulting metric space recovers the manifold exactly. Throughout, (M, g) is Riemannian; positive-definiteness is used everywhere here, since the length of a curve must be a nonnegative real number for the infimum to behave, and the indefinite case is excluded.

Recall that a piecewise-smooth curve on $[a, b]$ is a continuous map $\gamma: [a, b] \rightarrow M$ for which there is a partition $a = t_0 < t_1 < \dots < t_k = b$ such that γ restricted to each $[t_{i-1}, t_i]$ is smooth; at the partition points γ has a right and a left velocity, possibly different. The velocity $\dot{\gamma}(t)$ is defined for all but finitely many t , and $|\dot{\gamma}(t)|_g$ is a bounded function with at most finitely many discontinuities, hence integrable.

Definition 3.4.1: Length and the Riemannian Distance

Let (M, g) be a Riemannian manifold. The *length* of a piecewise-smooth curve $\gamma: [a, b] \rightarrow M$ is

$$L(\gamma) = \int_a^b |\dot{\gamma}(t)|_g dt = \int_a^b \sqrt{\langle \dot{\gamma}(t), \dot{\gamma}(t) \rangle_g} dt$$

Two points $p, q \in M$ are joined by a piecewise-smooth curve when M is connected. The *Riemannian distance* between p and q is

$$d_g(p, q) = \inf \{L(\gamma) \mid \gamma \text{ a piecewise-smooth curve from } p \text{ to } q\},$$

the infimum of the lengths of all piecewise-smooth curves γ with $\gamma(a) = p$ and $\gamma(b) = q$.

The length is defined for a piecewise-smooth curve, not only a smooth one, because the curves that compete to be shortest are built by concatenation: to go from p to q one might travel along one geodesic and then another, and the join is a corner. The integral does not see the corner, a finite set of points of measure zero, so length is additive over concatenation and unchanged by inserting extra partition points. Length is also invariant under reparametrization. If $\varphi: [c, d] \rightarrow [a, b]$ is a piecewise-smooth increasing bijection and $\tilde{\gamma} = \gamma \circ \varphi$, then $\dot{\tilde{\gamma}}(s) = \varphi'(s) \dot{\gamma}(\varphi(s))$ where $\varphi' > 0$, so $|\dot{\tilde{\gamma}}(s)|_g = \varphi'(s) |\dot{\gamma}(\varphi(s))|_g$, and the substitution $t = \varphi(s)$ gives $L(\tilde{\gamma}) = L(\gamma)$. Thus length is a property of the image curve together with its orientation, and the infimum defining d_g may as well range over curves parametrized on $[0, 1]$, or over unit-speed curves, without changing its value. The choice of parametrization that makes analysis cleanest is arc length: a curve γ on $[a, b]$ with $|\dot{\gamma}|_g \equiv 1$ has $L(\gamma) = b - a$, and every piecewise-smooth curve with nowhere-vanishing speed admits a unit-speed reparametrization.

That d_g is a nonnegative real number and not just a formal infimum requires that some curve join p to q and that lengths be bounded below, both of which hold: on a connected manifold any two points are joined by a piecewise-smooth curve, and lengths are nonnegative, so the infimum is a well-defined element of $[0, \infty)$.

Example 3.4.2: Length and Distance in Euclidean Space

On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, with $\bar{g} = \delta_{ij} dx^i dx^j$, the length of a piecewise-smooth curve $\gamma = (\gamma^1, \dots, \gamma^n)$ is

$$L(\gamma) = \int_a^b \sqrt{(\dot{\gamma}^1)^2 + \dots + (\dot{\gamma}^n)^2} dt,$$

the ordinary arc length of vector calculus. For the straight segment $\gamma(t) = p + t(q - p)$ on $[0, 1]$ one has $\dot{\gamma} \equiv q - p$, so $L(\gamma) = |q - p|_{\bar{g}}$, the Euclidean norm of the displacement. For any other piecewise-smooth curve σ from p to q , the fundamental theorem of calculus applied to each coordinate gives $q - p = \int_a^b \dot{\sigma}(t) dt$, and the triangle inequality for the vector-valued integral yields

$$|q - p|_{\bar{g}} = \left| \int_a^b \dot{\sigma}(t) dt \right|_{\bar{g}} \leq \int_a^b |\dot{\sigma}(t)|_{\bar{g}} dt = L(\sigma)$$

Hence $L(\sigma) \geq |q - p|_{\bar{g}}$ for every competitor, with equality for the straight segment, so

$$d_{\bar{g}}(p, q) = |q - p|_{\bar{g}}$$

The Riemannian distance on Euclidean space is exactly the Euclidean distance, and the infimum is attained precisely by the straight segments, that is, by the geodesics of $\bar{g}$ (whose geodesic equation $\ddot{\gamma}^k = 0$, from eq. (3.1) with vanishing Christoffel symbols, has the affine lines as solutions). This is the model that the local minimizing theorem generalizes: a geodesic is locally the unique shortest path, and the equality case of a triangle inequality is what pins down uniqueness.

The Euclidean computation is a template. The general local statement replaces the straight segment by a radial geodesic inside a normal ball and replaces the coordinatewise triangle inequality by the Gauss lemma. Fix $p \in M$ and a normal ball $B = \exp_p(\{v \in T_p M \mid |v|_g < \varepsilon\})$ on which $\exp_p$ is a diffeomorphism onto its image, as produced in section 3.2. For $q \in B$ with $q = \exp_p(v)$, the *radial geodesic* to q is $\gamma_v(t) = \exp_p(tv)$ on $[0, 1]$; it has constant speed $|v|_g$ by proposition 3.1.6, so its length is $L(\gamma_v) = |v|_g$, the radius at which q sits. The claim is that no piecewise-smooth path from p to q is shorter, and that the radial geodesic is the only one that ties.

Theorem 3.4.3: Radial Geodesics Locally Minimize

Let (M, g) be a Riemannian manifold, $p \in M$, and let $B = \exp_p(B_\varepsilon(0))$ be a geodesic ball about p , where $B_\varepsilon(0) = \{v \in T_p M \mid |v|_g < \varepsilon\}$ and $\exp_p$ is a diffeomorphism from $B_\varepsilon(0)$ onto B . Let $q \in B$, write $q = \exp_p(v)$ with $r = |v|_g < \varepsilon$, and let $\gamma_v(t) = \exp_p(tv)$, $t \in [0, 1]$, be the radial geodesic from p to q . Then for every piecewise-smooth curve $\sigma: [a, b] \rightarrow M$ with $\sigma(a) = p$ and $\sigma(b) = q$,

$$L(\sigma) \geq r = L(\gamma_v),$$

and equality holds if and only if σ is a monotone reparametrization of γ_v . Consequently the radial geodesic is the unique shortest piecewise-smooth path from p to q up to reparametrization, and $d_g(p, q) = r$.

Proof:

Write $q = \exp_p(v)$, $r = |v|_g$. Introduce geodesic polar coordinates on the punctured ball $B \setminus \{p\}$: by theorem 3.3.4 and proposition 3.3.5, on $B \setminus \{p\}$ the metric takes the form

$$g = dr^2 + h_r, \quad (3.7)$$

where r is the radial coordinate and h_r is for each fixed r a Riemannian metric on the geodesic sphere $\{x \in B \mid r(x) = r\}$; in particular h_r is positive semidefinite on the whole tangent space and vanishes exactly on the radial direction ∂_r . The radial coordinate function $r: B \setminus \{p\} \rightarrow (0, \varepsilon)$ is smooth, being $|\exp_p^{-1}(\cdot)|_g$, and its differential dr is dual to ∂_r ; the decomposition (3.7) is the content of the Gauss lemma, namely that ∂_r has unit length and is g -orthogonal to every vector tangent to a geodesic sphere.

Step 1: the length of any path dominates the change in radius. Let $\sigma: [a, b] \rightarrow M$ be piecewise smooth with $\sigma(a) = p$ and $\sigma(b) = q$. Reduce to the case that σ maps into the closed ball $\overline{B_r} = \exp_p(\{w \mid |w|_g \leq r\})$. If $\sigma([a, b]) \subseteq \overline{B_r}$ this holds already. Otherwise σ leaves $\overline{B_r}$, so there is a time t^* with $\sigma(t^*) \notin \overline{B_r}$. Choose a radius ρ_0 with $r < \rho_0 < \varepsilon$ small enough that $\sigma(t^*)$ lies outside the open geodesic ball $B_{\rho_0}^\circ := \exp_p(\{w \mid |w|_g < \rho_0\})$: if $\sigma(t^*) \in B$ it has some radius in (r, ε) and any ρ_0 below that radius serves, while if $\sigma(t^*) \notin B$ then $\sigma(t^*)$ lies outside every such ball with $\rho_0 < \varepsilon$. Let $c = \inf \{t \in [a, b] \mid \sigma(t) \notin B_{\rho_0}^\circ\}$ be the first time σ exits $B_{\rho_0}^\circ$; the defining set is nonempty (it contains t^*) and $\sigma(a) = p \in B_{\rho_0}^\circ$, so $a < c$. On $[a, c]$ the curve stays inside $B_{\rho_0}^\circ$, hence inside B , and by continuity $\sigma(c)$ lies in the closed ball $\overline{B_{\rho_0}}$ but not in $B_{\rho_0}^\circ$, so $\sigma(c)$ has radius exactly ρ_0 , that is, $\sigma(c)$ lies on the geodesic sphere $S_{\rho_0} = \exp_p(\{w \mid |w|_g = \rho_0\})$.

Thus $\sigma|_{[a, c]}$ is a path from p to a point of radius ρ_0 contained in $\overline{B_{\rho_0}}$. The estimate proved below, applied with ρ_0 in place of r to this portion, gives $L(\sigma) \geq L(\sigma|_{[a, c]}) \geq \rho_0 > r$, so the claimed inequality $L(\sigma) \geq r$ holds outright. It therefore suffices to prove the bound, and to analyze the equality case, for a path σ with $\sigma([a, b]) \subseteq \overline{B_r}$; assume this from now on.

Let $a_0 = \sup \{t \in [a, b] \mid \sigma(t) = p\}$ be the last time σ sits at p . On $(a_0, b]$ the curve avoids p , so $\rho(t) := r(\sigma(t))$ is defined and piecewise smooth there, with $\rho(t) \rightarrow 0$ as $t \downarrow a_0$ and $\rho(b) = r$. Where σ is smooth and away from p , decompose the velocity using (3.7):

$$|\dot{\sigma}|_g^2 = \langle \dot{\sigma}, \dot{\sigma} \rangle_g = dr(\dot{\sigma})^2 + h_r(\dot{\sigma}, \dot{\sigma}) = \dot{\rho}^2 + h_r(\dot{\sigma}, \dot{\sigma}),$$

using $dr(\dot{\sigma}) = \frac{d}{dt}r(\sigma(t)) = \dot{\rho}$. Since h_r is positive semidefinite, $h_r(\dot{\sigma}, \dot{\sigma}) \geq 0$, so

$$|\dot{\sigma}(t)|_g = \sqrt{\dot{\rho}(t)^2 + h_r(\dot{\sigma}, \dot{\sigma})} \geq \sqrt{\dot{\rho}(t)^2} = |\dot{\rho}(t)| \geq \dot{\rho}(t) \quad (3.8)$$

at every t where $\dot{\sigma}$ is defined and $\sigma(t) \neq p$.

Step 2: integrate. For $a_0 < c' < b$, integrating (3.8) over $[c', b]$ and using the fundamental theorem of calculus for the piecewise-smooth function ρ gives

$$\int_{c'}^b |\dot{\sigma}(t)|_g dt \geq \int_{c'}^b \dot{\rho}(t) dt = \rho(b) - \rho(c') = r - \rho(c')$$

Letting $c' \downarrow a_0$, the right side tends to $r - 0 = r$, since $\rho(c') \rightarrow 0$, while the left side increases to $\int_{a_0}^b |\dot{\sigma}|_g dt \leq L(\sigma)$. Hence

$$L(\sigma) \geq \int_{a_0}^b |\dot{\sigma}|_g dt \geq r = L(\gamma_v),$$

which is the asserted inequality for a path contained in $\overline{B_r}$; by the reduction of Step 1 the bound $L(\sigma) \geq r$ then holds for every competitor.

Step 3: the equality case. Suppose $L(\sigma) = r$. Then every inequality above is an equality. In particular $\int_{a_0}^b |\dot{\sigma}|_g dt = L(\sigma)$, forcing $|\dot{\sigma}(t)|_g = 0$ for almost every $t \in [a, a_0]$, so σ is constant at p on $[a, a_0]$ and may be discarded by reparametrization. On $(a_0, b]$ equality in (3.8) at almost every t requires, first, $h_r(\dot{\sigma}, \dot{\sigma}) = 0$, hence $\dot{\sigma}$ has no component tangent to the geodesic spheres, so $\dot{\sigma}$ is a radial vector, a multiple of ∂_r ; and second, $|\dot{\sigma}| = \dot{\rho}$, hence $\dot{\rho} \geq 0$, so ρ is nondecreasing. Thus wherever σ moves it moves radially and outward. A curve whose velocity is everywhere a nonnegative multiple of ∂_r stays on the radial line through q : in the coordinates $\exp_p^{-1}$, its direction $v/|v|_g$ is fixed and only its radius ρ varies, so $\sigma(t) = \exp_p(\rho(t)v/r)$ with ρ nondecreasing from 0 to r . This is precisely a monotone reparametrization of $\gamma_v(t) = \exp_p(tv)$, obtained by the increasing change of parameter $t \mapsto \rho(t)/r$. Conversely, any monotone reparametrization of γ_v has length r by reparametrization invariance. Therefore equality holds if and only if σ is a monotone reparametrization of γ_v .

Finally, the inequality $L(\sigma) \geq r$ over all competitors gives $d_g(p, q) \geq r$, while the radial geodesic itself has length r , so $d_g(p, q) = r$, completing the proof.

The theorem is the exact local analogue of the Euclidean computation of example 3.4.2, and the two proofs run in parallel. In both, one bounds the length of an arbitrary path by the length of the geodesic, using an inequality that becomes an equality only for the geodesic itself; in the Euclidean case the inequality is the triangle inequality for the integral of the velocity, and in the general case it is the pointwise inequality (3.8) coming from the Gauss-lemma decomposition $g = dr^2 + h_r$. The decomposition is what does the work: it says that the only length that counts toward reaching a farther sphere is the radial part $\dot{\rho}$, and any tangential motion is wasted length that a path could avoid by going straight out. A radial geodesic wastes nothing, which is why it, and it alone up to reparametrization, realizes the distance. The word *local* is not decorative. The conclusion holds only for q inside a normal ball on which $\exp_p$ is a diffeomorphism; beyond that ball the exponential map may fail to be injective, radial geodesics may cross, and a radial geodesic can stop being the shortest path, as the long great-circle arc on the sphere shows.

The round sphere makes both the local minimizing property and its limits concrete: on it the Riemannian distance is a closed-form function of the ambient inner product, the minimizer is a short great-circle arc, and the one place the minimizer fails to be unique is exactly the pair of antipodal points that no normal ball contains.

Example 3.4.4: The Distance Function on the Round Sphere

On the round sphere $S_r^n = \{x \in \mathbb{R}^{n+1} \mid |x| = r\}$ of example 1.3.4, with the metric g_r induced from $(\mathbb{R}^{n+1}, \bar{g})$, the Riemannian distance between two points is

$$d_{g_r}(p, q) = r \arccos\left(\frac{\langle p, q \rangle}{r^2}\right),$$

where $\langle -, - \rangle$ is the Euclidean inner product of $\mathbb{R}^{n+1}$. This is the length of the shorter of the two great-circle arcs joining p to q , and the minimizing geodesic is unique unless p and q are antipodal.

To compute it, first record the competitor whose length the formula reports. The geodesics of S_r^n are the great circles, the intersections of S_r^n with 2-planes through the origin, traced at constant

speed, as established in example 3.1.4. Fix $p, q \in S_r^n$ that are not antipodal, so $q \neq -p$, and set

$$\alpha = \arccos\left(\frac{\langle p, q \rangle}{r^2}\right) \in [0, \pi),$$

the angle at the origin subtended by p and q ; the restriction $\alpha < \pi$ is the non-antipodal hypothesis, since $\langle p, q \rangle = -r^2$ is the antipodal case. If $q = p$ then $\alpha = 0$ and both sides of the asserted formula vanish, so assume $q \neq p$. Then $q \neq \pm p$, and the vectors p, q span a 2-plane; let $u = p/r$ and let w be the unit vector in that plane obtained by orthonormalizing q against u , so that

$$q = r \cos \alpha u + r \sin \alpha w, \quad \langle u, u \rangle = \langle w, w \rangle = 1, \quad \langle u, w \rangle = 0$$

Define the unit-speed great-circle arc $\gamma: [0, r\alpha] \rightarrow S_r^n$ by

$$\gamma(s) = r \cos\left(\frac{s}{r}\right)u + r \sin\left(\frac{s}{r}\right)w$$

Then $|\dot{\gamma}(s)|^2 = r^2 \cos^2(s/r) + r^2 \sin^2(s/r) = r^2$, so γ lands in S_r^n ; it satisfies $\gamma(0) = ru = p$ and $\gamma(r\alpha) = r \cos \alpha u + r \sin \alpha w = q$. Its velocity is $\dot{\gamma}(s) = -\sin(s/r)u + \cos(s/r)w$, a unit vector by orthonormality of u, w , so γ is a geodesic of unit speed. Its length is

$$L(\gamma) = \int_0^{r\alpha} |\dot{\gamma}(s)|_{g_r} ds = \int_0^{r\alpha} 1 ds = r\alpha,$$

the length of the short great-circle arc from p to q . Hence $d_{g_r}(p, q) \leq r\alpha$.

The reverse inequality is where theorem 3.4.3 enters. The exponential map $\exp_p$ of S_r^n is a diffeomorphism from the open ball $\{v \in T_p S_r^n \mid |v|_{g_r} < \pi r\}$ onto $S_r^n \setminus \{-p\}$: a unit-speed geodesic from p returns to p after arc length $2\pi r$ and reaches the antipode $-p$ at arc length πr , so no two radial directions meet before that, and every point other than $-p$ is hit exactly once at a radius below πr . Thus for non-antipodal q the geodesic ball $B = \exp_p(\{v \mid |v|_{g_r} < \pi r\})$ contains q , and $q = \exp_p(v)$ with $r\alpha = |v|_{g_r} < \pi r$. The theorem applies verbatim: $d_{g_r}(p, q) = |v|_{g_r} = r\alpha$, and the radial geodesic, which is the short great-circle arc γ above, is the unique shortest path up to reparametrization. This proves the formula for $q \neq -p$.

For antipodal $q = -p$ the formula still reads $d_{g_r}(p, q) = r \arccos(-1) = r\pi$, and this too is correct, by continuity or directly: every great-circle arc from p to $-p$ is a half great circle of length $r\pi$, and any path attaining a smaller length would, by the estimate (3.8) applied on a normal ball about p out to radius approaching πr , be forced to have length at least πr in the limit, so $d_{g_r}(p, -p) = r\pi$. Uniqueness fails here in an extreme way: through p and $-p$ there pass infinitely many great circles (one for every 2-plane containing the line $\mathbb{R}p$), each contributing a half-circle of length $r\pi$, so infinitely many distinct geodesics realize the distance. The antipode $-p$ is the first point at which radial geodesics from p cease to minimize uniquely, the simplest instance of the *cut locus*, taken up in a later chapter; here it appears as the exact boundary of the normal ball, the one place the local theorem cannot reach.

An immediate consequence is a lower bound for d_g in terms of the exponential radius, the tool that turns the local minimizing property into a statement about the topology.

Corollary 3.4.5: Distance to Points of a Normal Ball

With the notation of theorem 3.4.3, for every $q = \exp_p(v) \in B$ one has $d_g(p, q) = |v|_g$. Consequently, for every δ with $0 < \delta < \varepsilon$ small enough that the metric ball $\{x \in M \mid d_g(p, x) < \delta\}$ is contained in B , the metric ball and the geodesic ball of radius δ coincide:

$$\{x \in M \mid d_g(p, x) < \delta\} = \exp_p(\{v \mid |v|_g < \delta\})$$

Proof:

The equality $d_g(p, q) = |v|_g$ is the last assertion of theorem 3.4.3, applied to q in place of the running point. Such a δ exists, and in fact any δ with $0 < \delta < \varepsilon$ works. To see this, it suffices to check that $d_g(p, x) \geq \varepsilon$ for every $x \in M \setminus B$, for then $d_g(p, x) < \delta < \varepsilon$ forces $x \in B$, so the metric ball of radius δ is contained in B . Fix $x \in M \setminus B$ and let γ be any piecewise-smooth curve from p to x . Since $x \notin B$, the curve γ leaves B , so for each $\rho < \varepsilon$ it reaches radius at least ρ ; as in Step 1 of theorem 3.4.3, the intermediate value theorem gives a first time at which γ meets the geodesic sphere $S_\rho = \exp_p(\{v \mid |v|_g = \rho\})$, and the estimate (3.8) integrated out to radius ρ shows that the portion of γ up to that meeting has length at least ρ , whence $L(\gamma) \geq \rho$. As this holds for every $\rho < \varepsilon$, $L(\gamma) \geq \varepsilon$; taking the infimum over γ gives $d_g(p, x) \geq \varepsilon$. Thus $d_g(p, M \setminus B) \geq \varepsilon > 0$ (the same argument used later in the diagonal-vanishing part of theorem 3.4.6), so any $\delta < \varepsilon$ works.

Fix such a δ , so that $\{x \mid d_g(p, x) < \delta\} \subseteq B$, and verify the two inclusions. If $d_g(p, x) < \delta$, then $x \in B$ by the choice of δ , so $x = \exp_p(w)$ for a unique $w \in B_\varepsilon(0)$, and the first part gives $|w|_g = d_g(p, x) < \delta$; hence $x \in \exp_p(\{v \mid |v|_g < \delta\})$. Conversely, if $x = \exp_p(w)$ with $|w|_g < \delta$, then $w \in B_\varepsilon(0)$ (as $\delta < \varepsilon$) and the first part gives $d_g(p, x) = |w|_g < \delta$; hence $x \in \{x \mid d_g(p, x) < \delta\}$. The two sets therefore coincide, as required.

With distances to nearby points controlled, the global object d_g can be shown to satisfy the axioms of a metric and to reproduce the manifold topology.

Theorem 3.4.6: The Riemannian Distance Is a Metric

Let (M, g) be a connected Riemannian manifold. Then d_g is a metric on M : for all $p, q, x \in M$,

1. $d_g(p, q) \geq 0$, with $d_g(p, q) = 0$ if and only if $p = q$;
2. $d_g(p, q) = d_g(q, p)$;
3. $d_g(p, x) \leq d_g(p, q) + d_g(q, x)$.

Moreover the metric-space topology induced by d_g coincides with the manifold topology of M . In particular (M, d_g) is the metric space anticipated in box 1.1.4, built from the tensor g yet recovering the original topology.

Proof:

Nonnegativity and symmetry. Lengths are integrals of the nonnegative function $|\dot{\gamma}|_g$, so $L(\gamma) \geq 0$ and hence $d_g \geq 0$. Symmetry follows from reparametrization: if $\gamma: [a, b] \rightarrow M$ runs from p to q , then $\gamma^-(t) = \gamma(a + b - t)$ runs from q to p with $|\dot{\gamma}^-(t)|_g = |\dot{\gamma}(a + b - t)|_g$, so $L(\gamma^-) = L(\gamma)$; the set of curves from p to q and the set from q to p are thus in length-preserving bijection, and their infima agree, giving $d_g(p, q) = d_g(q, p)$.

Triangle inequality. Fix p, q, x and $\eta > 0$. Choose a piecewise-smooth curve α from p to q with $L(\alpha) < d_g(p, q) + \eta/2$ and a piecewise-smooth curve β from q to x with $L(\beta) < d_g(q, x) + \eta/2$; these exist by definition of the infimum. The concatenation $\alpha * \beta$, which follows α and then β , is a piecewise-smooth curve from p to x (the join at q is at worst a corner, and length is additive over concatenation), so

$$d_g(p, x) \leq L(\alpha * \beta) = L(\alpha) + L(\beta) < d_g(p, q) + d_g(q, x) + \eta$$

As $\eta > 0$ was arbitrary, $d_g(p, x) \leq d_g(p, q) + d_g(q, x)$.

Vanishing exactly on the diagonal. If $p = q$ the constant curve has length 0, so $d_g(p, p) = 0$. The content is the converse: $d_g(p, q) = 0$ forces $p = q$. Suppose $p \neq q$. Choose a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p small enough that $q \notin B$; this is possible because B is an open neighborhood of p and M is Hausdorff, so shrinking ε excludes the fixed point $q \neq p$. Any piecewise-smooth curve γ from p to q must leave B , hence must cross the geodesic sphere $S_\delta = \exp_p(\{v \mid |v|_g = \delta\})$ for every $0 < \delta < \varepsilon$; in particular it crosses $S_{\varepsilon/2}$. By the argument of theorem 3.4.3 (the estimate (3.8) integrated out to radius $\varepsilon/2$), the portion of γ from p to its first meeting with $S_{\varepsilon/2}$ has length at least $\varepsilon/2$, so $L(\gamma) \geq \varepsilon/2$. Taking the infimum over γ gives $d_g(p, q) \geq \varepsilon/2 > 0$. Hence $d_g(p, q) = 0$ implies $p = q$, and d_g is positive off the diagonal. This is the one place positive-definiteness of g is used decisively: it makes $|\dot{\gamma}|_g$ a genuine norm, so that a nonconstant radial motion contributes strictly positive length and distinct points cannot be at distance zero.

The topologies agree. It must be shown that a set is open in the manifold topology if and only if it is open in the d_g -metric topology. It suffices to compare bases of neighborhoods at an arbitrary point p , that is, to show that every manifold-open neighborhood of p contains a metric ball about p , and every metric ball about p contains a manifold-open neighborhood of p .

Fix p and a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p on which corollary 3.4.5 applies. For $0 < \delta < \varepsilon$ small enough that the metric ball $\{x \mid d_g(p, x) < \delta\}$ is contained in B , that corollary gives

$$\{x \in M \mid d_g(p, x) < \delta\} = \exp_p(\{v \in T_p M \mid |v|_g < \delta\}) \quad (3.9)$$

The right-hand side is the image under the diffeomorphism $\exp_p$ of a Euclidean-open ball in $T_p M$, hence a manifold-open set; so (3.9) exhibits every sufficiently small metric ball about p as a manifold-open neighborhood of p . This shows every metric ball about p contains a manifold-open neighborhood of p (indeed small ones are manifold-open), giving one inclusion of topologies: every d_g -open set is manifold-open.

Conversely, let W be a manifold-open neighborhood of p . Since $\exp_p$ is a diffeomorphism on $B_\varepsilon(0)$ and continuous, $\exp_p^{-1}(W \cap B)$ is an open neighborhood of 0 in $T_p M$, so it contains a Euclidean ball $\{v \mid |v|_g < \delta\}$ for some $0 < \delta < \varepsilon$, which may be taken small enough that (3.9)

holds. Then, applying $\exp_p$,

$$\{x \mid d_g(p, x) < \delta\} = \exp_p(\{v \mid |v|_g < \delta\}) \subseteq W \cap B \subseteq W$$

Thus W contains the metric ball of radius δ about p , so W is a d_g -neighborhood of p ; every manifold-open set is therefore d_g -open. The two topologies have the same neighborhoods at every point and hence coincide, as we sought to show.

The theorem resolves the caution recorded at the very start of the book. In box 1.1.4 the word “metric” was flagged as carrying two incompatible meanings, the tensor g and a metric-space distance, and the passage from the first to the second was promised here. The promise is now kept: the tensor g produces, through the length functional and an infimum, a function d_g that satisfies the three axioms of a metric, and the metric space (M, d_g) carries exactly the topology M already had. Nothing about the point-set topology of the manifold is lost or gained by remembering the metric; what is gained is quantitative, a numerical distance between points where before there was only nearness. That the metric topology reproduces the manifold topology is what licenses the free interchange, in all later chapters, between analytic statements phrased with d_g (Cauchy sequences, completeness, bounded sets) and topological statements phrased with charts. Completeness of (M, d_g) and the equivalence of metric completeness with geodesic completeness are the subject of the Hopf–Rinow theorem later; here the point is only that d_g is a metric at all, and the correct one for the manifold.

One direction of the correspondence between geodesics and shortest paths has been established: radial geodesics minimize locally. The reverse direction asks whether a curve that minimizes length must be a geodesic. It is, once parametrized correctly, and the cleanest proof is variational.

Proposition 3.4.7: Minimizers Are Geodesics

Let (M, g) be a Riemannian manifold and let $\gamma: [a, b] \rightarrow M$ be a piecewise-smooth, unit-speed curve that minimizes length among piecewise-smooth curves with the same endpoints:

$$L(\gamma) = d_g(\gamma(a), \gamma(b))$$

Then γ is a geodesic; in particular γ is smooth (with no corners) and satisfies $D_t \dot{\gamma} = 0$.

Proof:

The argument has two parts: first, a length minimizer is locally a shortest path between nearby points of its image, hence locally a radial geodesic and so smooth with $D_t \dot{\gamma} = 0$; second, a global verification that the pieces fit with no corner. Both follow from theorem 3.4.3 and its uniqueness clause.

Step 1: local coincidence with a radial geodesic. Fix $t_0 \in [a, b]$ and put $p = \gamma(t_0)$. Choose a geodesic ball $B = \exp_p(B_\varepsilon(0))$ about p on which theorem 3.4.3 applies. By continuity of γ , there is $\tau > 0$ with $\gamma([t_0 - \tau, t_0 + \tau] \cap [a, b]) \subseteq B$. Take any $s_1 < s_2$ in this parameter interval and set $q_i = \gamma(s_i)$. The restriction $\gamma|_{[s_1, s_2]}$ is a piecewise-smooth curve from q_1 to q_2 , and it is length-minimizing among such curves: if some curve σ from q_1 to q_2 were shorter, replacing $\gamma|_{[s_1, s_2]}$ by σ inside γ would produce a piecewise-smooth curve from $\gamma(a)$ to $\gamma(b)$ of length

$$L(\gamma|_{[a, s_1]}) + L(\sigma) + L(\gamma|_{[s_2, b]}) < L(\gamma|_{[a, s_1]}) + L(\gamma|_{[s_1, s_2]}) + L(\gamma|_{[s_2, b]}) = L(\gamma),$$

contradicting minimality of γ . Hence $L(\gamma|_{[s_1, s_2]}) = d_g(q_1, q_2)$. Now $q_1, q_2 \in B$, and q_2 lies in a

geodesic ball about q_1 once $s_2 - s_1$ is small (shrink τ so that the image lies in a ball on which theorem 3.4.3 applies with center q_1); by the uniqueness clause of that theorem, the unique shortest path from q_1 to q_2 up to reparametrization is the radial geodesic. Therefore $\gamma|_{[s_1, s_2]}$ is a monotone reparametrization of a geodesic. Since γ is unit-speed by hypothesis and geodesics have constant speed by proposition 3.1.6, the reparametrization is affine with unit derivative, so $\gamma|_{[s_1, s_2]}$ is itself a geodesic. As $s_1 < s_2$ near t_0 were arbitrary, γ agrees near t_0 with a geodesic and hence is smooth there and satisfies $D_t \dot{\gamma} = 0$ on a neighborhood of t_0 .

Step 2: no corners, and global smoothness. Step 1 applies at every $t_0 \in [a, b]$, including the finitely many partition points where γ was only assumed piecewise smooth. At such a point t_0 the curve coincides on a two-sided neighborhood with a single geodesic $\tilde{\gamma}$: the local minimizer through t_0 is a radial geodesic from $\gamma(t_0)$, and by the argument of Step 1 applied on $[s_1, t_0]$ and on $[t_0, s_2]$, both the incoming and outgoing pieces are unit-speed geodesics through $\gamma(t_0)$ realizing the distance to the same nearby points; the uniqueness clause of theorem 3.4.3 forces them to be the two halves of one radial geodesic, so their velocities match at t_0 , $\dot{\gamma}(t_0^-) = \dot{\gamma}(t_0^+)$. Thus γ has no corner at t_0 , and being a solution of the geodesic equation on each side with matching position and velocity, it is by the uniqueness in theorem 3.1.3 the restriction of a single geodesic across t_0 . Hence γ is smooth on all of $[a, b]$ and satisfies $D_t \dot{\gamma} = 0$ throughout, so γ is a geodesic, as required.

Together with theorem 3.4.3 this closes the local circle between the two descriptions of a straight line. Straightness, the vanishing of covariant acceleration, and shortness, the realization of Riemannian distance, are locally the same condition, tied by the exponential map and the Gauss lemma: a radial geodesic is the unique local minimizer, and a unit-speed local minimizer is a geodesic. The proposition is stated for a curve minimizing over its full parameter interval, but its proof is entirely local, using only that short subarcs of a minimizer minimize; this is the property that makes the variational characterization of section 3.5 the natural next tool, where the same conclusion is obtained by differentiating the length and energy functionals and reading off that their critical points are exactly the geodesics. The two viewpoints, the direct distance estimate of this section and the variational calculus of the next, are complementary: the estimate gives uniqueness of the local minimizer outright, while the variation gives the geodesic equation as the vanishing of a derivative and extends without change to the indefinite setting where lengths alone no longer suffice.

3.5 First Variation of Energy and Length

The geodesic equation was introduced in section 3.1 as an equation of motion: a geodesic is a curve whose velocity does not turn, $D_t \dot{\gamma} = 0$. In section 3.4 geodesics acquired a second character, that of locally shortest paths, established inside a normal ball by the Gauss lemma. These two descriptions, autoparallel and length-minimizing, are logically distinct, and the bridge between them is variational. A shortest path is one that cannot be made shorter by any small deformation, and to make that idea precise means differentiating the length of a curve with respect to a parameter that deforms it. This section carries out that differentiation and reads off its consequence: the curves for which the derivative vanishes, the *critical points* of the length, are exactly the geodesics.

Length is not the most convenient functional to differentiate. Its integrand $|\dot{\gamma}|_g$ carries a square root, which is singular wherever the velocity vanishes and which obscures the algebra of the variation. The energy functional, whose integrand is $\frac{1}{2} |\dot{\gamma}|_g^2$, has no square root and no such singularity, and its critical points turn out to be the geodesics on the nose, with no reparametrization ambiguity.

The two functionals are tied together by a Cauchy–Schwarz inequality, $L(\gamma)^2 \leq 2(b-a)E(\gamma)$, with equality precisely for curves of constant speed. The strategy of the section is therefore to compute the first variation of energy first, extract the geodesic equation from it, and then transfer the conclusion to length by way of the inequality. Throughout, (M, g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ of definition 2.3.2, and D_t is the induced covariant derivative along a curve from definition 2.1.6; the Einstein summation convention is in force.

Definition 3.5.1: Energy, Length, and Admissible Variations

Let $\gamma: [a, b] \rightarrow M$ be a piecewise-smooth curve. The *length* of γ is

$$L(\gamma) = \int_a^b |\dot{\gamma}(t)|_g dt,$$

as in definition 3.4.1, and the *energy* of γ is

$$E(\gamma) = \frac{1}{2} \int_a^b |\dot{\gamma}(t)|_g^2 dt = \frac{1}{2} \int_a^b \langle \dot{\gamma}(t), \dot{\gamma}(t) \rangle dt$$

A *variation* of γ is a continuous map $\Gamma: (-\varepsilon, \varepsilon) \times [a, b] \rightarrow M$, for some $\varepsilon > 0$, that is piecewise smooth in the sense that there is a partition $a = t_0 < t_1 < \dots < t_m = b$ with Γ smooth on each rectangle $(-\varepsilon, \varepsilon) \times [t_{i-1}, t_i]$, and such that $\Gamma(0, t) = \gamma(t)$ for all t . Each curve $\gamma_s := \Gamma(s, \cdot)$ is a *longitudinal* curve of the variation. The *variation field* of Γ is the piecewise-smooth vector field along γ

$$V(t) = \left. \frac{\partial}{\partial s} \right|_{s=0} \Gamma(s, t) = \frac{\partial \Gamma}{\partial s}(0, t) \in T_{\gamma(t)}M$$

The variation is *proper*, or has *fixed endpoints*, if $\Gamma(s, a) = \gamma(a)$ and $\Gamma(s, b) = \gamma(b)$ for all s , equivalently if $V(a) = V(b) = 0$. A curve γ is a *critical point* of E (respectively of L) for proper variations if $\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = 0$ (respectively $\left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = 0$) for every proper variation Γ of γ .

A variation is a one-parameter family of curves through γ , and its variation field V records the initial velocity of that family: at each time t , the vector $V(t)$ is the direction in which the point $\gamma(t)$ begins to move as the parameter s is switched on. Deforming γ while holding its endpoints fixed is exactly the operation whose infinitesimal effect on length one wishes to measure, and the properness condition $V(a) = V(b) = 0$ encodes “holding the endpoints fixed” at the level of the variation field. The energy functional weights the curve by the square of its speed, so it penalizes both length and uneven parametrization; a curve that covers a fixed image but speeds up and slows down carries more energy than one that traverses the same image at constant speed. This double sensitivity is what makes energy the sharper of the two functionals, and it is the reason its critical points are geodesics without the reparametrization slack that length carries.

The variation field need not come from a prescribed variation to be useful; conversely, any vector field along γ is the variation field of some variation, as the following construction shows and as the criticality arguments below will use.

Example 3.5.2: Variations built from a vector field

Let $\gamma: [a, b] \rightarrow M$ be a smooth curve and let V be any piecewise-smooth vector field along γ . A variation of γ with variation field V can be produced from the exponential map of definition 3.2.1. For each t , the geodesic $s \mapsto \exp_{\gamma(t)}(sV(t))$ starts at $\gamma(t)$ with initial velocity $V(t)$; setting

$$\Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$$

defines, for $|s|$ small enough that $sV(t)$ lies in the domain of $\exp_{\gamma(t)}$ for every t in the compact interval $[a, b]$, a piecewise-smooth variation with $\Gamma(0, t) = \exp_{\gamma(t)}(0) = \gamma(t)$. Its variation field is

$$\left. \frac{\partial}{\partial s} \right|_{s=0} \exp_{\gamma(t)}(sV(t)) = d(\exp_{\gamma(t)})_0(V(t)) = V(t),$$

the last equality by proposition 3.2.5. If moreover $V(a) = V(b) = 0$, then $\Gamma(s, a) = \exp_{\gamma(a)}(0) = \gamma(a)$ and likewise at b , so the variation is proper. Thus every (piecewise-smooth) vector field along γ vanishing at the endpoints arises as the variation field of a proper variation, a fact that will let the first-variation formula be tested against an arbitrary such field.

As a concrete instance, take $\gamma(t) = (t, 0)$ in the plane $\mathbb{R}^2$ with its Euclidean metric, $t \in [0, 1]$, and $V(t) = (0, \sin(\pi t))$, which vanishes at both endpoints. Here $\exp_p(w) = p + w$, so $\Gamma(s, t) = (t, s \sin(\pi t))$: the segment is bowed into a family of arches, pinned at $(0, 0)$ and $(1, 0)$, and $\frac{\partial \Gamma}{\partial s}(0, t) = (0, \sin \pi t) = V(t)$ as required.

The computation of the first variation rests on one identity, the symmetry of the two covariant derivatives that a variation makes available. A smooth variation Γ has two coordinate directions, s and t , and hence two families of vector fields along it: $\partial_t \Gamma = \frac{\partial \Gamma}{\partial t}$, the velocities of the longitudinal curves, and $\partial_s \Gamma = \frac{\partial \Gamma}{\partial s}$, the variation fields of the transverse curves. Each can be differentiated covariantly in either direction. Because ∇ is torsion-free and the coordinate fields ∂_s, ∂_t commute, the two mixed covariant derivatives agree.

Lemma 3.5.3: Symmetry Lemma

Let $\Gamma: (-\varepsilon, \varepsilon) \times [a, b] \rightarrow M$ be a smooth map, and write D_t and D_s for the covariant derivatives along the t - and s -parameter curves, respectively. Then

$$D_s \partial_t \Gamma = D_t \partial_s \Gamma$$

Proof:

The statement is local in Γ , so it suffices to verify it in a coordinate chart $(U, (x^k))$ about a point $\Gamma(s_0, t_0)$. Write the coordinate expression $\Gamma^k(s, t) = x^k \circ \Gamma$. The two velocity fields have components

$$\partial_t \Gamma = \frac{\partial \Gamma^k}{\partial t} \partial_k, \quad \partial_s \Gamma = \frac{\partial \Gamma^k}{\partial s} \partial_k$$

Applying the coordinate formula for the covariant derivative along a curve from definition 2.1.6 to the field $\partial_t \Gamma$, differentiated in the s -direction, gives

$$D_s \partial_t \Gamma = \left(\frac{\partial^2 \Gamma^k}{\partial s \partial t} + \Gamma_{ij}^k(\Gamma) \frac{\partial \Gamma^i}{\partial s} \frac{\partial \Gamma^j}{\partial t} \right) \partial_k,$$

where Γ_{ij}^k are the Christoffel symbols and the argument Γ indicates evaluation at $\Gamma(s, t)$. Interchanging the roles of s and t ,

$$D_t \partial_s \Gamma = \left(\frac{\partial^2 \Gamma^k}{\partial t \partial s} + \Gamma_{ji}^k(\Gamma) \frac{\partial \Gamma^j}{\partial t} \frac{\partial \Gamma^i}{\partial s} \right) \partial_k$$

The mixed second partial derivatives of the smooth functions Γ^k are equal, so the first terms agree. In the second terms, relabelling the summed indices $i \leftrightarrow j$ in the display for $D_t \partial_s \Gamma$ turns its Christoffel factor into $\Gamma_{ij}^k \frac{\partial \Gamma^i}{\partial s} \frac{\partial \Gamma^j}{\partial t}$, which equals the second term of $D_s \partial_t \Gamma$ because the Levi-Civita connection is torsion-free, $\Gamma_{ij}^k = \Gamma_{ji}^k$, by definition 2.3.2 together with lemma 2.2.2. Hence the two expressions coincide, as we sought to show.

With the symmetry lemma in hand the first variation of energy is a direct calculation, differentiating under the integral sign and integrating by parts. The metric compatibility of ∇ enters at the step where the derivative of an inner product is distributed across its two slots; the symmetry lemma enters at the step where a derivative in the s -direction of the velocity is converted into a derivative in the t -direction of the variation field, so that an integration by parts in t becomes available.

Theorem 3.5.4: First Variation of Energy

Let $\gamma: [a, b] \rightarrow M$ be a piecewise-smooth curve, smooth on each subinterval $[t_{i-1}, t_i]$ of a partition $a = t_0 < t_1 < \dots < t_m = b$, and let Γ be a piecewise-smooth variation of γ with variation field V . Then the energy of the longitudinal curves is differentiable in s at $s = 0$, with

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i \dot{\gamma} \rangle + \left[\langle V, \dot{\gamma} \rangle \right]_a^b,$$

where $\Delta_i \dot{\gamma} := \dot{\gamma}(t_i^+) - \dot{\gamma}(t_i^-)$ is the jump in the velocity at the interior breakpoint t_i , and $\left[\langle V, \dot{\gamma} \rangle \right]_a^b = \langle V(b), \dot{\gamma}(b) \rangle - \langle V(a), \dot{\gamma}(a) \rangle$. In particular, if γ is smooth and Γ is proper, then

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt$$

Consequently a smooth curve γ is a critical point of E for proper variations if and only if $D_t \dot{\gamma} \equiv 0$, that is, if and only if γ is a geodesic.

Proof:

Step 1: differentiation under the integral. Fix first a subinterval $[t_{i-1}, t_i]$ on which Γ is smooth. On $(-\varepsilon, \varepsilon) \times [t_{i-1}, t_i]$ the function $(s, t) \mapsto \langle \partial_t \Gamma, \partial_t \Gamma \rangle$ is smooth, so the energy contribution

$$E_i(s) = \frac{1}{2} \int_{t_{i-1}}^{t_i} \langle \partial_t \Gamma(s, t), \partial_t \Gamma(s, t) \rangle dt$$

may be differentiated under the integral sign. Using the metric compatibility of the Levi-Civita connection (definition 2.2.4) to differentiate the inner product along the s -curve,

$$\frac{d}{ds} E_i(s) = \frac{1}{2} \int_{t_{i-1}}^{t_i} \frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle dt = \int_{t_{i-1}}^{t_i} \langle D_s \partial_t \Gamma, \partial_t \Gamma \rangle dt,$$

the factor $\frac{1}{2}$ cancelling against the two equal terms produced by the Leibniz rule in the symmetric inner product.

Step 2: the symmetry lemma and integration by parts. By lemma 3.5.3, $D_s \partial_t \Gamma = D_t \partial_s \Gamma$, so the integrand becomes $\langle D_t \partial_s \Gamma, \partial_t \Gamma \rangle$. Apply metric compatibility again, now to the t -derivative of the inner product $\langle \partial_s \Gamma, \partial_t \Gamma \rangle$:

$$\frac{d}{dt} \langle \partial_s \Gamma, \partial_t \Gamma \rangle = \langle D_t \partial_s \Gamma, \partial_t \Gamma \rangle + \langle \partial_s \Gamma, D_t \partial_t \Gamma \rangle$$

Solving for the first term on the right and substituting,

$$\frac{d}{ds} E_i(s) = \int_{t_{i-1}}^{t_i} \frac{d}{dt} \langle \partial_s \Gamma, \partial_t \Gamma \rangle dt - \int_{t_{i-1}}^{t_i} \langle \partial_s \Gamma, D_t \partial_t \Gamma \rangle dt$$

The first integral is elementary by the fundamental theorem of calculus, and the second is already in the desired form.

Step 3: evaluation at $s = 0$. At $s = 0$ the transverse field is the variation field, $\partial_s \Gamma(0, t) = V(t)$, and the longitudinal velocity is $\partial_t \Gamma(0, t) = \dot{\gamma}(t)$, whence $D_t \partial_t \Gamma(0, t) = D_t \dot{\gamma}(t)$. Setting $s = 0$ in the last display,

$$\left. \frac{d}{ds} \right|_{s=0} E_i = \left[\langle V, \dot{\gamma} \rangle \right]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle V, D_t \dot{\gamma} \rangle dt$$

Since $E(\gamma_s) = \sum_{i=1}^m E_i(s)$, summing over the subintervals gives

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = \sum_{i=1}^m \left[\langle V, \dot{\gamma} \rangle \right]_{t_{i-1}}^{t_i} - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt$$

The variation field V is continuous across the breakpoints, so $V(t_i^-) = V(t_i^+) = V(t_i)$, whereas $\dot{\gamma}$ may jump. Collecting the boundary contributions, the term at t_i appears once with a plus sign from the subinterval $[t_{i-1}, t_i]$ (evaluated at its right end, using $\dot{\gamma}(t_i^-)$) and once with a minus sign from $[t_i, t_{i+1}]$ (at its left end, using $\dot{\gamma}(t_i^+)$); their sum is $\langle V(t_i), \dot{\gamma}(t_i^-) \rangle - \langle V(t_i), \dot{\gamma}(t_i^+) \rangle = -\langle V(t_i), \Delta_i \dot{\gamma} \rangle$. The two extreme endpoints $t_0 = a$ and $t_m = b$ are not cancelled and contribute $\langle V(b), \dot{\gamma}(b) \rangle - \langle V(a), \dot{\gamma}(a) \rangle = \left[\langle V, \dot{\gamma} \rangle \right]_a^b$. Assembling,

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i \dot{\gamma} \rangle + \left[\langle V, \dot{\gamma} \rangle \right]_a^b,$$

which is the asserted formula.

Step 4: criticality. Suppose now γ is smooth, so that there are no interior breakpoints and the jump sum is absent, and suppose Γ is proper, so that $V(a) = V(b) = 0$ and the boundary term $\left[\langle V, \dot{\gamma} \rangle \right]_a^b$ vanishes. The formula reduces to

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt,$$

as claimed. If γ is a geodesic then $D_t \dot{\gamma} \equiv 0$ and the integral vanishes for every V , so γ is critical. Conversely, suppose γ is critical for all proper variations. By example 3.5.2, every smooth vector field V along γ with $V(a) = V(b) = 0$ is the variation field of a proper variation, so

$\int_a^b \langle V, D_t \dot{\gamma} \rangle dt = 0$ for all such V . Take in particular $V = \varphi D_t \dot{\gamma}$, where $\varphi: [a, b] \rightarrow [0, \infty)$ is smooth, positive on (a, b) , and vanishing at a and b ; this V is admissible. In the Riemannian case $\langle -, - \rangle$ is positive definite, so

$$0 = \int_a^b \langle V, D_t \dot{\gamma} \rangle dt = \int_a^b \varphi |D_t \dot{\gamma}|_g^2 dt,$$

an integral of a nonnegative continuous function that is strictly positive wherever $D_t \dot{\gamma} \neq 0$ on (a, b) . Its vanishing forces $D_t \dot{\gamma} \equiv 0$ on (a, b) , hence on $[a, b]$ by continuity, so γ is a geodesic. In the pseudo-Riemannian case, where the inner product is indefinite and the choice $V = \varphi D_t \dot{\gamma}$ need not give a positive integrand, argue componentwise instead: fix a point $t_* \in (a, b)$ and, working in a coordinate chart around $\gamma(t_*)$, apply the criticality condition to the fields $V = \varphi \partial_k|_\gamma$ for each k and each bump function φ supported near t_* . This yields $\int \varphi g_{kl}(\gamma)(D_t \dot{\gamma})^l dt = 0$ for all such φ , so by the fundamental lemma of the calculus of variations $g_{kl}(\gamma)(D_t \dot{\gamma})^l = 0$ near t_* for every k ; nondegeneracy of (g_{kl}) then gives $(D_t \dot{\gamma})^l = 0$ near t_* . As t_* was arbitrary, $D_t \dot{\gamma} \equiv 0$ on (a, b) , and again γ is a geodesic, completing the proof.

The formula isolates three ways in which a curve can fail to be a critical point of energy, and reading them off is the substance of the theorem. The interior integral $-\int \langle V, D_t \dot{\gamma} \rangle dt$ measures the curve's own acceleration $D_t \dot{\gamma}$, tested against the direction V in which it is deformed; it vanishes for every deformation exactly when the acceleration is zero, that is, when the curve is a geodesic. The jump sum penalizes corners: at a breakpoint where the velocity jumps by $\Delta_i \dot{\gamma}$, deforming in the direction of that jump decreases the energy, so a piecewise-smooth minimizer can have no genuine corners and must be smooth. The boundary term records what happens when the endpoints are allowed to move; for proper variations it is zero, but its form $[\langle V, \dot{\gamma} \rangle]_a^b$ already signals that a curve meeting a boundary condition freely will do so orthogonally to $\dot{\gamma}$, the transversality that governs free-endpoint problems. The clean equivalence “critical for energy $\iff$ geodesic” is the payoff: unlike length, energy has no reparametrization freedom, and its critical points are geodesics parametrized exactly as the equation $D_t \dot{\gamma} = 0$ dictates, at constant speed.

Example 3.5.5: First variation on the flat plane

Return to the segment $\gamma(t) = (t, 0)$ in $\mathbb{R}^2$, $t \in [0, 1]$, with the variation $\Gamma(s, t) = (t, s \sin \pi t)$ and variation field $V(t) = (0, \sin \pi t)$ from example 3.5.2. Since the metric is Euclidean and γ is a straight segment, $\dot{\gamma} = (1, 0)$ is constant and $D_t \dot{\gamma} = \bar{\nabla}_{\dot{\gamma}} \dot{\gamma} = 0$. The theorem predicts $\frac{d}{ds}|_{s=0} E(\gamma_s) = 0$, and this is confirmed directly: the longitudinal velocity is $\partial_t \Gamma = (1, \pi s \cos \pi t)$, so

$$E(\gamma_s) = \frac{1}{2} \int_0^1 (1 + \pi^2 s^2 \cos^2 \pi t) dt = \frac{1}{2} + \frac{\pi^2}{4} s^2,$$

whose derivative at $s = 0$ vanishes. The energy is minimized at $s = 0$: the straight segment is critical, consistent with its being a geodesic. The bowed curves γ_s all cost strictly more energy, the excess being quadratic in s , which foreshadows the second-variation analysis of later chapters. Had γ instead been a curved arc with $D_t \dot{\gamma} \neq 0$, the same variation field would generally produce a nonzero first variation, exhibiting the acceleration through the pairing $-\int \langle V, D_t \dot{\gamma} \rangle dt$.

Energy having been dispatched, the first variation of length is obtained by the same calculation with the integrand $|\dot{\gamma}|_g$ in place of $\frac{1}{2} |\dot{\gamma}|_g^2$. The square root introduces a factor $1/|\dot{\gamma}|_g$ in the integrand, which is harmless as long as the curve is regular, meaning $\dot{\gamma}$ never vanishes; the cleanest statement

assumes unit speed, $|\dot{\gamma}|_g \equiv 1$, which can always be arranged for a regular curve by reparametrizing by arc length, as recorded in section 3.4.

Theorem 3.5.6: First Variation of Length

Let $\gamma: [a, b] \rightarrow M$ be a piecewise-smooth, regular, unit-speed curve, $|\dot{\gamma}|_g \equiv 1$, in a Riemannian manifold, smooth on each subinterval of a partition $a = t_0 < \dots < t_m = b$, and let Γ be a piecewise-smooth variation of γ with variation field V . Then

$$\left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i \dot{\gamma} \rangle + \left[\langle V, \dot{\gamma} \rangle \right]_a^b$$

In particular, a smooth regular curve is a critical point of L for proper variations if and only if $D_t \dot{\gamma} \equiv 0$, that is, if and only if it is a geodesic after reparametrization by arc length. Moreover, for every piecewise-smooth curve $\gamma: [a, b] \rightarrow M$,

$$L(\gamma)^2 \leq 2(b-a) E(\gamma),$$

with equality if and only if γ has constant speed, $|\dot{\gamma}|_g$ constant on $[a, b]$.

Proof:

Step 1: the length variation. As in the proof of theorem 3.5.4, work first on a subinterval $[t_{i-1}, t_i]$ on which Γ is smooth, and write $|\partial_t \Gamma| = \langle \partial_t \Gamma, \partial_t \Gamma \rangle^{1/2}$, which is smooth and positive near $s = 0$ because γ is regular with $|\dot{\gamma}|_g \equiv 1 > 0$, so $\partial_t \Gamma$ stays nonzero for small $|s|$. Then

$$L_i(s) = \int_{t_{i-1}}^{t_i} |\partial_t \Gamma| dt, \quad \frac{d}{ds} L_i(s) = \int_{t_{i-1}}^{t_i} \frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle^{1/2} dt$$

By the chain rule and metric compatibility (definition 2.2.4),

$$\frac{\partial}{\partial s} \langle \partial_t \Gamma, \partial_t \Gamma \rangle^{1/2} = \frac{\langle D_s \partial_t \Gamma, \partial_t \Gamma \rangle}{|\partial_t \Gamma|}$$

Apply the symmetry lemma (lemma 3.5.3) to replace $D_s \partial_t \Gamma$ by $D_t \partial_s \Gamma$, and evaluate at $s = 0$, where $|\partial_t \Gamma| = |\dot{\gamma}|_g = 1$, $\partial_s \Gamma = V$, and $\partial_t \Gamma = \dot{\gamma}$:

$$\left. \frac{d}{ds} \right|_{s=0} L_i = \int_{t_{i-1}}^{t_i} \langle D_t V, \dot{\gamma} \rangle dt$$

The unit-speed hypothesis is what removes the denominator here; without it the factor $1/|\dot{\gamma}|_g$ would persist inside the integral.

Step 2: integration by parts. By metric compatibility again, $\frac{d}{dt} \langle V, \dot{\gamma} \rangle = \langle D_t V, \dot{\gamma} \rangle + \langle V, D_t \dot{\gamma} \rangle$, so $\langle D_t V, \dot{\gamma} \rangle = \frac{d}{dt} \langle V, \dot{\gamma} \rangle - \langle V, D_t \dot{\gamma} \rangle$ and

$$\left. \frac{d}{ds} \right|_{s=0} L_i = \left[\langle V, \dot{\gamma} \rangle \right]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle V, D_t \dot{\gamma} \rangle dt$$

Summing over the subintervals and collecting the boundary contributions exactly as in Step 3 of the proof of theorem 3.5.4, with the continuity of V across breakpoints and the possible jumps

of $\dot{\gamma}$, yields

$$\left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt - \sum_{i=1}^{m-1} \langle V(t_i), \Delta_i \dot{\gamma} \rangle + \left[\langle V, \dot{\gamma} \rangle \right]_a^b,$$

the asserted formula. For a smooth curve and a proper variation the jump sum and boundary term drop, leaving $\left. \frac{d}{ds} \right|_{s=0} L(\gamma_s) = - \int_a^b \langle V, D_t \dot{\gamma} \rangle dt$. The criticality equivalence follows exactly as in Step 4 of the proof of theorem 3.5.4: $D_t \dot{\gamma} \equiv 0$ makes the integral vanish for all V , and conversely testing against $V = \varphi D_t \dot{\gamma}$ with φ a positive interior bump function forces $D_t \dot{\gamma} \equiv 0$, using positive-definiteness of the Riemannian inner product. A unit-speed curve with $D_t \dot{\gamma} \equiv 0$ is a geodesic by definition 3.1.1; a general regular critical curve becomes such after arc-length reparametrization.

Step 3: the Cauchy–Schwarz inequality. For any piecewise-smooth $\gamma: [a, b] \rightarrow M$, apply the Cauchy–Schwarz inequality in $L^2([a, b])$ to the pair of functions $|\dot{\gamma}|_g$ and the constant function 1:

$$L(\gamma)^2 = \left(\int_a^b |\dot{\gamma}|_g \cdot 1 dt \right)^2 \leq \left(\int_a^b |\dot{\gamma}|_g^2 dt \right) \left(\int_a^b 1^2 dt \right) = (b-a) \int_a^b |\dot{\gamma}|_g^2 dt = 2(b-a) E(\gamma),$$

which is the claimed inequality. Equality in Cauchy–Schwarz holds if and only if the two functions are linearly dependent in L^2 , that is, if and only if $|\dot{\gamma}|_g$ is a constant multiple of 1, hence constant on $[a, b]$. Thus equality holds precisely for constant-speed curves, completing the proof.

The length variation carries the same interior term $-\int \langle V, D_t \dot{\gamma} \rangle dt$ as the energy variation, and for that reason the two functionals have the same critical points among regular curves, namely the geodesics; the difference is only that length, being insensitive to reparametrization, detects geodesics only up to how they are parametrized, whereas energy fixes the parametrization to be by constant speed. The inequality $L(\gamma)^2 \leq 2(b-a)E(\gamma)$ makes the relationship quantitative. Among all curves from p to q on a fixed parameter interval, the energy is at least $L^2/2(b-a)$, with equality exactly for the constant-speed ones; so minimizing energy over a fixed interval is minimizing length among constant-speed reparametrizations, and a minimizer of energy is automatically a length-minimizer traversed at constant speed. This is the technical reason energy is the functional of choice in the existence theory for geodesics: its critical points are exactly geodesics, and its minimizers are the shortest constant-speed paths.

Example 3.5.7: The inequality is strict for uneven parametrization

On the Euclidean interval, take the two curves in $\mathbb{R}$ that trace the segment from 0 to 1 over $t \in [0, 1]$: first $\alpha(t) = t$, at unit speed, and second $\beta(t) = t^2$, which speeds up. Both have length $L = 1$, since $\int_0^1 |\dot{\beta}| dt = \int_0^1 2t dt = 1$. Their energies differ:

$$E(\alpha) = \frac{1}{2} \int_0^1 1 dt = \frac{1}{2}, \quad E(\beta) = \frac{1}{2} \int_0^1 (2t)^2 dt = \frac{1}{2} \int_0^1 4t^2 dt = \frac{2}{3}$$

The inequality reads $L^2 = 1 \leq 2(b-a)E = 2E$ in both cases: for α it is $1 \leq 1$, equality, matching the constant speed $|\dot{\alpha}| \equiv 1$; for β it is $1 \leq \frac{4}{3}$, strict, matching the non-constant speed $|\dot{\beta}| = 2t$. The extra energy $\frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ is the penalty β pays for uneven parametrization while covering the same image. This is the sense in which energy sees more than length: the two curves are indistinguishable to L but separated by E .

The two theorems together settle the relationship between the differential-equation description of geodesics and the variational one. Restated for emphasis and cross-reference, the conclusion for energy is the following.

Corollary 3.5.8: Geodesics Are the Critical Points of Energy

Let (M, g) be a (pseudo-)Riemannian manifold and $\gamma: [a, b] \rightarrow M$ a smooth curve. Then γ is a critical point of the energy functional E for proper variations if and only if γ is a geodesic, that is, $D_t \dot{\gamma} \equiv 0$.

Proof:

This is the final assertion of theorem 3.5.4, recorded here for reference. For a smooth curve and proper variations the first-variation formula reduces to $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = -\int_a^b \langle V, D_t \dot{\gamma} \rangle dt$; this vanishes for every proper variation exactly when $D_t \dot{\gamma} \equiv 0$, by the argument given there, which is the geodesic equation of definition 3.1.1. Hence the critical points of E are precisely the geodesics, as claimed.

This variational characterization completes the promise made in section 3.4: a curve that minimizes length, once parametrized by arc length, is a critical point of length and hence, by theorem 3.5.6, satisfies $D_t \dot{\gamma} \equiv 0$, so it is a smooth geodesic. The jump term in the first-variation formula supplies the smoothness: a length-minimizing piecewise-smooth curve can have no corner, since a variation supported near a corner and pointing along the velocity jump would decrease the length. A minimizer is therefore a single smooth geodesic, not a broken one, which is the regularity statement that closes the loop between the metric and the differential-geometric pictures of shortest paths.

The Curvature Tensor

The geodesics and the exponential map of chapter 3 give a manifold its straight lines, but they do not yet measure how the manifold bends. On Euclidean space the second covariant derivatives of a vector field commute and parallel transport is path-independent; on a general Riemannian manifold neither holds, and the precise failure of both is a single tensor, the Riemann curvature tensor. This chapter defines that tensor, extracts the invariants built from it, and identifies the spaces in which it is as simple as possible.

The development moves from the general to the specific. The curvature endomorphism and its algebraic symmetries come first, together with the two Bianchi identities that every curvature tensor obeys. Contracting and restricting the tensor then produces the hierarchy of curvature invariants: the sectional curvature, which records the curvature of each tangent plane and in fact determines the whole tensor; the Ricci and scalar curvatures, the averaged quantities that govern volume and enter the Einstein condition; and the Weyl tensor, the trace-free remainder. The chapter closes by pinning down the extreme cases, the flat manifolds and the constant-curvature model spaces, and by computing curvature through the warped-product construction of chapter 1, which reproduces the sphere, hyperbolic space, and Euclidean space from a single formula. A fixed sign convention, stated at the outset, is in force throughout: with it, the sphere has positive curvature.

4.1 The Riemann Curvature Tensor

On Euclidean space the order of differentiation does not matter: the mixed second partial derivatives of a smooth function agree, and a vector field differentiated first along ∂_i and then along ∂_j gives the same result as the reverse. The Levi-Civita connection of definition 2.3.2 lets one differentiate vector fields on any manifold, and the question of whether the order matters becomes a question about the manifold rather than about calculus. On the round sphere it does matter, as the parallel-transport computation around a geodesic triangle in section 2.4 already showed: carrying a vector around

a closed loop and back returns it rotated, so the infinitesimal version, comparing $\nabla_X \nabla_Y Z$ with $\nabla_Y \nabla_X Z$, cannot vanish. The exact discrepancy is the object this chapter is built around.

That discrepancy is a tensor. The expression $\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z$ looks like a second-order differential operator applied to Z , and a differential operator is not tensorial in Z ; but subtracting the correction $\nabla_{[X,Y]} Z$ removes every derivative that falls on Z , and what remains depends only on the values of X , Y , and Z at a single point. The result is a $(1,3)$ -tensor field, the Riemann curvature endomorphism, and its vanishing is exactly the condition that second covariant derivatives commute. This section defines the tensor, proves that it is a tensor, records its coordinate components in terms of the Christoffel symbols, and checks that Euclidean space, where nothing bends, has curvature zero. Before any formula is written, the sign convention is fixed, because the various conventions in the literature differ by signs that propagate into every subsequent invariant.

4.1.1 Convention: The curvature sign convention Throughout this book the *Riemann curvature endomorphism* of a (pseudo-)Riemannian manifold (M, g) with Levi-Civita connection ∇ is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

The associated $(0,4)$ -tensor is $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$. The sectional curvature of a 2-plane $\Pi = \text{span}(X, Y)$ is

$$K(\Pi) = \frac{\text{Riem}(X, Y, Y, X)}{|X|^2 |Y|^2 - \langle X, Y \rangle^2}$$

the Ricci tensor is $\text{Ric}(X, Y) = \sum_i \langle R(e_i, X)Y, e_i \rangle$ for an orthonormal frame (e_i) , the scalar curvature is $\text{scal} = \text{tr}_g \text{Ric}$, and a manifold has constant sectional curvature c exactly when $R(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$. With these choices the round sphere S_r^n has $K \equiv 1/r^2 > 0$, hyperbolic space H^n has $K \equiv -1$, and Euclidean space has $K \equiv 0$.

Conventions in the literature diverge here. Some texts define the endomorphism with the opposite overall sign, and some place the arguments of the $(0,4)$ -tensor in a different order, so that a quoted formula for Ric or for the constant-curvature form may carry the reverse sign. Every curvature formula in this book is written relative to the convention stated above; a sign copied from another source should be reconciled with it before use.

The definition of the endomorphism repeats the display just fixed, now as a formal object attached to the connection.

Definition 4.1.2: Riemann curvature endomorphism

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2). The *Riemann curvature endomorphism* is the map

$$R: \mathfrak{X}(M) \times \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

defined for vector fields X, Y, Z by

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

For fixed X, Y the assignment $Z \mapsto R(X, Y)Z$ is written $R(X, Y)$ and called the curvature operator of the pair (X, Y) .

The three terms are read as a corrected commutator. The bracket $[\nabla_X, \nabla_Y]Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z$ measures the failure of the two second covariant derivatives to agree, and the third term $\nabla_{[X,Y]}Z$ is the derivative one must subtract because X and Y need not commute as vector fields. When X and Y are coordinate frame fields, $[X, Y] = [\partial_i, \partial_j] = 0$ and the correction disappears, so on a chart the tensor is literally the commutator of the two coordinate second derivatives; the general definition is what makes this a well-defined object independent of any chart. The interpretation that drives everything else is the one from the opening: R is the infinitesimal holonomy of the connection, the amount by which parallel transport around an infinitesimal loop spanned by X and Y fails to be the identity, and it vanishes identically precisely when parallel transport is path-independent, that is, when second covariant derivatives commute.

The expression for $R(X, Y)Z$ involves derivatives of Z , so it is not evident that its value at a point depends only on the values of X , Y , and Z at that point. The next proposition establishes exactly that: R is $C^\infty(M)$ -linear in each of its three arguments, and hence, by the tensor characterization lemma of [3], is a tensor field. This is the reason the corrected commutator, rather than the bare one, is the right object.

Proposition 4.1.3: Tensoriality of the curvature endomorphism

The Riemann curvature endomorphism R of definition 4.1.2 is $C^\infty(M)$ -linear in each of the three arguments X , Y , Z . Consequently R is a smooth $(1, 3)$ -tensor field: its value $R(X, Y)Z$ at a point p depends only on X_p , Y_p , Z_p , and it defines, for each p , a trilinear map $T_p M \times T_p M \times T_p M \rightarrow T_p M$.

Proof:

Additivity in each argument is immediate from the additivity of ∇ and of the Lie bracket, so it remains to check homogeneity over $C^\infty(M)$: that $R(fX, Y)Z = R(X, fY)Z = R(X, Y)(fZ) = fR(X, Y)Z$ for every $f \in C^\infty(M)$. The antisymmetry $R(X, Y)Z = -R(Y, X)Z$ is visible from the defining formula, since interchanging X and Y negates each of the three terms; hence linearity in the first argument follows from linearity in the second, and only the second and third arguments need separate treatment.

Step 1: linearity in Y . Fix $f \in C^\infty(M)$. Two identities for the Levi-Civita connection and the bracket are used. First, the Leibniz rule of definition 2.1.1,

$$\nabla_X(fW) = (Xf)W + f\nabla_X W$$

for any vector field W . Second, the bracket identity $[X, fY] = f[X, Y] + (Xf)Y$, established for the Lie bracket of vector fields in [3]. Compute the three terms of $R(X, fY)Z$ in turn. For the first,

$$\nabla_X \nabla_{fY} Z = \nabla_X (f \nabla_Y Z) = (Xf) \nabla_Y Z + f \nabla_X \nabla_Y Z$$

using $\nabla_{fY} Z = f \nabla_Y Z$ (the connection is $C^\infty(M)$ -linear in the differentiating slot, again by definition 2.1.1) and then the Leibniz rule. For the second term,

$$\nabla_{fY} \nabla_X Z = f \nabla_Y \nabla_X Z$$

directly. For the third term, using the bracket identity and then the linearity of ∇ in its lower slot,

$$\nabla_{[X, fY]} Z = \nabla_{f[X, Y] + (Xf)Y} Z = f \nabla_{[X, Y]} Z + (Xf) \nabla_Y Z$$

Assembling the three,

$$R(X, fY)Z = (Xf)\nabla_Y Z + f\nabla_X \nabla_Y Z - f\nabla_Y \nabla_X Z - f\nabla_{[X,Y]} Z - (Xf)\nabla_Y Z$$

The two $(Xf)\nabla_Y Z$ terms cancel, and the remaining three terms are f times the three terms of $R(X, Y)Z$, so $R(X, fY)Z = fR(X, Y)Z$. By the antisymmetry noted above, $R(fX, Y)Z = fR(X, Y)Z$ as well.

Step 2: linearity in Z . Fix $f \in C^\infty(M)$ and expand each term of $R(X, Y)(fZ)$. Applying the Leibniz rule twice to the first term,

$$\begin{aligned} \nabla_X \nabla_Y (fZ) &= \nabla_X ((Yf)Z + f\nabla_Y Z) \\ &= (X(Yf))Z + (Yf)\nabla_X Z + (Xf)\nabla_Y Z + f\nabla_X \nabla_Y Z \end{aligned}$$

Interchanging X and Y gives the second term,

$$\nabla_Y \nabla_X (fZ) = (Y(Xf))Z + (Xf)\nabla_Y Z + (Yf)\nabla_X Z + f\nabla_Y \nabla_X Z$$

and the Leibniz rule once gives the third,

$$\nabla_{[X,Y]}(fZ) = ([X, Y]f)Z + f\nabla_{[X,Y]} Z$$

Subtracting the second and third displays from the first, the terms $(Yf)\nabla_X Z$ cancel between the first and second lines, as do the terms $(Xf)\nabla_Y Z$. The coefficient of Z that survives is

$$X(Yf) - Y(Xf) - [X, Y]f = 0$$

which vanishes by the very definition of the Lie bracket, $[X, Y]f = X(Yf) - Y(Xf)$ from [3]. What remains is exactly f times the three terms of $R(X, Y)Z$, so $R(X, Y)(fZ) = fR(X, Y)Z$.

This is where the correction term earns its place: without $-\nabla_{[X,Y]} Z$ the coefficient of Z would be $X(Yf) - Y(Xf) \neq 0$ in general, and the bare commutator would fail to be tensorial in Z .

Step 3: from linearity to a tensor. A map that is $C^\infty(M)$ -multilinear in its vector-field arguments and takes values in vector fields is induced by a smooth tensor field, by the tensor characterization of [3]; concretely, its value at p depends only on the values of the arguments at p . Smoothness of the tensor field is inherited from the smoothness of ∇ and the bracket. Hence R is a smooth $(1, 3)$ -tensor field, and at each point p it restricts to a trilinear map on $T_p M$ with values in $T_p M$, as we sought to show.

Because R is a tensor, its value on vectors at a point may be computed by extending those vectors to any convenient vector fields, and the result does not depend on the extension. This is the license under which every later computation freely chooses coordinate or orthonormal frames. It is also what makes the pointwise trilinear form the object of study: the algebraic identities it satisfies, taken up in the next section, are constraints at each point, and the whole apparatus of sectional, Ricci, and scalar curvature is extracted from this single tensor.

Lowering the output index with the metric turns the endomorphism into a $(0, 4)$ -tensor, which is the form in which the symmetries are cleanest and in which sectional curvature is read off.

Definition 4.1.4: The (0,4) curvature tensor

Let R be the Riemann curvature endomorphism of definition 4.1.2. The (0,4) *Riemann curvature tensor* is the covariant 4-tensor field Riem defined by

$$\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$$

for vector fields X, Y, Z, W .

The passage from R to Riem is the metric contraction of the top index of the (1,3)-tensor against a fourth argument, an operation that is $C^\infty(M)$ -linear in W because g is bilinear over $C^\infty(M)$ and $C^\infty(M)$ -linear in X, Y, Z by proposition 4.1.3. Hence Riem is a genuine (0,4)-tensor field. It carries the same information as R : since g is nondegenerate, $\langle R(X, Y)Z, W \rangle = 0$ for all W forces $R(X, Y)Z = 0$, so Riem vanishes if and only if R does. The order of the four arguments is part of the convention; the sectional curvature in box 4.1.1 is $\text{Riem}(X, Y, Y, X)$ over the squared area of the parallelogram spanned by X and Y , and this pairing is what the sign convention was arranged to make positive for the sphere.

To compute with R in charts, and to connect the abstract tensor to the Christoffel symbols of definition 2.1.3, one records its components. The Einstein summation convention, summing over each repeated upper-lower index pair, is in force for the remainder of the section.

Definition 4.1.5: Components of the curvature tensor

Let $(U, (x^i))$ be a smooth chart on (M, g) , with coordinate frame (∂_i) . The *components* of the Riemann curvature endomorphism in this chart are the functions R^l_{ijk} defined by

$$R(\partial_i, \partial_j)\partial_k = R^l_{ijk} \partial_l$$

and the *components* of the (0,4)-tensor are

$$R_{lijk} = \text{Riem}(\partial_l, \partial_i, \partial_j, \partial_k) = \langle R(\partial_l, \partial_i)\partial_j, \partial_k \rangle$$

which are related to the endomorphism components by lowering the top index, $R_{mijk} = g_{ml} R^l_{ijk}$.

The components are computable directly from the Christoffel symbols. Since $[\partial_i, \partial_j] = 0$ for coordinate frame fields, the correction term in definition 4.1.2 drops out, and

$$R(\partial_i, \partial_j)\partial_k = \nabla_{\partial_i} \nabla_{\partial_j} \partial_k - \nabla_{\partial_j} \nabla_{\partial_i} \partial_k$$

Writing $\nabla_{\partial_j} \partial_k = \Gamma^m_{jk} \partial_m$ and applying ∇_{∂_i} with the Leibniz rule,

$$\nabla_{\partial_i} \nabla_{\partial_j} \partial_k = \nabla_{\partial_i} (\Gamma^m_{jk} \partial_m) = (\partial_i \Gamma^m_{jk}) \partial_m + \Gamma^m_{jk} \Gamma^l_{im} \partial_l = (\partial_i \Gamma^l_{jk} + \Gamma^l_{im} \Gamma^m_{jk}) \partial_l$$

after relabelling the summed index in the first term from m to l . Subtracting the same expression with i and j interchanged gives the component formula

$$R^l_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im} \Gamma^m_{jk} - \Gamma^l_{jm} \Gamma^m_{ik} \quad (4.1)$$

Every quantity on the right is built from the metric and its first and second derivatives, through the formula $\Gamma^k_{ij} = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$ of proposition 2.3.3. Thus the curvature is a second-order differential invariant of the metric: R^l_{ijk} depends on g_{ij} , ∂g_{ij} , and $\partial^2 g_{ij}$. The antisymmetry

$R^l_{ijk} = -R^l_{jik}$ in the first two lower indices is visible in (4.1), matching $R(\partial_i, \partial_j) = -R(\partial_j, \partial_i)$; the deeper symmetries of R_{lijk} are proved in section 4.2.

A remark on index placement is in order, because it interacts with the sign convention. The two lower indices i, j that are antisymmetrized are the first two, the arguments of the operator $R(\partial_i, \partial_j)$; the third lower index k is the vector being acted on. Different books order these differently, and reordering them changes the sign of (4.1). The placement here matches definition 4.1.2 and box 4.1.1 exactly, and it is the placement used in every component computation that follows.

The first test of the definition is the flat model, where the answer must be zero.

Example 4.1.6: Curvature of Euclidean space

Consider Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, with $\bar{g} = \delta_{ij} dx^i dx^j$ in the standard chart. The metric components are the constants $g_{ij} = \delta_{ij}$, so all their partial derivatives vanish, and by the Christoffel formula of proposition 2.3.3,

$$\Gamma^k_{ij} = \frac{1}{2} \delta^{kl} (\partial_i \delta_{jl} + \partial_j \delta_{il} - \partial_l \delta_{ij}) = 0$$

identically. Substituting $\Gamma \equiv 0$ into the component formula (4.1), every term is either a derivative of a Christoffel symbol or a product of two, so

$$R^l_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im} \Gamma^m_{jk} - \Gamma^l_{jm} \Gamma^m_{ik} = 0$$

for all indices. Hence $R \equiv 0$ on $\mathbb{R}^n$, and therefore $\text{Riem} \equiv 0$ as well. Every invariant built from R vanishes: the sectional curvature of every 2-plane is 0, and $\mathbb{R}^n$ is flat, in agreement with the value $K \equiv 0$ recorded in box 4.1.1.

The computation also exhibits, in the simplest case, why R measures the manifold and not the chart. In polar coordinates on $\mathbb{R}^2$ the induced metric is $dr^2 + r^2 d\theta^2$, and its Christoffel symbols are not all zero: as computed in proposition 2.3.3 one has $\Gamma^r_{\theta\theta} = -r$ and $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = 1/r$, with the rest vanishing. Feeding these into (4.1) and using the one relevant nonzero derivative $\partial_r \Gamma^r_{\theta\theta} = -1$ gives, for the component with $(l, i, j, k) = (r, r, \theta, \theta)$,

$$R^r_{r\theta\theta} = \partial_r \Gamma^r_{\theta\theta} - \partial_\theta \Gamma^r_{r\theta} + \Gamma^r_{rm} \Gamma^m_{\theta\theta} - \Gamma^r_{\theta m} \Gamma^m_{r\theta} = (-1) - 0 + 0 - \Gamma^r_{\theta\theta} \Gamma^\theta_{r\theta} = -1 - (-r)\left(\frac{1}{r}\right) = 0$$

and every other independent component vanishes by the same cancellation. The nonzero Christoffel symbols conspire so that R still vanishes: the plane is flat regardless of the coordinates used to describe it, and it is the curvature tensor, not the Christoffel symbols, that detects this.

The Euclidean computation isolates the two features that the rest of the chapter builds on. First, curvature is a second-order quantity in the metric, so it cannot be removed by a first-order coordinate change; the normal coordinates of proposition 3.2.8 arrange $g_{ij}(p) = \delta_{ij}$, $\Gamma^k_{ij}(p) = 0$, and $\partial_k g_{ij}(p) = 0$ at a single point, yet on a curved manifold the second derivatives $\partial_k \partial_l g_{ij}(p)$ cannot all be made to vanish, and (4.1) extracts exactly the combination of them that is coordinate-independent. Second, the vanishing of R is the precise flatness criterion: a manifold with $R \equiv 0$ is locally isometric to Euclidean space, a statement proved in section 4.5.

4.2 Symmetries and the Bianchi Identities

The curvature endomorphism $R(X, Y)Z$ of definition 4.1.2 and the four-tensor $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$ of definition 4.1.4 are, as they stand, three vector inputs and one covector input away from being unconstrained. In dimension n a general $(0, 4)$ -tensor at a point has n^4 independent components; the Riemann tensor of a (pseudo-)Riemannian manifold has far fewer, because it inherits rigid algebraic relations from the way it was built. Two of those relations, antisymmetry in the first two slots and antisymmetry in the last two, express that $R(X, Y)$ is a bundle map that swaps sign under exchange of X, Y and takes values in the skew-symmetric endomorphisms of each tangent space. A third, the symmetry that exchanges the first pair of arguments with the second, is less transparent and turns out to be a consequence of the other relations together with the algebraic Bianchi identity. These are the symmetries that make sectional curvature well defined in section 4.3 and that cut the number of independent components down to $n^2(n^2 - 1)/12$.

Alongside the pointwise (algebraic) symmetries there is a single relation among the covariant derivatives of R , the differential Bianchi identity. Where the algebraic symmetries are statements about R at one point, the differential identity constrains how R changes from point to point; contracting it produces the divergence identity that forces Schur's lemma and, in general relativity, singles out the Einstein tensor as the divergence-free combination of curvature. This section proves all of these, first the four algebraic symmetries in one proposition, then the differential identity, and finally records the same relations in index notation for later coordinate computations. Throughout, the sign convention fixed in box 4.1.1 is in force, and Einstein's summation convention (sum over any index appearing once up and once down) is used in the component statements.

The starting point is a small observation that removes all the Lie-bracket clutter from the proofs. Every identity below is $C^\infty(M)$ -linear in each argument, hence tensorial, hence pointwise: to check it at a point p it suffices to check it for *any* convenient choice of vector fields extending prescribed vectors at p . Choosing extensions whose pairwise Lie brackets vanish at p collapses the bracket term $\nabla_{[X, Y]}Z$ and reduces each proof to a computation with the two double-derivative terms. Such extensions exist: in the normal coordinates at p of proposition 3.2.8, the coordinate vector fields ∂_i have identically vanishing brackets $[\partial_i, \partial_j] = 0$, and they also satisfy $\nabla_{\partial_i} \partial_j(p) = \Gamma_{ij}^k(p) \partial_k = 0$ because the Christoffel symbols vanish at the center. Both features are exploited below.

Proposition 4.2.1: Algebraic symmetries of the curvature tensor

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ and Riemann tensor Riem . For all vector fields X, Y, Z, W on M :

1. (antisymmetry in the first pair) $\text{Riem}(X, Y, Z, W) = -\text{Riem}(Y, X, Z, W)$;
2. (antisymmetry in the second pair) $\text{Riem}(X, Y, Z, W) = -\text{Riem}(X, Y, W, Z)$;
3. (pair symmetry) $\text{Riem}(X, Y, Z, W) = \text{Riem}(Z, W, X, Y)$;
4. (first Bianchi identity) $\text{Riem}(X, Y, Z, W) + \text{Riem}(Y, Z, X, W) + \text{Riem}(Z, X, Y, W) = 0$.

Proof:

Each identity is $\mathbb{R}$ -linear in all four arguments, and is $C^\infty(M)$ -linear because R is a tensor (proposition 4.1.3) and the metric pairing $\langle -, - \rangle$ is $C^\infty(M)$ -bilinear. Hence each is a pointwise statement about the values X_p, Y_p, Z_p, W_p , and it suffices to prove it at an arbitrary $p \in M$ using any vector fields realizing those values.

1. Exchanging X and Y in the defining formula

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$$

negates the first two terms and, since $[Y, X] = -[X, Y]$, the third as well, so $R(Y, X)Z = -R(X, Y)Z$. Pairing with W gives $\text{Riem}(Y, X, Z, W) = -\text{Riem}(X, Y, Z, W)$, which is (1).

2. It suffices to prove the equivalent statement $\text{Riem}(X, Y, Z, Z) = 0$ for all X, Y, Z ; the general form follows by polarizing $Z \mapsto Z + W$ and using the bilinearity of Riem in the last two slots. Indeed, expanding $0 = \text{Riem}(X, Y, Z + W, Z + W) = \text{Riem}(X, Y, Z, Z) + \text{Riem}(X, Y, Z, W) + \text{Riem}(X, Y, W, Z) + \text{Riem}(X, Y, W, W)$ and cancelling the two vanishing diagonal terms leaves $\text{Riem}(X, Y, Z, W) = -\text{Riem}(X, Y, W, Z)$, which is (2).

So fix p and, by the reduction above, choose vector fields X, Y, Z with $[X, Y]_p = 0$ (for instance coordinate fields of a chart at p , whose brackets vanish identically). At p then

$$\text{Riem}(X, Y, Z, Z) = \langle \nabla_X \nabla_Y Z, Z \rangle - \langle \nabla_Y \nabla_X Z, Z \rangle$$

since the bracket term drops out. Metric compatibility (definition 2.2.4), $U \langle A, B \rangle = \langle \nabla_U A, B \rangle + \langle A, \nabla_U B \rangle$, applied twice gives

$$\begin{aligned} XY \langle Z, Z \rangle &= X(2 \langle \nabla_Y Z, Z \rangle) \\ &= 2 \langle \nabla_X \nabla_Y Z, Z \rangle + 2 \langle \nabla_Y Z, \nabla_X Z \rangle \end{aligned}$$

and, symmetrically, $YX \langle Z, Z \rangle = 2 \langle \nabla_Y \nabla_X Z, Z \rangle + 2 \langle \nabla_X Z, \nabla_Y Z \rangle$. Subtracting, the two mixed terms $\langle \nabla_Y Z, \nabla_X Z \rangle$ and $\langle \nabla_X Z, \nabla_Y Z \rangle$ coincide and cancel, so

$$(XY - YX) \langle Z, Z \rangle = 2 \langle \nabla_X \nabla_Y Z, Z \rangle - 2 \langle \nabla_Y \nabla_X Z, Z \rangle = 2 \text{Riem}(X, Y, Z, Z)$$

using the previous display for the right-hand side. But $(XY - YX)f = [X, Y]f$ for any function f , and $[X, Y]_p = 0$, so the left-hand side vanishes at p . Hence $\text{Riem}(X, Y, Z, Z)_p = 0$, and since p was arbitrary, $\text{Riem}(X, Y, Z, Z) \equiv 0$, proving (2).

3. Pair symmetry follows formally from (1), (2), and the first Bianchi identity (4), which is proved in part (4) below independently of (3). Write $b(A, B, C, D)$ for the first-Bianchi cyclic sum in the first three arguments,

$$b(A, B, C, D) := \text{Riem}(A, B, C, D) + \text{Riem}(B, C, A, D) + \text{Riem}(C, A, B, D)$$

so that $b \equiv 0$ by (4). Form the specific combination

$$\Sigma := b(X, Y, Z, W) + b(Y, Z, W, X) - b(Z, W, X, Y) - b(W, X, Y, Z)$$

which is 0 because each of the four summands is 0. Expanding the four b -terms gives twelve summands:

$$\begin{aligned} \Sigma &= \text{Riem}(X, Y, Z, W) + \text{Riem}(Y, Z, X, W) + \text{Riem}(Z, X, Y, W) \\ &\quad + \text{Riem}(Y, Z, W, X) + \text{Riem}(Z, W, Y, X) + \text{Riem}(W, Y, Z, X) \\ &\quad - \text{Riem}(Z, W, X, Y) - \text{Riem}(W, X, Z, Y) - \text{Riem}(X, Z, W, Y) \\ &\quad - \text{Riem}(W, X, Y, Z) - \text{Riem}(X, Y, W, Z) - \text{Riem}(Y, W, X, Z) \end{aligned}$$

Reduce each of the twelve summands to a canonical form by ordering its first pair with (1) and its second pair with (2), each reordering contributing a sign, and collect the results. The twelve summands fall into six canonical classes, each fed by two of them. Four of the six classes cancel outright, by the following identities, in which (1) and (2) bring both members to a common canonical form:

$$\begin{aligned} \text{Riem}(Y, Z, X, W) + \text{Riem}(Y, Z, W, X) &= \text{Riem}(Y, Z, X, W) - \text{Riem}(Y, Z, X, W) = 0, \\ \text{Riem}(Z, X, Y, W) - \text{Riem}(X, Z, W, Y) &= -\text{Riem}(X, Z, Y, W) + \text{Riem}(X, Z, Y, W) = 0, \\ \text{Riem}(W, Y, Z, X) - \text{Riem}(Y, W, X, Z) &= \text{Riem}(Y, W, X, Z) - \text{Riem}(Y, W, X, Z) = 0, \\ -\text{Riem}(W, X, Z, Y) - \text{Riem}(W, X, Y, Z) &= -\text{Riem}(X, W, Y, Z) + \text{Riem}(X, W, Y, Z) = 0 \end{aligned}$$

The two remaining classes are the diagonal ones. The class of $\text{Riem}(X, Y, Z, W)$ is fed by $\text{Riem}(X, Y, Z, W)$ and by $-\text{Riem}(X, Y, W, Z) = +\text{Riem}(X, Y, Z, W)$, contributing $+2 \text{Riem}(X, Y, Z, W)$; the class of $\text{Riem}(Z, W, X, Y)$ is fed by $-\text{Riem}(Z, W, X, Y)$ and by $\text{Riem}(Z, W, Y, X) = -\text{Riem}(Z, W, X, Y)$, contributing $-2 \text{Riem}(Z, W, X, Y)$. Summing the six classes,

$$0 = \Sigma = 2 \text{Riem}(X, Y, Z, W) - 2 \text{Riem}(Z, W, X, Y)$$

so $\text{Riem}(X, Y, Z, W) = \text{Riem}(Z, W, X, Y)$, which is (3). The reduction is purely algebraic, using only (1), (2), and $b \equiv 0$; no additional hypothesis on X, Y, Z, W is needed.

4. By tensoriality the identity is pointwise; fix p and choose vector fields X, Y, Z with vanishing pairwise brackets at p , so that every $\nabla[\cdot, \cdot]$ term below vanishes at p . Then

$$\begin{aligned} R(X, Y)Z + R(Y, Z)X + R(Z, X)Y &= (\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z) \\ &\quad + (\nabla_Y \nabla_Z X - \nabla_Z \nabla_Y X) + (\nabla_Z \nabla_X Y - \nabla_X \nabla_Z Y) \end{aligned}$$

at p . Regroup the six terms so that each covariant derivative ∇_U is applied to a difference:

$$= \nabla_X(\nabla_Y Z - \nabla_Z Y) + \nabla_Y(\nabla_Z X - \nabla_X Z) + \nabla_Z(\nabla_X Y - \nabla_Y X)$$

Because the Levi-Civita connection is torsion-free (definition 2.2.1, theorem 2.3.1), $\nabla_U V - \nabla_V U = [U, V]$, so this equals

$$\nabla_X[Y, Z] + \nabla_Y[Z, X] + \nabla_Z[X, Y]$$

Now use torsion-freeness once more, in the form $\nabla_U[V, W] = [U, [V, W]] + \nabla_{[V, W]}U$, together with the vanishing of the bracket terms at p : at p the fields $[Y, Z], [Z, X], [X, Y]$ all vanish (the pairwise brackets were chosen to vanish at p), so $\nabla_{[V, W]}U(p) = 0$, and

$$\nabla_X[Y, Z] + \nabla_Y[Z, X] + \nabla_Z[X, Y] = [X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

by the Jacobi identity for the Lie bracket of vector fields. Hence $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$ at p , and pairing with any W gives (4). Since p is arbitrary, (4) holds everywhere.

This establishes all four symmetries, completing the proof.

The four relations are not independent. Antisymmetry in each pair, (1) and (2), and the first Bianchi identity (4) are the primitive facts; the pair symmetry (3) is their formal consequence, as the proof shows. Their provenance is worth separating: (1) is purely combinatorial, built into the antisymmetric definition of $R(X, Y)$; (4) is the signature of torsion-freeness, and would fail for a connection with torsion; (2) is the one place metric compatibility enters, which is why it is special to the Levi-Civita connection among all torsion-free connections. A general affine connection has a curvature satisfying (1) always, and (4) when it is torsion-free, but (2) and hence (3) require the metric. This is the algebraic shadow of the fact that the Riemann tensor of (M, g) takes values in a smaller space than the curvature of an arbitrary connection.

A restatement clarifies (1) and (2). For fixed X, Y the map $Z \mapsto R(X, Y)Z$ is an endomorphism of each tangent space $T_p M$, and (2) says $\langle R(X, Y)Z, W \rangle = -\langle R(X, Y)W, Z \rangle$, that is, this endomorphism is skew-adjoint with respect to g : $R(X, Y) \in \mathfrak{so}(T_p M, g)$, the Lie algebra of the orthogonal (or, in the pseudo-Riemannian case, the indefinite-orthogonal) group of $(T_p M, g_p)$. Meanwhile (1) says the assignment $(X, Y) \mapsto R(X, Y)$ is itself alternating. So R is an alternating bilinear map into skew-adjoint endomorphisms, exactly the object that will reappear in section 4.3 as the curvature operator on two-forms.

The flat model exhibits all four symmetries in the empty way. On Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2, every component of R vanishes (example 4.1.6), so each identity of proposition 4.2.1 reads $0 = 0$. A first nontrivial check comes on the round sphere, computed in section 4.3, where $\text{Riem}(X, Y, Z, W) = r^{-2}(\langle X, W \rangle \langle Y, Z \rangle - \langle X, Z \rangle \langle Y, W \rangle)$; the reader may verify directly from this closed form that (1)–(4) all hold, and that the pattern of signs is forced by them.

Example 4.2.2: The constant-curvature tensor obeys the symmetries

Any (pseudo-)Riemannian manifold whose curvature has the constant-curvature form

$$R(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$$

of box 4.1.1, for a constant $c \in \mathbb{R}$, provides a fully explicit test of proposition 4.2.1. Lowering an index,

$$\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle = c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle)$$

Each symmetry is now a symmetry of this bracketed expression in the four inner products.

1. Swapping $X \leftrightarrow Y$ turns the bracket into $\langle X, Z \rangle \langle Y, W \rangle - \langle Y, Z \rangle \langle X, W \rangle$, the negative of the original; so $\text{Riem}(X, Y, Z, W) = -\text{Riem}(Y, X, Z, W)$.
2. Swapping $Z \leftrightarrow W$ turns it into $\langle Y, W \rangle \langle X, Z \rangle - \langle X, W \rangle \langle Y, Z \rangle$, again the negative; so $\text{Riem}(X, Y, Z, W) = -\text{Riem}(X, Y, W, Z)$.
3. Swapping the pairs $(X, Y) \leftrightarrow (Z, W)$ sends the bracket to $\langle W, X \rangle \langle Z, Y \rangle - \langle Z, X \rangle \langle W, Y \rangle$, which equals the original by symmetry of the inner product; so $\text{Riem}(X, Y, Z, W) = \text{Riem}(Z, W, X, Y)$.
4. The cyclic sum in the first three slots is

$$\begin{aligned} & c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle) \\ & + c(\langle Z, X \rangle \langle Y, W \rangle - \langle Y, X \rangle \langle Z, W \rangle) \\ & + c(\langle X, Y \rangle \langle Z, W \rangle - \langle Z, Y \rangle \langle X, W \rangle) \end{aligned}$$

in which the six products cancel in pairs ($\langle Y, Z \rangle \langle X, W \rangle$ against $-\langle Z, Y \rangle \langle X, W \rangle$, and so on), giving 0; so the first Bianchi identity holds.

The round sphere S_r^n of example 1.3.4 realizes this with $c = 1/r^2$ and hyperbolic space H^n of example 1.3.5 with $c = -1$, so both model spaces carry curvature tensors of exactly this form; the computation above is then their curvature symmetries in one stroke. That $R(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$ is in fact the *only* curvature-type tensor compatible with constant sectional curvature is proved in section 4.5; here it serves only as a worked instance of the algebraic symmetries.

The four algebraic identities constrain the value of R at each point. The next identity constrains its derivative and is the one relation among the components of ∇R . It is the geometric analogue of the fact that the exterior derivative of a curvature two-form is controlled by the connection, and its contraction is the source of the conservation law behind Einstein's equations. The covariant derivative ∇R appearing below is the $(1, 4)$ -tensor obtained by differentiating the $(1, 3)$ -tensor R ; recall from definition 2.4.6 that ∇ extends to all tensor bundles and obeys the tensor Leibniz rule, so that

$$(\nabla_X R)(Y, Z)W = \nabla_X(R(Y, Z)W) - R(\nabla_X Y, Z)W - R(Y, \nabla_X Z)W - R(Y, Z)\nabla_X W$$

is the value of the differentiated tensor on the arguments Y, Z, W .

Theorem 4.2.3: Second (differential) Bianchi identity

Let (M, g) be a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ and Riemann curvature R . For all vector fields X, Y, Z, W ,

$$(\nabla_X R)(Y, Z)W + (\nabla_Y R)(Z, X)W + (\nabla_Z R)(X, Y)W = 0$$

that is, the cyclic sum over X, Y, Z of $(\nabla_X R)(Y, Z)$ vanishes.

Proof:

Both sides are $C^\infty(M)$ -linear in each of X, Y, Z, W : the tensor ∇R is $C^\infty(M)$ -linear in all its arguments by construction (definition 2.4.6), and a cyclic sum of tensors is a tensor. Hence the identity is pointwise, and it suffices to verify it at an arbitrary point $p \in M$.

Fix p and work in normal coordinates (x^i) centered at p (definition 3.2.6), applied to the coordinate frame $X = \partial_a, Y = \partial_b, Z = \partial_c, W = \partial_d$. By $C^\infty(M)$ -multilinearity it suffices to prove the identity for these frame fields, since general X, Y, Z, W are C^∞ -combinations of them and both sides are multilinear. Two features of normal coordinates are used at the single point p , from proposition 3.2.8:

- $[\partial_i, \partial_j] = 0$ identically (coordinate fields commute); and
- $\Gamma_{ij}^k(p) = 0$, so $\nabla_{\partial_i} \partial_j(p) = 0$ for all i, j .

The first feature simplifies the curvature endomorphism on the frame to $R(\partial_i, \partial_j) = \nabla_{\partial_i} \nabla_{\partial_j} - \nabla_{\partial_j} \nabla_{\partial_i}$, with no bracket term, everywhere in the chart. The second collapses the covariant derivative of a tensor to the plain directional derivative of its frame components *at* p : for any tensor T built from the frame, since all $\nabla_{\partial_a} \partial_i(p) = 0$, the correction terms in the tensor Leibniz rule vanish at p , so

$$(\nabla_{\partial_a} R)(\partial_b, \partial_c) \partial_d \Big|_p = \nabla_{\partial_a} (R(\partial_b, \partial_c) \partial_d) \Big|_p$$

with all four frame-argument correction terms absent at p .

Compute the last quantity. Using $R(\partial_b, \partial_c) = \nabla_{\partial_b} \nabla_{\partial_c} - \nabla_{\partial_c} \nabla_{\partial_b}$,

$$(\nabla_{\partial_a} R)(\partial_b, \partial_c) \partial_d \big|_p = \nabla_{\partial_a} (\nabla_{\partial_b} \nabla_{\partial_c} \partial_d - \nabla_{\partial_c} \nabla_{\partial_b} \partial_d) \big|_p = (\nabla_{\partial_a} \nabla_{\partial_b} \nabla_{\partial_c} - \nabla_{\partial_a} \nabla_{\partial_c} \nabla_{\partial_b}) \partial_d \big|_p$$

Now form the cyclic sum over (a, b, c) , that is, over $(\partial_a, \partial_b, \partial_c)$ in the pattern $(a, b, c) \rightarrow (b, c, a) \rightarrow (c, a, b)$. Writing ∇_{abc} for $\nabla_{\partial_a} \nabla_{\partial_b} \nabla_{\partial_c} \partial_d \big|_p$, the three terms of the cyclic sum are

$$(\nabla_{abc} - \nabla_{acb}) + (\nabla_{bca} - \nabla_{bac}) + (\nabla_{cab} - \nabla_{cba})$$

These six third-order derivatives cancel in pairs. To see this, note that at p the operators ∇_{∂_i} and ∇_{∂_j} applied in the first two slots may be interchanged with an error measured by curvature: for any vector field V ,

$$\nabla_{\partial_i} \nabla_{\partial_j} V - \nabla_{\partial_j} \nabla_{\partial_i} V = R(\partial_i, \partial_j) V$$

since $[\partial_i, \partial_j] = 0$. Apply this to the outer two derivatives of each triple, taking $V = \nabla_{\partial_c} \partial_d$ and its permutations. Pairing ∇_{abc} with ∇_{bac} : their difference is

$$\nabla_{abc} - \nabla_{bac} = (\nabla_{\partial_a} \nabla_{\partial_b} - \nabla_{\partial_b} \nabla_{\partial_a}) \nabla_{\partial_c} \partial_d \big|_p = R(\partial_a, \partial_b) (\nabla_{\partial_c} \partial_d) \big|_p = 0$$

because $\nabla_{\partial_c} \partial_d(p) = 0$ and $R(\partial_a, \partial_b)$ is a pointwise-linear (tensorial) operator, so it annihilates a vector field vanishing at p . The same argument, using $\nabla_{\partial_i} \partial_j(p) = 0$ for the appropriate inner index, gives

$$\nabla_{bca} - \nabla_{cba} = R(\partial_b, \partial_c) (\nabla_{\partial_a} \partial_d) \big|_p = 0, \quad \nabla_{cab} - \nabla_{acb} = R(\partial_c, \partial_a) (\nabla_{\partial_b} \partial_d) \big|_p = 0$$

each vanishing for the same reason: the innermost covariant derivative $\nabla_{\partial_c} \partial_d$, $\nabla_{\partial_a} \partial_d$, or $\nabla_{\partial_b} \partial_d$ is a vector field vanishing at p , and the tensorial curvature operator sends it to 0 at p . The six terms of the cyclic sum therefore partition into the three vanishing differences

$$(\nabla_{abc} - \nabla_{bac}) + (\nabla_{bca} - \nabla_{cba}) + (\nabla_{cab} - \nabla_{acb}) = 0$$

which is exactly the cyclic sum rearranged. Hence

$$(\nabla_{\partial_a} R)(\partial_b, \partial_c) \partial_d + (\nabla_{\partial_b} R)(\partial_c, \partial_a) \partial_d + (\nabla_{\partial_c} R)(\partial_a, \partial_b) \partial_d \big|_p = 0$$

Since the frame vectors $\partial_a, \partial_b, \partial_c, \partial_d$ span $T_p M$ and both sides are multilinear over $\mathbb{R}$, the identity holds for arbitrary X, Y, Z, W at p . As $p \in M$ was arbitrary, the second Bianchi identity holds on all of M , as we sought to show.

The normal-coordinate device is what makes this proof short. In an arbitrary chart the third covariant derivatives of ∂_d would carry the derivatives of the Christoffel symbols, and the cancellations would be buried under those terms; centering normal coordinates at p kills the first-order data $\Gamma_{ij}^k(p)$ and leaves only the second-order data, in which the identity is manifest. The identity itself is the covariant form of a differential relation the reader may already know for connection and curvature forms: it is $dR = 0$ in the appropriate exterior-covariant sense. Its most consequential corollary is obtained by contracting twice, which produces the *contracted* second Bianchi identity governing the divergence of the Ricci tensor and of the Einstein tensor $\text{Ric} - \frac{1}{2} \text{scal } g$. That contraction, and the Schur lemma it yields, are carried out in section 4.4 once the Ricci tensor has been defined.

For Euclidean space the identity is again vacuous, since $R \equiv 0$ makes $\nabla R \equiv 0$ and every term is zero (example 4.1.6). More informatively, any manifold whose curvature is *parallel*, $\nabla R \equiv 0$ (a

locally symmetric space), satisfies the second Bianchi identity trivially term by term; the constant-curvature models of example 4.2.2 are of this kind, because differentiating $\text{Riem}(X, Y, Z, W) = c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle)$ with c constant and $\nabla g = 0$ gives $\nabla \text{Riem} = 0$. Thus for the sphere and for hyperbolic space the differential Bianchi identity reduces, correctly, to $0 = 0$, and the content of theorem 4.2.3 is that the same cyclic relation persists even where ∇R does not vanish.

For coordinate computations, and for the contractions of section 4.4 and the classification of section 4.5, it is convenient to have the symmetries recorded in components. Fix a chart with coordinate frame (∂_i) . Following the component convention of definition 4.1.5, write the endomorphism components R^l_{ijk} through $R(\partial_i, \partial_j)\partial_k = R^l_{ijk} \partial_l$, and lower the free upper index with the metric to obtain the fully covariant components

$$R_{ijkl} = \text{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) = \langle R(\partial_i, \partial_j)\partial_k, \partial_l \rangle = g_{lm} R^m_{ijk}$$

so that R_{ijkl} carries the first pair (i, j) in the endomorphism slots and the second pair (k, l) in the two Riemann arguments, matching the four-argument order of Riem .

Proposition 4.2.4: Curvature symmetries in components

In any coordinate frame the fully covariant curvature components satisfy

$$R_{ijkl} = -R_{jikl}, \quad R_{ijkl} = -R_{ijlk}, \quad R_{ijkl} = R_{klij}$$

and the first Bianchi identity

$$R_{ijkl} + R_{jkil} + R_{kijl} = 0$$

while the second Bianchi identity reads, in terms of the covariant-derivative components $\nabla_m R_{ijkl}$,

$$\nabla_m R_{ijkl} + \nabla_i R_{jmkl} + \nabla_j R_{mikl} = 0$$

the cyclic sum being over the three indices (m, i, j) in the first two curvature slots.

Proof:

The three equalities and the first Bianchi identity are the componentwise transcriptions of parts (1), (2), (3), and (4) of proposition 4.2.1, obtained by evaluating each on the coordinate frame (∂_i) . For the first equality, $R_{ijkl} = \text{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) = -\text{Riem}(\partial_j, \partial_i, \partial_k, \partial_l) = -R_{jikl}$ by part (1). For the second, part (2) gives $R_{ijkl} = -\text{Riem}(\partial_i, \partial_j, \partial_l, \partial_k) = -R_{ijlk}$. For the third, part (3) gives $R_{ijkl} = \text{Riem}(\partial_k, \partial_l, \partial_i, \partial_j) = R_{klij}$. For the Bianchi relation, part (4) applied to $(\partial_i, \partial_j, \partial_k)$ in the first three slots and paired with ∂_l gives

$$\text{Riem}(\partial_i, \partial_j, \partial_k, \partial_l) + \text{Riem}(\partial_j, \partial_k, \partial_i, \partial_l) + \text{Riem}(\partial_k, \partial_i, \partial_j, \partial_l) = 0$$

that is $R_{ijkl} + R_{jkil} + R_{kijl} = 0$.

For the differential identity, evaluate theorem 4.2.3 on the coordinate frame, taking $X = \partial_m$, $Y = \partial_i$, $Z = \partial_j$ in the derivative and first two curvature slots and pairing the endomorphism value with ∂_l while its argument is ∂_k :

$$(\nabla_{\partial_m} R)(\partial_i, \partial_j)\partial_k + (\nabla_{\partial_i} R)(\partial_j, \partial_m)\partial_k + (\nabla_{\partial_j} R)(\partial_m, \partial_i)\partial_k = 0$$

Pairing each term with ∂_l under $\langle -, - \rangle$ and writing $\nabla_m R_{ijkl}$ for the components of the tensor ∇R (so that $\langle (\nabla_{\partial_m} R)(\partial_i, \partial_j)\partial_k, \partial_l \rangle = \nabla_m R_{ijkl}$, the metric being parallel) turns the three terms

into $\nabla_m R_{ijkl}$, $\nabla_i R_{jmk l}$, and $\nabla_j R_{mik l}$, giving the stated cyclic relation. This proves all the component identities, as required.

The component antisymmetries $R_{ijkl} = -R_{jikl} = -R_{ijlk}$ say that R_{ijkl} is skew in (i, j) and skew in (k, l) , so it descends to a symmetric bilinear form on the space $\Lambda^2 T_p^* M$ of two-forms, the pair symmetry $R_{ijkl} = R_{klij}$ being exactly this symmetry; this is the curvature operator, and it is the natural home for the eigenvalue estimates of later chapters. Counting independent components with these relations, together with the first Bianchi identity, gives $\frac{1}{12}n^2(n^2 - 1)$, which is 0 for $n = 1$, 1 for $n = 2$, 6 for $n = 3$, and 20 for $n = 4$; the single component in dimension two is the Gauss curvature, and the twenty in dimension four are the arena of general relativity. The decomposition of this space into scalar, Ricci, and Weyl pieces is taken up in section 4.4, and the fact that the scalar quantity K on two-planes already recovers the whole of R_{ijkl} is proved in section 4.3.

4.3 Sectional Curvature

The Riemann tensor R is a formidable object: at each point it is a $(1, 3)$ -tensor, and even after the algebraic symmetries of proposition 4.2.1 cut down its independent components, it records $\frac{1}{12}n^2(n^2 - 1)$ numbers at every point of an n -manifold. That is a large amount of data, and none of it is visibly geometric. The purpose of this section is to repack the same information as a single real-valued function on a concrete geometric object, the set of tangent 2-planes, in a way that connects the curvature tensor to the shape of the manifold one can actually see.

The idea is old and comes from surfaces. On a surface the Gaussian curvature at a point is one number, and it measures how the surface bends: positive where the surface is dome-like, negative at a saddle, zero on a plane or cylinder. In higher dimensions there is no single bending number at a point, because the manifold can bend differently in different two-dimensional directions. The remedy is to attach a curvature number to each of those directions. A tangent 2-plane $\Pi \subseteq T_p M$ is spanned by two vectors, and the quantity $\text{Riem}(X, Y, Y, X)$ built from a spanning pair, once normalized so that it depends only on the plane and not on the pair, is the *sectional curvature* of Π . It equals the Gaussian curvature of the little surface swept out by geodesics leaving p in the directions of Π , a statement that will be made precise once the second fundamental form is available. Two facts justify the reduction. First, the sectional curvature is well defined: it is a function of the plane alone, independent of the pair used to span it. Second, and less obviously, it loses nothing: the numbers $K(\Pi)$, as Π ranges over all tangent 2-planes, determine the full curvature tensor. Sectional curvature is therefore not a coarsening of R but a faithful re-encoding of it, and it is the encoding through which the model spaces S^n , H^n , and $\mathbb{R}^n$ reveal themselves as the spaces of constant curvature $+1/r^2$, -1 , and 0.

Throughout, (M, g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ , R is the curvature endomorphism of definition 4.1.2, and $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$ is the associated $(0, 4)$ -tensor of definition 4.1.4. The algebraic symmetries recorded in proposition 4.2.1 are used constantly; they are (i) antisymmetry in the first pair, (ii) antisymmetry in the second pair, (iii) symmetry under exchange of the two pairs, and (iv) the first Bianchi identity. Sectional curvature in the stated normalization requires the metric to be nondegenerate on the plane in question, so the definition is given first in the Riemannian case, where every 2-plane is admissible.

Definition 4.3.1: Sectional curvature

Let (M, g) be a Riemannian manifold, $p \in M$, and let $\Pi \subseteq T_p M$ be a 2-dimensional linear subspace spanned by vectors $X, Y \in T_p M$. The *sectional curvature* of Π is

$$K(\Pi) = \frac{\text{Riem}(X, Y, Y, X)}{|X|^2 |Y|^2 - \langle X, Y \rangle^2}$$

The denominator is the squared area of the parallelogram spanned by X and Y ; by the Cauchy–Schwarz inequality it is positive precisely because X, Y are linearly independent, so the quotient is defined. That $K(\Pi)$ depends only on Π , and not on the chosen spanning pair, is the content of proposition 4.3.2. When emphasis on the point is needed the value is written $K_p(\Pi)$; the notations $\text{sec}(\Pi)$ and $K(X, Y)$ are also used, the latter for the sectional curvature of the plane spanned by X and Y .

The denominator deserves a name. Write

$$G(X, Y) = |X|^2 |Y|^2 - \langle X, Y \rangle^2$$

for the *Gram determinant* of the pair (X, Y) , the determinant of the matrix $\begin{pmatrix} \langle X, X \rangle & \langle X, Y \rangle \\ \langle Y, X \rangle & \langle Y, Y \rangle \end{pmatrix}$. It is the square of the area of the parallelogram on X and Y , vanishes exactly when the two vectors are dependent, and transforms under a change of spanning pair by the square of the determinant of the change-of-basis matrix. That last property is what makes the quotient in definition 4.3.1 basis-independent, as the next proposition shows. In the pseudo-Riemannian setting $G(X, Y)$ can vanish or be negative for independent vectors (a plane on which g degenerates, or a timelike plane), and the definition is restricted to *nondegenerate* 2-planes, those on which g restricts to a nondegenerate form so that $G(X, Y) \neq 0$; the well-definedness argument below is unchanged in that case.

A single worked instance fixes the meaning before the general theory. Take Euclidean space $(\mathbb{R}^n, \bar{g})$. Its curvature tensor vanishes identically, since in standard coordinates the metric components are constant, every Christoffel symbol is zero, and R is built from the Γ_{ij}^k and their derivatives (example 4.1.6). Hence $\text{Riem}(X, Y, Y, X) = 0$ for every pair, and $K(\Pi) = 0/G(X, Y) = 0$ for every 2-plane Π at every point. Euclidean space has vanishing sectional curvature in all directions, the analytic statement of its flatness. The two model spaces with nonzero curvature, S_r^n and H^n , are computed at the end of the section.

Proposition 4.3.2: Sectional curvature is independent of the spanning pair

Let (M, g) be Riemannian, $p \in M$, and $\Pi \subseteq T_p M$ a 2-plane. If (X, Y) and (X', Y') are two bases of Π , then

$$\frac{\text{Riem}(X', Y', Y', X')}{G(X', Y')} = \frac{\text{Riem}(X, Y, Y, X)}{G(X, Y)}$$

so $K(\Pi)$ in definition 4.3.1 is well defined. The same holds for a nondegenerate 2-plane in the pseudo-Riemannian case.

Proof:

Since (X, Y) and (X', Y') are bases of the same 2-dimensional space Π , there is an invertible

matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{R})$ with

$$X' = aX + bY, \quad Y' = cX + dY$$

It suffices to prove that both the numerator and the denominator of the quotient scale by the same factor $(\det A)^2 = (ad - bc)^2$ under this substitution; the ratio is then unchanged.

The numerator. Write $Q(U, V) = \text{Riem}(U, V, V, U)$ for the numerator functional. Expanding by multilinearity of Riem in each slot,

$$Q(X', Y') = \text{Riem}(aX + bY, cX + dY, cX + dY, aX + bY)$$

is a sum of terms $\text{Riem}(\cdot, \cdot, \cdot, \cdot)$ in which each of the four arguments is one of X, Y with a scalar coefficient. By the antisymmetry $\text{Riem}(U, V, \cdot, \cdot) = -\text{Riem}(V, U, \cdot, \cdot)$ of proposition 4.2.1(i), any term whose first two arguments are equal (both X or both Y) vanishes; likewise by antisymmetry in the last pair, proposition 4.2.1(ii), any term whose last two arguments are equal vanishes. The surviving terms have first pair a permutation of $\{X, Y\}$ and last pair a permutation of $\{X, Y\}$. Reading off the scalar coefficient of each from the two factors $cX + dY$ in the middle and the outer factors $aX + bY$, and collecting,

$$Q(X', Y') = abcd \text{Riem}(X, Y, X, Y) + a^2 d^2 \text{Riem}(X, Y, Y, X) + b^2 c^2 \text{Riem}(Y, X, X, Y) + abcd \text{Riem}(Y, X, Y, X)$$

Using the two antisymmetries, $\text{Riem}(X, Y, X, Y) = -\text{Riem}(X, Y, Y, X)$, $\text{Riem}(Y, X, X, Y) = -\text{Riem}(X, Y, X, Y) = \text{Riem}(X, Y, Y, X)$, and $\text{Riem}(Y, X, Y, X) = -\text{Riem}(X, Y, Y, X)$, so every surviving term is a multiple of $\text{Riem}(X, Y, Y, X)$:

$$Q(X', Y') = [a^2 d^2 - abcd - abcd + b^2 c^2] \text{Riem}(X, Y, Y, X) = (ad - bc)^2 Q(X, Y)$$

The denominator. The Gram determinant $G(X, Y) = \det(\langle X_i, X_j \rangle)$ with $(X_1, X_2) = (X, Y)$ is the determinant of the matrix of inner products. Under the substitution $(X', Y') = (X, Y)A^T$ in the sense above, the Gram matrix transforms as $(\langle X'_i, X'_j \rangle) = A(\langle X_i, X_j \rangle)A^T$, so

$$G(X', Y') = \det(A(\langle X_i, X_j \rangle)A^T) = (\det A)^2 G(X, Y) = (ad - bc)^2 G(X, Y)$$

This uses only bilinearity and symmetry of $\langle -, - \rangle$, which hold for any (pseudo-)Riemannian g ; for a nondegenerate plane $G(X, Y) \neq 0$, so the quotient is defined.

Dividing, the factor $(ad - bc)^2$ cancels:

$$\frac{Q(X', Y')}{G(X', Y')} = \frac{(ad - bc)^2 Q(X, Y)}{(ad - bc)^2 G(X, Y)} = \frac{Q(X, Y)}{G(X, Y)}$$

Hence the quotient is the same for the two bases, and $K(\Pi)$ depends only on Π , as we sought to show.

In practice the formula simplifies whenever a convenient spanning pair is available. If X, Y are orthonormal, then $|X|^2 = |Y|^2 = 1$ and $\langle X, Y \rangle = 0$, so $G(X, Y) = 1$ and $K(\Pi) = \text{Riem}(X, Y, Y, X)$ with no denominator at all. Proposition 4.3.2 guarantees that this stripped-down value agrees with the general one, so orthonormal pairs may always be used to compute sectional curvature, and this is how the examples below proceed.

The heart of the section is that no information is lost in passing from R to K . The precise statement concerns algebraic curvature tensors: multilinear maps carrying the four symmetries of the Riemann tensor, of which the value Riem_p at a point is the geometric example.

Theorem 4.3.3: Sectional curvature determines the curvature tensor

Let V be a finite-dimensional real vector space with a nondegenerate symmetric bilinear form $\langle -, - \rangle$, and let $\mathcal{R}, \mathcal{R}': V^4 \rightarrow \mathbb{R}$ be two *algebraic curvature tensors*, that is, quadrilinear maps each satisfying the symmetries of proposition 4.2.1:

1. $\mathcal{R}(x, y, z, w) = -\mathcal{R}(y, x, z, w)$;
2. $\mathcal{R}(x, y, z, w) = -\mathcal{R}(x, y, w, z)$;
3. $\mathcal{R}(x, y, z, w) = \mathcal{R}(z, w, x, y)$;
4. $\mathcal{R}(x, y, z, w) + \mathcal{R}(y, z, x, w) + \mathcal{R}(z, x, y, w) = 0$.

If $\mathcal{R}(x, y, y, x) = \mathcal{R}'(x, y, y, x)$ for all $x, y \in V$, then $\mathcal{R} = \mathcal{R}'$. Consequently, on a Riemannian manifold the sectional curvature function $\Pi \mapsto K(\Pi)$ on 2-planes determines the full curvature tensor Riem_p at each point, and two Riemannian manifolds whose curvature tensors have the same sectional curvatures at corresponding points have equal curvature tensors there.

Proof:

Set $D = \mathcal{R} - \mathcal{R}'$. The symmetries (i)–(iv) are preserved under subtraction, so D is again an algebraic curvature tensor, and the hypothesis says $D(x, y, y, x) = 0$ for all $x, y \in V$. The claim $\mathcal{R} = \mathcal{R}'$ is equivalent to $D \equiv 0$, and this is what will be shown: an algebraic curvature tensor whose associated quadratic function $Q(x, y) := D(x, y, y, x)$ vanishes identically is itself zero. The argument is polarization in three steps, each extracting more of D from the vanishing of Q .

Step 1: linearize the first argument. Fix $y \in V$ and expand $Q(x + z, y)$ using multilinearity of D :

$$D(x + z, y, y, x + z) = D(x, y, y, x) + D(x, y, y, z) + D(z, y, y, x) + D(z, y, y, z)$$

The outer terms are $Q(x, y)$ and $Q(z, y)$. The two middle terms are equal: by the pair symmetry (iii), $D(z, y, y, x) = D(y, x, z, y)$; then the first-pair antisymmetry (i) gives $D(y, x, z, y) = -D(x, y, z, y)$, and the second-pair antisymmetry (ii) gives $-D(x, y, z, y) = D(x, y, y, z)$. Chaining these, $D(z, y, y, x) = D(y, x, z, y) = D(x, y, y, z)$. Therefore

$$Q(x + z, y) - Q(x, y) - Q(z, y) = 2D(x, y, y, z)$$

Since $Q \equiv 0$ by hypothesis, the left side vanishes for all x, y, z , and hence

$$D(x, y, y, z) = 0 \quad \text{for all } x, y, z \in V \tag{4.2}$$

Step 2: linearize the repeated middle argument. In (4.2) the second and third arguments coincide. Replace y by $y + w$ and expand:

$$0 = D(x, y + w, y + w, z) = D(x, y, y, z) + D(x, y, w, z) + D(x, w, y, z) + D(x, w, w, z)$$

The first and last terms vanish by (4.2), leaving

$$D(x, y, w, z) + D(x, w, y, z) = 0 \quad \text{for all } x, y, w, z \in V \tag{4.3}$$

so D is antisymmetric in its second and third arguments.

Step 3: apply the first Bianchi identity. Fix $x, y, z, w \in V$. The first Bianchi identity (iv), applied with the four arguments x, z, y in the cyclic slots and w in the last slot, reads

$$D(x, z, y, w) + D(z, y, x, w) + D(y, x, z, w) = 0$$

Rewrite two of these terms. By the first-pair antisymmetry (i), $D(y, x, z, w) = -D(x, y, z, w)$; and by the pair symmetry (iii), $D(z, y, x, w) = D(x, w, z, y)$, which by the second-pair antisymmetry (ii) equals $-D(x, w, y, z)$, and by (4.3) equals $+D(x, y, w, z)$, which by (ii) equals $-D(x, y, z, w)$. Substituting,

$$D(x, z, y, w) - D(x, y, z, w) - D(x, y, z, w) = 0$$

that is $D(x, z, y, w) = 2D(x, y, z, w)$. On the other hand, applying (4.3) directly with the roles of the second and third arguments gives $D(x, z, y, w) = -D(x, y, z, w)$. Comparing,

$$2D(x, y, z, w) = D(x, z, y, w) = -D(x, y, z, w)$$

so $3D(x, y, z, w) = 0$, and therefore $D(x, y, z, w) = 0$ for all $x, y, z, w \in V$.

Thus $D \equiv 0$, i.e. $\mathcal{R} = \mathcal{R}'$. For the geometric consequence, apply this to $V = T_p M$ with $\langle -, - \rangle = g_p$: the value $\mathcal{R} = \text{Riem}_p$ is an algebraic curvature tensor by proposition 4.2.1, and for linearly independent x, y the number $\text{Riem}_p(x, y, y, x) = G(x, y) K(\Pi)$ is recovered from the sectional curvature of $\Pi = \text{span}(x, y)$, while for dependent x, y it vanishes by antisymmetry. Hence the function K on 2-planes determines $\text{Riem}_p(x, y, y, x)$ for all x, y , and by the theorem determines Riem_p completely, as claimed.

The theorem is the reason sectional curvature is the primary invariant of Riemannian geometry. It says that the seemingly modest data of one number per tangent plane carry every part of the curvature tensor, so any curvature condition can in principle be phrased as a condition on sectional curvatures, and conversely a hypothesis such as "all sectional curvatures are positive" or "all sectional curvatures equal c " is a full constraint on the tensor R . The constant case is the sharpest illustration. Consider the $(1, 3)$ -tensor $R_c(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$ and its $(0, 4)$ -form $\mathcal{R}_c(X, Y, Z, W) = \langle R_c(X, Y)Z, W \rangle = c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle)$; this $\mathcal{R}_c$ is an algebraic curvature tensor, and $\mathcal{R}_c(x, y, y, x) = c(|x|^2 |y|^2 - \langle x, y \rangle^2) = cG(x, y)$, so its sectional curvature is c on every plane. If a Riemannian manifold has $K(\Pi) = c$ for every 2-plane at every point, then Riem_p and $\mathcal{R}_c$ agree on the quantity $(x, y) \mapsto \text{Riem}_p(x, y, y, x)$ at each point, and by theorem 4.3.3 the two tensors coincide: $\text{Riem}_p = \mathcal{R}_c$. This is how the constant-curvature form of R is pinned down in the classification of space forms, taken up later in the chapter; here it already lets the model spaces be identified from a single sectional curvature computation, since each is isotropic and homogeneous.

The first computation is the round sphere, the constant-positive-curvature benchmark. Its Christoffel symbols were found in example 2.3.5; feeding them into the coordinate curvature formula of definition 4.1.5 gives the sectional curvature directly.

Example 4.3.4: The round sphere has constant sectional curvature $1/r^2$

Work first on the unit sphere S^2 with the coordinates (θ, φ) of example 2.3.5, in which

$$g = d\theta^2 + \sin^2 \theta d\varphi^2, \quad 0 < \theta < \pi$$

and the only nonzero Christoffel symbols are $\Gamma_{\varphi\varphi}^\theta = -\sin \theta \cos \theta$ and $\Gamma_{\theta\varphi}^\varphi = \Gamma_{\varphi\theta}^\varphi = \cot \theta$. The curvature endomorphism of definition 4.1.2 on the coordinate frame is $R(\partial_i, \partial_j)\partial_k = R^l_{ijk}\partial_l$, and

by the coordinate formula of definition 4.1.5,

$$R^l_{ijk} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im} \Gamma^m_{jk} - \Gamma^l_{jm} \Gamma^m_{ik}$$

(the sign convention is that of definition 4.1.2; the coordinate vector fields commute, so the $[\partial_i, \partial_j]$ term is absent). The single sectional plane at each point is $\Pi = \text{span}(\partial_\theta, \partial_\varphi)$, and the number that enters the sectional curvature is $\text{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta) = \langle R(\partial_\theta, \partial_\varphi) \partial_\varphi, \partial_\theta \rangle$, the ∂_θ -coefficient of $R(\partial_\theta, \partial_\varphi) \partial_\varphi$ times $g_{\theta\theta}$. That coefficient is the component $R^\theta_{\theta\varphi\varphi}$ (upper index $l = \theta$; operator pair $i = \theta, j = \varphi$; acted-on index $k = \varphi$):

$$R^\theta_{\theta\varphi\varphi} = \partial_\theta \Gamma^\theta_{\varphi\varphi} - \partial_\varphi \Gamma^\theta_{\theta\varphi} + \Gamma^\theta_{\theta m} \Gamma^m_{\varphi\varphi} - \Gamma^\theta_{\varphi m} \Gamma^m_{\theta\varphi}$$

Here $\partial_\theta \Gamma^\theta_{\varphi\varphi} = \partial_\theta(-\sin \theta \cos \theta) = -\cos^2 \theta + \sin^2 \theta = \sin^2 \theta - \cos^2 \theta$; the term $\partial_\varphi \Gamma^\theta_{\theta\varphi} = 0$; the third sum vanishes because $\Gamma^\theta_{\theta m} = 0$ for every m ; and the fourth sum keeps only $m = \varphi$, giving $-\Gamma^\theta_{\varphi\varphi} \Gamma^\varphi_{\theta\varphi} = -(-\sin \theta \cos \theta)(\cot \theta) = \sin \theta \cos \theta \cdot \frac{\cos \theta}{\sin \theta} = \cos^2 \theta$. Adding,

$$R^\theta_{\theta\varphi\varphi} = (\sin^2 \theta - \cos^2 \theta) + \cos^2 \theta = \sin^2 \theta$$

The component $R^\theta_{\theta\varphi\varphi}$ is by definition the ∂_θ -coefficient of $R(\partial_\theta, \partial_\varphi) \partial_\varphi$. Lowering that index with $g_{\theta\theta} = 1$ gives the $(0, 4)$ -component in the order needed for the sectional curvature,

$$\text{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta) = \langle R(\partial_\theta, \partial_\varphi) \partial_\varphi, \partial_\theta \rangle = g_{\theta\theta} R^\theta_{\theta\varphi\varphi} = \sin^2 \theta$$

The Gram determinant of the coordinate frame is

$$G(\partial_\theta, \partial_\varphi) = g_{\theta\theta} g_{\varphi\varphi} - g_{\theta\varphi}^2 = 1 \cdot \sin^2 \theta - 0 = \sin^2 \theta$$

so the sectional curvature of the only 2-plane at each point is

$$K = \frac{\text{Riem}(\partial_\theta, \partial_\varphi, \partial_\varphi, \partial_\theta)}{G(\partial_\theta, \partial_\varphi)} = \frac{\sin^2 \theta}{\sin^2 \theta} = 1$$

independent of the point. Thus the unit round sphere has $K \equiv 1$. The blow-up of $\cot \theta$ at the poles, noted in example 2.3.5, cancels out of K , as it must: the curvature is a geometric quantity and the poles are ordinary points of S^2 .

For the sphere of radius r , the metric of S_r^n scales as $g_r = r^2 g_1$ relative to the unit metric on the same underlying manifold. Under a constant rescaling $g \mapsto \lambda^2 g$ the Christoffel symbols are unchanged (they depend only on ratios and derivatives of metric components, and the λ^2 cancels in $\frac{1}{2} g^{kl}(\partial_i g_{jl} + \dots)$), so the $(1, 3)$ -curvature R^l_{ijk} is unchanged, and the $(0, 4)$ -tensor Riem scales by λ^2 from the single lowered index. The Gram determinant scales by λ^4 . Hence K scales by $\lambda^2/\lambda^4 = \lambda^{-2}$, and with $\lambda = r$ the round S_r^n has

$$K \equiv \frac{1}{r^2}$$

For $n \geq 3$ this value holds for *every* tangent 2-plane, not only the coordinate one: by example 1.3.4 the sphere S_r^n is homogeneous and isotropic, so its isometry group carries any 2-plane at any point to any other 2-plane at any other point, and sectional curvature is invariant under isometries (isometries commute with ∇ and preserve g , hence preserve Riem and G). The sectional curvature is therefore the same on all planes, equal to the value $1/r^2$ computed on one of them. The round sphere is the space of constant curvature $+1/r^2$.

The second computation is the constant-negative-curvature model. The upper half-plane realization of H^2 carries the Christoffel symbols already found in example 3.1.5, and the sectional curvature follows from the same coordinate formula.

Example 4.3.5: The hyperbolic plane has constant sectional curvature -1

Take the upper half-plane model $H^2 = \{(x, y) \mid y > 0\}$ of example 1.3.5 with the metric of example 3.1.5,

$$g = \frac{1}{y^2}(dx^2 + dy^2)$$

whose nonzero Christoffel symbols are $\Gamma_{xy}^x = \Gamma_{yx}^x = -\frac{1}{y}$, $\Gamma_{xx}^y = \frac{1}{y}$, and $\Gamma_{yy}^y = -\frac{1}{y}$. As in the spherical case the coordinate vector fields commute, so

$$R_{ijk}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{im}^l \Gamma_{jk}^m - \Gamma_{jm}^l \Gamma_{ik}^m$$

by definition 4.1.5, with the sign convention of definition 4.1.2. The one 2-plane at each point is $\text{span}(\partial_x, \partial_y)$; compute R_{xyy}^x , taking $(i, j) = (x, y)$ acting on ∂_y with upper index x :

$$R_{xyy}^x = \partial_x \Gamma_{yy}^x - \partial_y \Gamma_{xy}^x + \Gamma_{xm}^x \Gamma_{yy}^m - \Gamma_{ym}^x \Gamma_{xy}^m$$

Evaluate term by term. First, $\Gamma_{yy}^x = 0$, so $\partial_x \Gamma_{yy}^x = 0$. Second, $\partial_y \Gamma_{xy}^x = \partial_y(-\frac{1}{y}) = y^{-2}$. Third, the sum $\Gamma_{xm}^x \Gamma_{yy}^m$ over m : the factor Γ_{xm}^x is nonzero only for $m = y$ (value $\Gamma_{xy}^x = -\frac{1}{y}$), and then $\Gamma_{yy}^y = -\frac{1}{y}$, contributing $(-\frac{1}{y})(-\frac{1}{y}) = y^{-2}$. Fourth, the sum $\Gamma_{ym}^x \Gamma_{xy}^m$ over m : the factor Γ_{ym}^x is nonzero only for $m = x$ (value $\Gamma_{yx}^x = -\frac{1}{y}$), and then $\Gamma_{xy}^y = -\frac{1}{y}$, contributing $(-\frac{1}{y})(-\frac{1}{y}) = y^{-2}$, entering with a minus sign. Collecting,

$$R_{xyy}^x = 0 - y^{-2} + y^{-2} - y^{-2} = -y^{-2}$$

Lower the index with $g_{xx} = y^{-2}$: the $(0, 4)$ -component is

$$\text{Riem}(\partial_x, \partial_y, \partial_y, \partial_x) = \langle R(\partial_x, \partial_y)\partial_y, \partial_x \rangle = g_{xx} R_{xyy}^x = y^{-2} \cdot (-y^{-2}) = -y^{-4}$$

The Gram determinant of the coordinate frame is

$$G(\partial_x, \partial_y) = g_{xx}g_{yy} - g_{xy}^2 = y^{-2} \cdot y^{-2} - 0 = y^{-4}$$

so the sectional curvature is

$$K = \frac{\text{Riem}(\partial_x, \partial_y, \partial_y, \partial_x)}{G(\partial_x, \partial_y)} = \frac{-y^{-4}}{y^{-4}} = -1$$

at every point of H^2 , independent of (x, y) . The hyperbolic plane has constant curvature -1 .

The extension to H^n follows the same isotropy argument as for the sphere. By example 1.3.5, hyperbolic space H^n is homogeneous and isotropic, its isometry group $O^+(1, n)$ acting transitively on tangent 2-planes; and the several models (hyperboloid, upper half-space, Poincaré ball) of example 1.3.5 are mutually isometric, so the value of K is model-independent. Since sectional curvature is preserved by isometries and computed to be -1 on one 2-plane, it equals -1 on all of them:

$$K \equiv -1$$

Rescaling the metric by λ^2 multiplies K by λ^{-2} , exactly as for the sphere, so the rescaled hyperbolic space of "radius" ρ has constant curvature $-1/\rho^2$; the value -1 corresponds to the standard normalization. Hyperbolic space is the space of constant negative curvature, completing the trio of constant-curvature models alongside S_r^n with $K = +1/r^2$ and Euclidean space $\mathbb{R}^n$ with $K \equiv 0$.

4.4 Ricci and Scalar Curvature; Einstein Metrics

The Riemann curvature tensor of an n -dimensional manifold is an intricate object: at each point it is a section of a bundle whose fibers, once the algebraic symmetries of proposition 4.2.1 are imposed, have dimension $\frac{1}{12}n^2(n^2 - 1)$. For $n = 4$ that is already 20 independent components at every point. Much of Riemannian geometry proceeds not by confronting Riem in its entirety but by extracting from it coarser tensors that retain part of the geometric information while being far more tractable. The sectional curvature of the previous section is one such extract, and it loses nothing: by theorem 4.3.3 the sectional curvatures determine Riem completely. The tensors introduced here, the Ricci and scalar curvatures, are genuine simplifications. They are traces of Riem, and passing to a trace discards information. What one gains in exchange is a symmetric 2-tensor and a scalar function that couple directly to the geometry of volumes, to the Laplacian, and to the equations of general relativity.

The Ricci curvature measures how the volume of a small geodesic ball differs from its Euclidean counterpart in a given direction, and the scalar curvature aggregates this over all directions into a single number at each point. Both arise by contracting the Riemann tensor. The order and the slots in which one contracts are governed by the sign convention fixed for this chapter, and the payoff of that convention is that the round sphere and hyperbolic space acquire the expected signs: positive Ricci and scalar curvature for the sphere, negative for hyperbolic space. This section defines these traces, records their symmetry, decomposes the full curvature tensor into the pieces that Ricci and scalar curvature see together with the trace-free remainder (the Weyl tensor), and singles out the metrics whose Ricci tensor is a constant multiple of g , the Einstein metrics. The section closes with Schur's lemma, a rigidity statement showing that in dimension at least three a pointwise proportionality $\text{Ric} = f g$ forces f to be constant.

Throughout, (M, g) is an n -dimensional (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), and R , Riem denote the curvature endomorphism and its $(0, 4)$ -form as fixed in section 4.1. The Einstein summation convention is in force: an index repeated once up and once down is summed from 1 to n . Recall the sign convention:

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]}Z, \quad \text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$$

Definition 4.4.1: Ricci Curvature

Let (M, g) be a (pseudo-)Riemannian manifold. The *Ricci curvature* Ric is the covariant 2-tensor field obtained from the curvature endomorphism R of definition 4.1.2 by tracing over its first and last arguments: for $X, Y \in T_p M$,

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(Z, X)Y)$$

the trace of the linear endomorphism $Z \mapsto R(Z, X)Y$ of $T_p M$. Equivalently, if $(e_i)_{i=1}^n$ is a local orthonormal frame for a Riemannian metric, then

$$\text{Ric}(X, Y) = \sum_{i=1}^n \langle R(e_i, X)Y, e_i \rangle = \sum_{i=1}^n \text{Riem}(e_i, X, Y, e_i)$$

In local coordinates (x^i) , writing the components of the curvature endomorphism as $R(\partial_i, \partial_j)\partial_k = R^l_{ijk}\partial_l$ as in definition 4.1.5, the components $R_{ij} := \text{Ric}(\partial_i, \partial_j)$ are the contraction

$$R_{ij} = R^k_{kij}$$

the trace taken over the upper index and the first lower index.

Two remarks fix the bookkeeping, because tracing a $(1, 3)$ -tensor over one of its four index slots is exactly where sign and normalization errors enter. First, the coordinate formula $R_{ij} = R^k_{kij}$ is the same contraction as the frame formula: with $Z = \partial_k$ playing the role summed against e_i , the endomorphism $Z \mapsto R(Z, \partial_i)\partial_j$ has ∂_l -component R^l_{kij} , and its trace is $\sum_k R^k_{kij}$. Second, the orthonormal-frame expression is the definition one uses in practice: it makes manifest that Ric is obtained from Riem by summing $\text{Riem}(e_i, X, Y, e_i)$ over an orthonormal basis, so that Ric is a metric contraction of Riem over its first and fourth slots. In the pseudo-Riemannian setting the frame sum acquires the signs $\varepsilon_i = \langle e_i, e_i \rangle = \pm 1$ of the orthonormal frame, $\text{Ric}(X, Y) = \sum_i \varepsilon_i \text{Riem}(e_i, X, Y, e_i)$; the coordinate formula $R_{ij} = R^k_{kij}$ is unchanged and signature-agnostic, which is why it is the cleaner one to carry.

Interpreting the definition: $\text{Ric}(X, X)$ is a sum of sectional-curvature-like quantities. If X is a unit vector and (e_i) is an orthonormal basis with $e_n = X$, then $\text{Riem}(e_i, X, X, e_i)$ is, for $i \neq n$, exactly the numerator of the sectional curvature $K(\text{span}(e_i, X))$ of the plane spanned by e_i and X (the denominator is 1 since the vectors are orthonormal), while the $i = n$ term vanishes by the antisymmetry of proposition 4.2.1. Hence

$$\text{Ric}(X, X) = \sum_{i=1}^{n-1} K(\text{span}(e_i, X))$$

the sum of the sectional curvatures of the $n - 1$ coordinate planes containing X . The Ricci curvature in the direction X is the total sectional curvature of the planes through X ; up to the normalization $n - 1$ it is their average. This is the precise sense in which $\text{Ric}(X, X)$ controls the spreading of nearby geodesics leaving p in directions near X , and thereby the infinitesimal volume of a geodesic cone about X : positive Ricci curvature means geodesics converge on average, negative means they spread.

Before recording the first structural property, a computation on the model spaces shows the definition in action and pins down the signs.

Example 4.4.2: Ricci Curvature of the Space Forms

Suppose (M, g) has constant sectional curvature c , meaning $K(\Pi) = c$ for every 2-plane Π . By theorem 4.3.3 the curvature endomorphism is then forced to be

$$R(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$$

the constant-curvature form of the sign convention (this identity is derived in full in section 4.5; here it is used only to read off the trace). Fix a point and an orthonormal basis (e_i) . For unit $X = e_a$,

$$\begin{aligned} \text{Ric}(X, X) &= \sum_{i=1}^n \langle R(e_i, X)X, e_i \rangle = \sum_{i=1}^n c(\langle X, X \rangle \langle e_i, e_i \rangle - \langle e_i, X \rangle \langle X, e_i \rangle) \\ &= c \sum_{i=1}^n (1 - \langle e_i, X \rangle^2) = c(n-1) \end{aligned}$$

since $\sum_i \langle e_i, X \rangle^2 = |X|^2 = 1$. By polarization $\text{Ric}(X, Y) = c(n-1) \langle X, Y \rangle$ for all X, Y , that is,

$$\text{Ric} = (n-1)c g$$

For the round sphere S_r^n of example 1.3.4 the sectional curvature is $c = 1/r^2$ (example 4.3.4), so $\text{Ric} = \frac{n-1}{r^2} g$, a positive multiple of g . For Euclidean space $c = 0$ and $\text{Ric} \equiv 0$, consistent with $R \equiv 0$ from example 4.1.6. For hyperbolic space H^n the sectional curvature is $c = -1$ (example 4.3.5), so $\text{Ric} = -(n-1)g$, a negative multiple. The signs are exactly those the convention was chosen to produce.

The Ricci tensor is symmetric. This is not automatic from the definition, which distinguishes the slots X and Y ; it is a consequence of the pair symmetry of Riem .

Proposition 4.4.3: Symmetry of the Ricci Tensor

The Ricci curvature of a (pseudo-)Riemannian manifold is a symmetric 2-tensor:

$$\text{Ric}(X, Y) = \text{Ric}(Y, X) \quad \text{for all } X, Y \in T_p M$$

Equivalently, in coordinates $R_{ij} = R_{ji}$.

Proof:

Fix $p \in M$ and an orthonormal basis $(e_i)_{i=1}^n$ of $T_p M$ (Riemannian case; the pseudo-Riemannian case is identical after inserting the signs ε_i , which are common to the two expressions below and cancel in the comparison). By definition,

$$\text{Ric}(X, Y) = \sum_{i=1}^n \text{Riem}(e_i, X, Y, e_i)$$

Apply the pair symmetry $\text{Riem}(A, B, C, D) = \text{Riem}(C, D, A, B)$ from proposition 4.2.1(iii) to each summand, with $(A, B, C, D) = (e_i, X, Y, e_i)$:

$$\text{Riem}(e_i, X, Y, e_i) = \text{Riem}(Y, e_i, e_i, X)$$

Now apply the two antisymmetries of proposition 4.2.1(i)–(ii): swapping the first pair introduces a sign, $\text{Riem}(Y, e_i, e_i, X) = -\text{Riem}(e_i, Y, e_i, X)$, and swapping the last pair introduces a second sign, $-\text{Riem}(e_i, Y, e_i, X) = \text{Riem}(e_i, Y, X, e_i)$. The two signs cancel, giving

$$\text{Riem}(e_i, X, Y, e_i) = \text{Riem}(e_i, Y, X, e_i)$$

Summing over i ,

$$\text{Ric}(X, Y) = \sum_{i=1}^n \text{Riem}(e_i, X, Y, e_i) = \sum_{i=1}^n \text{Riem}(e_i, Y, X, e_i) = \text{Ric}(Y, X)$$

as we sought to show.

Symmetry of Ric is what makes it a legitimate candidate to be proportional to g , and it is what allows one to take its metric trace unambiguously. That trace is the scalar curvature.

Definition 4.4.4: Scalar Curvature

Let (M, g) be a (pseudo-)Riemannian manifold with Ricci tensor Ric . The *scalar curvature* is the function $\text{scal}: M \rightarrow \mathbb{R}$ obtained as the metric trace of Ric :

$$\text{scal} = \text{tr}_g \text{Ric} = g^{ij} R_{ij}$$

Equivalently, for a local orthonormal frame (e_i) (Riemannian case),

$$\text{scal} = \sum_{i=1}^n \text{Ric}(e_i, e_i) = \sum_{i,j=1}^n \text{Riem}(e_i, e_j, e_j, e_i)$$

the sum of the sectional curvatures over all ordered pairs of distinct frame directions.

Unwinding the frame expression makes the geometric content of scal transparent. Using $\text{Ric}(e_i, e_i) = \sum_j \text{Riem}(e_j, e_i, e_i, e_j)$ and the antisymmetry that kills the diagonal terms $i = j$, the double sum runs over ordered pairs $i \neq j$, and each unordered pair $\{i, j\}$ contributes twice the sectional curvature $K(\text{span}(e_i, e_j))$ of the plane it spans. Thus

$$\text{scal} = 2 \sum_{i < j} K(\text{span}(e_i, e_j))$$

The scalar curvature at p is twice the sum of the sectional curvatures of the $\binom{n}{2}$ coordinate planes at p ; equivalently, up to the constant $n(n-1)$ it is the average sectional curvature over all 2-planes. It carries the least pointwise information of the three traces of Riem , one number per point, but it is the trace that appears in the Gauss–Bonnet integrand in dimension two, in the total-scalar-curvature (Einstein–Hilbert) functional, and as the coefficient governing the leading volume defect of small geodesic balls.

Example 4.4.5: Scalar Curvature of the Space Forms

Continuing example 4.4.2, suppose (M, g) has constant sectional curvature c , so that $\text{Ric} = (n-1)c g$. Taking the metric trace and using $\text{tr}_g g = g^{ij} g_{ij} = n$,

$$\text{scal} = \text{tr}_g \text{Ric} = (n-1)c \text{tr}_g g = n(n-1)c$$

For the round sphere S_r^n with $c = 1/r^2$ this gives $\text{scal} = \frac{n(n-1)}{r^2}$, a positive constant. For Euclidean space $\text{scal} \equiv 0$, and for H^n with $c = -1$ one gets $\text{scal} = -n(n-1)$. In dimension $n = 2$ the formula reads $\text{scal} = 2c$, so for the unit sphere S^2 the scalar curvature is 2 while the Gauss curvature (the single sectional curvature) is 1: the scalar curvature is twice the Gauss curvature in dimension two, matching the frame formula $\text{scal} = 2 \sum_{i < j} K = 2K(\text{span}(e_1, e_2))$.

The passage from Riem to Ric to scal can be organized as an orthogonal decomposition of the full curvature tensor. The Ricci and scalar curvatures capture the “trace part” of Riem; the trace-free remainder is a tensor with the same algebraic symmetries as Riem but with all of its contractions vanishing, the Weyl tensor. To state this cleanly it is convenient to use the Kulkarni–Nomizu product, which builds a curvature-type tensor out of two symmetric 2-tensors.

Definition 4.4.6: Ricci Decomposition and the Weyl Tensor

Let (M, g) be a (pseudo-)Riemannian manifold of dimension $n \geq 3$. For symmetric 2-tensors h, k , define their *Kulkarni–Nomizu product* $h \odot k$, a $(0, 4)$ -tensor, by

$$(h \odot k)(X, Y, Z, W) = h(X, W)k(Y, Z) + h(Y, Z)k(X, W) - h(X, Z)k(Y, W) - h(Y, W)k(X, Z)$$

It shares the algebraic symmetries of Riem in proposition 4.2.1: it is antisymmetric in (X, Y) and in (Z, W) , symmetric under exchange of the pairs, and satisfies the first Bianchi identity. Write the *trace-free Ricci tensor* as

$$\mathring{\text{Ric}} = \text{Ric} - \frac{\text{scal}}{n} g$$

The *Ricci decomposition* of the curvature tensor is

$$\text{Riem} = \frac{\text{scal}}{2n(n-1)} g \odot g + \frac{1}{n-2} \mathring{\text{Ric}} \odot g + W$$

which defines the *Weyl tensor* W as the remainder. The three summands are the *scalar part*, the *trace-free Ricci part*, and the *Weyl (conformal) part*.

The defining property of the Weyl tensor, and the reason the coefficients above are what they are, is that W is *totally trace-free*: every contraction of W over a pair of its indices vanishes. This is proved directly.

Proposition 4.4.7: The Weyl Tensor Is Trace-Free

For $n \geq 3$ the Weyl tensor W defined in definition 4.4.6 satisfies $g^{il}W_{ijkl} = 0$; that is, the metric contraction of W over its first and last slots vanishes, and hence (by the symmetries) every contraction of W vanishes.

Proof:

Fix $p \in M$ and an orthonormal basis (e_i) , so that the metric contraction over the first and fourth slots of a $(0, 4)$ -tensor T is the map $(Y, Z) \mapsto \sum_i T(e_i, Y, Z, e_i)$; for $T = \text{Riem}$ this is Ric by definition. It suffices to compute this contraction on each of the three summands of definition 4.4.6 and check that the sum of the first two equals Ric, forcing the contraction of W

to vanish.

Step 1: contraction of a Kulkarni–Nomizu product. Let h, k be symmetric 2-tensors. Viewing h, k as symmetric endomorphisms of $T_p M$ via the metric, write hk and kh for the two composite endomorphisms and $\text{tr}_g h, \text{tr}_g k$ for their traces. Contracting $h \otimes k$ over its first and fourth slots term by term,

$$\begin{aligned} \sum_i (h \otimes k)(e_i, Y, Z, e_i) &= \sum_i \left[h(e_i, e_i)k(Y, Z) + h(Y, Z)k(e_i, e_i) \right. \\ &\quad \left. - h(e_i, Z)k(Y, e_i) - h(Y, e_i)k(e_i, Z) \right] \\ &= (\text{tr}_g h)k(Y, Z) + (\text{tr}_g k)h(Y, Z) - (kh)(Y, Z) - (hk)(Y, Z) \end{aligned}$$

the third and fourth terms being the two endomorphism composites $(kh)(Y, Z) = \sum_i k(Y, e_i)h(e_i, Z)$ and $(hk)(Y, Z) = \sum_i h(Y, e_i)k(e_i, Z)$. Two special cases are needed, and in each the two composites coincide because one factor is g .

Taking $h = k = g$: then $\text{tr}_g g = n$ and $gg = g$ as an endomorphism, so

$$\sum_i (g \otimes g)(e_i, Y, Z, e_i) = n g(Y, Z) + n g(Y, Z) - g(Y, Z) - g(Y, Z) = 2(n-1)g(Y, Z)$$

Taking $h = \mathring{\text{Ric}}$ and $k = g$: then $\text{tr}_g g = n$, $\text{tr}_g \mathring{\text{Ric}} = \text{scal} - \frac{\text{scal}}{n} \cdot n = 0$, and both composites $\mathring{\text{Ric}}g$ and $g\mathring{\text{Ric}}$ equal $\mathring{\text{Ric}}$ as endomorphisms, so

$$\sum_i (\mathring{\text{Ric}} \otimes g)(e_i, Y, Z, e_i) = 0 \cdot g(Y, Z) + n \mathring{\text{Ric}}(Y, Z) - \mathring{\text{Ric}}(Y, Z) - \mathring{\text{Ric}}(Y, Z) = (n-2)\mathring{\text{Ric}}(Y, Z)$$

Step 2: assembling the trace of the decomposition. Contract the identity

$$\text{Riem} = \frac{\text{scal}}{2n(n-1)} g \otimes g + \frac{1}{n-2} \mathring{\text{Ric}} \otimes g + W$$

over the first and fourth slots. The left side gives Ric . Using Step 1, the first summand contributes

$$\frac{\text{scal}}{2n(n-1)} \cdot 2(n-1)g = \frac{\text{scal}}{n}g$$

and the second contributes

$$\frac{1}{n-2} \cdot (n-2)\mathring{\text{Ric}} = \mathring{\text{Ric}} = \text{Ric} - \frac{\text{scal}}{n}g$$

Write $C(Y, Z) := \sum_i W(e_i, Y, Z, e_i)$ for the first-and-fourth contraction of the Weyl summand. Adding the three contributions gives

$$\text{Ric} = \frac{\text{scal}}{n}g + \left(\text{Ric} - \frac{\text{scal}}{n}g \right) + C = \text{Ric} + C$$

Hence $C = 0$: the contraction of W over its first and fourth slots vanishes. Because W inherits the antisymmetries and pair symmetry of Riem (each summand of the decomposition satisfies them, and Riem does by proposition 4.2.1), every other pairwise contraction of W reduces to C up to sign: contracting the first two or the last two slots gives zero by antisymmetry, and contracting any mixed pair reduces to C by the pair symmetry together with the antisymmetries. Therefore W is totally trace-free, as we sought to show.

The decomposition is an orthogonal one with respect to the natural inner product on curvature-type tensors, and it is the pointwise decomposition of the space of algebraic curvature tensors into irreducible representations of the orthogonal group $O(n)$: the scalar part, the trace-free Ricci part, and the Weyl part are the three irreducible summands for $n \geq 4$. Two low-dimensional cases collapse this and are worth recording, since they explain why curvature is so much more rigid in low dimensions.

For $n = 2$ the space of algebraic curvature tensors is one-dimensional, and Riem is completely determined by the scalar curvature: explicitly $\text{Riem} = \frac{\text{scal}}{2} \frac{1}{2} g \otimes g$ reduces to $\text{Riem}(X, Y, Z, W) = \frac{\text{scal}}{2} (\langle X, W \rangle \langle Y, Z \rangle - \langle X, Z \rangle \langle Y, W \rangle)$, so a surface has a single curvature function, the Gauss curvature $K = \frac{\text{scal}}{2}$. (The decomposition of definition 4.4.6 was stated for $n \geq 3$ because of the $\frac{1}{n-2}$; in dimension two the Weyl and trace-free-Ricci pieces are absent.) For $n = 3$ the Weyl tensor vanishes identically: the dimension of the space of algebraic curvature tensors is $\frac{1}{12} \cdot 9 \cdot 8 = 6$, which is exactly the dimension of the space of symmetric 2-tensors on $\mathbb{R}^3$, so the map $\text{Riem} \mapsto \text{Ric}$ is injective on curvature tensors and $W \equiv 0$. Consequently, in dimension three Riem is determined by Ric through

$$\text{Riem} = \frac{\text{scal}}{12} g \otimes g + \overset{\circ}{\text{Ric}} \otimes g$$

For $n \geq 4$ the Weyl tensor is an independent piece of the curvature, invisible to Ric and scal; it is the obstruction to a metric being conformally flat.

Among all metrics, those for which the Ricci tensor is proportional to the metric occupy a distinguished place, both because they are the critical points of the total scalar curvature functional and because they are the natural higher-dimensional analogues of constant-curvature surfaces.

Definition 4.4.8: Einstein Metric

A (pseudo-)Riemannian metric g on M is an *Einstein metric*, and (M, g) is an *Einstein manifold*, if there is a constant $\lambda \in \mathbb{R}$ with

$$\text{Ric} = \lambda g$$

The constant λ is the *Einstein constant* of g .

Taking the metric trace of $\text{Ric} = \lambda g$ gives $\text{scal} = \lambda \text{tr}_g g = n\lambda$, so the Einstein constant is determined by the scalar curvature, $\lambda = \text{scal}/n$, and the scalar curvature of an Einstein metric is constant. An Einstein metric is exactly one whose Ricci tensor is a constant multiple of g , equivalently one whose trace-free Ricci tensor $\overset{\circ}{\text{Ric}}$ of definition 4.4.6 vanishes identically. Geometrically the condition says that the Ricci curvature is the same in every direction: $\text{Ric}(X, X) = \lambda |X|^2$ for all X , so the average sectional curvature of the planes through any unit vector is the constant $\lambda/(n-1)$, independent of the vector. This is a far weaker requirement than constant sectional curvature, which asks that *every* sectional curvature be equal; an Einstein metric only asks that the directional averages agree. In dimensions two and three the two notions collapse (in dimension three by the vanishing of the Weyl tensor, $\text{Ric} = 0$ together with $W = 0$ forces the constant-curvature form of example 4.4.2), but from dimension four onward Einstein metrics that are not of constant curvature exist in abundance.

The running example makes the definition concrete and reconnects with the space forms.

Example 4.4.9: The Sphere Is Einstein

Consider the round sphere S_r^n of example 1.3.4, of constant sectional curvature $c = 1/r^2$ (example 4.3.4). By the computation of example 4.4.2,

$$\text{Ric} = (n-1)c g = \frac{n-1}{r^2} g$$

so S_r^n is an Einstein manifold with Einstein constant $\lambda = \frac{n-1}{r^2} > 0$, and by example 4.4.5 its scalar curvature is the constant

$$\text{scal} = n(n-1)c = \frac{n(n-1)}{r^2}$$

Both are positive, as the sign convention requires for the sphere. The same computation shows the other two space forms are Einstein: Euclidean space is Einstein with $\lambda = 0$ (indeed $\text{Ric} \equiv 0$), and H^n is Einstein with $\lambda = -(n-1) < 0$ and $\text{scal} = -n(n-1)$. For $n = 2$ the sphere gives $\text{Ric} = \frac{1}{r^2} g$ and $\text{scal} = \frac{2}{r^2}$, so every surface is trivially Einstein: the Ricci tensor of a surface is always $\frac{\text{scal}}{2} g$, whence the Einstein condition in dimension two carries no information and the interesting range is $n \geq 3$.

In defining an Einstein metric the constancy of λ was imposed by hand. It is worth asking whether it must be imposed at all: if the Ricci tensor is pointwise a multiple $f(p) g_p$ of the metric, with f a function that is a priori allowed to vary, is f automatically constant? For surfaces the answer is no, since $\text{Ric} = \frac{\text{scal}}{2} g$ always and scal can be any function. For $n = 3$ the same phenomenon is blocked, and in fact for every $n \geq 3$ pointwise proportionality forces constancy on a connected manifold. This is Schur's lemma, and its proof is the first substantial application of the second Bianchi identity.

Theorem 4.4.10: Schur's Lemma

Let (M, g) be a connected (pseudo-)Riemannian manifold of dimension $n \geq 3$. Suppose there is a smooth function $f: M \rightarrow \mathbb{R}$ with

$$\text{Ric} = f g$$

at every point. Then f is constant; consequently g is an Einstein metric and $\text{scal} = n f$ is constant.

Proof:

The argument contracts the second Bianchi identity twice and reads off a first-order equation for scal that forces df to vanish. Throughout, fix $p \in M$ and work in a local orthonormal frame (e_i) (Riemannian case; the pseudo-Riemannian case is identical with the frame signs ε_i inserted, which cancel as before). All contractions below are metric contractions, written as sums over the frame.

Step 1: the once-contracted second Bianchi identity. Recall first that the covariant derivative ∇Riem inherits the algebraic symmetries of Riem (proposition 4.2.1): since $\nabla g = 0$, differentiating an identity such as $\text{Riem}(A, B, C, D) = -\text{Riem}(B, A, C, D)$ gives $(\nabla_E \text{Riem})(A, B, C, D) = -(\nabla_E \text{Riem})(B, A, C, D)$, and likewise for the other symmetries. The differential Bianchi identity (theorem 4.2.3), written for the $(0, 4)$ -tensor Riem and evaluated on frame vectors, reads

$$(\nabla_{e_a} \text{Riem})(e_b, e_c, e_d, e_e) + (\nabla_{e_b} \text{Riem})(e_c, e_a, e_d, e_e) + (\nabla_{e_c} \text{Riem})(e_a, e_b, e_d, e_e) = 0$$

the sum being cyclic in the differentiation direction together with the first two arguments, (a, b, c) . Contract this over the differentiation slot and the last argument, that is, set $e_a = e_c = e_i$ and sum over i . Each summand is evaluated by the first-and-fourth-slot contraction $\sum_i (\nabla_E \text{Riem})(e_i, \cdot, \cdot, e_i)$, which by definition 4.4.1 is the covariant derivative of Ric. The first summand is

$$\sum_i (\nabla_{e_i} \text{Riem})(e_b, e_c, e_d, e_i)$$

the metric contraction of ∇Riem over its differentiation direction and its last argument. The second summand becomes

$$\sum_i (\nabla_{e_b} \text{Riem})(e_c, e_i, e_d, e_i) = - \sum_i (\nabla_{e_b} \text{Riem})(e_i, e_c, e_d, e_i) = -(\nabla_{e_b} \text{Ric})(e_c, e_d)$$

using the antisymmetry proposition 4.2.1(i) in the first two arguments and then the first-and-fourth contraction. The third summand becomes

$$\sum_i (\nabla_{e_c} \text{Riem})(e_i, e_b, e_d, e_i) = (\nabla_{e_c} \text{Ric})(e_b, e_d)$$

directly by the first-and-fourth contraction. Adding the three and rearranging,

$$(\nabla_{e_c} \text{Ric})(e_b, e_d) - (\nabla_{e_b} \text{Ric})(e_c, e_d) = - \sum_i (\nabla_{e_i} \text{Riem})(e_b, e_c, e_d, e_i)$$

Relabel $(e_c, e_b, e_d) = (X, Y, Z)$ and use the antisymmetry $\text{Riem}(Y, X, Z, e_i) = -\text{Riem}(X, Y, Z, e_i)$ on the right to obtain the once-contracted differential Bianchi identity

$$(\nabla_X \text{Ric})(Y, Z) - (\nabla_Y \text{Ric})(X, Z) = \sum_i (\nabla_{e_i} \text{Riem})(X, Y, Z, e_i) \quad (4.4)$$

valid for all X, Y, Z since both sides are tensorial.

Step 2: the twice-contracted identity $\text{div Ric} = \frac{1}{2} d \text{scal}$. Define the divergence of the symmetric 2-tensor Ric by $(\text{div Ric})(X) := \sum_i (\nabla_{e_i} \text{Ric})(e_i, X)$. Contract (4.4) a second time, over the pair (Y, Z) : put $Y = Z = e_j$ and sum over j . On the left,

$$\sum_j (\nabla_X \text{Ric})(e_j, e_j) - \sum_j (\nabla_{e_j} \text{Ric})(X, e_j) = \nabla_X \text{scal} - (\text{div Ric})(X)$$

using $\sum_j (\nabla_X \text{Ric})(e_j, e_j) = \nabla_X (\text{tr}_g \text{Ric}) = \nabla_X \text{scal}$ (since ∇ commutes with the metric trace) and the symmetry of Ric in the second sum. On the right,

$$\sum_{i,j} (\nabla_{e_i} \text{Riem})(X, e_j, e_j, e_i) = \sum_i (\nabla_{e_i} \text{Ric})(X, e_i) = (\text{div Ric})(X)$$

where the inner sum over j is evaluated by the antisymmetries: $\text{Riem}(X, e_j, e_j, e_i) = -\text{Riem}(e_j, X, e_j, e_i) = \text{Riem}(e_j, X, e_i, e_j)$ by proposition 4.2.1(i)–(ii), and summing over j this is the first-and-fourth-slot contraction, giving $\text{Ric}(X, e_i)$ (the derivative $\nabla_{e_i} \text{Riem}$ inherits the algebraic symmetries of Riem because $\nabla g = 0$). Equating the two sides,

$$\nabla_X \text{scal} - (\text{div Ric})(X) = (\text{div Ric})(X)$$

that is $\nabla_X \text{scal} = 2(\text{div Ric})(X)$ for every X , which is the twice-contracted differential Bianchi identity

$$\text{div Ric} = \frac{1}{2} d \text{scal} \quad (4.5)$$

The factor $\frac{1}{2}$ is fixed by the sign and contraction conventions of this chapter, and (4.5) is the identity underlying the divergence-freeness of the Einstein tensor $\text{Ric} - \frac{1}{2} \text{scal } g$.

Step 3: imposing $\text{Ric} = f g$. Now use the hypothesis. Since $\nabla g = 0$ (metric compatibility, definition 2.2.4), the covariant derivative of $\text{Ric} = f g$ is

$$(\nabla_X \text{Ric})(Y, Z) = (\nabla_X f) g(Y, Z) + f (\nabla_X g)(Y, Z) = (Xf) g(Y, Z)$$

Taking the divergence, for any Y ,

$$(\text{div Ric})(Y) = \sum_i (\nabla_{e_i} \text{Ric})(e_i, Y) = \sum_i (e_i f) g(e_i, Y) = Yf = df(Y)$$

since $\sum_i (e_i f) g(e_i, Y) = \sum_i (e_i f) \langle e_i, Y \rangle$ is the expansion of Y against the orthonormal frame applied to df , namely $df(Y)$. On the other hand the scalar curvature of $\text{Ric} = f g$ is $\text{scal} = \text{tr}_g \text{Ric} = f \text{tr}_g g = nf$, so $d \text{scal} = n df$. Substituting both into (4.5),

$$df = \text{div Ric} = \frac{1}{2} d \text{scal} = \frac{n}{2} df$$

Hence $(1 - \frac{n}{2}) df = 0$, that is $\frac{2-n}{2} df = 0$. Because $n \geq 3$ the scalar factor $\frac{2-n}{2}$ is nonzero, so $df = 0$ identically. Therefore f is locally constant, and since M is connected, f is constant on M . With f constant, $\text{Ric} = f g$ exhibits g as an Einstein metric with Einstein constant f , and $\text{scal} = nf$ is constant, as we sought to show.

Schur's lemma is the reason the definition of an Einstein metric may safely be stated with a constant λ rather than a function: in dimension at least three, pointwise proportionality of Ric to g already delivers constancy. The mechanism is worth isolating. The single input beyond the algebraic symmetries is the differential Bianchi identity, contracted to (4.5); the hypothesis $\text{Ric} = f g$ then turns that identity into $df = \frac{n}{2} df$, and the dimensional restriction $n \geq 3$ is exactly what makes the coefficient $1 - \frac{n}{2}$ invertible. The failure in dimension two is visible in the same equation: there $1 - \frac{n}{2} = 0$, the identity $df = \frac{n}{2} df$ is vacuous, and f is free. The contracted Bianchi identity (4.5) recorded along the way, $\text{div Ric} = \frac{1}{2} d \text{scal}$, is itself one of the load-bearing identities of the subject: rearranged as $\text{div}(\text{Ric} - \frac{1}{2} \text{scal } g) = 0$, it says the Einstein tensor is divergence-free, the identity that makes the field equations of general relativity consistent with local conservation of energy and momentum.

4.5 Flat Manifolds and Space Forms

The curvature tensor was introduced as the obstruction to commuting second covariant derivatives. The extreme case in which the obstruction vanishes deserves separate treatment: a manifold on which $R \equiv 0$ is called *flat*, and the first result of this section shows that flatness is exactly local indistinguishability from Euclidean space. This is the sharpest possible converse to the computation of example 4.1.6, where the vanishing of the Euclidean curvature followed from the constancy of the metric coefficients. Here the implication runs the other way: if the curvature vanishes then

coordinates can be manufactured in which the metric coefficients are constant, and the manifold looks, patch by patch, like a piece of $\mathbb{R}^n$.

Between the flat case and the general case sits a class of manifolds whose curvature is as symmetric as it can be without vanishing: those whose sectional curvature $K(\Pi)$ takes a single value c on every 2-plane Π at every point. These are the *space forms*, and they generalize the three model geometries that have run through the book. The plane $\mathbb{R}^n$ has $c = 0$, the round sphere S_r^n has $c = 1/r^2$ (example 4.3.4), and hyperbolic space H^n has $c = -1$ (example 4.3.5). The main structural fact is that these three, after rescaling, exhaust the local possibilities: any manifold of constant sectional curvature c is locally isometric to exactly one of them. Constant sectional curvature is a scalar condition on a 2-plane, yet it pins down the full $(0, 4)$ curvature tensor and, with more work, the local geometry.

Throughout, (M, g) is a (pseudo-)Riemannian manifold of dimension n with Levi-Civita connection ∇ (definition 2.3.2) and curvature endomorphism R (definition 4.1.2); the associated $(0, 4)$ tensor is $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$ (definition 4.1.4). The sign convention of box 4.1.1 is in force. The Einstein summation convention is used where indices appear. The results on flatness and the local classification are stated for the Riemannian case; the frame constructions go through verbatim in the pseudo-Riemannian setting with orthonormal replaced by pseudo-orthonormal, and this is noted where it matters.

Theorem 4.5.1: Flat Manifolds are Locally Euclidean

Let (M, g) be a Riemannian manifold. Then the curvature endomorphism satisfies $R \equiv 0$ if and only if every point of M has a neighborhood isometric to an open subset of Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2. A manifold with $R \equiv 0$ is called *flat*.

Proof:

One direction is immediate from example 4.1.6. If a neighborhood U of p is isometric to an open subset of $(\mathbb{R}^n, \bar{g})$, then the curvature endomorphism of U agrees under the isometry with that of Euclidean space, which vanishes; since R is a tensor determined by the metric (proposition 4.1.3, definition 2.3.2), it vanishes on U , and as p was arbitrary, $R \equiv 0$ on M . The content is the converse.

Assume $R \equiv 0$ and fix $p \in M$. The strategy is to build, on a neighborhood of p , a frame $(E_1, \dots, E_n)$ that is simultaneously orthonormal and parallel, and then to show that this frame is the coordinate frame of a chart in which g is the Euclidean metric.

Step 1: a parallel orthonormal frame on a coordinate cube. Choose a chart $x = (x^1, \dots, x^n): W \rightarrow (-\varepsilon, \varepsilon)^n$ centered at p , so $x(p) = 0$, and let $(e_1, \dots, e_n)$ be an orthonormal basis of $(T_p M, g_p)$. Fix a and drop the subscript, writing $v = e_a$; the field $E = E_a$ is built on the cube by parallel transport along coordinate directions, and the vanishing of curvature is what makes it parallel in every direction. For $0 \leq m \leq n$ let

$$S_m = \{q \in W : x^{m+1}(q) = \dots = x^n(q) = 0\}$$

be the m -dimensional coordinate slab through p , so $S_0 = \{p\}$ and $S_n = W$. The construction proceeds by induction on m , producing E on S_m with the property

$$P(m): \quad \nabla_{\partial_j} E = 0 \text{ on } S_m \text{ for all } j \leq m$$

Set $E(p) = v$, which is $P(0)$ vacuously. Given E on S_m satisfying $P(m)$, extend it to S_{m+1} as

follows: each point of S_{m+1} lies on a unique line segment in the x^{m+1} -direction issuing from a point of S_m , and E is defined on S_{m+1} by parallel transporting its value at that base point along the segment (theorem 2.4.2). This makes

$$\nabla_{\partial_{m+1}} E = 0 \text{ on } S_{m+1} \quad (4.6)$$

by construction, and it remains only to recover the earlier conditions $\nabla_{\partial_j} E = 0$ for $j \leq m$ on the enlarged slab S_{m+1} .

Fix $j \leq m$ and consider the field $F = \nabla_{\partial_j} E$ on S_{m+1} . Its covariant derivative in the ∂_{m+1} -direction is controlled by curvature: since $[\partial_{m+1}, \partial_j] = 0$, the definition of R (definition 4.1.2) gives

$$\nabla_{\partial_{m+1}} F = \nabla_{\partial_{m+1}} \nabla_{\partial_j} E = \nabla_{\partial_j} \nabla_{\partial_{m+1}} E + R(\partial_{m+1}, \partial_j) E$$

The first term on the right vanishes because $\nabla_{\partial_{m+1}} E = 0$ throughout S_{m+1} by (4.6), and the second vanishes because $R \equiv 0$. Hence $\nabla_{\partial_{m+1}} F = 0$: the field F is parallel along each x^{m+1} -segment. At the base point of the segment, on S_m , one has $F = \nabla_{\partial_j} E = 0$ by $P(m)$. By uniqueness of parallel transport (theorem 2.4.2), $F \equiv 0$ along the whole segment, so $\nabla_{\partial_j} E = 0$ on all of S_{m+1} . Together with (4.6) this is $P(m+1)$.

By induction $P(n)$ holds: $\nabla_{\partial_j} E = 0$ on $S_n = W$ for every j . Since $(\partial_1, \dots, \partial_n)$ frames TW and $\nabla_Y E$ is $C^\infty(W)$ -linear in Y , this gives $\nabla_Y E = 0$ for every vector field Y , so E is parallel on W . Carrying out the construction for each $a = 1, \dots, n$ produces parallel fields $E_1, \dots, E_n$ with $E_a(p) = e_a$.

Step 2: the frame is orthonormal. Metric compatibility (definition 2.2.4) gives, for any Y ,

$$Y \langle E_a, E_b \rangle = \langle \nabla_Y E_a, E_b \rangle + \langle E_a, \nabla_Y E_b \rangle = 0$$

because both E_a and E_b are parallel. Hence each function $\langle E_a, E_b \rangle$ is constant on the (connected) cube, equal to its value $\langle e_a, e_b \rangle = \delta_{ab}$ at p . So $(E_1, \dots, E_n)$ is orthonormal at every point.

Step 3: the frame commutes and gives Euclidean coordinates. Because ∇ is torsion-free (definition 2.2.1) and each E_a is parallel,

$$[E_a, E_b] = \nabla_{E_a} E_b - \nabla_{E_b} E_a = 0$$

A frame of pairwise-commuting vector fields is, on a possibly smaller neighborhood U of p , the coordinate frame of a chart: by the canonical form for commuting vector fields ([3], the simultaneous-flow-box theorem), there are coordinates $y = (y^1, \dots, y^n)$ on U with $\partial/\partial y^a = E_a$ for each a . In these coordinates the metric components are

$$g_{ab} = \left\langle \frac{\partial}{\partial y^a}, \frac{\partial}{\partial y^b} \right\rangle = \langle E_a, E_b \rangle = \delta_{ab}$$

constant and equal to the Euclidean values by Step 2. Thus $y: U \rightarrow y(U) \subset \mathbb{R}^n$ pulls the Euclidean metric $\bar{g} = \delta_{ab} dy^a dy^b$ back to g , so y is an isometry of (U, g) onto the open subset $y(U)$ of $(\mathbb{R}^n, \bar{g})$ (definition 1.3.1). As p was arbitrary, every point of M has a Euclidean neighborhood, completing the proof.

The theorem isolates curvature as the sole local obstruction to being Euclidean. Two flat manifolds are locally identical no matter how different they look globally: the flat torus $\mathbb{R}^n/\mathbb{Z}^n$ of example 1.3.6 is compact and has closed geodesics, yet each of its points has a neighborhood indistinguishable from

a neighborhood in the plane, precisely because it is a quotient of $\mathbb{R}^n$ by isometries and so inherits $R \equiv 0$. The distinction between local and global is therefore genuine, and it is topology, not curvature, that separates the plane from the torus. The same phenomenon organizes the constant-curvature case, treated next: constant sectional curvature is a purely local condition, and it too leaves the global classification to topology.

The parallel-frame argument used only that the direction of R was zero. When the curvature is instead a fixed nonzero constant on every plane, the algebraic content of that constancy must first be extracted. The following definition names the class, and the proposition after it converts the scalar constancy of K into a rigid formula for the full curvature tensor.

Definition 4.5.2: Constant Sectional Curvature

A Riemannian manifold (M, g) has *constant sectional curvature* $c \in \mathbb{R}$ if $K(\Pi) = c$ for every point $p \in M$ and every 2-plane $\Pi \subset T_p M$, where K is the sectional curvature of definition 4.3.1. When M is understood, one says M is a *space form* of curvature c .

Proposition 4.5.3: The Curvature Tensor of a Space of Constant Curvature

Let (M, g) be a Riemannian manifold and $c \in \mathbb{R}$. Then (M, g) has constant sectional curvature c if and only if the curvature endomorphism has the form

$$R(X, Y)Z = c(\langle Y, Z \rangle X - \langle X, Z \rangle Y) \quad (4.7)$$

for all vector fields X, Y, Z , equivalently

$$\text{Riem}(X, Y, Z, W) = c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle) \quad (4.8)$$

Proof:

The equivalence of (4.7) and (4.8) is a matter of pairing with W and using $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$ (definition 4.1.4); the two are the same statement.

The formula implies constant curvature. Suppose (4.8) holds. For a 2-plane Π spanned by X, Y ,

$$\text{Riem}(X, Y, Y, X) = c(\langle Y, Y \rangle \langle X, X \rangle - \langle X, Y \rangle \langle Y, X \rangle) = c(|X|^2 |Y|^2 - \langle X, Y \rangle^2)$$

The denominator in the definition of sectional curvature (definition 4.3.1) is exactly $|X|^2 |Y|^2 - \langle X, Y \rangle^2$, which is positive because X, Y are linearly independent (the Cauchy–Schwarz inequality is strict for a nonproportional pair). Dividing gives $K(\Pi) = c$, for every plane at every point.

Constant curvature implies the formula. Suppose $K(\Pi) = c$ for all planes. Define a $(0, 4)$ tensor T by the right-hand side of (4.8):

$$T(X, Y, Z, W) = c(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle)$$

A direct check shows T satisfies the algebraic symmetries of a curvature tensor. It is antisymmetric in the first pair (X, Y) and in the last pair (Z, W) , symmetric under interchange of the two pairs, and satisfies the first Bianchi identity:

$$T(X, Y, Z, W) + T(Y, Z, X, W) + T(Z, X, Y, W)$$

expands to

$$c[(\langle Y, Z \rangle \langle X, W \rangle - \langle X, Z \rangle \langle Y, W \rangle) + (\langle Z, X \rangle \langle Y, W \rangle - \langle Y, X \rangle \langle Z, W \rangle) + (\langle X, Y \rangle \langle Z, W \rangle - \langle Z, Y \rangle \langle X, W \rangle)] = 0$$

since the six terms cancel in pairs. Thus both Riem (by proposition 4.2.1) and T are algebraic curvature tensors on each $T_p M$. By the computation in the previous paragraph applied to T in place of Riem, the sectional curvatures of T are also constant equal to c ; that is, T and Riem induce the same sectional-curvature function on every 2-plane. By theorem 4.3.3, an algebraic curvature tensor is determined by its sectional curvatures, so $\text{Riem} = T$, which is (4.8). Passing back to the endomorphism form gives (4.7), as required.

The proposition is the algebraic heart of the theory of space forms. The condition $K \equiv c$ names one number on each 2-plane, a function on the Grassmannian of planes, and a priori this is far less data than the whole tensor Riem; the polarization result theorem 4.3.3 is what closes the gap, forcing the entire curvature to be the universal expression (4.7). The right-hand side of (4.7) depends on the metric alone once c is fixed, so all local curvature information in a space form is carried by the single scalar c . The three model spaces realize the three signs of c , as the next example records.

Example 4.5.4: The Model Space Forms

The three running model geometries have constant sectional curvature and so satisfy (4.7); they are the canonical representatives of the three signs of c .

Euclidean space $(\mathbb{R}^n, \bar{g})$ (example 1.1.2) has $R \equiv 0$ by example 4.1.6, so it is a space form with $c = 0$, and (4.7) reads $R(X, Y)Z = 0$.

The round sphere S_r^n (example 1.3.4) has $K \equiv 1/r^2$: for S^2 this was computed from the Christoffel symbols in example 4.3.4, and the value extends to all S_r^n by the homogeneity and isotropy recorded in example 1.3.4. Thus for the sphere (4.7) becomes

$$R(X, Y)Z = \frac{1}{r^2}(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$$

As a check, for orthonormal X, Y this gives $\langle R(X, Y)Y, X \rangle = \frac{1}{r^2}(\langle Y, Y \rangle \langle X, X \rangle - 0) = \frac{1}{r^2}$, matching $K = 1/r^2$.

Hyperbolic space H^n (example 1.3.5) has $K \equiv -1$: for the hyperbolic plane this was computed in example 4.3.5, and again it extends to all dimensions by the homogeneity and isotropy of the model. Here (4.7) reads

$$R(X, Y)Z = -(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$$

so that $\langle R(X, Y)Y, X \rangle = -1$ on an orthonormal pair, matching $K = -1$. Rescaling the hyperbolic metric by the constant factor $1/(-c)$ for $c < 0$ produces the model of curvature c , in the same way that S_r^n realizes curvature $1/r^2$; the sign of c is the invariant that cannot be scaled away.

The models settle the three signs of c one manifold at a time. The classification theorem asserts that these are the only local possibilities: a space form of curvature c is, near each point, an isometric copy of the model with the same c (after normalizing the scale). The proof produces the local isometry explicitly through geodesic normal coordinates, in which the metric of a constant-curvature manifold is forced into a universal form depending only on c .

Theorem 4.5.5: Local Classification of Space Forms

Let (M, g) be a Riemannian n -manifold of constant sectional curvature c . Then every point of M has a neighborhood isometric to an open subset of the model space of curvature c , namely:

1. Euclidean space $(\mathbb{R}^n, \bar{g})$ if $c = 0$;
2. the round sphere $S_{1/\sqrt{c}}^n$ of radius $1/\sqrt{c}$ if $c > 0$;
3. hyperbolic space H^n rescaled to curvature c if $c < 0$.

In particular any two Riemannian manifolds of the same constant sectional curvature c are locally isometric to each other.

Proof:

The case $c = 0$ is theorem 4.5.1: constant curvature 0 means $K \equiv 0$, hence $R \equiv 0$ by proposition 4.5.3, hence local isometry to $(\mathbb{R}^n, \bar{g})$. Assume $c \neq 0$.

Fix $p \in M$ and let $(x^1, \dots, x^n)$ be geodesic normal coordinates centered at p on a normal ball B , arising from an orthonormal basis of $(T_p M, g_p)$ (definition 3.2.6). Introduce geodesic polar coordinates (ρ, θ) , where $\rho = |x|$ is the geodesic distance from p and θ ranges over the unit sphere $\Sigma = \{u \in T_p M : |u| = 1\}$; a point of the punctured ball is $\exp_p(\rho\theta)$. By the Gauss lemma (theorem 3.3.4) and its corollary on the polar form of the metric (proposition 3.3.5), the metric on the punctured normal ball is

$$g = d\rho^2 + h_\rho \quad (4.9)$$

where h_ρ is, for each fixed $\rho > 0$, a metric on the geodesic sphere $\{|x| = \rho\}$, degenerating to the point p as $\rho \rightarrow 0$. The strategy is to compute h_ρ from the constant-curvature hypothesis and to recognize the result as the model metric.

Step 1: the radial variation equation. The metric h_ρ is read off from the differential of the exponential map. Fix a unit vector $\theta \in \Sigma$ and let $\gamma(\rho) = \exp_p(\rho\theta)$ be the radial geodesic in direction θ , so $\dot{\gamma}(\rho) = (d\exp_p)_{\rho\theta}(\theta)$; it has constant speed $|\dot{\gamma}| \equiv |\theta| = 1$ (proposition 3.1.6). For a tangent vector $w \in T_\theta \Sigma \subset T_p M$ (so $w \perp \theta$, $|w| = 1$), consider the variation of geodesics

$$\alpha(\rho, s) = \exp_p(\rho\theta(s)), \quad \theta(0) = \theta, \quad \frac{d}{ds}\Big|_0 \theta(s) = w$$

where $\theta(s)$ is a curve in Σ with the stated initial data. For each fixed s the curve $\rho \mapsto \alpha(\rho, s)$ is a geodesic, and the variation field

$$J(\rho) = \frac{\partial \alpha}{\partial s}\Big|_{s=0}$$

is a vector field along γ measuring the infinitesimal spreading of the radial geodesics. Its value $J(\rho) = (d\exp_p)_{\rho\theta}(\rho w)$ is exactly the image under $d\exp_p$ of the vector ρw tangent to the sphere of radius ρ in $T_p M$, so the norm $|J(\rho)|$ is the factor by which h_ρ stretches the direction w : for $w_1, w_2 \in T_\theta \Sigma$,

$$h_\rho(w_1, w_2) = \langle (d\exp_p)_{\rho\theta}(\rho w_1), (d\exp_p)_{\rho\theta}(\rho w_2) \rangle = \langle J_1(\rho), J_2(\rho) \rangle \quad (4.10)$$

with J_i the variation field associated to w_i . By the Gauss lemma $J(\rho) \perp \dot{\gamma}(\rho)$ for all ρ .

The field J satisfies a second-order equation involving curvature. Let $T = \partial\alpha/\partial\rho$ and $S = \partial\alpha/\partial s$, so along γ (at $s = 0$) one has $T = \dot{\gamma}$ and $S = J$. Since ∂_ρ and ∂_s are coordinate fields on the

parameter rectangle, the symmetry lemma (lemma 3.5.3) gives $D_s T = D_\rho S$, where D_ρ, D_s are the covariant derivatives along the ρ - and s -curves (definition 2.1.6). Each ρ -curve is a geodesic, so $D_\rho T = 0$. The curvature endomorphism governs the failure of D_ρ and D_s to commute on a field along α : with the sign convention of box 4.1.1, and because the parameter fields commute, the definition of R (definition 4.1.2) yields for any field V along α

$$D_\rho D_s V - D_s D_\rho V = R(T, S)V$$

Applying this to $V = T$ and using $D_s T = D_\rho S$ together with $D_\rho T = 0$,

$$D_\rho D_\rho S = D_\rho D_s T = D_s D_\rho T + R(T, S)T = R(T, S)T$$

Evaluated at $s = 0$, where $T = \dot{\gamma}$ and $S = J$, this is $D_\rho^2 J = R(\dot{\gamma}, J)\dot{\gamma}$, equivalently

$$D_\rho^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0 \quad (4.11)$$

by the antisymmetry $R(\dot{\gamma}, J) = -R(J, \dot{\gamma})$ (proposition 4.2.1). This is the radial variation equation, derived here directly from torsion-freeness and the definition of R .

Now insert the constant-curvature form (4.7). Since $|\dot{\gamma}| = 1$ and $J \perp \dot{\gamma}$,

$$R(J, \dot{\gamma})\dot{\gamma} = c(\langle \dot{\gamma}, \dot{\gamma} \rangle J - \langle J, \dot{\gamma} \rangle \dot{\gamma}) = cJ$$

so (4.11) becomes the linear equation $D_\rho^2 J + cJ = 0$. Its initial data are fixed by the behavior of J at p . From $J(\rho) = (d \exp_p)_{\rho\theta}(\rho w)$ one has $J(0) = 0$; and since $J(\rho) = (d \exp_p)_0(\rho w) + O(\rho^2) = \rho w + O(\rho^2)$ by $d(\exp_p)_0 = \text{id}$ (proposition 3.2.5), the covariant initial velocity is $D_\rho J|_0 = w$, a unit vector. Let $e(\rho)$ be the parallel field along γ with $e(0) = w$, obtained by parallel transport of w ; it is a unit field ($|e| \equiv |w| = 1$) and stays orthogonal to $\dot{\gamma}$, because parallel transport preserves inner products (proposition 2.4.4), $\dot{\gamma}$ is itself parallel, and $\langle w, \dot{\gamma}(0) \rangle = \langle w, \theta \rangle = 0$. Let ψ solve the scalar initial value problem

$$\psi'' + c\psi = 0, \quad \psi(0) = 0, \quad \psi'(0) = 1 \quad (4.12)$$

and set $\tilde{J}(\rho) = \psi(\rho) e(\rho)$. Then $D_\rho^2 \tilde{J} = \psi'' e = -c\psi e = -c\tilde{J}$, so $\tilde{J}$ solves $D_\rho^2 \tilde{J} + c\tilde{J} = 0$ with $\tilde{J}(0) = 0$ and $D_\rho \tilde{J}|_0 = \psi'(0)e(0) = w$; the same initial value problem as J . By uniqueness of solutions of this linear second-order equation along γ (a linear ODE in components, theorem 2.4.2 providing the parallel frame in which it is a constant-coefficient system), $J = \tilde{J} = \psi e$. The equation (4.12) is independent of the direction θ and of the choice of unit $w \perp \theta$, which is the analytic form of the isotropy forced by constant sectional curvature.

Step 2: solving the radial equation. The solution of (4.12) is

$$\psi(\rho) = \begin{cases} \frac{1}{\sqrt{c}} \sin(\sqrt{c} \rho) & c > 0 \\ \frac{1}{\sqrt{-c}} \sinh(\sqrt{-c} \rho) & c < 0 \end{cases}$$

as is verified by substitution. For $c > 0$, $\psi'(\rho) = \cos(\sqrt{c} \rho)$ and $\psi''(\rho) = -\sqrt{c} \sin(\sqrt{c} \rho) = -c\psi$, so $\psi'' + c\psi = 0$, with $\psi(0) = 0$ and $\psi'(0) = 1$; for $c < 0$, $\psi''(\rho) = (-c) \cdot \frac{1}{\sqrt{-c}} \sinh(\sqrt{-c} \rho) = (-c)\psi$, that is $\psi'' + c\psi = 0$, again with $\psi(0) = 0$, $\psi'(0) = 1$. The metric h_ρ is now read off on an orthonormal basis. Fix an orthonormal basis $w_1, \dots, w_{n-1}$ of $T_\theta \Sigma$, and let J_i be the variation field with $D_\rho J_i|_0 = w_i$; by Step 1, $J_i(\rho) = \psi(\rho) e_i(\rho)$ where e_i is the parallel field with $e_i(0) = w_i$.

Since parallel transport is a linear isometry (proposition 2.4.4) and $e_i(0) = w_i$ are orthonormal, $e_1(\rho), \dots, e_{n-1}(\rho)$ are orthonormal for every ρ . Hence by (4.10),

$$h_\rho(w_i, w_j) = \langle J_i(\rho), J_j(\rho) \rangle = \psi(\rho)^2 \langle e_i(\rho), e_j(\rho) \rangle = \psi(\rho)^2 \delta_{ij} = \psi(\rho)^2 \langle w_i, w_j \rangle$$

As both h_ρ and $\psi(\rho)^2 g_\Sigma$ are bilinear forms on $T_\theta \Sigma$ agreeing on a basis, where g_Σ denotes the round metric of radius 1 on the unit sphere $\Sigma \subset T_p M$ (whose value on w_i, w_j is $\langle w_i, w_j \rangle$),

$$h_\rho = \psi(\rho)^2 g_\Sigma \quad (4.13)$$

Combining (4.9) and (4.13),

$$g = d\rho^2 + \psi(\rho)^2 g_\Sigma \quad (4.14)$$

Step 3: recognizing the model. The form (4.14) is exactly the geodesic polar form of the model space of curvature c . For $c > 0$, set $r = 1/\sqrt{c}$; then $\psi(\rho) = r \sin(\rho/r)$, and (4.14) becomes

$$g = d\rho^2 + r^2 \sin^2(\rho/r) g_\Sigma$$

which is the round metric of S_r^n written in geodesic polar coordinates about a point, as recorded for the sphere in example 1.3.4 and section 3.2. For $c < 0$, set $k = \sqrt{-c}$; then $\psi(\rho) = \frac{1}{k} \sinh(k\rho)$, and

$$g = d\rho^2 + \frac{1}{k^2} \sinh^2(k\rho) g_\Sigma$$

which is the metric of hyperbolic space of curvature $c = -k^2$ in geodesic polar coordinates, matching the model of example 1.3.5 after the rescaling by $1/k$. The point is that the model space M_c of curvature c also has constant sectional curvature c , so Steps 1 and 2 apply to it verbatim: about any of its points q_0 its metric takes, in geodesic polar coordinates on a normal ball, the same expression (4.14) with the same radial factor ψ , since ψ is determined by c alone through (4.12).

The isometry is now explicit. Fix a point q_0 of the model M_c and a linear isometry $\iota: (T_p M, g_p) \rightarrow (T_{q_0} M_c, g_{q_0})$; such an ι exists because both are inner-product spaces of dimension n . On the normal ball B about p , define

$$\Phi = \exp_{q_0} \circ \iota \circ (\exp_p)^{-1}$$

which is a diffeomorphism of B onto a normal ball about q_0 (composition of the diffeomorphism $(\exp_p)^{-1}$, the linear isomorphism ι , and the diffeomorphism $\exp_{q_0}$ on the appropriate domains, by proposition 3.2.5). Because ι is a linear isometry, it carries the unit sphere $\Sigma \subset T_p M$ isometrically to the unit sphere in $T_{q_0} M_c$ and preserves the radial coordinate ρ ; hence in polar coordinates Φ is $(\rho, \theta) \mapsto (\rho, \iota\theta)$. Both metrics are $d\rho^2 + \psi(\rho)^2 g_\Sigma$ in these coordinates with the identical ψ , and ι pulls back g_Σ on the model unit sphere to g_Σ on Σ , so $\Phi^* g_{M_c} = g$. Thus Φ is an isometry (definition 1.3.1) of (B, g) onto an open subset of the model. As p was arbitrary this gives the local isometry to the model space of curvature c ; for two manifolds of the same c the corresponding maps to the common model compose to a local isometry between them, completing the proof.

The classification reduces the local geometry of a constant-curvature manifold to a single scalar and one of three universal metrics. Every such manifold, viewed through a small enough window, is a piece of a sphere, a plane, or a hyperbolic space; there is no further local invariant. The role of the Gauss lemma and the radial variation equation (4.11) is to convert the algebraic rigidity

of proposition 4.5.3 into an explicit metric normal form (4.14), whose radial factor ψ satisfies the elementary equation (4.12) and so is completely pinned down by c . The variation field J of Step 1 and its equation (4.11) are the first appearance of a construction taken up systematically under the name of Jacobi fields in a later chapter; here it is derived from scratch, using only the definition of curvature and torsion-freeness, so the argument rests on the tools of the present chapter.

What the theorem does not settle is the global shape. Local isometry to the round sphere does not make a manifold a sphere: the real projective space $\mathbb{RP}^n$ inherits constant curvature 1 from S^n through the antipodal quotient (example 1.2.3), yet it is not homeomorphic to S^n , and the flat torus is locally Euclidean without being $\mathbb{R}^n$. The passage from local to global requires completeness and simple connectivity, and it is the content of the Killing–Hopf theorem, stated next as the natural companion to the local classification.

4.5.6 Foreshadowing: The Killing–Hopf Theorem The local classification theorem 4.5.5 has a global counterpart. A *complete, simply connected* Riemannian n -manifold of constant sectional curvature c is (globally) isometric to the model space of curvature c : to the sphere $S_{1/\sqrt{c}}^n$ for $c > 0$, to Euclidean space $\mathbb{R}^n$ for $c = 0$, and to hyperbolic space H^n (rescaled) for $c < 0$. This is the *Killing–Hopf theorem*. A general complete space form of curvature c is then a quotient $\widetilde{M}/\Gamma$ of the corresponding model $\widetilde{M}$ by a discrete group Γ of isometries acting freely and properly discontinuously, so that the classification of complete space forms becomes a problem about such groups; the flat tori $\mathbb{R}^n/\mathbb{Z}^n$ and $\mathbb{RP}^n = S^n/\{\pm 1\}$ are the first instances. The proof requires two tools not yet available. Completeness, in the sense that $\exp_p$ is defined on all of $T_p M$, is developed through the Hopf–Rinow theorem, and the globalization of a local isometry into a covering map uses the completeness of both source and model together with a monodromy argument along paths. Both belong to the completeness machinery of a later chapter, and the theorem is invoked there once that machinery is in place. A complete self-contained account may be found in do Carmo, *Riemannian Geometry*, and in John M. Lee, *Introduction to Riemannian Manifolds*, where the Killing–Hopf theorem is proved in full from the local normal form (4.14) established above.

4.6 Curvature of Warped Products

The warped products of section 1.6 were built at the level of the metric tensor, and the promise made there was that their curvature is computable in closed form from the geometry of the two factors and the derivatives of the single warping function. That promise is redeemed here. The point of the construction is not that it produces new manifolds, but that it produces manifolds whose curvature can be read off a one-dimensional differential expression: a warped product $I \times_f N$ over an interval has, in the radial direction, sectional curvature exactly $-f''/f$, so a metric of the form $dr^2 + f(r)^2 g_N$ has constant curvature κ in that direction precisely when f solves the linear equation $f'' + \kappa f = 0$. This is the mechanism, foreshadowed in example 1.6.5, by which the three warping functions r , $\sin r$, and $\sinh r$ deliver the three constant-curvature model spaces; the present section makes it a theorem and then reads the models off it.

The route to the curvature runs through the Levi-Civita connection of the warped product, which is itself determined by the two factor connections and the gradient of f . The connection of Chapter 2 is now available, so the derivation can proceed without deferral: a short lemma computes ∇ on horizontal and vertical vectors, and the curvature theorem then substitutes those formulas into

the defining expression $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$ of definition 4.1.2. The sign convention of box 4.1.1 is in force throughout, so that the round sphere carries $K = +1/r^2$ and hyperbolic space $K = -1$; the model computations at the end confirm both signs. The Einstein summation convention is not needed here, since the entire computation is carried out invariantly in terms of vector fields.

Fix, once and for all, a warped product $M \times_f N$ with base (M, g_M) , fiber (N, g_N) , and smooth warping function $f: M \rightarrow (0, \infty)$, carrying the metric $g = g_M \oplus f^2 g_N$ of definition 1.6.2. Its Levi-Civita connection is written ∇ , and the Levi-Civita connections of the two factors are ∇^M and ∇^N . Two classes of vector field on $M \times N$ organize every computation. A vector field is *horizontal* if it is everywhere tangent to the base slices $M \times \{q\}$, and *vertical* if it is everywhere tangent to the fibers $\{p\} \times N$; by the orthogonal splitting of proposition 1.6.3 these are the g -orthogonal complements of one another, and every vector field is uniquely the sum of a horizontal and a vertical one. A vector field X on M has a unique horizontal lift to $M \times N$, characterized by $d\pi_M \tilde{X} = X$ and $d\pi_N \tilde{X} = 0$; the lift is again written X where no confusion results, and such lifts are called *basic horizontal* fields. Likewise a vector field on N lifts to a *basic vertical* field. Basic lifts are π_M - and π_N -related to the originals, so their brackets and the metric pairings among them are controlled by the factors: for basic horizontal X, Y (lifts of base fields) and basic vertical V, W (lifts of fiber fields),

$$g(X, Y) = g_M(X, Y) \circ \pi_M, \quad g(V, W) = (f \circ \pi_M)^2 g_N(V, W) \circ \pi_N, \quad g(X, V) = 0$$

and the bracket of a basic horizontal with a basic vertical field vanishes, $[X, V] = 0$, while $[X, Y]$ is the horizontal lift of the base bracket and $[V, W]$ the vertical lift of the fiber bracket. These facts are the standard behaviour of lifts under the projections π_M, π_N ([3]); they are used without further comment below.

One further object enters, the Hessian of the warping function on the base. For a smooth function f on (M, g_M) the gradient $\text{grad } f = (df)^\sharp$ of definition 1.4.3 and the Levi-Civita connection ∇^M combine into the *Hessian*

$$\text{Hess}^M f(X, Y) = \langle \nabla_X^M \text{grad } f, Y \rangle_{g_M} = X(Yf) - (\nabla_X^M Y)f$$

a symmetric covariant 2-tensor field on M ; its metric trace $\Delta_M f = \text{tr}_{g_M} \text{Hess}^M f = \text{div}^M(\text{grad } f)$ is the Laplacian met in definition 1.4.7. The gradient here is the base gradient, so $Xf = \langle \text{grad } f, X \rangle_{g_M}$ for horizontal X , and $|\text{grad } f|^2$ means the base norm $|\text{grad } f|_{g_M}^2 = g_M(\text{grad } f, \text{grad } f)$. These are the only invariants of f that appear in the curvature.

Lemma 4.6.1: The connection of a warped product

Let $M \times_f N$ be a warped product with Levi-Civita connection ∇ , and let X, Y be basic horizontal fields and V, W basic vertical fields. Then

1. $\nabla_X Y = \nabla_X^M Y$, again basic horizontal;
2. $\nabla_X V = \nabla_V X = \frac{Xf}{f} V$, vertical;
3. $\nabla_V W = \nabla_V^N W - \frac{g(V, W)}{f} \text{grad } f$, whose vertical part is the fiber connection $\nabla_V^N W$ and whose horizontal part is $-g_N(V, W) f \text{grad } f$.

Here ∇^M and ∇^N are the Levi-Civita connections of the factors, $Xf = \langle \text{grad } f, X \rangle_{g_M}$, and $\text{grad } f$ is the base gradient, lifted horizontally.

Proof:

The Levi-Civita connection is characterized by the Koszul formula (2.3) of theorem 2.3.1: for vector fields A, B, C ,

$$2 \langle \nabla_A B, C \rangle = A \langle B, C \rangle + B \langle A, C \rangle - C \langle A, B \rangle + \langle [A, B], C \rangle - \langle [A, C], B \rangle - \langle [B, C], A \rangle$$

where all inner products are those of g . Each claim is proved by evaluating $2 \langle \nabla_A B, C \rangle$ from this formula as C ranges over horizontal and vertical fields separately, using the pairings and brackets recorded above; two vectors with the same inner product against every horizontal and every vertical field are equal, since these span the tangent space g -orthogonally.

1. Take $A = X, B = Y$ horizontal. Test first against a horizontal $C = Z$. Every term of the Koszul formula then involves only horizontal fields and the function $g(-, -) = g_M(-, -) \circ \pi_M$, and the brackets $[X, Y], [X, Z], [Y, Z]$ are horizontal lifts of the corresponding base brackets. The formula is therefore literally the base Koszul formula for $2 g_M(\nabla_X^M Y, Z)$ pulled back to $M \times N$, so

$$2 \langle \nabla_X Y, Z \rangle = 2 g_M(\nabla_X^M Y, Z) \circ \pi_M = 2 \langle \nabla_X^M Y, Z \rangle$$

Now test against a vertical $C = V$. Then $\langle X, V \rangle = \langle Y, V \rangle = 0$, so the first three terms vanish after noting that $X \langle Y, V \rangle, Y \langle X, V \rangle$, and $V \langle X, Y \rangle$ are derivatives of a horizontal pairing: the first two vanish because $\langle Y, V \rangle = \langle X, V \rangle = 0$ identically, and the third is $V(g_M(X, Y) \circ \pi_M) = 0$ because $g_M(X, Y) \circ \pi_M$ is constant along fibers, being a function of the base point alone, and V is vertical. Of the bracket terms, $\langle [X, Y], V \rangle = 0$ since $[X, Y]$ is horizontal, while $[X, V] = [Y, V] = 0$, so $\langle [X, V], Y \rangle = \langle [Y, V], X \rangle = 0$. Hence $2 \langle \nabla_X Y, V \rangle = 0$ for every vertical V . Thus $\nabla_X Y$ has the horizontal value $\nabla_X^M Y$ and no vertical part, which is claim (1).

2. Take $A = X$ horizontal, $B = V$ vertical. First test against a horizontal $C = Z$. The Koszul terms are

$$X \langle V, Z \rangle + V \langle X, Z \rangle - Z \langle X, V \rangle + \langle [X, V], Z \rangle - \langle [X, Z], V \rangle - \langle [V, Z], X \rangle$$

Here $\langle V, Z \rangle = \langle X, V \rangle = 0$, so the first and third terms vanish; $[X, V] = 0$; $[X, Z]$ is horizontal so $\langle [X, Z], V \rangle = 0$; and $[V, Z] = -[Z, V] = 0$ since Z is horizontal and V vertical, so the last term vanishes. The surviving term is $V \langle X, Z \rangle$, and $\langle X, Z \rangle = g_M(X, Z) \circ \pi_M$ is constant along fibers, so $V \langle X, Z \rangle = 0$. Therefore $2 \langle \nabla_X V, Z \rangle = 0$ for all horizontal Z : the field $\nabla_X V$ is vertical. Now test against a vertical $C = W$. The Koszul terms are

$$X \langle V, W \rangle + V \langle X, W \rangle - W \langle X, V \rangle + \langle [X, V], W \rangle - \langle [X, W], V \rangle - \langle [V, W], X \rangle$$

The terms with $\langle X, W \rangle = \langle X, V \rangle = 0$ drop, as do $\langle [X, V], W \rangle = 0$ (since $[X, V] = 0$) and $\langle [X, W], V \rangle = 0$ (since $[X, W] = 0$); and $\langle [V, W], X \rangle = 0$ because $[V, W]$ is vertical. There remains $X \langle V, W \rangle$. By the fiber pairing $\langle V, W \rangle = f^2 g_N(V, W) \circ \pi_N$, and $g_N(V, W) \circ \pi_N$ is constant along the base direction while f^2 is a function of the base, so

$$X \langle V, W \rangle = (X f^2) g_N(V, W) \circ \pi_N = 2f (X f) g_N(V, W) \circ \pi_N = \frac{2 X f}{f} \langle V, W \rangle$$

Comparing, $2 \langle \nabla_X V, W \rangle = \frac{2 X f}{f} \langle V, W \rangle$ for every vertical W , and since $\nabla_X V$ is vertical this forces $\nabla_X V = \frac{X f}{f} V$. Because X and V are basic and $[X, V] = 0$, the connection is symmetric there: $\nabla_V X - \nabla_X V = [V, X] = 0$, so $\nabla_V X = \nabla_X V$, which is claim (2).

3. Take $A = V$, $B = W$ vertical. Test first against a vertical $C = U$. The bracket $[V, U]$, $[W, U]$ and $[V, W]$ are all vertical, and the pairings $\langle V, W \rangle$, $\langle V, U \rangle$, $\langle W, U \rangle$ are $f^2 g_N(-, -) \circ \pi_N$. Because f^2 is constant along fibers, a vertical field differentiates it to zero, so for the three derivative terms

$$V \langle W, U \rangle + W \langle V, U \rangle - U \langle V, W \rangle = f^2 (V g_N(W, U) + W g_N(V, U) - U g_N(V, W)) \circ \pi_N$$

and, combined with the three vertical bracket terms, the whole right side is f^2 times the fiber Koszul formula for $2 g_N(\nabla_V^N W, U)$. Thus

$$2 \langle \nabla_V W, U \rangle = f^2 \cdot 2 g_N(\nabla_V^N W, U) \circ \pi_N = 2 \langle \nabla_V^N W, U \rangle$$

the last equality by the vertical pairing again. So the vertical part of $\nabla_V W$ is $\nabla_V^N W$. Now test against a horizontal $C = X$. The Koszul terms are

$$V \langle W, X \rangle + W \langle V, X \rangle - X \langle V, W \rangle + \langle [V, W], X \rangle - \langle [V, X], W \rangle - \langle [W, X], V \rangle$$

The pairings $\langle W, X \rangle = \langle V, X \rangle = 0$ kill the first two terms; $[V, W]$ is vertical so $\langle [V, W], X \rangle = 0$; and $[V, X] = [W, X] = 0$ kill the last two. There remains $-X \langle V, W \rangle$, and by the computation of part (2),

$$-X \langle V, W \rangle = -\frac{2 X f}{f} \langle V, W \rangle = -2 (X f) f g_N(V, W) \circ \pi_N$$

Writing $X f = \langle \text{grad } f, X \rangle_{g_M} = \langle \text{grad } f, X \rangle$ (the base and warped pairings agree on horizontal vectors), this is $-2 g_N(V, W) f \langle \text{grad } f, X \rangle$. Hence $2 \langle \nabla_V W, X \rangle = -2 g_N(V, W) f \langle \text{grad } f, X \rangle$ for all horizontal X , so the horizontal part of $\nabla_V W$ is $-g_N(V, W) f \text{grad } f$. Using $g(V, W) = f^2 g_N(V, W)$ this horizontal part is $-\frac{g(V, W)}{f} \text{grad } f$, and combining with the vertical part gives

$$\nabla_V W = \nabla_V^N W - \frac{g(V, W)}{f} \text{grad } f$$

which is claim (3), as we sought to show.

The three formulas say something geometric about how the fiber sits inside the warped product. The base slices are totally geodesic: by (1) the connection of $M \times_f N$ restricted to horizontal fields is exactly the base connection, so a geodesic of the base, lifted into a slice, remains a geodesic of the whole. The fibers, by contrast, are generally not totally geodesic; the horizontal correction $-\frac{g(V,W)}{f} \text{grad } f$ in (3) is the second fundamental form of the fiber, and it vanishes exactly where $\text{grad } f = 0$, that is where the warping is momentarily stationary. Formula (2) is the one that will drive the curvature: it records that transporting a vertical vector in a horizontal direction rescales it by the logarithmic derivative Xf/f of the warping, the infinitesimal version of the fact that fibers over nearby base points are rescaled copies of one another.

With the connection in hand the curvature is a substitution. The next theorem is the O'Neill formula for the sectional curvatures of a warped product, organized by the three types of tangent 2-plane: a plane spanned by two horizontal vectors, a mixed plane spanned by one horizontal and one vertical vector, and a plane spanned by two vertical vectors.

Theorem 4.6.2: Curvature of a warped product

Let $M \times_f N$ be a warped product with the sign convention of box 4.1.1, and write K , K^M , K^N for the sectional curvatures of $M \times_f N$, of the base, and of the fiber respectively. Let X, Y be g -orthonormal horizontal vectors and V, W be g -orthonormal vertical vectors at a point.

1. (*Base planes.*) The plane $\text{span}(X, Y)$ has

$$K(X, Y) = K^M(X, Y)$$

the base sectional curvature of the corresponding base 2-plane.

2. (*Mixed planes.*) The plane $\text{span}(X, V)$ has

$$K(X, V) = -\frac{\text{Hess}^M f(X, X)}{f}$$

independent of V ; here $\text{Hess}^M f$ is the base Hessian and X is unit.

3. (*Fiber planes.*) The plane $\text{span}(V, W)$ has

$$K(V, W) = \frac{K^N(\bar{V}, \bar{W}) - |\text{grad } f|^2}{f^2}$$

where $\bar{V}, \bar{W}$ are the images of V, W in the fiber and $|\text{grad } f|^2 = g_M(\text{grad } f, \text{grad } f)$ is the squared base gradient; $K^N(\bar{V}, \bar{W})$ is the sectional curvature of the unscaled fiber (N, g_N) .

Consequently the Ricci curvature, with $d = \dim M$ and $m = \dim N$, satisfies, for X, Y horizontal and V, W vertical,

$$\text{Ric}(X, Y) = \text{Ric}^M(X, Y) - \frac{m}{f} \text{Hess}^M f(X, Y), \quad \text{Ric}(X, V) = 0$$

$$\text{Ric}(V, W) = \text{Ric}^N(\bar{V}, \bar{W}) - g(V, W) \left(\frac{\Delta_M f}{f} + (m-1) \frac{|\text{grad } f|^2}{f^2} \right)$$

where $\text{Ric}^M, \text{Ric}^N$ are the factor Ricci tensors and $\Delta_M f = \text{tr}_{g_M} \text{Hess}^M f$. In particular, for a one-dimensional base interval I with the metric $g = dr^2 + f(r)^2 g_N$, a plane containing the radial direction ∂_r has sectional curvature

$$K = -\frac{f''}{f}$$

and a plane tangent to the fiber has $K = \frac{K^N - (f')^2}{f^2}$.

Proof:

It suffices to compute the $(0, 4)$ curvature $\text{Riem}(A, B, B, A) = \langle R(A, B)B, A \rangle$ of definition 4.1.4 on basic fields whose values at the point are the given orthonormal vectors, since for a g -orthonormal pair the sectional curvature of definition 4.3.1 is $K(A, B) = \text{Riem}(A, B, B, A)$, the denominator $|A|^2 |B|^2 - \langle A, B \rangle^2$ being 1. Extend X, Y to basic horizontal fields and V, W to basic vertical fields; because curvature is tensorial (proposition 4.1.3), the value at the point depends only on the values there. Throughout, R is the curvature endomorphism of definition 4.1.2, and the connection formulas are those of lemma 4.6.1. It is convenient to abbreviate the logarithmic derivative Xf/f and to record that $\text{grad } f$ is horizontal, so $g(\text{grad } f, V) = 0$ for vertical V .

Step 1: base planes. Let X, Y be basic horizontal. By lemma 4.6.1(1) each of $\nabla_X Y$, $\nabla_Y X$ is horizontal and equals the base covariant derivative, and $[X, Y]$ is a basic horizontal field, so $\nabla_{[X, Y]} Y$ is again the base covariant derivative by (1). Every term in $R(X, Y)Y = \nabla_X \nabla_Y Y - \nabla_Y \nabla_X Y - \nabla_{[X, Y]} Y$ is thus computed entirely within the horizontal fields by the base connection, and reproduces the base curvature endomorphism $R^M(X, Y)Y$ lifted horizontally. Pairing with the horizontal X and using $\langle -, - \rangle = g_M(-, -) \circ \pi_M$ on horizontal fields,

$$\text{Riem}(X, Y, Y, X) = \langle R(X, Y)Y, X \rangle = g_M(R^M(X, Y)Y, X) \circ \pi_M = \text{Riem}^M(X, Y, Y, X)$$

For g -orthonormal X, Y (which are then g_M -orthonormal, the two metrics agreeing on horizontal vectors) this is $K(X, Y) = K^M(X, Y)$, establishing (1).

Step 2: mixed planes. Let X be basic horizontal and V basic vertical, with $[X, V] = 0$. Compute $R(X, V)V = \nabla_X \nabla_V V - \nabla_V \nabla_X V$ (the bracket term drops). For the first term, by lemma 4.6.1(3),

$$\nabla_V V = \nabla_V^N V - \frac{g(V, V)}{f} \text{grad } f$$

Differentiate horizontally. The vertical part $\nabla_V^N V$ is a basic vertical field, so by (2) $\nabla_X(\nabla_V^N V) = \frac{Xf}{f} \nabla_V^N V$, which is vertical. The horizontal part is $-\frac{g(V, V)}{f} \text{grad } f$; since V is basic, $g(V, V) = f^2 g_N(V, V)$ with $g_N(V, V)$ constant along the base, so $\frac{g(V, V)}{f} = f g_N(V, V)$ and

$$\nabla_X \left(-\frac{g(V, V)}{f} \text{grad } f \right) = -g_N(V, V) \nabla_X (f \text{grad } f) = -g_N(V, V) ((Xf) \text{grad } f + f \nabla_X^M \text{grad } f)$$

using that on horizontal fields ∇ is the base connection (part (1)) and that $\text{grad } f$ is a basic horizontal field. Only the horizontal component of $R(X, V)V$ contributes to $\langle R(X, V)V, X \rangle$, so the vertical pieces may be discarded from here on. The first term of $R(X, V)V$ contributes the horizontal part

$$(\nabla_X \nabla_V V)^h = -g_N(V, V) ((Xf) \text{grad } f + f \nabla_X^M \text{grad } f)$$

For the second term, $\nabla_X V = \frac{Xf}{f} V$ by lemma 4.6.1(2), a vertical field, so applying (3),

$$\nabla_V \left(\frac{Xf}{f} V \right) = \frac{Xf}{f} \nabla_V V = \frac{Xf}{f} \left(\nabla_V^N V - \frac{g(V, V)}{f} \text{grad } f \right)$$

the coefficient Xf/f being a function of the base and hence constant along the fiber, so $V(Xf/f) = 0$. Its horizontal part is $-\frac{Xf}{f} \cdot \frac{g(V, V)}{f} \text{grad } f = -(Xf) g_N(V, V) \text{grad } f$. Subtract-

ing,

$$(R(X, V)V)^h = -g_N(V, V)((Xf) \operatorname{grad} f + f \nabla_X^M \operatorname{grad} f) + (Xf) g_N(V, V) \operatorname{grad} f = -g_N(V, V) f \nabla_X^M \operatorname{grad} f$$

the two $(Xf) \operatorname{grad} f$ terms cancelling. Pairing with the horizontal X , and using $\langle \nabla_X^M \operatorname{grad} f, X \rangle = \operatorname{Hess}^M f(X, X)$ and $g(V, V) = f^2 g_N(V, V)$,

$$\operatorname{Riem}(X, V, V, X) = \langle R(X, V)V, X \rangle = -g_N(V, V) f \operatorname{Hess}^M f(X, X) = -\frac{g(V, V)}{f} \operatorname{Hess}^M f(X, X)$$

For g -orthonormal X, V this is $g(V, V) = 1$ and $|X| = 1$, giving $K(X, V) = -\frac{\operatorname{Hess}^M f(X, X)}{f}$, independent of the choice of unit vertical V , which is (2).

Step 3: fiber planes. Let V, W be basic vertical with $[V, W]$ vertical (the vertical lift of the fiber bracket). Compute

$$R(V, W)W = \nabla_V \nabla_W W - \nabla_W \nabla_V W - \nabla_{[V, W]} W$$

and take the vertical part, which is what pairs with the vertical V . By lemma 4.6.1(3), $\nabla_W W = \nabla_W^N W - \frac{g(W, W)}{f} \operatorname{grad} f$. Apply ∇_V . The vertical field $\nabla_W^N W$ differentiates by

(3) to $\nabla_V \nabla_W^N W = \nabla_V^N \nabla_W^N W - \frac{g(V, \nabla_W^N W)}{f} \operatorname{grad} f$, whose vertical part is $\nabla_V^N \nabla_W^N W$. The horizontal field $-\frac{g(W, W)}{f} \operatorname{grad} f = -g_N(W, W) f \operatorname{grad} f$ differentiates by (2), since $\operatorname{grad} f$ is basic horizontal and $g_N(W, W)$ is constant along the fiber:

$$\nabla_V(-g_N(W, W) f \operatorname{grad} f) = -g_N(W, W) f \nabla_V(\operatorname{grad} f) = -g_N(W, W) f \cdot \frac{(\operatorname{grad} f)f}{f} V = -g_N(W, W) |\operatorname{grad} f|^2 V$$

where $(\operatorname{grad} f)f = \langle \operatorname{grad} f, \operatorname{grad} f \rangle = |\operatorname{grad} f|^2$ and the factor of f cancels the $1/f$ from (2). This term is already vertical. Hence

$$(\nabla_V \nabla_W W)^v = \nabla_V^N \nabla_W^N W - g_N(W, W) |\operatorname{grad} f|^2 V$$

By the same computation with V, W exchanged in the appropriate slots,

$$(\nabla_W \nabla_V W)^v = \nabla_W^N \nabla_V^N W - g_N(V, W) |\operatorname{grad} f|^2 W$$

and the bracket term, $[V, W]$ being vertical, gives by (3) the vertical part $\nabla_{[V, W]} W = \nabla_{[V, W]}^N W - \frac{g([V, W], W)}{f} \operatorname{grad} f$, whose vertical part is $\nabla_{[V, W]}^N W$. Assembling the vertical part of $R(V, W)W$,

$$(R(V, W)W)^v = \underbrace{\nabla_V^N \nabla_W^N W - \nabla_W^N \nabla_V^N W - \nabla_{[V, W]}^N W}_{= R^N(V, W)W} - |\operatorname{grad} f|^2 (g_N(W, W)V - g_N(V, W)W)$$

The underbraced part is the fiber curvature endomorphism $R^N(\bar{V}, \bar{W})\bar{W}$, lifted vertically. Pair with V . The metric pairing of vertical fields is $\langle -, - \rangle = f^2 g_N(-, -)$, so

$$\langle R^N(V, W)W, V \rangle = f^2 g_N(R^N(\bar{V}, \bar{W})\bar{W}, \bar{V}) = f^2 \operatorname{Riem}^N(\bar{V}, \bar{W}, \bar{W}, \bar{V})$$

and

$$\left\langle -|\operatorname{grad} f|^2 (g_N(W, W)V - g_N(V, W)W), V \right\rangle = -|\operatorname{grad} f|^2 f^2 (g_N(W, W)g_N(V, V) - g_N(V, W)^2)$$

Therefore

$$\operatorname{Riem}(V, W, W, V) = f^2 \operatorname{Riem}^N(\bar{V}, \bar{W}, \bar{W}, \bar{V}) - |\operatorname{grad} f|^2 f^2 (g_N(V, V)g_N(W, W) - g_N(V, W)^2)$$

Now take V, W to be g -orthonormal. Then $f^2 g_N(V, V) = f^2 g_N(W, W) = 1$ and $g_N(V, W) = 0$, so $g_N(V, V) = g_N(W, W) = 1/f^2$ and the Gram determinant $g_N(V, V)g_N(W, W) - g_N(V, W)^2 = 1/f^4$. The fiber vectors $\bar{V}, \bar{W}$ have g_N -Gram determinant $1/f^4$ as well, so

$$\operatorname{Riem}^N(\bar{V}, \bar{W}, \bar{W}, \bar{V}) = K^N(\bar{V}, \bar{W}) \cdot \frac{1}{f^4}$$

by the definition of fiber sectional curvature (definition 4.3.1). Substituting,

$$K(V, W) = \operatorname{Riem}(V, W, W, V) = f^2 \cdot \frac{K^N(\bar{V}, \bar{W})}{f^4} - |\operatorname{grad} f|^2 f^2 \cdot \frac{1}{f^4} = \frac{K^N(\bar{V}, \bar{W}) - |\operatorname{grad} f|^2}{f^2}$$

which is (3).

Step 4: the Ricci curvature. By definition $\operatorname{Ric}(A, A) = \sum_i \langle R(e_i, A)A, e_i \rangle$ over a g -orthonormal frame $\{e_i\}$ (definition 4.4.1), and it is convenient to take a frame adapted to the splitting: d horizontal unit fields $X_1, \dots, X_d$ forming a g_M -orthonormal base frame, and m vertical unit fields $E_1, \dots, E_m$ forming a g -orthonormal fiber frame, so that $g(E_\alpha, E_\beta) = \delta_{\alpha\beta}$ and the images $\bar{E}_\alpha$ are g_N -orthonormal up to the scaling $g_N(\bar{E}_\alpha, \bar{E}_\beta) = \delta_{\alpha\beta}/f^2$. For X horizontal unit,

$$\operatorname{Ric}(X, X) = \sum_{i=1}^d \operatorname{Riem}(X_i, X, X, X_i) + \sum_{\alpha=1}^m \operatorname{Riem}(E_\alpha, X, X, E_\alpha)$$

The horizontal terms sum by Step 1 to the base Ricci curvature $\operatorname{Ric}^M(X, X)$. Each mixed term is, by the (0, 4) identity of Step 2 (with the roles of the two arguments arranged so that the horizontal X is the differentiated direction), $\operatorname{Riem}(E_\alpha, X, X, E_\alpha) = -\frac{g(E_\alpha, E_\alpha)}{f} \operatorname{Hess}^M f(X, X) = -\frac{1}{f} \operatorname{Hess}^M f(X, X)$, using the symmetry $\operatorname{Riem}(E_\alpha, X, X, E_\alpha) = \operatorname{Riem}(X, E_\alpha, E_\alpha, X)$ of proposition 4.2.1. There are m such terms, so

$$\operatorname{Ric}(X, X) = \operatorname{Ric}^M(X, X) - \frac{m}{f} \operatorname{Hess}^M f(X, X)$$

and polarization in X gives the stated horizontal formula for $\operatorname{Ric}(X, Y)$. The mixed component $\operatorname{Ric}(X, V)$ vanishes, but this needs the curvature endomorphisms $R(e_i, X)V$ that Steps 1–3 did not compute, so it is verified directly. Take X horizontal and V vertical, both basic and unit, and evaluate

$$\operatorname{Ric}(X, V) = \sum_{i=1}^d \operatorname{Riem}(X_i, X, V, X_i) + \sum_{\alpha=1}^m \operatorname{Riem}(E_\alpha, X, V, E_\alpha)$$

where $\operatorname{Riem}(e_i, X, V, e_i) = \langle R(e_i, X)V, e_i \rangle$, over the same adapted frame. Consider first a horizontal frame vector X_j . Since X_j and X are basic horizontal and V basic vertical, and

$\nabla_A V = \frac{Af}{f}V$ for horizontal A by lemma 4.6.1(2), each covariant derivative appearing in $R(X_j, X)V = \nabla_{X_j}\nabla_X V - \nabla_X\nabla_{X_j}V - \nabla_{[X_j, X]}V$ is vertical: indeed $\nabla_X V = \frac{Xf}{f}V$ is vertical, and differentiating a vertical field in the horizontal direction X_j keeps it vertical by (2), as does the bracket term with the horizontal $[X_j, X]$. Hence $R(X_j, X)V$ is vertical and pairs to zero with the horizontal X_j :

$$\text{Riem}(X_j, X, V, X_j) = \langle R(X_j, X)V, X_j \rangle = 0$$

for every j , so the horizontal sum vanishes. Now consider a vertical frame vector E_α . Here $[E_\alpha, X] = 0$, so $R(E_\alpha, X)V = \nabla_{E_\alpha}\nabla_X V - \nabla_X\nabla_{E_\alpha}V$, and only the vertical part of this contributes to the pairing with the vertical E_α . For the first term, $\nabla_X V = \frac{Xf}{f}V$ by (2), and Xf/f is a function of the base and hence constant along the fiber, so $E_\alpha(Xf/f) = 0$ and

$$\nabla_{E_\alpha}\nabla_X V = \frac{Xf}{f}\nabla_{E_\alpha}V = \frac{Xf}{f}\left(\nabla_{E_\alpha}^N V - \frac{g(E_\alpha, V)}{f}\text{grad } f\right)$$

by lemma 4.6.1(3), whose vertical part is $\frac{Xf}{f}\nabla_{E_\alpha}^N V$. For the second term, applying (3) to $\nabla_{E_\alpha}V = \nabla_{E_\alpha}^N V - \frac{g(E_\alpha, V)}{f}\text{grad } f$ and then ∇_X : the vertical field $\nabla_{E_\alpha}^N V$ is basic, so by (2) $\nabla_X(\nabla_{E_\alpha}^N V) = \frac{Xf}{f}\nabla_{E_\alpha}^N V$, which is vertical, while the horizontal field $-\frac{g(E_\alpha, V)}{f}\text{grad } f = -g_N(E_\alpha, V)f\text{grad } f$ differentiates by part (1) to $-g_N(E_\alpha, V)((Xf)\text{grad } f + f\nabla_X^M \text{grad } f)$, which is horizontal. So the vertical part of $\nabla_X\nabla_{E_\alpha}V$ is again $\frac{Xf}{f}\nabla_{E_\alpha}^N V$. Subtracting, the vertical part of $R(E_\alpha, X)V$ is

$$\frac{Xf}{f}\nabla_{E_\alpha}^N V - \frac{Xf}{f}\nabla_{E_\alpha}^N V = 0$$

so $R(E_\alpha, X)V$ has no vertical component and pairs to zero with E_α :

$$\text{Riem}(E_\alpha, X, V, E_\alpha) = \langle R(E_\alpha, X)V, E_\alpha \rangle = 0$$

for every α . Both sums vanish, so $\text{Ric}(X, V) = 0$, and by bilinearity the mixed Ricci component vanishes on all horizontal-vertical pairs. For V vertical unit,

$$\text{Ric}(V, V) = \sum_{i=1}^d \text{Riem}(X_i, V, V, X_i) + \sum_{\alpha=1}^m \text{Riem}(E_\alpha, V, V, E_\alpha)$$

Each horizontal term is a mixed plane, equal by Step 2 to $-\frac{1}{f}\text{Hess}^M f(X_i, X_i)$, and summing over the g_M -orthonormal base frame gives $-\frac{1}{f}\text{tr}_{g_M} \text{Hess}^M f = -\frac{\Delta_M f}{f}$. Each fiber term with α such that $E_\alpha \neq V$ is a fiber plane, equal by Step 3 to $\frac{K^N(\bar{E}_\alpha, \bar{V}) - |\text{grad } f|^2}{f^2}$ (the term $E_\alpha = V$ contributes nothing, being $\text{Riem}(V, V, V, V) = 0$). Summing over the $m - 1$ fiber directions

orthogonal to V , the sectional curvatures K^N assemble into the fiber Ricci curvature $\text{Ric}^N(\bar{V}, \bar{V})$ evaluated on the unit fiber vector $\bar{V}$ (which for the scaled fiber frame equals the intrinsic $\text{Ric}^N(\bar{V}, \bar{V})$ since Ricci is a sum of sectional curvatures over an orthonormal complement), and the $-|\text{grad } f|^2/f^2$ contributes $m-1$ times. Collecting,

$$\text{Ric}(V, V) = \text{Ric}^N(\bar{V}, \bar{V}) - \frac{\Delta_M f}{f} - (m-1) \frac{|\text{grad } f|^2}{f^2}$$

and since $g(V, V) = 1$ this is the stated formula, polarizing to general vertical V, W .

Step 5: the one-dimensional base. When $M = I$ is an interval with $g_M = dr^2$, the unit horizontal field is ∂_r , the base is flat so $K^M = 0$, and $\text{grad } f = f' \partial_r$ with $|\text{grad } f|^2 = (f')^2$. The base Hessian of a function on a 1-manifold with the flat metric is $\text{Hess}^M f(\partial_r, \partial_r) = \partial_r(\partial_r f) - (\nabla_{\partial_r}^M \partial_r) f = f''$, since $\nabla_{\partial_r}^M \partial_r = 0$ for the flat metric dr^2 . A plane containing ∂_r is a mixed plane, so by (2) its sectional curvature is

$$K = -\frac{\text{Hess}^M f(\partial_r, \partial_r)}{f} = -\frac{f''}{f}$$

and a plane tangent to the fiber is a fiber plane, so by (3) its sectional curvature is $\frac{K^N - (f')^2}{f^2}$, completing the proof.

The theorem separates the curvature of a warped product into three mechanisms. Base planes see only the base: warping the fiber does not change how the base is curved among itself, consistent with the base slices being totally geodesic. Mixed planes see only the second derivative of the warping, through the base Hessian, and are blind to the fiber: bending the fiber sizes creates curvature in the directions that couple base and fiber, and the sign is opposite to that of $\text{Hess}^M f$, so a warping function that is concave along a base geodesic produces positive mixed curvature there. Fiber planes see a competition between the intrinsic fiber curvature K^N and the steepness $|\text{grad } f|^2$ of the warping: even a flat fiber acquires nonzero curvature once f has nonzero gradient, and a fiber of positive curvature can be flattened, or driven negative, by a sufficiently steep warping. The one-dimensional specialization $K = -f''/f$ is the case that recovers the models, and it is the reason the constant curvature condition is the linear equation $f'' + \kappa f = 0$ announced in example 1.6.5.

Example 4.6.3: The model spaces from the warped-product formula

The three constant-curvature models were presented in example 1.6.4 and example 1.6.5 as warped products $(0, \cdot) \times_f S^{n-1}$ over a half-line, with unit-sphere fibers and metric $g = dr^2 + f(r)^2 g_{S^{n-1}}$, distinguished by their warping functions $f(r) = r, \sin r, \sinh r$. Their sectional curvatures follow from the one-dimensional case of theorem 4.6.2, and reproduce the values computed independently in example 4.3.4 and example 4.3.5. Here $\dim M = 1$, so the only plane types are the radial (mixed) planes containing ∂_r and the tangential (fiber) planes; there are no base planes.

Radial planes. By the one-dimensional formula, a plane containing ∂_r has $K = -f''/f$. For the three warpings,

$$\begin{aligned} f(r) = r: \quad -\frac{f''}{f} &= -\frac{0}{r} = 0, & f(r) = \sin r: \quad -\frac{f''}{f} &= -\frac{-\sin r}{\sin r} = 1, \\ f(r) = \sinh r: \quad -\frac{f''}{f} &= -\frac{\sinh r}{\sinh r} = -1 \end{aligned}$$

so the radial sectional curvatures are 0, +1, and -1 for the Euclidean, spherical, and hyperbolic models respectively.

Tangential planes. The fiber is the unit sphere S^{n-1} , which by example 4.3.4 has constant sectional curvature $K^N = 1$. The fiber formula gives $K = \frac{K^N - (f')^2}{f^2} = \frac{1 - (f')^2}{f^2}$. For the three warpings, using $f' = 1, \cos r, \cosh r$ and the identities $1 - \cos^2 r = \sin^2 r$, $1 - \cosh^2 r = -\sinh^2 r$,

$$f(r) = r: \quad \frac{1 - 1}{r^2} = 0, \quad f(r) = \sin r: \quad \frac{1 - \cos^2 r}{\sin^2 r} = \frac{\sin^2 r}{\sin^2 r} = 1,$$

$$f(r) = \sinh r: \quad \frac{1 - \cosh^2 r}{\sinh^2 r} = \frac{-\sinh^2 r}{\sinh^2 r} = -1$$

so the tangential sectional curvatures are again 0, +1, and -1 . In each model the radial and tangential curvatures coincide, so the sectional curvature is constant over all 2-planes at every point: $K \equiv 0$ for $\mathbb{R}^n = (0, \infty) \times_r S^{n-1}$, $K \equiv +1$ for the unit sphere $S^n = (0, \pi) \times_{\sin r} S^{n-1}$, and $K \equiv -1$ for hyperbolic space $H^n = (0, \infty) \times_{\sinh r} S^{n-1}$. This matches the sign convention of box 4.1.1, under which S^n has $K = +1$ and H^n has $K = -1$, and reproduces example 4.3.4 and example 4.3.5 from a single formula.

The three warpings are exactly the solutions of $f'' + \kappa f = 0$ with $f(0) = 0$ and $f'(0) = 1$ for $\kappa = 0, 1, -1$, as anticipated in example 1.6.5. The radial curvature $-f''/f = \kappa$ is constant by construction; that the tangential curvature is also κ is the content of the identity $(1 - (f')^2)/f^2 = \kappa$, which follows by differentiating the first integral $(f')^2 + \kappa f^2 = 1$ of the equation $f'' + \kappa f = 0$ subject to $f(0) = 0$, $f'(0) = 1$. Thus a warped product $(0, \cdot) \times_f S^{n-1}$ has constant sectional curvature κ precisely when f solves $f'' + \kappa f = 0$, and the sphere, Euclidean space, and hyperbolic space are the three solutions capped smoothly at $r = 0$, where the collapsing fibers close up the missing center point.

For a sphere of radius ρ the warping is $f(r) = \rho \sin(r/\rho)$ on $(0, \pi\rho)$, giving metric $dr^2 + \rho^2 \sin^2(r/\rho) g_{S^{n-1}}$; then $-f''/f = 1/\rho^2$ and $(1 - (f')^2)/f^2 = (1 - \cos^2(r/\rho))/(\rho^2 \sin^2(r/\rho)) = 1/\rho^2$, so $K \equiv 1/\rho^2$, recovering the value $K(S_\rho^n) = 1/\rho^2$ of example 4.3.4 for the sphere of radius ρ .

5

Jacobi Fields and Conjugate Points

The first variation of chapter 3 identified geodesics as the critical points of the energy, and the curvature tensor of chapter 4 measured how the manifold bends. This chapter joins the two. Differentiating the energy a second time along a geodesic produces a quadratic form in which the curvature appears explicitly, and the sign of that curvature term decides whether a geodesic is a genuine minimum or merely a critical point. Curvature, in other words, controls stability, and the vehicle for the analysis is the Jacobi field.

A Jacobi field is the infinitesimal variation of a geodesic through nearby geodesics; equivalently, it is a solution of a single linear second-order equation, the Jacobi equation, in which the curvature enters as the coefficient. When the curvature is positive the neighboring geodesics are pulled back together and may refocus at a conjugate point, where the exponential map degenerates; when it is nonpositive they spread and never refocus. The chapter develops the second variation, the Jacobi equation and its solutions, the conjugate points and their identification with the singularities of the exponential map, and the index form whose positivity governs minimization. The recurring computation is the constant-curvature model, where the Jacobi equation is $f'' + cf = 0$ and every statement becomes explicit: the round sphere refocuses geodesics at the antipode, and nonpositively curved spaces never do. These results are the analytic core on which the comparison theorems of the next chapter rest.

5.1 The Second Variation of Energy

The first variation of section 3.5 located the geodesics as the critical points of the energy: a smooth curve γ is a geodesic exactly when $\frac{d}{ds}\big|_{s=0} E(\gamma_s) = 0$ for every proper variation. Criticality alone, however, does not distinguish a minimum from a saddle. A straight segment in the plane minimizes energy, but a great-circle arc on the sphere that runs past the antipode is still a geodesic and so still critical, yet it can be shortened, so it is not a minimum. To separate these cases one differentiates the energy a second time and asks whether the resulting quadratic form is positive.

That second derivative is the subject of this section. Differentiating the first-variation formula along a critical geodesic produces a quadratic form in the variation field V , and curvature enters it explicitly. The formula

$$E''(0) = \int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle) dt$$

carries all the geometry of the chapter in its second term. The pairing $\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle$ equals $\text{Riem}(V, \dot{\gamma}, \dot{\gamma}, V)$, which for V orthogonal to $\dot{\gamma}$ is the sectional curvature of the plane spanned by V and $\dot{\gamma}$ times $|V|^2 |\dot{\gamma}|^2$. It enters with a minus sign, so positive curvature drives $E''(0)$ down and can make it negative, destabilizing the geodesic, while nonpositive curvature keeps $E''(0)$ nonnegative and preserves minimization. The remainder of the chapter is a study of this competition between the bending term $|D_t V|^2$, which resists deformation, and the curvature term, which may abet it. Throughout, (M, g) is a Riemannian manifold with Levi-Civita connection ∇ of definition 2.3.2, curvature endomorphism R of definition 4.1.2 in the sign convention of box 4.1.1, and D_t the induced covariant derivative along a curve from definition 2.1.6; geodesics are parametrized by constant speed as in definition 3.1.1, and $[0, b]$ is the parameter interval.

Theorem 5.1.1: Second Variation of Energy

Let $\gamma: [0, b] \rightarrow M$ be a geodesic, and let $\Gamma: (-\varepsilon, \varepsilon) \times [0, b] \rightarrow M$ be a smooth proper variation of γ , with variation field $V(t) = \partial_s \Gamma(0, t)$ satisfying $V(0) = V(b) = 0$. Then the energy $E(\gamma_s)$ of the longitudinal curves is twice differentiable in s at $s = 0$, and

$$E''(0) := \left. \frac{d^2}{ds^2} \right|_{s=0} E(\gamma_s) = \int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle) dt$$

Equivalently, integrating the first term by parts,

$$E''(0) = - \int_0^b \langle D_t^2 V + R(V, \dot{\gamma})\dot{\gamma}, V \rangle dt$$

Proof:

The variation Γ is smooth on the whole rectangle $(-\varepsilon, \varepsilon) \times [0, b]$, so all the derivatives below may be taken under the integral sign. Write $T = \partial_t \Gamma$ and $S = \partial_s \Gamma$ for the coordinate velocity fields along Γ ; at $s = 0$ these restrict to $T = \dot{\gamma}$ and $S = V$ along γ . The energy of the longitudinal curve $\gamma_s = \Gamma(s, \cdot)$ is

$$E(\gamma_s) = \frac{1}{2} \int_0^b \langle T, T \rangle dt$$

Step 1: the first s -derivative. Metric compatibility of ∇ (definition 2.2.4), applied to the s -derivative of the inner product $\langle T, T \rangle$, gives

$$\frac{\partial}{\partial s} \langle T, T \rangle = 2 \langle D_s T, T \rangle, \quad \text{so} \quad E'(s) = \frac{d}{ds} E(\gamma_s) = \int_0^b \langle D_s T, T \rangle dt$$

By the symmetry lemma (lemma 3.5.3), $D_s T = D_s \partial_t \Gamma = D_t \partial_s \Gamma = D_t S$, so

$$E'(s) = \int_0^b \langle D_t S, T \rangle dt$$

Step 2: the second s -derivative. Differentiate $E'(s)$ again in s , using metric compatibility once more:

$$E''(s) = \int_0^b \frac{\partial}{\partial s} \langle D_t S, T \rangle dt = \int_0^b \left(\langle D_s D_t S, T \rangle + \langle D_t S, D_s T \rangle \right) dt$$

The second summand is already in usable form: $D_s T = D_t S$ by the symmetry lemma, so $\langle D_t S, D_s T \rangle = \langle D_t S, D_t S \rangle = |D_t S|^2$. The first summand requires commuting the two covariant derivatives D_s and D_t acting on the field S along Γ . This is exactly what the curvature endomorphism measures. Since ∂_s and ∂_t are coordinate fields on the parameter rectangle, their bracket vanishes, and the definition of R (definition 4.1.2) in the convention of box 4.1.1 gives, for any field W along Γ ,

$$D_s D_t W - D_t D_s W = R(\partial_s \Gamma, \partial_t \Gamma) W = R(S, T) W$$

Applying this with $W = S$,

$$D_s D_t S = D_t D_s S + R(S, T) S$$

Therefore

$$E''(s) = \int_0^b \left(\langle D_t D_s S, T \rangle + \langle R(S, T) S, T \rangle + |D_t S|^2 \right) dt$$

Step 3: evaluation at $s = 0$ and integration by parts. At $s = 0$ one has $S = V$, $T = \dot{\gamma}$, and $D_t S = D_t V$. The first integrand is rewritten by metric compatibility, moving one D_t off of $D_s S$ and onto $T = \dot{\gamma}$:

$$\langle D_t D_s S, T \rangle = \frac{d}{dt} \langle D_s S, T \rangle - \langle D_s S, D_t T \rangle$$

Along the base curve, $D_t T = D_t \dot{\gamma} = 0$ because γ is a geodesic (definition 3.1.1), so the last term vanishes at $s = 0$ and the first integrand becomes the total derivative $\frac{d}{dt} \langle D_s S, \dot{\gamma} \rangle$. Its integral over $[0, b]$ is the boundary difference

$$\int_0^b \frac{d}{dt} \langle D_s S, \dot{\gamma} \rangle dt = \left[\langle D_s S, \dot{\gamma} \rangle \right]_0^b$$

This boundary term vanishes. Indeed, the variation is proper, so $\Gamma(s, 0) = \gamma(0)$ and $\Gamma(s, b) = \gamma(b)$ are constant in s ; hence $\partial_s \Gamma(s, 0) = 0$ and $\partial_s \Gamma(s, b) = 0$ for all s , and the covariant s -derivative $D_s S = D_s \partial_s \Gamma$ is the covariant derivative along the constant s -curves $t = 0$ and $t = b$, which is 0 there. Thus $\langle D_s S, \dot{\gamma} \rangle = 0$ at both endpoints, and the boundary difference is 0. Collecting the surviving terms at $s = 0$,

$$E''(0) = \int_0^b \left(\langle R(V, \dot{\gamma}) V, \dot{\gamma} \rangle + |D_t V|^2 \right) dt$$

Finally, the symmetries of the curvature tensor (proposition 4.2.1) convert the curvature term into the stated form. By the definition of the $(0, 4)$ -tensor (definition 4.1.4) and antisymmetry in the second pair,

$$\langle R(V, \dot{\gamma}) V, \dot{\gamma} \rangle = \text{Riem}(V, \dot{\gamma}, V, \dot{\gamma}) = -\text{Riem}(V, \dot{\gamma}, \dot{\gamma}, V) = -\langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle$$

Substituting,

$$E''(0) = \int_0^b \left(|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt,$$

which is the asserted formula.

Step 4: the integrated-by-parts form. The field V is smooth on $[0, b]$ with $V(0) = V(b) = 0$. Metric compatibility gives $\frac{d}{dt} \langle D_t V, V \rangle = \langle D_t^2 V, V \rangle + |D_t V|^2$, so

$$\int_0^b |D_t V|^2 dt = \left[\langle D_t V, V \rangle \right]_0^b - \int_0^b \langle D_t^2 V, V \rangle dt = - \int_0^b \langle D_t^2 V, V \rangle dt,$$

the boundary term vanishing because $V(0) = V(b) = 0$. Substituting into the first form,

$$E''(0) = - \int_0^b \langle D_t^2 V, V \rangle dt - \int_0^b \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle dt = - \int_0^b \langle D_t^2 V + R(V, \dot{\gamma}) \dot{\gamma}, V \rangle dt,$$

completing the proof.

The two forms of the formula read off two different aspects of the same quadratic form. The first, $\int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle) dt$, displays the tension that governs the whole chapter: the term $|D_t V|^2$ is a nonnegative bending cost, always working to raise $E''(0)$, while the curvature term enters with a minus sign, so a plane of positive sectional curvature lowers $E''(0)$ and a plane of negative sectional curvature raises it. Writing $V = V^\perp + V^\top$ into components normal and tangent to $\dot{\gamma}$, and using $\langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle = \text{Riem}(V, \dot{\gamma}, \dot{\gamma}, V)$, the curvature term sees only the normal part: with $V^\perp$ orthogonal to $\dot{\gamma}$,

$$\langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle = \text{Riem}(V^\perp, \dot{\gamma}, \dot{\gamma}, V^\perp) = K(V^\perp, \dot{\gamma}) (|V^\perp|^2 |\dot{\gamma}|^2 - \langle V^\perp, \dot{\gamma} \rangle^2) = K(V^\perp, \dot{\gamma}) |V^\perp|^2 |\dot{\gamma}|^2,$$

by definition 4.3.1 together with the tangential vanishing $\text{Riem}(\dot{\gamma}, \dot{\gamma}, \dot{\gamma}, V) = 0$ from the antisymmetry of proposition 4.2.1. Positive sectional curvature in the plane of $V^\perp$ and $\dot{\gamma}$ therefore subtracts a positive quantity from $E''(0)$; enough of it, spread over a long enough geodesic, can outweigh the bending term and make $E''(0)$ negative. The second form, $-\int_0^b \langle D_t^2 V + R(V, \dot{\gamma}) \dot{\gamma}, V \rangle dt$, exposes the operator $V \mapsto D_t^2 V + R(V, \dot{\gamma}) \dot{\gamma}$ acting on the variation field; the fields on which this operator vanishes are the Jacobi fields of section 5.2, the directions in which γ can be deformed through geodesics at no second-order cost.

The bilinear form obtained by polarizing $E''(0)$ is called the *index form* of γ ; on piecewise-smooth fields V, W along γ vanishing at the endpoints it is

$$I(V, W) = \int_0^b (\langle D_t V, D_t W \rangle - \langle R(V, \dot{\gamma}) \dot{\gamma}, W \rangle) dt,$$

so that $I(V, V) = E''(0)$. Its systematic study, including the index lemma and the connection between its signature and the conjugate points of γ , is carried out in section 5.4; the present section records only the formula and its first consequence for minimization.

The immediate consequence is a one-sided sign condition. If a geodesic actually minimizes energy, then perturbing it in any admissible direction cannot decrease the energy, and this constrains the sign of the second variation.

Corollary 5.1.2: Minimizers Have Nonnegative Second Variation

Let $\gamma: [0, b] \rightarrow M$ be a geodesic that locally minimizes energy among fixed-endpoint curves, in the sense that $E(\gamma) \leq E(c)$ for every piecewise-smooth curve $c: [0, b] \rightarrow M$ with $c(0) = \gamma(0)$, $c(b) = \gamma(b)$ sufficiently close to γ . Then $E''(0) \geq 0$ for every smooth proper variation of γ ; equivalently, $I(V, V) \geq 0$ for every smooth field V along γ with $V(0) = V(b) = 0$.

Proof:

Let Γ be a smooth proper variation of γ with variation field V , and set $E(s) = E(\gamma_s)$. Each longitudinal curve γ_s has the same endpoints as γ , and $\gamma_s \rightarrow \gamma$ uniformly as $s \rightarrow 0$ (indeed in C^1 , by smoothness of Γ on the compact rectangle), so for $|s|$ small γ_s lies in the range where the minimizing hypothesis applies. Hence $E(s) \geq E(0)$ for all $|s|$ small, so $s = 0$ is a local minimum of the smooth function $E(s)$. Since γ is a geodesic, $E'(0) = 0$ by corollary 3.5.8, and the second-derivative test for a smooth function with a local minimum at an interior critical point gives $E''(0) \geq 0$. By theorem 5.1.1 this is exactly $\int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle) dt = I(V, V) \geq 0$, as claimed.

The contrapositive is the tool for detecting non-minimizing geodesics: if some proper variation has $E''(0) < 0$, then γ does not minimize energy, because the energy strictly decreases in that direction. Producing such a variation is a matter of finding a field V vanishing at the endpoints for which the negative curvature contribution overwhelms the bending term. On a positively curved space a geodesic long enough for the curvature to accumulate will admit one, and this is what fails minimization past the antipode on the sphere.

Example 5.1.3: A long great-circle arc does not minimize

On the round sphere S_r^n of radius r , which has constant sectional curvature $c = 1/r^2$ by example 4.3.4, take a unit-speed great-circle geodesic $\gamma: [0, b] \rightarrow S_r^n$ of length b , one of the great circles identified in example 3.1.4. The claim is that if $b > \pi r$, so that γ runs strictly past the antipode of its starting point, then γ does not minimize energy, and a proper variation with $E''(0) < 0$ can be written down.

Choose a parallel unit vector field $E(t)$ along γ with $E(t) \perp \dot{\gamma}(t)$ for all t ; such a field exists because parallel transport preserves both length and orthogonality to the parallel field $\dot{\gamma}$, so it suffices to parallel-transport one unit vector orthogonal to $\dot{\gamma}(0)$, which is available since $n \geq 2$. Define the variation field

$$V(t) = \sin\left(\frac{\pi t}{b}\right) E(t),$$

which vanishes at $t = 0$ and $t = b$, hence is the field of a proper variation, for instance $\Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ as in example 3.5.2. Since E is parallel, $D_t E = 0$, so

$$D_t V = \frac{\pi}{b} \cos\left(\frac{\pi t}{b}\right) E(t), \quad |D_t V|^2 = \frac{\pi^2}{b^2} \cos^2\left(\frac{\pi t}{b}\right)$$

using $|E| \equiv 1$. For the curvature term, $V(t) = \sin(\pi t/b) E(t)$ is orthogonal to $\dot{\gamma}$ and γ has unit speed, so by the constant-curvature reduction above,

$$\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = c |V|^2 = \frac{1}{r^2} \sin^2\left(\frac{\pi t}{b}\right)$$

Feeding both into the second-variation formula of theorem 5.1.1 and using $\int_0^b \cos^2(\pi t/b) dt = \int_0^b \sin^2(\pi t/b) dt = b/2$,

$$E''(0) = \int_0^b \left(\frac{\pi^2}{b^2} \cos^2 \frac{\pi t}{b} - \frac{1}{r^2} \sin^2 \frac{\pi t}{b} \right) dt = \frac{\pi^2}{b^2} \cdot \frac{b}{2} - \frac{1}{r^2} \cdot \frac{b}{2} = \frac{b}{2} \left(\frac{\pi^2}{b^2} - \frac{1}{r^2} \right) = \frac{\pi^2 r^2 - b^2}{2b r^2}$$

This is negative precisely when $b^2 > \pi^2 r^2$, that is, when $b > \pi r$. Thus a great-circle arc longer than πr has $E''(0) < 0$ for this variation, so by the contrapositive of corollary 5.1.2 it does not minimize energy, and by the length-energy inequality of theorem 3.5.6 it does not minimize length either: the arc can be shortened by sliding it off toward the shorter way around. At the threshold $b = \pi r$, exactly the antipodal distance, this particular variation gives $E''(0) = 0$, the borderline where the great circle stops minimizing; the same value πr reappears in section 5.3 as the first conjugate distance on S_r^n . For $b < \pi r$ the variation gives $E''(0) > 0$, consistent with the fact that a great-circle arc shorter than the antipodal distance is the unique shortest path between its endpoints.

5.2 The Jacobi Equation

The second variation of energy, computed for a geodesic γ and a proper variation with field V , produced the quadratic form

$$E''(0) = \int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle) dt$$

in which the curvature enters through the term $\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = \text{Riem}(V, \dot{\gamma}, \dot{\gamma}, V)$. Whether a geodesic minimizes energy against a given deformation is decided by the sign of this integral, and the direction V that makes it smallest is the direction along which the geodesic is closest to failing to minimize. Setting the associated Euler–Lagrange equation to zero singles out those variation fields, and the equation they satisfy is the object of this section.

There is a second route to the same equation, and it is the one that gives the fields their geometric meaning. A geodesic can be deformed not through arbitrary curves but through other geodesics: a one-parameter family γ_s of geodesics, all issuing from a common construction, sweeps out a variation whose field $J(t) = \partial_s \gamma_s(t)|_{s=0}$ records how neighboring geodesics separate or converge. Differentiating the geodesic equation of the whole family in the transverse direction, and invoking the symmetry lemma and the definition of curvature, turns $D_t \dot{\gamma}_s = 0$ into a linear second-order equation for J . The two derivations meet: the fields that arise from variations through geodesics are exactly the critical fields of the second variation, and both are the solutions of the Jacobi equation. Throughout, (M, g) is a (pseudo-)Riemannian manifold with Levi-Civita connection ∇ (definition 2.3.2), curvature endomorphism R of definition 4.1.2 under the sign convention of box 4.1.1, D_t is the covariant derivative along a curve (definition 2.1.6), and $\gamma: [0, b] \rightarrow M$ is a geodesic (definition 3.1.1). The Einstein summation convention is in force where indices appear.

Definition 5.2.1: Jacobi field

Let $\gamma: [0, b] \rightarrow M$ be a geodesic. A *Jacobi field* along γ is a smooth vector field J along γ satisfying the *Jacobi equation*

$$D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0 \quad (5.1)$$

where R is the curvature endomorphism of definition 4.1.2 and $D_t^2 J = D_t(D_t J)$ is the second covariant derivative of J along γ .

The equation (5.1) is a linear second-order ordinary differential equation for the field J , and the curvature term is what couples it to the geometry of the ambient manifold. Read as an equation of motion, it describes an infinitesimal separation vector J subject to an acceleration $-R(J, \dot{\gamma})\dot{\gamma}$ produced entirely by curvature; the field $D_t^2 J$ is the relative acceleration of two neighboring geodesics, and the curvature dictates its sign. Pairing (5.1) with J and using metric compatibility to write $\langle D_t^2 J, J \rangle = \frac{d}{dt} \langle D_t J, J \rangle - |D_t J|^2$ ties the equation to the second variation of the opening: the curvature coefficient there and the curvature coefficient here are the same object, so a Jacobi field is precisely a critical field of the second-variation integral. The interpretation to keep in mind is the one the next theorem makes exact: J is the velocity, at the base geodesic, of a family of geodesics fanning out from γ , and the Jacobi equation is the linearization of the geodesic equation along that fan.

The variational meaning is established first, because it justifies calling (5.1) the equation of a variation through geodesics and supplies both the fields that appear in the second variation and the geometric picture of geodesic spreading.

Theorem 5.2.2: Jacobi fields are the variation fields of geodesic variations

Let $\gamma: [0, b] \rightarrow M$ be a geodesic.

1. If $\Gamma: (-\varepsilon, \varepsilon) \times [0, b] \rightarrow M$ is a smooth variation of γ (so $\Gamma(0, t) = \gamma(t)$) such that every longitudinal curve $\gamma_s = \Gamma(s, \cdot)$ is a geodesic, then the variation field $J(t) = \partial_s \Gamma(0, t)$ is a Jacobi field along γ .
2. Conversely, every Jacobi field J along γ arises as the variation field of some smooth variation of γ through geodesics.

Proof:

Write $T = \partial_t \Gamma$ and $S = \partial_s \Gamma$ for the two coordinate velocity fields of the variation, and D_t, D_s for the covariant derivatives along the t - and s -parameter curves (definition 2.1.6). At $s = 0$ one has $T = \dot{\gamma}$ and $S = J$.

1. Since each longitudinal curve γ_s is a geodesic, its velocity T is autoparallel along it: $D_t T = 0$ for every s . Differentiating this identity covariantly in the s -direction gives $D_s D_t T = 0$ identically. The plan is to convert $D_s D_t T$ into an expression in $D_t^2 S$ and curvature, evaluated at $s = 0$.

Step 1: commuting the derivatives. The parameter fields ∂_s, ∂_t on the rectangle $(-\varepsilon, \varepsilon) \times [0, b]$ commute, and Γ carries them to $S = \partial_s \Gamma$ and $T = \partial_t \Gamma$. For any smooth vector field W along Γ , the definition of the curvature endomorphism (definition 4.1.2), applied with

the commuting parameter fields so that the bracket term drops, gives

$$D_s D_t W - D_t D_s W = R(S, T)W \quad (5.2)$$

This is the standard passage from the endomorphism $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$ to the covariant derivatives along the two parameter directions of a variation; it holds with the sign of box 4.1.1 because S and T are the images of commuting coordinate fields, so no $\nabla_{[S, T]}$ correction survives.

Step 2: applying the identity. Take $W = T$ in (5.2):

$$D_s D_t T = D_t D_s T + R(S, T)T$$

The left-hand side vanishes because $D_t T = 0$ for every s . By the symmetry lemma (lemma 3.5.3), $D_s T = D_s \partial_t \Gamma = D_t \partial_s \Gamma = D_t S$, so $D_t D_s T = D_t D_t S = D_t^2 S$. Hence

$$0 = D_t^2 S + R(S, T)T$$

identically on the rectangle.

Step 3: evaluation at $s = 0$. Set $s = 0$, where $S = J$ and $T = \dot{\gamma}$:

$$D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$$

This is the Jacobi equation (5.1), so J is a Jacobi field along γ .

2. Let J be a Jacobi field along γ ; a variation of γ through geodesics with variation field J is constructed by exponentiating a suitable family of initial data. Set $p = \gamma(0)$ and $v = \dot{\gamma}(0)$, so $\gamma(t) = \exp_p(tv)$ (definition 3.2.1). Choose a smooth curve $c: (-\varepsilon, \varepsilon) \rightarrow M$ with $c(0) = p$ and $\dot{c}(0) = J(0)$, and let $W(s)$ be a smooth vector field along c with $W(0) = v$ and $D_s W(0) = D_t J(0)$; such a c and W exist because $J(0) \in T_p M$ and $D_t J(0) \in T_p M$ are prescribed vectors, realized for instance by taking c a short curve with the given initial velocity and W any field along it matching the two conditions at $s = 0$. Define

$$\Gamma(s, t) = \exp_{c(s)}(tW(s))$$

for $|s| < \varepsilon$ small enough that $tW(s)$ lies in the domain of $\exp_{c(s)}$ for all $t \in [0, b]$, which is possible by smoothness of the exponential map and compactness of $[0, b]$. For each fixed s the curve $t \mapsto \Gamma(s, t)$ is the geodesic through $c(s)$ with initial velocity $W(s)$, so Γ is a variation of γ through geodesics with $\Gamma(0, t) = \exp_p(tv) = \gamma(t)$. By part (1), its variation field $\tilde{J}(t) = \partial_s \Gamma(0, t)$ is a Jacobi field. Its initial data are

$$\tilde{J}(0) = \partial_s \Gamma(s, 0)|_{s=0} = \dot{c}(0) = J(0)$$

since $\Gamma(s, 0) = \exp_{c(s)}(0) = c(s)$, and, using the symmetry lemma once more,

$$D_t \tilde{J}(0) = D_t \partial_s \Gamma(0, 0) = D_s \partial_t \Gamma(0, 0) = D_s W(0) = D_t J(0)$$

because $\partial_t \Gamma(s, 0) = \frac{d}{dt}|_{t=0} \exp_{c(s)}(tW(s)) = W(s)$. Thus $\tilde{J}$ and J are Jacobi fields with the same value and the same covariant derivative at $t = 0$; by the uniqueness statement of theorem 5.2.3 below they coincide, $\tilde{J} = J$. Hence Γ is a variation through geodesics whose variation field is J , as required.

Both directions are established, completing the proof.

The theorem gives the Jacobi field its name and its picture. A Jacobi field is the derivative of a family of geodesics with respect to the family parameter, so it measures the rate at which two infinitesimally close geodesics drift apart or draw together; the Jacobi equation is what remains of the geodesic equation after this differentiation, the curvature term being the trace left by the noncommutativity of covariant derivatives. The converse half is the more useful for computation: it says that no generality is lost in thinking of a Jacobi field as an actual spread of geodesics, since every solution of (5.1) is realized by one. The construction in part (2) also isolates the two pieces of initial data that determine a Jacobi field, the initial value $J(0)$ (how far the neighboring geodesic starts) and the initial covariant derivative $D_t J(0)$ (how fast its initial velocity turns away), which is the content of the existence and uniqueness theorem taken up next.

Because the Jacobi equation is a linear second-order equation, its solution theory is the linear-ODE theory already used for geodesics and for parallel transport. Choosing a parallel orthonormal frame along γ converts (5.1) into a constant-form linear system, from which existence, uniqueness, and the dimension count follow.

Theorem 5.2.3: The Jacobi fields form a $2n$ -dimensional space

Let $\gamma: [0, b] \rightarrow M$ be a geodesic in an n -dimensional manifold. For each pair of vectors $u, w \in T_{\gamma(0)}M$ there is a unique Jacobi field J along γ with $J(0) = u$ and $D_t J(0) = w$. Consequently the set $\mathcal{J}(\gamma)$ of Jacobi fields along γ is a real vector space of dimension $2n$, and the map

$$\mathcal{J}(\gamma) \longrightarrow T_{\gamma(0)}M \oplus T_{\gamma(0)}M, \quad J \longmapsto (J(0), D_t J(0))$$

is a linear isomorphism.

Proof:

Let $(E_1, \dots, E_n)$ be a parallel orthonormal frame along γ , obtained by parallel transporting an orthonormal basis of $T_{\gamma(0)}M$ (theorem 2.4.2); it is orthonormal at every t because parallel transport is a linear isometry (proposition 2.4.4). Write a vector field J along γ in this frame as $J(t) = J^i(t)E_i(t)$ with scalar coefficients $J^i: [0, b] \rightarrow \mathbb{R}$. Since the E_i are parallel, $D_t E_i = 0$, so covariant differentiation of J reduces to ordinary differentiation of its coefficients:

$$D_t J = \dot{J}^i E_i, \quad D_t^2 J = \ddot{J}^i E_i$$

The curvature term is linear in J , so its frame coefficients are linear in the J^i . Define the smooth functions

$$a^k_i(t) = \langle R(E_i(t), \dot{\gamma}(t))\dot{\gamma}(t), E_k(t) \rangle$$

the components of the endomorphism $J \mapsto R(J, \dot{\gamma})\dot{\gamma}$ in the frame; they are smooth because R , $\dot{\gamma}$, and the frame are smooth. Then $R(J, \dot{\gamma})\dot{\gamma} = J^i a^k_i E_k$, and the Jacobi equation (5.1) becomes, coefficient by coefficient, the linear system

$$\ddot{J}^k(t) + a^k_i(t) J^i(t) = 0, \quad k = 1, \dots, n \quad (5.3)$$

This is a homogeneous linear second-order system of n equations with smooth coefficients on the compact interval $[0, b]$. Rewriting it as a first-order system in the $2n$ unknowns $(J^1, \dots, J^n, \dot{J}^1, \dots, \dot{J}^n)$, the standard existence and uniqueness theorem for linear ordinary differential equations (as used for the geodesic equation in theorem 3.1.3 and for the parallel-transport equation in theorem 2.4.2) provides, for each initial datum $(J^i(0), \dot{J}^i(0))$, a unique

solution on all of $[0, b]$. The initial datum corresponds exactly to $(J(0), D_t J(0))$ through the frame: $J(0) = J^i(0)E_i(0)$ and $D_t J(0) = \dot{J}^i(0)E_i(0)$. Hence for each $u, w \in T_{\gamma(0)}M$ there is a unique Jacobi field with $J(0) = u$, $D_t J(0) = w$.

Linearity of (5.1) in J makes $\mathcal{J}(\gamma)$ a real vector space, and the evaluation map $J \mapsto (J(0), D_t J(0))$ is linear. It is injective by the uniqueness just proved (a Jacobi field with zero initial data is the zero solution) and surjective by the existence, so it is a linear isomorphism onto $T_{\gamma(0)}M \oplus T_{\gamma(0)}M$, which has dimension $2n$. Therefore $\dim \mathcal{J}(\gamma) = 2n$, as claimed.

The theorem records that a Jacobi field is pinned down by exactly the same amount of data as a geodesic through a point, but now transverse to the base geodesic: its starting displacement and the rate at which that displacement is turning. The $2n$ -dimensional count will be split in the next proposition into the part along the geodesic, which is geometrically inert, and the part transverse to it, which carries all the information about how geodesics focus. Two of the $2n$ dimensions are visibly present already, and they are the least interesting: multiples of $\dot{\gamma}$ and of $t\dot{\gamma}$ solve the Jacobi equation trivially, because the curvature term annihilates $\dot{\gamma}$. Separating these off is the content of the following splitting.

Proposition 5.2.4: Tangential and normal splitting of Jacobi fields

Let $\gamma: [0, b] \rightarrow M$ be a geodesic, and suppose γ has constant speed $|\dot{\gamma}| \equiv \ell$.

1. A vector field of the form $J(t) = (a + bt)\dot{\gamma}(t)$ with $a, b \in \mathbb{R}$ is a Jacobi field, and these are exactly the Jacobi fields everywhere tangent to γ .
2. Every Jacobi field J splits uniquely as $J = J^\top + J^\perp$, where $J^\top$ is tangent to $\dot{\gamma}$ and $J^\perp$ is everywhere orthogonal to $\dot{\gamma}$, and each of $J^\top$, $J^\perp$ is again a Jacobi field. If γ is nonconstant ($\ell \neq 0$), the tangential part is $J^\top(t) = (a + bt)\dot{\gamma}(t)$ with $a = \langle J(0), \dot{\gamma}(0) \rangle / \ell^2$ and $b = \langle D_t J(0), \dot{\gamma}(0) \rangle / \ell^2$.

In particular the space $\mathcal{J}(\gamma)$ decomposes as an internal direct sum $\mathcal{J}(\gamma) = \mathcal{J}^\top(\gamma) \oplus \mathcal{J}^\perp(\gamma)$ with $\dim \mathcal{J}^\top(\gamma) = 2$ and $\dim \mathcal{J}^\perp(\gamma) = 2n - 2$.

Proof:

1. Let $J(t) = (a + bt)\dot{\gamma}(t)$. Since γ is a geodesic, $D_t \dot{\gamma} = 0$ (definition 3.1.1), so

$$D_t J = b \dot{\gamma} + (a + bt)D_t \dot{\gamma} = b \dot{\gamma}, \quad D_t^2 J = b D_t \dot{\gamma} = 0$$

The curvature term vanishes because R is antisymmetric in its first pair of arguments, $R(\dot{\gamma}, \dot{\gamma}) = 0$ (definition 4.1.2), so $R(J, \dot{\gamma})\dot{\gamma} = (a + bt)R(\dot{\gamma}, \dot{\gamma})\dot{\gamma} = 0$. Hence $D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$ and J is a Jacobi field. These form a two-dimensional space, spanned by $\dot{\gamma}$ and $t\dot{\gamma}$, corresponding to initial data $(J(0), D_t J(0)) \in \{(a\dot{\gamma}(0), b\dot{\gamma}(0)) : a, b \in \mathbb{R}\}$. Conversely a Jacobi field tangent to γ at every t may be written $J(t) = \varphi(t)\dot{\gamma}(t)$ for a smooth scalar φ ; then $D_t^2 J = \ddot{\varphi}\dot{\gamma}$ as above and the curvature term vanishes, so (5.1) reduces to $\ddot{\varphi}\dot{\gamma} = 0$, forcing $\ddot{\varphi} \equiv 0$ (for nonconstant γ , where $\dot{\gamma} \neq 0$), hence $\varphi(t) = a + bt$. Thus the tangential Jacobi fields are exactly the $(a + bt)\dot{\gamma}$.

2. Consider the scalar function $f(t) = \langle J(t), \dot{\gamma}(t) \rangle$ for a Jacobi field J . Using metric compati-

bility (definition 2.2.4) and $D_t \dot{\gamma} = 0$,

$$\dot{f} = \langle D_t J, \dot{\gamma} \rangle, \quad \ddot{f} = \langle D_t^2 J, \dot{\gamma} \rangle$$

By the Jacobi equation $D_t^2 J = -R(J, \dot{\gamma})\dot{\gamma}$, so

$$\ddot{f} = -\langle R(J, \dot{\gamma})\dot{\gamma}, \dot{\gamma} \rangle = -\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, \dot{\gamma}) = 0$$

the pairing vanishing by the antisymmetry of Riem in its last pair of arguments (definition 4.1.2 gives $R(J, \dot{\gamma})\dot{\gamma} \perp \dot{\gamma}$ directly, since $\langle R(X, Y)Z, Z \rangle = 0$ from the antisymmetry $\text{Riem}(X, Y, Z, W) = -\text{Riem}(X, Y, W, Z)$). Hence f is affine, $f(t) = f(0) + \dot{f}(0)t = \langle J(0), \dot{\gamma}(0) \rangle + \langle D_t J(0), \dot{\gamma}(0) \rangle t$. Assume γ nonconstant, $\ell \neq 0$, and define the tangential part

$$J^\top(t) = \frac{f(t)}{\ell^2} \dot{\gamma}(t) = \left(\frac{\langle J(0), \dot{\gamma}(0) \rangle}{\ell^2} + \frac{\langle D_t J(0), \dot{\gamma}(0) \rangle}{\ell^2} t \right) \dot{\gamma}(t)$$

which is of the form $(a + bt)\dot{\gamma}$ with the stated a, b , and so is a tangential Jacobi field by part (1). Set $J^\perp = J - J^\top$. Since $|\dot{\gamma}|^2 = \ell^2$ is constant, $\langle J^\top(t), \dot{\gamma}(t) \rangle = \frac{f(t)}{\ell^2} \ell^2 = f(t) = \langle J(t), \dot{\gamma}(t) \rangle$, whence $\langle J^\perp(t), \dot{\gamma}(t) \rangle = 0$ for all t ; thus $J^\perp$ is everywhere orthogonal to $\dot{\gamma}$. As a difference of two Jacobi fields, $J^\perp$ is a Jacobi field by linearity of (5.1). The decomposition $J = J^\top + J^\perp$ is unique, since a field simultaneously tangent to and orthogonal to the nonzero $\dot{\gamma}$ is zero. (When γ is constant the statement is vacuous, as there is no distinguished direction; the substantive case is $\ell \neq 0$, and one may take $\ell = 1$ by unit-speed reparametrization.)

For the dimension count, the tangential Jacobi fields $\mathcal{J}^\top(\gamma)$ form the two-dimensional space of part (1). The normal Jacobi fields $\mathcal{J}^\perp(\gamma)$ are those with $J \perp \dot{\gamma}$ throughout; by the splitting, every Jacobi field is uniquely a sum of one of each type, so $\mathcal{J}(\gamma) = \mathcal{J}^\top(\gamma) \oplus \mathcal{J}^\perp(\gamma)$, and $\dim \mathcal{J}^\perp(\gamma) = 2n - \dim \mathcal{J}^\top(\gamma) = 2n - 2$ by theorem 5.2.3, completing the proof.

The splitting quarantines the trivial part of the Jacobi equation. A tangential field $(a + bt)\dot{\gamma}$ is nothing more than a reparametrization of the geodesic itself: sliding γ along its own image, or speeding it up linearly, produces a variation through geodesics whose field points along $\dot{\gamma}$ and carries no transverse information. All the geometry of how nearby geodesics focus is in the normal part $J^\perp$, whose $2n - 2$ dimensions match the $n - 1$ transverse directions at each of the two ends of the interval. From this point on the object of interest is the normal Jacobi field, and the model computation that follows treats exactly that case, with $J \perp \dot{\gamma}$ and $J(0) = 0$, the initial condition that will detect conjugate points in the next section.

The Jacobi equation becomes fully explicit on a space of constant curvature, where the curvature endomorphism is the universal expression of proposition 4.5.3. There the equation collapses to a scalar equation $f'' + cf = 0$, and its solutions are the trigonometric or hyperbolic functions whose zeros locate the conjugate points.

Example 5.2.5: Normal Jacobi fields on a space of constant curvature

Let (M, g) have constant sectional curvature c (definition 4.5.2), and let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic, $|\dot{\gamma}| \equiv 1$. The normal Jacobi fields with $J(0) = 0$ are computed here in closed form; they are the fields governing the focusing of geodesics issuing from $\gamma(0)$.

Let E be a parallel unit field along γ with $E \perp \dot{\gamma}$, obtained by parallel transporting a unit vector

orthogonal to $\dot{\gamma}(0)$; it stays orthogonal to $\dot{\gamma}$ and of unit length because parallel transport is a linear isometry and $\dot{\gamma}$ is itself parallel (proposition 2.4.4). Seek a Jacobi field of the form $J(t) = f(t)E(t)$ with f scalar. Since E is parallel, $D_t J = \dot{f}E$ and $D_t^2 J = \ddot{f}E$. For the curvature term, the constant-curvature form (4.7) of proposition 4.5.3 gives, using $|\dot{\gamma}|^2 = 1$ and $\langle E, \dot{\gamma} \rangle = 0$,

$$R(J, \dot{\gamma})\dot{\gamma} = f R(E, \dot{\gamma})\dot{\gamma} = f c (\langle \dot{\gamma}, \dot{\gamma} \rangle E - \langle E, \dot{\gamma} \rangle \dot{\gamma}) = c f E$$

Substituting into the Jacobi equation (5.1),

$$D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = (\ddot{f} + c f) E = 0$$

and since E is nowhere zero this is the scalar equation

$$f'' + c f = 0 \tag{5.4}$$

With the initial conditions $J(0) = 0$ and $D_t J(0) = E(0)$, that is $f(0) = 0$ and $f'(0) = 1$, the solution of (5.4) is

$$f_c(t) = \begin{cases} \frac{\sin(\sqrt{c}t)}{\sqrt{c}} & c > 0 \\ t & c = 0 \\ \frac{\sinh(\sqrt{-c}t)}{\sqrt{-c}} & c < 0 \end{cases}$$

as is checked by substitution: for $c > 0$, $f_c'' = -\sqrt{c} \sin(\sqrt{c}t) = -c f_c$; for $c = 0$, $f_c'' = 0$; for $c < 0$, $f_c'' = \sqrt{-c} \sinh(\sqrt{-c}t) = -c f_c$, and in every case $f_c(0) = 0$, $f_c'(0) = 1$. The Jacobi field is $J(t) = f_c(t)E(t)$, and the general normal Jacobi field with $J(0) = 0$ is a sum $\sum_{i=1}^{n-1} f_c(t) c_i E_i(t)$ over a parallel orthonormal frame $E_1, \dots, E_{n-1}$ of the orthogonal complement of $\dot{\gamma}$, with constants c_i ; all $n - 1$ such fields share the single radial factor f_c , the analytic signature of isotropy.

The three cases carry the geometry directly. On the round sphere S_r^n , which has constant curvature $c = 1/r^2$ by example 4.3.4 and example 1.3.4, the field is

$$J(t) = r \sin(t/r) E(t)$$

whose norm $|J(t)| = r \sin(t/r)$ starts at 0, grows to its maximum r at $t = \pi r/2$, and returns to 0 at $t = \pi r$: geodesics leaving a point of the sphere spread apart and then refocus, meeting again at the antipodal point a distance πr away. On Euclidean space $\mathbb{R}^n$ ($c = 0$) the field is $J(t) = t E(t)$, with $|J(t)| = t$ growing without bound, so geodesics spread linearly and never refocus. On hyperbolic space H^n (curvature $c = -1$, example 4.3.5) the field is $J(t) = \sinh(t) E(t)$, with $|J(t)| = \sinh t$ growing exponentially, so geodesics spread even faster and again never meet again. The vanishing of f_c at a positive time, which happens only for $c > 0$, is precisely the event that the next section names a conjugate point.

5.3 Conjugate Points

The Jacobi equation of section 5.2 governs how a family of geodesics fans out from a common starting point. A Jacobi field J that vanishes at $t = 0$ is the infinitesimal record of a pencil of geodesics all leaving $\gamma(0)$, and its value $J(t)$ measures, to first order, how far the neighboring geodesics have

drifted from γ by time t . On a nonpositively curved space these fields grow and the geodesics keep spreading. On a positively curved space the curvature term in $D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$ acts as a restoring force, pulling the fanned geodesics back toward one another, and a Jacobi field that started at zero can return to zero. The instant at which it does is a *conjugate point*: a point along γ at which the nearby geodesics from $\gamma(0)$ attempt to refocus.

The phenomenon is visible on the round sphere. Every great circle through the north pole passes through the south pole, so all the geodesics leaving the north pole in every direction reconverge at the antipode. The antipode is the point at which the exponential map, which spread the tangent space out over the sphere, folds it back to a single point. This is the second face of a conjugate point, and the central result of the section identifies the two: q is conjugate to p along a geodesic exactly when the exponential map degenerates in the direction of q , and the number of independent geodesic pencils that refocus there is the dimension of the kernel of its differential. Conjugate points are therefore where the linear model of the manifold provided by $\exp_p$ breaks down, and their absence, guaranteed by nonpositive curvature, is what makes that model global. Both directions of this dictionary are used repeatedly in the sequel: conjugate points obstruct minimization, studied through the index form in section 5.4, and their nonexistence under $K \leq 0$ is the analytic input to the Cartan–Hadamard theorem of the next chapter.

Throughout, (M, g) is a Riemannian manifold with Levi-Civita connection ∇ , curvature endomorphism R (definition 4.1.2), and sectional curvature K (definition 4.3.1); γ is a geodesic (definition 3.1.1) and D_t the covariant derivative along it. Jacobi fields and the Jacobi equation (5.1) are those of section 5.2. The sign conventions fixed there and in Chapter 4 are in force.

Definition 5.3.1: Conjugate Point

Let $\gamma: [0, b] \rightarrow M$ be a geodesic and set $p = \gamma(0)$. A point $\gamma(t_0)$ with $0 < t_0 \leq b$ is *conjugate to p along γ* if there exists a Jacobi field J along γ , not identically zero, with

$$J(0) = 0 \quad \text{and} \quad J(t_0) = 0$$

The *multiplicity* of the conjugate point $\gamma(t_0)$ is the dimension of the vector space

$$\mathcal{J}_{0,t_0} = \{J \text{ a Jacobi field along } \gamma \mid J(0) = 0, J(t_0) = 0\}$$

of Jacobi fields vanishing at both endpoints. The relation is symmetric: if $\gamma(t_0)$ is conjugate to p along γ , then $p = \gamma(0)$ is conjugate to $\gamma(t_0)$ along the reversed geodesic $t \mapsto \gamma(t_0 - t)$, with the same multiplicity, since $J(t_0 - t)$ is again a Jacobi field vanishing at both ends. One therefore speaks of the two points as *conjugate along γ* .

The definition isolates the geometric event described in the opening: a nonzero Jacobi field vanishing at 0 and at t_0 is the infinitesimal generator of a family of geodesics that leave $\gamma(0)$ together and, to first order, meet again at $\gamma(t_0)$. That the multiplicity is a dimension, and not a bare count, reflects that several independent families may refocus at once, as on the sphere, where every direction transverse to a great circle produces a Jacobi field vanishing at the antipode. Two structural remarks bound the multiplicity. A Jacobi field vanishing at 0 and t_0 has no tangential part: the tangential Jacobi fields are $(a + bt)\dot{\gamma}$ (proposition 5.2.4), and such a field vanishes at two distinct times only if $a = b = 0$. Hence every field in $\mathcal{J}_{0,t_0}$ is normal to γ , and the space of normal Jacobi fields vanishing at 0 is already $(n - 1)$ -dimensional; imposing $J(t_0) = 0$ can only cut this down. The multiplicity of a conjugate point therefore lies between 1 and $n - 1$.

A first concrete instance grounds the definition before the general theory. On the unit sphere S^n a geodesic γ is a unit-speed great circle, and by example 5.2.5, applied with curvature $c = 1$, a normal Jacobi field J with $J(0) = 0$ has the form $J(t) = \sin(t) E(t)$ with E a parallel unit normal field, so $|J(t)| = |\sin t|$. This vanishes at $t = 0$ and next at $t = \pi$, and at no interior time $0 < t < \pi$. Consequently no point $\gamma(t_0)$ with $0 < t_0 < \pi$ is conjugate to p : any Jacobi field vanishing at 0 is normal (its tangential part $(a + bt)\dot{\gamma}$ must vanish at $t = 0$, forcing $a = 0$, and it is a sum of terms $\sin(t)E_i(t)$, none of which vanishes on $(0, \pi)$), so it cannot vanish again before $t = \pi$. The first conjugate point is the antipode $\gamma(\pi)$, computed in full in example 5.3.4. Below the antipodal distance the geodesics from p have not yet had time to refocus, and $\exp_p$ remains a diffeomorphism on the corresponding ball, as example 3.2.4 recorded.

The sphere calculation already suggests that conjugate points are where the exponential map ceases to be a local diffeomorphism. The connection is exact, and it is the organizing theorem of the section. It rests on a single computation: the Jacobi fields along a radial geodesic that vanish at the base point are precisely the images, under the differential of $\exp_p$, of the radial rays in the tangent space. The idea goes back to the definition of the exponential map as the time-one flow of geodesics, and to the fact from section 5.2 that Jacobi fields are the variation fields of variations through geodesics.

Theorem 5.3.2: Conjugate Points are the Singularities of the Exponential Map

Let $p \in M$, let $v \in T_p M$ lie in the domain of $\exp_p$, and let $\gamma(t) = \exp_p(tv)$ be the geodesic with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$, defined for t in an interval containing $[0, 1]$. Fix $t_0 > 0$ with $t_0 v$ in the domain of $\exp_p$, and set $q = \exp_p(t_0 v) = \gamma(t_0)$. Then:

1. For every $w \in T_p M$, the field

$$J_w(t) = d(\exp_p)_{tv}(tw) \quad (5.5)$$

is the Jacobi field along γ with $J_w(0) = 0$ and $D_t J_w(0) = w$, where on the right $tw \in T_{tv}(T_p M)$ is identified with an element of $T_p M$ in the usual way. Every Jacobi field along γ vanishing at 0 arises this way.

2. The point $q = \gamma(t_0)$ is conjugate to p along γ if and only if the differential $d(\exp_p)_{t_0 v} : T_{t_0 v}(T_p M) \rightarrow T_q M$ is singular (not injective).
3. The multiplicity of the conjugate point q equals $\dim \ker d(\exp_p)_{t_0 v}$.

Proof:

The tangent space $T_a(T_p M)$ to the vector space $T_p M$ at a point a is canonically identified with $T_p M$ itself, as in proposition 3.2.5; this identification is used without further comment, so $d(\exp_p)_a$ is read as a linear map $T_p M \rightarrow T_{\exp_p(a)} M$.

1. Fix $w \in T_p M$. Build a variation of γ through geodesics whose variation field is J_w . In the tangent space $T_p M$, consider the two-parameter family of vectors

$$(s, t) \mapsto t(v + sw), \quad s \in (-\varepsilon, \varepsilon), \quad t \in [0, t_0 + \delta]$$

For ε and δ small the vectors $t(v + sw)$ all lie in the (open) domain of $\exp_p$, because $t_0 v$ lies in that open set and the map $(s, t) \mapsto t(v + sw)$ is continuous with $(0, t) \mapsto tv$ in the

domain for $t \in [0, t_0]$. Define

$$\Gamma(s, t) = \exp_p(t(v + sw))$$

For each fixed s , the rescaling identity (proposition 3.2.2) shows that $t \mapsto \Gamma(s, t) = \exp_p(t(v + sw))$ is the geodesic with initial point p and initial velocity $v + sw$; thus Γ is a variation of $\gamma = \Gamma(0, \cdot)$ through geodesics. By theorem 5.2.2, its variation field

$$J(t) = \partial_s \Gamma(s, t) \big|_{s=0}$$

is a Jacobi field along γ . Compute it from the chain rule. The map $s \mapsto t(v + sw)$ is a straight line in $T_p M$ through tv with velocity tw at $s = 0$, so

$$J(t) = \frac{\partial}{\partial s} \bigg|_{s=0} \exp_p(t(v + sw)) = d(\exp_p)_{tv}(tw) = J_w(t)$$

which is (5.5). It remains to identify the initial data of J_w . At $t = 0$ the differential $d(\exp_p)_0(0 \cdot w) = 0$, so $J_w(0) = 0$. For the initial derivative, all curves $s \mapsto \Gamma(s, 0) = \exp_p(0) = p$ are constant, and each $t \mapsto \Gamma(s, t)$ is a geodesic starting at p with velocity $v + sw$, so $\partial_t \Gamma(s, 0) = v + sw$. Using the symmetry lemma (lemma 3.5.3) and evaluating at $t = 0$,

$$D_t J_w(0) = D_t \partial_s \Gamma \big|_{(0,0)} = D_s \partial_t \Gamma \big|_{(0,0)} = D_s \big|_{s=0}(v + sw) = w,$$

the last covariant derivative being computed along the constant curve $s \mapsto \Gamma(s, 0) = p$, where covariant differentiation of the $T_p M$ -valued curve $s \mapsto v + sw$ reduces to the ordinary derivative w . Thus J_w is the Jacobi field with $J_w(0) = 0$ and $D_t J_w(0) = w$.

Conversely, let J be any Jacobi field along γ with $J(0) = 0$. Set $w = D_t J(0)$. Then J_w is a Jacobi field with the same initial data $J_w(0) = 0 = J(0)$ and $D_t J_w(0) = w = D_t J(0)$, so $J = J_w$ by the uniqueness of Jacobi fields from their initial data (theorem 5.2.3). Hence every Jacobi field vanishing at 0 has the form (5.5).

2. By part (1), the Jacobi fields along γ vanishing at 0 are exactly the fields J_w , $w \in T_p M$, and the assignment $w \mapsto J_w$ is linear and injective (distinct w give distinct initial derivatives $D_t J_w(0) = w$, hence distinct fields). Evaluating at $t = t_0$,

$$J_w(t_0) = d(\exp_p)_{t_0 v}(t_0 w) = t_0 d(\exp_p)_{t_0 v}(w),$$

using linearity of the differential. Since $t_0 > 0$, this vanishes if and only if $d(\exp_p)_{t_0 v}(w) = 0$. Now $q = \gamma(t_0)$ is conjugate to p if and only if there is some $w \neq 0$ with J_w not identically zero and $J_w(t_0) = 0$. A field J_w with $w \neq 0$ is not identically zero, because $D_t J_w(0) = w \neq 0$. Therefore q is conjugate to p if and only if there exists $w \neq 0$ with $d(\exp_p)_{t_0 v}(w) = 0$, that is, if and only if $d(\exp_p)_{t_0 v}$ has nontrivial kernel, which is exactly the statement that it is singular.

3. The multiplicity of q is $\dim \mathcal{J}_{0,t_0}$ (definition 5.3.1). Every field in $\mathcal{J}_{0,t_0}$ vanishes at 0, so by part (1) it equals J_w for a unique w , and the map $J \mapsto D_t J(0)$ is a linear isomorphism from $\mathcal{J}_{0,t_0}$ onto

$$\{w \in T_p M \mid J_w(t_0) = 0\} = \{w \in T_p M \mid d(\exp_p)_{t_0 v}(w) = 0\} = \ker d(\exp_p)_{t_0 v},$$

where the first equality is the computation of part (2) and the injectivity of $w \mapsto J_w$ gives the isomorphism. Hence $\dim \mathcal{J}_{0,t_0} = \dim \ker d(\exp_p)_{t_0 v}$, as claimed.

This establishes all three parts, completing the proof.

The theorem converts a statement about Jacobi fields, which are solutions of a differential equation, into a statement about the rank of a single linear map, the differential of the exponential map. Both viewpoints earn their keep. The Jacobi side is where the curvature enters and where the constant-curvature models are computed; the exponential side is where conjugate points connect to the failure of $\exp_p$ to be a chart, and hence to the geometry of geodesic balls. The formula (5.5) is the bridge, and it says something concrete: to differentiate the exponential map in a direction w at the point t_0v , follow the Jacobi field that starts at $\gamma(0)$ with zero value and initial velocity w , and read off its value at time t_0 , up to the factor t_0 . A conjugate point is where some such Jacobi field, having started at zero, has returned to zero, so that the exponential map has collapsed that infinitesimal direction. The absence of conjugate points on $[0, b]$ is thus equivalent to $\exp_p$ being an immersion, hence a local diffeomorphism, along the whole radial geodesic, a fact used constantly in the minimization theory of section 5.4.

One consequence is worth recording before the curvature hypotheses are imposed. A geodesic has only finitely many conjugate points on any compact parameter interval, and so, whenever conjugate points exist at all, there is a well-defined *first* one. The mechanism is the linear structure of the Jacobi equation, not any general property of smooth functions: a C^∞ function can vanish on a whole subinterval without being identically zero, so isolation of zeros must come from the differential equation, and it does.

Set up the count in a frame. Let $(E_1, \dots, E_{n-1})$ be a parallel orthonormal frame of the normal bundle $\dot{\gamma}^\perp$ along γ , obtained by parallel transport of an orthonormal basis of $\dot{\gamma}(0)^\perp$, and let $J_1, \dots, J_{n-1}$ be the normal Jacobi fields with $J_i(0) = 0$ and $D_t J_i(0) = E_i(0)$. These are linearly independent (their initial derivatives $E_1(0), \dots, E_{n-1}(0)$ are), and by the discussion following definition 5.3.1 every Jacobi field vanishing at 0 and at a second time is normal, so $J_1, \dots, J_{n-1}$ span the space of normal Jacobi fields vanishing at 0, which is the only space that can produce conjugate points. Let $A(t)$ be the $(n-1) \times (n-1)$ matrix whose i th column records the components of $J_i(t)$ in the frame, $J_i(t) = \sum_k A_{ki}(t) E_k(t)$; then $A(0) = 0$ and, since the E_k are parallel, $A'(0) = I$. A point $\gamma(t_0)$ with $t_0 \in (0, b]$ is conjugate to p exactly when some nonzero combination $\sum_i c_i J_i$ vanishes at t_0 , that is, exactly when $A(t_0)c = 0$ for some $c \neq 0$, so the conjugate parameters are the zeros of $t \mapsto \det A(t)$ on $(0, b]$.

The kernel data at a conjugate parameter is constrained by a conserved quantity. Writing the curvature operator $S(t): \dot{\gamma}^\perp \rightarrow \dot{\gamma}^\perp$, $S(t)X = R(X, \dot{\gamma})\dot{\gamma}$, in the frame gives a symmetric matrix $S(t)$, symmetric because the pair symmetry of the curvature tensor (proposition 4.2.1) yields $\langle R(X, \dot{\gamma})\dot{\gamma}, Y \rangle = \langle R(Y, \dot{\gamma})\dot{\gamma}, X \rangle$; and since the E_k are parallel, the Jacobi equation for each column reads $A''(t) + S(t)A(t) = 0$ in matrix form. The Wronskian-type matrix $W(t) = A'(t)^\top A(t) - A(t)^\top A'(t)$ has derivative

$$W'(t) = A''^\top A - A^\top A'' = (-SA)^\top A - A^\top (-SA) = -A^\top S^\top A + A^\top SA = 0,$$

using $S^\top = S$, so W is constant; and $W(0) = 0$ because $A(0) = 0$. Hence $A'(t)^\top A(t) = A(t)^\top A'(t)$ for all t : the matrix $A^\top A'$ is symmetric throughout.

Each conjugate parameter is an isolated zero. Fix $t_0 \in (0, b]$ with $\det A(t_0) = 0$ and choose $0 \neq c \in \ker A(t_0)$. The field $J = \sum_i c_i J_i$ is a nonzero normal Jacobi field with $J(0) = 0$ and $J(t_0) = 0$; because the frame (E_k) is parallel, its components in the frame are the entries of $A(t)c$ and those of $D_t J$ are the entries of $A'(t)c$. Were $A'(t_0)c = 0$ as well, the field J would satisfy $J(t_0) = 0$ and $D_t J(t_0) = 0$, forcing $J \equiv 0$ by the uniqueness of Jacobi fields from their value and covariant derivative at t_0 (theorem 5.2.3, whose linear-ODE proof determines a solution equally from data at any initial time), contradicting $c \neq 0$; hence $A'(t_0)c \neq 0$, so the restriction of $A'(t_0)$ to $\ker A(t_0)$ is injective.

The conserved symmetry pins down its image: for $c \in \ker A(t_0)$ and any $d \in \mathbb{R}^{n-1}$,

$$\langle A'(t_0)c, A(t_0)d \rangle = c^\top A'(t_0)^\top A(t_0)d = c^\top A(t_0)^\top A'(t_0)d = (A(t_0)c)^\top A'(t_0)d = 0,$$

so $A'(t_0)(\ker A(t_0))$ is orthogonal to $\text{im } A(t_0)$. Let $m = \dim \ker A(t_0) \geq 1$, so $\dim \text{im } A(t_0) = n-1-m$. Choose a basis $c^{(1)}, \dots, c^{(n-1)}$ of $\mathbb{R}^{n-1}$ whose first m vectors span $\ker A(t_0)$. For $j \leq m$ the column $A(t)c^{(j)}$ vanishes at t_0 , so $A(t)c^{(j)} = (t-t_0)B_j(t)$ with B_j smooth and $B_j(t_0) = A'(t_0)c^{(j)}$. Expanding the determinant multilinearly in the columns,

$$\det(A(t)c^{(1)}, \dots, A(t)c^{(n-1)}) = (t-t_0)^m \varphi(t),$$

where

$$\varphi(t) = \det(B_1(t), \dots, B_m(t), A(t)c^{(m+1)}, \dots, A(t)c^{(n-1)})$$

The left side is $\det A(t)$ times the fixed nonzero factor $\det(c^{(1)}, \dots, c^{(n-1)})$. At $t = t_0$ the first m columns of φ are $A'(t_0)c^{(1)}, \dots, A'(t_0)c^{(m)}$, a basis of $A'(t_0)(\ker A(t_0))$ by injectivity, and the last $n-1-m$ columns are $A(t_0)c^{(m+1)}, \dots, A(t_0)c^{(n-1)}$, a basis of $\text{im } A(t_0)$; by the orthogonality just shown these two spaces meet only in 0, and their dimensions sum to $n-1$, so the $n-1$ columns are linearly independent and $\varphi(t_0) \neq 0$. Thus $\det A(t) = (t-t_0)^m \psi(t)$ with ψ smooth and $\psi(t_0) \neq 0$, so $\det A$ is nonzero on a punctured neighborhood of t_0 and t_0 is an isolated zero of finite order m , the order equal to the multiplicity of the conjugate point.

Isolated zeros in the compact interval $[0, b]$ are finite in number: an infinite set of zeros would accumulate at some $\bar{t} \in [0, b]$ by the Bolzano–Weierstrass theorem, and $\det A(\bar{t}) = 0$ by continuity, but then $\bar{t}$ would be a non-isolated zero, contradicting the previous paragraph (the argument there applies to every zero, including $\bar{t}$). Hence the conjugate points along γ are finite in number and do not accumulate, and there is a well-defined *first* conjugate point, the smallest zero of $\det A$ in $(0, b]$, whenever any exist at all. This recovers the exponential-map picture. By part (1) of theorem 5.3.2 the columns satisfy $J_i(t) = t d(\exp_p)_{tv}(E_i(0))$, so $A(t)$ is t times the matrix of $d(\exp_p)_{tv}$ restricted to $\dot{\gamma}^\perp$, and $\det A(t) = t^{n-1} \det(d(\exp_p)_{tv}|_{\dot{\gamma}^\perp})$. For $t > 0$ this vanishes exactly when $d(\exp_p)_{tv}$ is singular, since $d(\exp_p)_{tv}$ carries the radial direction v to the nonzero radial geodesic velocity and preserves the orthogonal splitting into radial and normal parts (theorem 3.3.4), so its singularity is that of its normal block. Thus the conjugate parameters are precisely the zeros of $t \mapsto \det(d(\exp_p)_{tv})$ on $(0, b]$, in agreement with theorem 5.3.2, and the finiteness just established is the finiteness of those zeros.

Whether any exist is decided by the sign of the curvature. The restoring term in the Jacobi equation has the sign of the sectional curvature, and where that sign is nonpositive the term cannot bend a nonzero Jacobi field back to the zero it started from. The next proposition makes this precise, and it is the mechanism behind the global rigidity of nonpositively curved spaces.

Proposition 5.3.3: Nonpositive Curvature Excludes Conjugate Points

Let (M, g) be a Riemannian manifold with sectional curvature $K \leq 0$, that is, $K(\Pi) \leq 0$ for every 2-plane Π at every point (definition 4.3.1). Then no geodesic in M has a pair of conjugate points. Equivalently, for every $p \in M$ the differential $d(\exp_p)_v$ is nonsingular at every v in the domain of $\exp_p$.

Proof:

Let $\gamma: [0, b] \rightarrow M$ be a geodesic and J a Jacobi field along γ with $J(0) = 0$. It suffices to show that $J(t) \neq 0$ for all $t \in (0, b]$ unless $J \equiv 0$; then no nonzero Jacobi field vanishes at two points, so γ has no conjugate points, and by theorem 5.3.2 this is the asserted nonsingularity of $d(\exp_p)_v$.

Consider the smooth function $f(t) = |J(t)|^2 = \langle J, J \rangle$ on $[0, b]$. Differentiating with metric compatibility (definition 2.2.4), and writing D_t for the covariant derivative along γ ,

$$f'(t) = \frac{d}{dt} \langle J, J \rangle = 2 \langle D_t J, J \rangle$$

Differentiating once more,

$$f''(t) = 2 \langle D_t^2 J, J \rangle + 2 \langle D_t J, D_t J \rangle = 2 \langle D_t^2 J, J \rangle + 2 |D_t J|^2$$

The Jacobi equation (5.1) gives $D_t^2 J = -R(J, \dot{\gamma})\dot{\gamma}$, so

$$\langle D_t^2 J, J \rangle = -\langle R(J, \dot{\gamma})\dot{\gamma}, J \rangle = -\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J),$$

using the $(0, 4)$ notation $\text{Riem}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle$. The quantity $\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J)$ is related to sectional curvature: for a geodesic $\dot{\gamma}$ is nowhere zero (a geodesic has constant speed, and $J(0) = 0$ with $J \not\equiv 0$ forces the speed to be positive on the interval of interest), and where J and $\dot{\gamma}$ are linearly independent, spanning a 2-plane Π ,

$$\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J) = K(\Pi)(|J|^2 |\dot{\gamma}|^2 - \langle J, \dot{\gamma} \rangle^2)$$

by the definition of sectional curvature (definition 4.3.1); where J and $\dot{\gamma}$ are dependent, the Gram determinant $|J|^2 |\dot{\gamma}|^2 - \langle J, \dot{\gamma} \rangle^2$ vanishes and, by the same identity read as an equality of quadratic expressions in J , so does $\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J)$. In either case

$$\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J) = K(\Pi)(|J|^2 |\dot{\gamma}|^2 - \langle J, \dot{\gamma} \rangle^2) \leq 0,$$

because $K \leq 0$ by hypothesis and the Gram determinant is nonnegative by the Cauchy-Schwarz inequality. Therefore $\langle D_t^2 J, J \rangle = -\text{Riem}(J, \dot{\gamma}, \dot{\gamma}, J) \geq 0$, and

$$f''(t) = 2 \langle D_t^2 J, J \rangle + 2 |D_t J|^2 \geq 2 |D_t J|^2 \geq 0 \quad (5.6)$$

So $f = |J|^2$ is a convex function on $[0, b]$.

Now suppose, for contradiction, that $J(t_0) = 0$ for some $t_0 \in (0, b]$, while $J \not\equiv 0$. Then $f(0) = 0$ and $f(t_0) = 0$, and $f \geq 0$; a convex nonnegative function vanishing at two points $0 < t_0$ vanishes on the whole interval between them, so $f \equiv 0$ on $[0, t_0]$, that is, $J \equiv 0$ on $[0, t_0]$. In particular $J(0) = 0$ and $D_t J(0) = 0$ (the derivative of the identically zero field), so by uniqueness of Jacobi fields from initial data (theorem 5.2.3) $J \equiv 0$ on all of $[0, b]$, contradicting $J \not\equiv 0$. Hence no nonzero Jacobi field with $J(0) = 0$ vanishes again on $(0, b]$, so γ has no conjugate points, completing the proof.

The proof extracts the whole force of the hypothesis from the single inequality (5.6): nonpositive curvature makes $|J|^2$ convex, and a nonnegative convex function that starts at zero and is not identically zero can never return to zero. Geometrically, the geodesics emanating from p spread

at least as fast as in Euclidean space and never refocus. The equivalent statement, that $\exp_p$ is everywhere nonsingular, is the local half of the Cartan–Hadamard theorem, which upgrades it on a complete simply connected manifold of nonpositive curvature to the conclusion that $\exp_p$ is a diffeomorphism of all of $T_p M$ onto M ; the completeness argument belongs to the next chapter and is flagged here only as the destination of this proposition. The flat case $K \equiv 0$ is the boundary: there $f'' = 2|D_t J|^2$, and a Jacobi field with $J(0) = 0$ has $J(t) = t D_t J(0)$ growing linearly, again with no second zero, consistent with $\mathbb{R}^n$ having no conjugate points.

The complementary picture, positive curvature forcing refocusing, is exhibited in its sharpest form by the sphere, where the constant-curvature computation of section 5.2 makes every conjugate point explicit.

Example 5.3.4: Conjugate points on the round sphere

Let S_r^n be the round sphere of radius r , with constant sectional curvature $c = 1/r^2$ (example 4.3.4), and let γ be a unit-speed geodesic, a great circle traversed at unit speed. Fix $p = \gamma(0)$. By example 5.2.5, applied with $c = 1/r^2$, the normal Jacobi fields along γ with $J(0) = 0$ are exactly

$$J(t) = f(t) E(t), \quad f(t) = \frac{\sin(t/r)}{1/r} = r \sin\left(\frac{t}{r}\right),$$

where E is any parallel unit field along γ orthogonal to $\dot{\gamma}$ (the coefficient function solves $f'' + cf = 0$, $f(0) = 0$, $f'(0) = 1$). Choose a parallel orthonormal frame $(E_1, \dots, E_{n-1})$ of the normal bundle along γ , obtained by parallel transport of an orthonormal basis of $\dot{\gamma}(0)^\perp$; each E_i is parallel and unit, and

$$J_i(t) = r \sin\left(\frac{t}{r}\right) E_i(t), \quad i = 1, \dots, n-1,$$

are $n-1$ linearly independent normal Jacobi fields vanishing at $t = 0$.

Location of the conjugate points. The common coefficient $r \sin(t/r)$ vanishes precisely when t/r is an integer multiple of π , that is, at $t = \pi r, 2\pi r, 3\pi r, \dots$. At each such t_0 all $n-1$ fields J_i vanish, so each is a nonzero Jacobi field vanishing at 0 and at t_0 , and $\gamma(t_0)$ is conjugate to p . The first conjugate point is at

$$t_0 = \pi r,$$

the point $\gamma(\pi r)$, which is the antipode $-p$ of p : the great circle of length $2\pi r$ reaches the point diametrically opposite p after arc length πr . This is exactly the refocusing described at the start of the section, where all great circles from p meet again at $-p$.

Multiplicity. At $t_0 = \pi r$ the space $\mathcal{J}_{0,t_0}$ of Jacobi fields vanishing at both ends contains the $n-1$ independent fields $J_1, \dots, J_{n-1}$, so its dimension is at least $n-1$. It is at most $n-1$ by the general bound following definition 5.3.1: every field in $\mathcal{J}_{0,t_0}$ is normal, and the normal Jacobi fields vanishing at 0 form an $(n-1)$ -dimensional space. Hence the multiplicity of the first conjugate point $\gamma(\pi r)$ is exactly

$$\dim \mathcal{J}_{0,\pi r} = n-1$$

The same count holds at every conjugate point $t_0 = k\pi r$, $k \geq 1$. For the two-sphere S_r^2 the multiplicity is 1, the single normal direction; for S_r^n it is $n-1$, the full transverse space, which is why the antipode is a point of total refocusing.

The exponential-map reading. By theorem 5.3.2, the differential $d(\exp_p)_v$ at $|v| = \pi r$ is singular with $(n-1)$ -dimensional kernel: the exponential map on the sphere of radius πr in $T_p S_r^n$ collapses the entire coordinate sphere to the single point $-p$, matching the direct computation $\exp_p(v) =$

$(\cos(|v|/r))p + \cdots$ of example 3.2.4, which sends every v with $|v| = \pi r$ to $-p$. Below the first conjugate radius, on $|v| < \pi r$, the differential is everywhere nonsingular and $\exp_p$ is a diffeomorphism, matching the Jacobi-field computation that no J_i vanishes on $(0, \pi r)$.

The contrast with the other two model space forms completes the trichotomy. Hyperbolic space H^n has $c = -1 < 0$ and Euclidean space $\mathbb{R}^n$ has $c = 0$, so both satisfy $K \leq 0$ and have *no* conjugate points along any geodesic, by proposition 5.3.3; explicitly, the Jacobi coefficient of example 5.2.5 is $\sinh(t)$ for H^n and t for $\mathbb{R}^n$, neither of which returns to zero for $t > 0$. Positive curvature refocuses the geodesics at finite distance, zero curvature lets them spread linearly, and negative curvature drives them apart exponentially; the sphere, the plane, and the hyperbolic space realize the three cases, with the location of the first conjugate point $\pi r = \pi/\sqrt{c}$ on the sphere decreasing as the curvature grows.

5.4 The Index Form and the Index Lemma

The second variation of energy in section 5.1 attached to every geodesic a quadratic form on the vector fields that deform it: the value $E''(0)$ produced by a proper variation with variation field V . That form is the true measure of whether a geodesic minimizes. A geodesic is a critical point of the energy, so its first variation vanishes for every proper variation; what separates a minimum from a saddle is the sign of the second variation, and a single field V with $E''(0) < 0$ is enough to show that the geodesic is beaten by a nearby competitor. This section isolates the quadratic form itself, the *index form* I , studies it as an object in its own right, and proves the two results that convert Jacobi-field information into minimization conclusions.

The center of the analysis is a comparison. Among all fields V with prescribed endpoint values, which one makes $I(V, V)$ smallest? The answer, when the geodesic carries no interior conjugate point, is the Jacobi field with those endpoint values, and it wins strictly against every other competitor. This is the index lemma, and it is the exact statement needed to run the two directions of the minimization theory. In one direction it shows that before the first conjugate point a geodesic cannot be shortened by a small deformation that keeps its endpoints; in the other it shows that past the first conjugate point the geodesic is no longer even a local minimum, because a deformation exists that strictly decreases the energy. The Jacobi fields of section 5.2 and the conjugate points of section 5.3 thereby acquire their decisive variational meaning. Throughout, (M, g) is a Riemannian manifold with Levi-Civita connection ∇ , $\gamma: [0, b] \rightarrow M$ is a unit-speed geodesic, D_t is the covariant derivative along γ , and R is the curvature endomorphism of definition 4.1.2; the curvature sign convention of box 4.1.1 is in force.

Definition 5.4.1: The index form

Let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic. Denote by $\mathcal{V}(\gamma)$ the real vector space of piecewise-smooth vector fields V along γ with $V(0) = V(b) = 0$: fields that are continuous on $[0, b]$, smooth on each subinterval of some partition $0 = t_0 < t_1 < \cdots < t_m = b$, and vanishing at the endpoints. The *index form* of γ is the symmetric bilinear form $I: \mathcal{V}(\gamma) \times \mathcal{V}(\gamma) \rightarrow \mathbb{R}$ given by

$$I(V, W) = \int_0^b \left(\langle D_t V, D_t W \rangle - \langle R(V, \dot{\gamma}) \dot{\gamma}, W \rangle \right) dt$$

the integral being taken over each smooth subinterval and summed, since $D_t V$ and $D_t W$ may jump at the finitely many breakpoints while the integrand stays bounded.

The index form is exactly the second variation of energy, now regarded as a bilinear pairing rather than a value on a single deformation. Its diagonal reproduces the formula of section 5.1:

$$I(V, V) = \int_0^b \left(|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt = E''(0)$$

For a smooth field V vanishing at the endpoints this is precisely the second-variation formula of theorem 5.1.1. For a merely piecewise-smooth field the identity still holds, but theorem 5.1.1 does not deliver it directly, since that theorem is stated for a smooth variation with a smooth variation field; the extension to the piecewise-smooth fields of $\mathcal{V}(\gamma)$ is recorded in lemma 5.4.2 below, whose proof splits the variation over the smooth subintervals and checks that the corners contribute no boundary term. Symmetry in V and W is visible in the first term and, for the second, follows from the pair symmetry of the curvature tensor: $\langle R(V, \dot{\gamma}) \dot{\gamma}, W \rangle = \text{Riem}(V, \dot{\gamma}, \dot{\gamma}, W) = \text{Riem}(W, \dot{\gamma}, \dot{\gamma}, V) = \langle R(W, \dot{\gamma}) \dot{\gamma}, V \rangle$ by proposition 4.2.1. Bilinearity is immediate from the linearity of D_t and of R in the relevant slots. The two terms pull in opposite directions. The first, the derivative energy $\int |D_t V|^2$, is a nonnegative penalty on how fast V turns, and it alone would make I positive definite. The second subtracts a curvature term whose sign is that of the sectional curvature of the plane spanned by V and $\dot{\gamma}$, since $\langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle = K(V, \dot{\gamma}) |V \wedge \dot{\gamma}|^2$; positive curvature therefore lowers I and can make it negative, while nonpositive curvature only raises it. The whole minimization theory is the contest between these two terms.

The identification of $I(V, V)$ with a second variation of energy is needed below precisely for the piecewise-smooth fields of $\mathcal{V}(\gamma)$, since the field that defeats minimization past a conjugate point has a corner. The following lemma supplies that identification. Its content is that the corner costs nothing: the second-variation computation is carried out on each smooth piece, and the boundary terms produced at the breakpoints cancel.

Lemma 5.4.2: Second variation for a piecewise-smooth field

Let $\gamma: [0, b] \rightarrow M$ be a geodesic and let $V \in \mathcal{V}(\gamma)$ be a piecewise-smooth field vanishing at the endpoints, smooth on each subinterval of a partition $0 = t_0 < t_1 < \cdots < t_m = b$. Let $\Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ be the proper variation built from V as in example 3.5.2, defined for $|s|$ small. Then $s \mapsto E(\gamma_s)$ is twice differentiable at $s = 0$ and

$$E''(0) = \int_0^b \left(|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt = I(V, V)$$

Proof:

The map Γ is smooth on each strip $(-\varepsilon, \varepsilon) \times [t_{i-1}, t_i]$, being the composition of the exponential map with the smooth field $sV(t)$ on that strip, and continuous across the breakpoints because V is continuous. Write $T = \partial_t \Gamma$ and $S = \partial_s \Gamma$ for the coordinate velocity fields; at $s = 0$ they restrict to $T = \dot{\gamma}$ and $S = V$ along γ . The energy decomposes over the smooth pieces,

$$E(\gamma_s) = \sum_{i=1}^m E_i(s), \quad E_i(s) = \frac{1}{2} \int_{t_{i-1}}^{t_i} \langle T, T \rangle dt$$

and each E_i is a smooth function of s because Γ is smooth on the closed strip over $[t_{i-1}, t_i]$.

Fix one subinterval and differentiate E_i twice in s , exactly as in the proof of theorem 5.1.1, whose steps use only the smoothness of Γ on that strip and not any endpoint condition. Metric compatibility of ∇ , the symmetry lemma $D_s T = D_t S$ (lemma 3.5.3), and the definition of R (definition 4.1.2) give, at $s = 0$ where $S = V$, $T = \dot{\gamma}$, and $D_t \dot{\gamma} = 0$,

$$E_i''(0) = \int_{t_{i-1}}^{t_i} \left(|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt + \left[\langle D_s S, \dot{\gamma} \rangle \right]_{t_{i-1}}^{t_i}$$

the boundary term arising from the same integration by parts as in Step 3 there, namely $\int_{t_{i-1}}^{t_i} \langle D_t D_s S, \dot{\gamma} \rangle dt = \int_{t_{i-1}}^{t_i} \frac{d}{dt} \langle D_s S, \dot{\gamma} \rangle dt$ once $\langle D_s S, D_t \dot{\gamma} \rangle = 0$ is used along the geodesic γ . Unlike the smooth proper case, this boundary term is not discarded by an endpoint condition; instead it vanishes for a structural reason. For each fixed t the curve $s \mapsto \Gamma(s, t) = \exp_{\gamma(t)}(sV(t))$ is a geodesic, so its covariant acceleration vanishes for all s : $D_s S = D_s \partial_s \Gamma \equiv 0$ on the whole rectangle. Hence $\langle D_s S, \dot{\gamma} \rangle \equiv 0$ at $s = 0$, and the boundary term $[\langle D_s S, \dot{\gamma} \rangle]_{t_{i-1}}^{t_i}$ is zero on every subinterval.

Summing over $i = 1, \dots, m$, the interval integrals combine into a single integral over $[0, b]$, whose integrand $|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle$ is bounded and has at most the finitely many jumps of $D_t V$ at the breakpoints, so the sum is exactly $\int_0^b (|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle) dt$. Therefore

$$E''(0) = \sum_{i=1}^m E_i''(0) = \int_0^b \left(|D_t V|^2 - \langle R(V, \dot{\gamma}) \dot{\gamma}, V \rangle \right) dt = I(V, V)$$

the last equality by definition 5.4.1, as required.

A useful first computation records what the index form does to a field that is proportional to a fixed profile in a parallel direction, on a space of constant curvature. This is the setting in which every quantity is explicit, and it will let the abstract comparison below be checked against a known answer.

Example 5.4.3: The index form on a constant-curvature space

Let (M, g) have constant sectional curvature c , and let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic. Fix a parallel unit field E along γ with $E \perp \dot{\gamma}$, which exists because parallel transport preserves both length and the angle with $\dot{\gamma}$ (the latter is parallel along γ). For a scalar function $f: [0, b] \rightarrow \mathbb{R}$ with $f(0) = f(b) = 0$, smooth, consider the normal field $V = fE \in \mathcal{V}(\gamma)$. Since E is parallel, $D_t V = f' E$, so $|D_t V|^2 = (f')^2$. For the curvature term, the constant-curvature form of proposition 4.5.3 gives

$$R(V, \dot{\gamma}) \dot{\gamma} = c(\langle \dot{\gamma}, \dot{\gamma} \rangle V - \langle V, \dot{\gamma} \rangle \dot{\gamma}) = c f E$$

using $\langle \dot{\gamma}, \dot{\gamma} \rangle = 1$ and $\langle V, \dot{\gamma} \rangle = f \langle E, \dot{\gamma} \rangle = 0$, so $\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = cf^2$. Hence

$$I(V, V) = \int_0^b ((f')^2 - cf^2) dt$$

On the round sphere S_r^n , where $c = 1/r^2$, take $b = \pi r$ and $f(t) = \sin(t/r)$, which vanishes at $t = 0$ and $t = \pi r$. Then $f'(t) = \frac{1}{r} \cos(t/r)$, and

$$I(V, V) = \int_0^{\pi r} \left(\frac{1}{r^2} \cos^2(t/r) - \frac{1}{r^2} \sin^2(t/r) \right) dt = \frac{1}{r^2} \int_0^{\pi r} \cos(2t/r) dt = 0$$

so this particular field lies in the null space of I . Indeed $V = \sin(t/r)E$ is the Jacobi field of example 5.2.5 vanishing at both ends of the arc to the antipode, and the vanishing of $I(V, V)$ is the borderline case the index lemma describes at a conjugate point. On hyperbolic space, where $c = -1$, the curvature term $-cf^2 = f^2$ adds to the derivative energy, so $I(V, V) = \int_0^b ((f')^2 + f^2) dt > 0$ for every nonzero f vanishing at the endpoints; the index form is positive definite and there is no competing minimizer, in agreement with the absence of conjugate points in nonpositive curvature.

The comparison at the heart of the section rests on an integration by parts that rewrites $I(V, W)$ so that the Jacobi operator appears. For a field V that is smooth on a subinterval $[t_{i-1}, t_i]$, metric compatibility of ∇ gives $\frac{d}{dt} \langle D_t V, W \rangle = \langle D_t^2 V, W \rangle + \langle D_t V, D_t W \rangle$, so

$$\int_{t_{i-1}}^{t_i} \langle D_t V, D_t W \rangle dt = \left[\langle D_t V, W \rangle \right]_{t_{i-1}}^{t_i} - \int_{t_{i-1}}^{t_i} \langle D_t^2 V, W \rangle dt$$

Summing over the subintervals and using $W(0) = W(b) = 0$, the interior boundary terms collect into a sum over breakpoints, and the index form becomes

$$I(V, W) = - \int_0^b \langle D_t^2 V + R(V, \dot{\gamma})\dot{\gamma}, W \rangle dt - \sum_{i=1}^{m-1} \langle \Delta_i D_t V, W(t_i) \rangle \quad (5.7)$$

where $\Delta_i D_t V = D_t V(t_i^+) - D_t V(t_i^-)$ is the jump in the covariant derivative of V at the interior breakpoint t_i . Formula (5.7) holds whenever V is piecewise smooth and $W \in \mathcal{V}(\gamma)$; it makes the connection between the index form and the Jacobi operator $J \mapsto D_t^2 J + R(J, \dot{\gamma})\dot{\gamma}$ of definition 5.2.1 explicit. A Jacobi field is smooth, so it has no jumps, and the operator applied to it vanishes; both terms on the right then drop, a fact used repeatedly below.

Theorem 5.4.4: The Index Lemma

Let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic with no interior point conjugate to $\gamma(0)$ along γ ; that is, $\gamma(t)$ is not conjugate to $\gamma(0)$ for any $t \in (0, b]$. Fix $w \in T_{\gamma(b)}M$ with $w \perp \dot{\gamma}(b)$, and let J be the unique Jacobi field along γ with $J(0) = 0$ and $J(b) = w$. Then for every piecewise-smooth normal field V along γ with $V(0) = 0$ and $V(b) = w$,

$$I(J, J) \leq I(V, V)$$

with equality if and only if $V = J$ on $[0, b]$.

Before the proof, the hypotheses deserve comment. The absence of interior conjugate points is what guarantees both that the boundary-value Jacobi field J with $J(0) = 0$, $J(b) = w$ exists and is unique,

and that the Jacobi fields vanishing at 0 furnish a moving frame in which any competitor V can be expanded. The restriction to normal fields (perpendicular to $\dot{\gamma}$) costs nothing: the tangential part of any field in $\mathcal{V}(\gamma)$ contributes to I exactly the derivative energy of a scalar function vanishing at both ends, which is strictly positive unless that function is zero, so tangential components only increase I and the sharpest comparison is the normal one. The conclusion says that the Jacobi field is the unique minimizer of the index form subject to its boundary values: no other admissible field bends as economically.

Proof:

The proof proceeds in three steps: first the existence of a Jacobi frame adapted to the no-conjugate-point hypothesis, then the expansion of an arbitrary competitor in that frame, and finally the algebraic identity that exhibits the difference $I(V, V) - I(J, J)$ as a nonnegative quantity vanishing only when $V = J$.

Step 1: a Jacobi frame with no interior zeros. Let $\{e_1, \dots, e_{n-1}\}$ be an orthonormal basis of the orthogonal complement of $\dot{\gamma}(0)$ in $T_{\gamma(0)}M$, and extend each e_α to a parallel field E_α along γ ; the E_α remain orthonormal and normal to $\dot{\gamma}$ for all t , since parallel transport is a linear isometry preserving the angle with the parallel field $\dot{\gamma}$. For each $\alpha = 1, \dots, n-1$ let J_α be the Jacobi field with $J_\alpha(0) = 0$ and $D_t J_\alpha(0) = E_\alpha(0) = e_\alpha$. Each J_α is normal to $\dot{\gamma}$: a Jacobi field with $J_\alpha(0) = 0$ and $D_t J_\alpha(0) \perp \dot{\gamma}$ stays normal by the tangential-normal splitting of proposition 5.2.4. The $n-1$ fields $J_1, \dots, J_{n-1}$ are linearly independent as solutions of the Jacobi equation, because their initial data $(0, e_\alpha)$ are linearly independent.

The no-conjugate-point hypothesis makes these fields a frame away from $t = 0$. Fix $t \in (0, b]$. A relation $\sum_\alpha c_\alpha J_\alpha(t) = 0$ would exhibit the nonzero normal Jacobi field $\sum_\alpha c_\alpha J_\alpha$ (nonzero because the c_α are not all zero and the fields are independent) as vanishing at both 0 and t , making $\gamma(t)$ conjugate to $\gamma(0)$; this is excluded for $t \in (0, b]$ by hypothesis. Hence $\{J_1(t), \dots, J_{n-1}(t)\}$ is a basis of the $(n-1)$ -dimensional space of normal vectors at $\gamma(t)$ for every $t \in (0, b]$. In particular the target w is in their span at $t = b$, so there are constants $a_1, \dots, a_{n-1}$ with $w = \sum_\alpha a_\alpha J_\alpha(b)$, and $J := \sum_\alpha a_\alpha J_\alpha$ is the Jacobi field with $J(0) = 0$, $J(b) = w$. This proves existence; uniqueness holds because the difference of two such fields is a Jacobi field vanishing at 0 and b , hence zero by the no-conjugate-point hypothesis.

Step 2: expansion of a competitor. Let V be any piecewise-smooth normal field with $V(0) = 0$, $V(b) = w$. On the half-open interval $(0, b]$ the frame $\{J_\alpha\}$ is defined and pointwise a basis of the normal space, so V can be written

$$V(t) = \sum_{\alpha=1}^{n-1} \varphi^\alpha(t) J_\alpha(t), \quad t \in (0, b]$$

for scalar coefficient functions φ^α . Each φ^α is piecewise smooth on $(0, b]$ wherever V is smooth, being obtained from V and the smooth invertible frame $\{J_\alpha(t)\}$ by Cramer's rule. Two facts about the behavior at the endpoints are needed. At $t = b$, the equality $V(b) = w = \sum_\alpha a_\alpha J_\alpha(b)$ and the independence of the $J_\alpha(b)$ force $\varphi^\alpha(b) = a_\alpha$. At $t = 0$, the coefficients extend continuously with a finite limit: expanding $J_\alpha(t) = tE_\alpha(0) + O(t^2)$ near 0 (since $J_\alpha(0) = 0$, $D_t J_\alpha(0) = e_\alpha$) and $V(t) = O(t)$ (since $V(0) = 0$ and V is Lipschitz near 0, being piecewise smooth), the matrix relating V to the J_α shows that each $\varphi^\alpha(t)$ has a finite limit as $t \rightarrow 0^+$, and moreover $t(\varphi^\alpha)'(t) \rightarrow 0$; these are recorded now and used in the boundary evaluation of Step 3. The field $J = \sum_\alpha a_\alpha J_\alpha$ corresponds to the constant coefficients $\varphi^\alpha \equiv a_\alpha$.

Step 3: the key identity. The computation rests on two auxiliary quantities built from the frame.

Define, on any subinterval where V is smooth, the normal fields

$$A = \sum_{\alpha} (\varphi^{\alpha})' J_{\alpha}, \quad B = \sum_{\alpha} \varphi^{\alpha} D_t J_{\alpha}$$

so that $D_t V = A + B$ because $D_t V = \sum_{\alpha} ((\varphi^{\alpha})' J_{\alpha} + \varphi^{\alpha} D_t J_{\alpha})$. Two identities among the J_{α} are used. First, the *Lagrange identity* for Jacobi fields: for any two Jacobi fields J_{α}, J_{β} the quantity $\langle D_t J_{\alpha}, J_{\beta} \rangle - \langle J_{\alpha}, D_t J_{\beta} \rangle$ is constant in t , because its derivative is

$$\langle D_t^2 J_{\alpha}, J_{\beta} \rangle - \langle J_{\alpha}, D_t^2 J_{\beta} \rangle = -\langle R(J_{\alpha}, \dot{\gamma})\dot{\gamma}, J_{\beta} \rangle + \langle J_{\alpha}, R(J_{\beta}, \dot{\gamma})\dot{\gamma} \rangle = 0$$

the last equality by the pair symmetry $\langle R(J_{\alpha}, \dot{\gamma})\dot{\gamma}, J_{\beta} \rangle = \langle R(J_{\beta}, \dot{\gamma})\dot{\gamma}, J_{\alpha} \rangle$ from proposition 4.2.1. Evaluating the constant at $t = 0$, where every $J_{\alpha}(0) = 0$, gives

$$\langle D_t J_{\alpha}, J_{\beta} \rangle = \langle J_{\alpha}, D_t J_{\beta} \rangle \quad \text{for all } t \quad (5.8)$$

Second, $\langle A, B \rangle$ is symmetric under this identity:

$$\langle A, B \rangle = \sum_{\alpha, \beta} (\varphi^{\alpha})' \varphi^{\beta} \langle J_{\alpha}, D_t J_{\beta} \rangle = \sum_{\alpha, \beta} (\varphi^{\alpha})' \varphi^{\beta} \langle D_t J_{\alpha}, J_{\beta} \rangle$$

by (5.8), so $\langle A, B \rangle = \sum_{\alpha, \beta} \varphi^{\alpha} \varphi^{\beta} (\varphi^{\alpha})' \langle D_t J_{\alpha}, J_{\beta} \rangle$.

Now expand $|D_t V|^2 = |A|^2 + 2\langle A, B \rangle + |B|^2$ and treat the curvature term. Using the Jacobi equation $D_t^2 J_{\alpha} = -R(J_{\alpha}, \dot{\gamma})\dot{\gamma}$ and metric compatibility,

$$\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = \sum_{\alpha, \beta} \varphi^{\alpha} \varphi^{\beta} \langle R(J_{\alpha}, \dot{\gamma})\dot{\gamma}, J_{\beta} \rangle = -\sum_{\alpha, \beta} \varphi^{\alpha} \varphi^{\beta} \langle D_t^2 J_{\alpha}, J_{\beta} \rangle$$

The claim is that the integrand of I satisfies, on each smooth subinterval,

$$|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = |A|^2 + \frac{d}{dt} \langle B, V \rangle \quad (5.9)$$

To verify (5.9), compute the total derivative on the right using metric compatibility:

$$\frac{d}{dt} \langle B, V \rangle = \langle D_t B, V \rangle + \langle B, D_t V \rangle = \langle D_t B, V \rangle + \langle B, A \rangle + |B|^2$$

Here $D_t B = \sum_{\alpha} ((\varphi^{\alpha})' D_t J_{\alpha} + \varphi^{\alpha} D_t^2 J_{\alpha})$, so

$$\langle D_t B, V \rangle = \sum_{\alpha, \beta} \varphi^{\beta} (\varphi^{\alpha})' \langle D_t J_{\alpha}, J_{\beta} \rangle + \sum_{\alpha, \beta} \varphi^{\alpha} \varphi^{\beta} \langle D_t^2 J_{\alpha}, J_{\beta} \rangle = \langle A, B \rangle - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle$$

the first sum being $\langle A, B \rangle$ by the symmetric form derived above and the second being $-\langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle$ by the curvature computation. Substituting,

$$\frac{d}{dt} \langle B, V \rangle = \langle A, B \rangle - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle + \langle B, A \rangle + |B|^2 = 2\langle A, B \rangle + |B|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle$$

Adding $|A|^2$ to both sides yields $|A|^2 + \frac{d}{dt} \langle B, V \rangle = |A|^2 + 2\langle A, B \rangle + |B|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle = |D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle$, which is (5.9).

Integrate (5.9) over $[0, b]$. The field $\langle B, V \rangle$ is continuous across the interior breakpoints of V : V is continuous there by definition of $\mathcal{V}(\gamma)$, and $B = \sum_{\alpha} \varphi^{\alpha} D_t J_{\alpha}$ is continuous because the J_{α} are smooth and the φ^{α} are continuous (they are recovered from the continuous field V by the smooth invertible frame). Hence the total-derivative term telescopes across breakpoints, contributing only the endpoint values:

$$I(V, V) = \int_0^b |A|^2 dt + \left[\langle B, V \rangle \right]_0^b$$

At $t = b$, $V(b) = w$; at $t = 0$, $V(0) = 0$ and $B(0) = \sum_{\alpha} \varphi^{\alpha}(0) D_t J_{\alpha}(0) = \sum_{\alpha} \varphi^{\alpha}(0) e_{\alpha}$ is finite by Step 2, so $\langle B, V \rangle \rightarrow 0$ as $t \rightarrow 0^+$. Therefore the boundary term is $\langle B(b), w \rangle$, and

$$I(V, V) = \int_0^b |A|^2 dt + \langle B(b), w \rangle$$

The boundary term depends only on the endpoint data. At $t = b$ one has $\varphi^{\alpha}(b) = a_{\alpha}$ from Step 2, so $B(b) = \sum_{\alpha} a_{\alpha} D_t J_{\alpha}(b) = D_t J(b)$, and $\langle B(b), w \rangle = \langle D_t J(b), J(b) \rangle$, a quantity fixed by w alone and independent of which competitor V realizes w . Consequently

$$I(V, V) - I(J, J) = \int_0^b |A|^2 dt - \int_0^b |A_J|^2 dt$$

where A_J is the field A built from the coefficients of J . But J has constant coefficients $\varphi^{\alpha} \equiv a_{\alpha}$, so $(\varphi^{\alpha})' \equiv 0$ and $A_J \equiv 0$; hence $I(J, J) = \langle D_t J(b), J(b) \rangle$ and

$$I(V, V) - I(J, J) = \int_0^b |A|^2 dt = \int_0^b \left| \sum_{\alpha} (\varphi^{\alpha})' J_{\alpha} \right|^2 dt \geq 0$$

This proves $I(J, J) \leq I(V, V)$. Equality holds if and only if $A \equiv 0$ on $[0, b]$, that is $\sum_{\alpha} (\varphi^{\alpha})' J_{\alpha} \equiv 0$; since the $J_{\alpha}(t)$ are independent for $t \in (0, b]$, this forces $(\varphi^{\alpha})' \equiv 0$, so each φ^{α} is constant, equal to its value a_{α} at b . Then $V = \sum_{\alpha} a_{\alpha} J_{\alpha} = J$ on $(0, b]$, and by continuity at 0 on all of $[0, b]$. Thus equality holds exactly when $V = J$, completing the proof.

The index lemma is the technical engine of the minimization theory, and its two consequences run in opposite directions. Read one way, it says that on a geodesic without interior conjugate points the index form is as small as the endpoint data allow when the field is Jacobi, and strictly larger otherwise; this positivity underlies the fact that such geodesics do minimize among nearby curves. Read the other way, it isolates the failure mode: a Jacobi field vanishing at both endpoints is a nonzero element on which the two competing terms of the index form balance exactly, $I(J, J) = \langle D_t J(b), J(b) \rangle = 0$, and this borderline field is the seed of a strictly energy-decreasing deformation once the geodesic is pushed past the conjugate point. That second reading is the next theorem.

The construction that produces a negative direction of the index form past the first conjugate point is a broken-field argument. A Jacobi field J that vanishes at the conjugate parameter t_0 is truncated there and continued by zero; the resulting field lies in $\mathcal{V}(\gamma)$ but is not smooth at t_0 , and its index-form value is exactly zero. Adding a small multiple of a smooth field concentrated at t_0 , in the direction dictated by the jump in $D_t J$, then lowers the value below zero. The resulting competitor is still a piecewise-smooth field, so its second variation is computed through lemma 5.4.2. The following theorem carries this out.

Theorem 5.4.5: No minimization past a conjugate point

Let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic, and suppose $\gamma(t_0)$ is conjugate to $\gamma(0)$ along γ for some $t_0 \in (0, b)$ strictly interior. Then there is a proper variation of γ whose variation field $W \in \mathcal{V}(\gamma)$ satisfies $I(W, W) = E''(0) < 0$. Consequently γ is not a local minimum of the energy among fixed-endpoint curves, and γ does not minimize length from $\gamma(0)$ to $\gamma(b)$.

Proof:

Let J be a nonzero Jacobi field along γ with $J(0) = 0$ and $J(t_0) = 0$, which exists by the definition of conjugate point (definition 5.3.1). Its derivative at t_0 is nonzero: were $D_t J(t_0) = 0$ as well, then J would be the solution of the Jacobi equation with vanishing data $J(t_0) = 0$, $D_t J(t_0) = 0$, hence identically zero by the uniqueness in theorem 5.2.3, contradicting $J \neq 0$. Put $u := D_t J(t_0) \neq 0$. The field J is normal to $\dot{\gamma}$: it vanishes at 0 with $D_t J(0)$ necessarily normal, since a tangential Jacobi field has the form $(a + bt)\dot{\gamma}$ by proposition 5.2.4 and cannot vanish twice unless it is zero, so the tangential part of J is zero and J , together with u , is normal to $\dot{\gamma}$.

Step 1: the broken field has zero index. Define $Z \in \mathcal{V}(\gamma)$ by truncating J at t_0 and extending by zero:

$$Z(t) = \begin{cases} J(t), & 0 \leq t \leq t_0 \\ 0, & t_0 \leq t \leq b \end{cases}$$

This Z is continuous (both pieces give 0 at t_0 since $J(t_0) = 0$), piecewise smooth with a single breakpoint at t_0 , normal to $\dot{\gamma}$, and vanishing at 0 and b ; so $Z \in \mathcal{V}(\gamma)$. Its index-form value is computed from (5.7). On $[0, t_0]$ the field $Z = J$ is a smooth Jacobi field, so $D_t^2 Z + R(Z, \dot{\gamma})\dot{\gamma} = 0$ there; on $[t_0, b]$ it is identically zero, so the operator vanishes there too. The integral term in (5.7) thus vanishes. The only breakpoint is t_0 , where $D_t Z(t_0^-) = D_t J(t_0) = u$ and $D_t Z(t_0^+) = 0$, giving the jump $\Delta Z := D_t Z(t_0^+) - D_t Z(t_0^-) = -u$. Since $Z(t_0) = J(t_0) = 0$, the breakpoint term $\langle \Delta Z, Z(t_0) \rangle$ also vanishes. Hence

$$I(Z, Z) = 0$$

The field Z is nonzero, so this exhibits a nonzero element of $\mathcal{V}(\gamma)$ on which the index form is zero; the geodesic γ sits exactly at the boundary between minimizing and non-minimizing, and the next step tips it over.

Step 2: a rounding field that lowers the index. Choose a smooth field $Y \in \mathcal{V}(\gamma)$ concentrated near t_0 with

$$\langle Y(t_0), u \rangle = 1, \quad Y(0) = Y(b) = 0$$

Such a Y exists: let U be the parallel field along γ with $U(t_0) = u$, let $\rho: [0, b] \rightarrow \mathbb{R}$ be a smooth bump with $\rho(t_0) = 1$ and $\rho(0) = \rho(b) = 0$, and set $Y = \rho U$; then $\langle Y(t_0), u \rangle = \rho(t_0) |u|^2 = |u|^2 > 0$, and rescaling Y by $|u|^{-2}$ arranges $\langle Y(t_0), u \rangle = 1$. Now Y is smooth, hence has no jump in $D_t Y$, so by (5.7) the mixed value $I(Z, Y)$ receives no contribution from the integral over the pieces where Z satisfies the Jacobi equation or is zero, and picks up only the breakpoint term of Z at t_0 :

$$I(Z, Y) = -\langle \Delta Z, Y(t_0) \rangle = -\langle -u, Y(t_0) \rangle = \langle u, Y(t_0) \rangle = 1$$

Here (5.7) is applied with $V = Z$ (piecewise smooth, with its single jump at t_0) and $W = Y \in \mathcal{V}(\gamma)$; the integral term vanishes as in Step 1 and only the t_0 jump survives. By symmetry of I , $I(Y, Z) = 1$ as well.

Consider the one-parameter family of fields $W_\varepsilon := Z + \varepsilon Y \in \mathcal{V}(\gamma)$. By bilinearity and Step 1,

$$I(W_\varepsilon, W_\varepsilon) = I(Z, Z) + 2\varepsilon I(Z, Y) + \varepsilon^2 I(Y, Y) = 2\varepsilon + \varepsilon^2 I(Y, Y)$$

Since $I(Y, Y)$ is a fixed finite number, choosing $\varepsilon < 0$ with $|\varepsilon|$ small makes the linear term dominate: for $\varepsilon < 0$ small enough that $|\varepsilon| |I(Y, Y)| < 2$, one has $I(W_\varepsilon, W_\varepsilon) = \varepsilon(2 + \varepsilon I(Y, Y)) < 0$. Fix such an ε and set $W := W_\varepsilon$, a nonzero normal field in $\mathcal{V}(\gamma)$ with $I(W, W) < 0$.

Step 3: conclusion. The field W vanishes at both endpoints, so the construction of example 3.5.2 makes it the variation field of the proper variation $\Gamma(s, t) = \exp_{\gamma(t)}(sW(t))$ of γ . This W is piecewise smooth with a single corner at t_0 , coming from the corner of Z ; it is not a smooth field, so the smooth second-variation theorem does not apply to it directly. The bridge is lemma 5.4.2, which gives the second variation of energy of this variation as $E''(0) = I(W, W) < 0$. Because γ is a geodesic, the first variation vanishes for every proper variation (theorem 3.5.4), so along the variation Γ the energy $s \mapsto E(\gamma_s)$ has vanishing first derivative and negative second derivative at $s = 0$; hence $E(\gamma_s) < E(\gamma)$ for all small $s \neq 0$, and γ is not a local minimum of the energy among fixed-endpoint curves. Finally, minimizing length would force minimizing energy: for a fixed-endpoint competitor σ on $[0, b]$ one has $L(\sigma)^2 \leq 2b E(\sigma)$ with equality for constant speed, and the unit-speed geodesic γ satisfies $L(\gamma)^2 = 2b E(\gamma)$, so a curve γ_s with $E(\gamma_s) < E(\gamma)$ has $L(\gamma_s)^2 \leq 2b E(\gamma_s) < 2b E(\gamma) = L(\gamma)^2$, whence $L(\gamma_s) < L(\gamma)$. Thus γ does not minimize length either, as required.

The two theorems close the circle opened in section 5.1. Together with the local minimization of theorem 3.4.3 and the identification of conjugate points with singularities of the exponential map in theorem 5.3.2, they give a clean dichotomy along a geodesic emanating from $p = \gamma(0)$: up to the first conjugate point the geodesic is a local minimizer of energy and length, and immediately past it the geodesic fails to minimize, beaten by an explicit deformation. The constant-curvature computation of example 5.2.5 makes this quantitative on the round sphere S_r^n , where the first conjugate point sits at arc length πr : a great-circle arc shorter than πr minimizes, and one longer than πr does not, recovering the elementary fact that the long way around a great circle is not the shortest path. On a manifold of nonpositive curvature there are no conjugate points at all (proposition 5.3.3), the index form is positive definite on every geodesic segment, and no geodesic ever loses its minimizing property to a conjugate point.

The count of exactly how negative the index form becomes, past several conjugate points, is the content of a theorem too large to prove here.

5.4.6 Foreshadowing: The Morse Index Theorem The index form I of a geodesic $\gamma: [0, b] \rightarrow M$ is a symmetric bilinear form on the infinite-dimensional space $\mathcal{V}(\gamma)$, and its *index* is the maximal dimension of a subspace of $\mathcal{V}(\gamma)$ on which I is negative definite. The Morse index theorem asserts that this index is finite and equals the number of interior points $\gamma(t)$, $t \in (0, b)$, conjugate to $\gamma(0)$ along γ , each counted with its multiplicity as in definition 5.3.1. The theorem contains theorem 5.4.5 as its first increment: the appearance of one conjugate point in $(0, b)$ raises the index from 0 to at least 1, so I acquires a negative direction, which is exactly the field W constructed above. It also refines proposition 5.3.3: when there are no interior conjugate points the index is 0, so I is positive semidefinite, and by the index lemma theorem 5.4.4 in fact positive definite on normal fields, which is the analytic form of local minimization. The proof organizes $\mathcal{V}(\gamma)$ by cutting $[0, b]$ into subintervals short enough to carry no conjugate points, reducing I on each piece to a finite-dimensional form via the index lemma, and tracking how the index jumps as b increases through each conjugate value; the jump at a conjugate point equals its multiplicity. A complete account is given in John Milnor, *Morse Theory*, and in John M. Lee, *Introduction to Riemannian Manifolds*, where the theorem is proved from the index lemma established above. It is the bridge from this chapter to Morse theory proper, in which the conjugate points of geodesics become the critical-point data of the energy functional on the path space.

Completeness and Comparison Theorems

The local theory of the previous chapters becomes global here. Jacobi fields measured how neighboring geodesics spread, and conjugate points marked where they refocus; the present chapter turns those infinitesimal statements into theorems about the manifold as a whole, about its diameter, its topology, and its diffeomorphism type. The bridge is completeness, the hypothesis under which geodesics run forever and any two points can be joined by a shortest one.

The chapter opens with the Hopf–Rinow theorem, which reconciles the two meanings completeness could have, that geodesics extend indefinitely and that the distance function makes M a complete metric space, and shows they coincide and guarantee minimizing geodesics. This is a place where the Riemannian hypothesis is not cosmetic: the theorem fails for Lorentzian metrics, and the failure is flagged where it occurs. With minimizers in hand, the cut locus records exactly how far a geodesic stays shortest, and the injectivity radius quantifies it. The curvature then reasserts control, now globally and through comparison. Positive Ricci curvature bounds the diameter and forces compactness, by Bonnet–Myers; nonpositive sectional curvature unwraps the manifold to Euclidean space, by Cartan–Hadamard, completing the noncompact space forms left open in chapter 4. The Rauch comparison theorem is the general engine behind these results, comparing the growth of Jacobi fields under one-sided curvature bounds against the constant-curvature models, and it returns the conjugate-point statements of chapter 5 as special cases. Throughout, the recurring benchmark is the trichotomy of model spaces: the sphere refocuses and is compact, Euclidean space is flat and open, and hyperbolic space spreads and is a Cartan–Hadamard manifold.

6.1 Geodesic Completeness and the Hopf–Rinow Theorem

The exponential map of section 3.2 was defined on a domain $\mathcal{E}_p \subseteq T_p M$ that need not be all of $T_p M$: a geodesic may run off the manifold in finite time, as the geodesic aimed at the deleted origin does on $\mathbb{R}^n \setminus \{0\}$, reaching the puncture at a finite parameter and having nowhere to continue. This possibility has been flagged since definition 3.1.1 and set aside; the present section confronts it. The manifolds on which geodesics never fall off the edge are the *geodesically complete* ones, and they are the manifolds on which the local geometry of the preceding chapters becomes global. Only when geodesics run forever can one ask whether *every* pair of points is joined by a shortest one, and only then do curvature bounds control the manifold as a whole rather than a neighborhood at a time.

There is a second sense in which a manifold can be complete, older and purely metric. The distance function d_g of definition 3.4.1 makes M a metric space, and a metric space is complete when every Cauchy sequence converges. These two completeness conditions, geodesic and metric, are stated in different languages, one about the domain of an ODE and one about the convergence of sequences, and there is no obvious reason they should agree. The Hopf–Rinow theorem is the statement that they do agree, and that either of them supplies the conclusion one actually wants: that any two points of a connected complete manifold are joined by a minimizing geodesic, a global shortest path where the local theory of section 3.4 produced only local ones. The proof turns on a single construction, growing a minimizing geodesic outward from a point and showing it cannot stop short of its target, and this construction is where positive-definiteness of g is used decisively. The theorem is false for Lorentzian metrics, and the failure is not a technicality; it is recorded at the end of the section.

Throughout, (M, g) is a connected Riemannian manifold with its Levi-Civita connection, and d_g is the Riemannian distance of definition 3.4.1. Positive-definiteness is in force from the outset and will be flagged where it is the load-bearing hypothesis.

Definition 6.1.1: Geodesic Completeness

A Riemannian manifold (M, g) is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$; equivalently, if for every $p \in M$ the exponential map $\exp_p$ of definition 3.2.1 is defined on all of $T_p M$, that is $\mathcal{E}_p = T_p M$.

The two formulations agree by the rescaling identity of proposition 3.2.2. If every maximal geodesic is defined on all of $\mathbb{R}$, then for any p and any $v \in T_p M$ the geodesic γ_v is defined at $t = 1$, so $v \in \mathcal{E}_p$; conversely, if $\mathcal{E}_p = T_p M$ for every p , then given a maximal geodesic γ and any time t_1 in its domain, the vector $t_1 \dot{\gamma}(0) \in T_p M$ lies in $\mathcal{E}_p$, so $\gamma_{t_1 \dot{\gamma}(0)}(1) = \gamma(t_1)$ is defined, and letting t_1 range over all of $\mathbb{R}$ shows the maximal geodesic extends to all of $\mathbb{R}$. Completeness is thus a property one can test one point at a time, through the domain of a single exponential map, or globally, through the extension of every geodesic; the equivalence of these tests with metric completeness is the theorem below, and the further equivalence with testing $\exp_p$ at a *single* point is one of its more useful consequences.

The three model spaces sort themselves immediately, and they exhibit completeness alongside the three curvature signs.

Example 6.1.2: Completeness of the model spaces

Euclidean space $(\mathbb{R}^n, \bar{g})$ of example 1.1.2 is geodesically complete: its geodesics are the affine lines $\gamma_v(t) = p + tv$ of example 3.2.3, defined for all $t \in \mathbb{R}$, so $\mathcal{E}_p = T_p \mathbb{R}^n$ for every p . The round sphere S_r^n of example 1.3.4 is geodesically complete: its geodesics are the great circles traversed

at constant speed, periodic and hence defined for all $t \in \mathbb{R}$, as recorded in example 3.2.4, so $\mathcal{E}_p = T_p S_r^n$. Hyperbolic space H^n is geodesically complete as well: in the hyperboloid model of example 1.3.5 its geodesics are the constant-speed intersections with 2-planes through the origin of $\mathbb{R}^{1,n}$, unbounded curves defined for all real time. Thus all three space forms, of curvature $+1/r^2$, 0, and -1 , are complete, and none exhibits the finite-time breakdown of a punctured or open example.

By contrast, the open unit ball $B \subseteq \mathbb{R}^n$ with the Euclidean metric is *not* complete: a geodesic aimed at the boundary sphere reaches $|x| = 1$ at a finite parameter and cannot continue, so its maximal domain is a bounded interval and $\mathcal{E}_p \neq T_p B$. The same manifold with the hyperbolic (Poincaré ball) metric of example 1.3.5 *is* complete, the metric stretching the radial distance to the boundary to infinity so that no geodesic reaches it in finite time. Completeness is a joint property of the smooth manifold and the metric, not of the manifold alone.

The mismatch between the ball and hyperbolic space warns that the definition of completeness has content only relative to g . The distance d_g of example 3.4.4 and its Euclidean counterpart provide the metric side of the story. On $\mathbb{R}^n$ the Riemannian distance is the ordinary Euclidean distance, and $(\mathbb{R}^n, d_g)$ is a complete metric space in the classical sense; on the round sphere the distance is bounded by πr , yet (S_r^n, d_{g_r}) is again complete because it is compact. The open ball with the Euclidean metric fails metric completeness too: a sequence marching toward the boundary is Cauchy in d_g but has no limit in B . That the geodesic and metric failures coincide in every one of these examples is the pattern the theorem explains.

The proof of Hopf–Rinow rests on one local fact, that geodesics can always be continued for a short additional time from any point they reach, together with one global construction, that a minimizing geodesic can be grown outward from a point until it arrives at any prescribed target. The following theorem assembles these into the equivalence of every reasonable notion of completeness and, as its operative conclusion, the existence of minimizing geodesics between arbitrary points.

Theorem 6.1.3: Hopf–Rinow

Let (M, g) be a connected Riemannian manifold and let d_g be its Riemannian distance (definition 3.4.1). The following are equivalent.

1. The metric space (M, d_g) is complete: every d_g -Cauchy sequence converges.
2. M is geodesically complete (definition 6.1.1): every maximal geodesic is defined on all of $\mathbb{R}$.
3. There is a point $p \in M$ for which $\exp_p$ is defined on all of $T_p M$.
4. Every closed and d_g -bounded subset of M is compact.

Moreover, if any one of these holds, then for every pair of points $p, q \in M$ there is a minimizing geodesic from p to q : a geodesic γ with $\gamma(0) = p$, $\gamma(1) = q$, and $L(\gamma) = d_g(p, q)$.

Proof:

The scheme is (iii) $\Rightarrow$ minimizers from $p \Rightarrow$ (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), with the last implication trivial and the minimizing-geodesic conclusion produced along the way. The heart of the argument is the first implication, a construction due to Hopf and Rinow that builds a minimizing

geodesic from p to an arbitrary target assuming only that geodesics from p run forever.

Step 1: from (iii), any point is joined to p by a minimizing geodesic. Suppose $\exp_p$ is defined on all of $T_p M$, so every geodesic from p is defined for all time. Fix $q \in M$ with $q \neq p$ and set $\rho = d_g(p, q) > 0$; the goal is a geodesic of length ρ from p to q . Choose a geodesic sphere about p of a small radius δ on which theorem 3.4.3 and corollary 3.4.5 apply, with $0 < \delta < \rho$. The geodesic sphere $S_\delta = \exp_p(\{v \in T_p M \mid |v|_g = \delta\})$ is the continuous image of a Euclidean sphere, hence compact, and the function $x \mapsto d_g(x, q)$ is continuous on it, so it attains a minimum at some point $x_0 \in S_\delta$. Write $x_0 = \exp_p(\delta u)$ for a unit vector $u \in T_p M$, and let

$$\gamma(s) = \exp_p(su), \quad s \in [0, \infty)$$

be the unit-speed geodesic from p in the direction u , defined for all $s \geq 0$ by hypothesis (iii). The claim is that γ reaches q at parameter $s = \rho$, that is, $\gamma(\rho) = q$; since γ is unit-speed, its length from 0 to ρ is then $\rho = d_g(p, q)$, so $\gamma|_{[0, \rho]}$ is the desired minimizing geodesic (reparametrized to $[0, 1]$ if one insists on the endpoint convention).

First, the point x_0 realizes the distance from p to q through the sphere: every curve from p to q must cross S_δ , because q lies outside the geodesic ball of radius δ (as $\delta < \rho = d_g(p, q)$ and corollary 3.4.5 identifies that ball with the metric ball of radius δ), so

$$d_g(p, q) = \min_{x \in S_\delta} (d_g(p, x) + d_g(x, q)) = \delta + \min_{x \in S_\delta} d_g(x, q) = \delta + d_g(x_0, q) \quad (6.1)$$

Here $d_g(p, x) = \delta$ for every $x \in S_\delta$ by corollary 3.4.5; the first equality holds because a shortest route to q passes through the sphere and its length is at least the distance to the crossing point plus the distance from there to q , while conversely concatenating a radial geodesic to any $x \in S_\delta$ with a near-minimizing curve from x to q realizes the right-hand side up to any ε . Rearranging (6.1) gives $d_g(x_0, q) = \rho - \delta$, and since $x_0 = \gamma(\delta)$ this reads

$$d_g(\gamma(\delta), q) = \rho - \delta \quad (6.2)$$

Now consider the set of parameters along γ at which this exact accounting persists:

$$A = \{s \in [\delta, \rho] \mid d_g(\gamma(s), q) = \rho - s\}$$

The identity $d_g(\gamma(s), q) = \rho - s$ says that the geodesic has spent length s getting from p to $\gamma(s)$ and has exactly $\rho - s$ of length left to reach q , with no waste: it stays on a shortest route. By (6.2), $\delta \in A$, so A is nonempty. It is closed in $[\delta, \rho]$, being defined by the vanishing of the continuous function $s \mapsto d_g(\gamma(s), q) - (\rho - s)$. Let $s_0 = \sup A$; by closedness $s_0 \in A$, so

$$d_g(\gamma(s_0), q) = \rho - s_0 \quad (6.3)$$

The claim is $s_0 = \rho$. Suppose instead $s_0 < \rho$, and write $p_0 = \gamma(s_0)$. Then $p_0 \neq q$, since $d_g(p_0, q) = \rho - s_0 > 0$. Repeat the sphere construction at p_0 : choose a geodesic sphere $S_{\delta'} = \exp_{p_0}(\{w \mid |w|_g = \delta'\})$ of radius δ' small enough that theorem 3.4.3 and corollary 3.4.5 hold at p_0 and $\delta' < \rho - s_0$. As before, $x \mapsto d_g(x, q)$ attains its minimum on the compact sphere $S_{\delta'}$ at some x'_0 , and the analogue of (6.1) gives

$$d_g(p_0, q) = \delta' + d_g(x'_0, q), \quad \text{hence} \quad d_g(x'_0, q) = (\rho - s_0) - \delta' \quad (6.4)$$

using (6.3). Combining (6.4) with the triangle inequality,

$$d_g(p, x'_0) \geq d_g(p, q) - d_g(x'_0, q) = \rho - ((\rho - s_0) - \delta') = s_0 + \delta'$$

On the other hand, the concatenation of $\gamma|_{[0, s_0]}$ (length s_0 , a curve from p to p_0) with the radial geodesic of length δ' from p_0 to x'_0 is a curve from p to x'_0 of length $s_0 + \delta'$, so $d_g(p, x'_0) \leq s_0 + \delta'$. Therefore $d_g(p, x'_0) = s_0 + \delta'$, and this concatenated curve is length-minimizing from p to x'_0 . By proposition 3.4.7, a length-minimizing piecewise-smooth curve, once reparametrized by arc length, is a smooth geodesic with no corner; in particular the concatenation has no corner at p_0 , so the outgoing radial geodesic from p_0 to x'_0 continues γ smoothly, with the same unit velocity. By uniqueness of geodesics from given initial data (theorem 3.1.3), the radial geodesic from p_0 to x'_0 is exactly γ continued, so $x'_0 = \gamma(s_0 + \delta')$. Substituting into (6.4),

$$d_g(\gamma(s_0 + \delta'), q) = d_g(x'_0, q) = (\rho - s_0) - \delta' = \rho - (s_0 + \delta')$$

which says $s_0 + \delta' \in A$, contradicting $s_0 = \sup A$ since $\delta' > 0$. Hence $s_0 = \rho$, and (6.3) becomes $d_g(\gamma(\rho), q) = 0$, so $\gamma(\rho) = q$ by theorem 3.4.6. The geodesic $\gamma|_{[0, \rho]}$ is unit-speed of length $\rho = d_g(p, q)$, hence a minimizing geodesic from p to q . This proves the minimizing-geodesic conclusion from the base point p whose exponential map is everywhere defined.

Step 2: (iii) implies (iv). Let p be as in (iii), so every point is joined to p by a minimizing geodesic by Step 1. Let $C \subseteq M$ be closed and d_g -bounded, say $C \subseteq \{x \mid d_g(p, x) \leq R\}$ for some R . Every $x \in C$ satisfies $x = \exp_p(v)$ for a vector $v = v(x)$ with $|v|_g = d_g(p, x) \leq R$, namely the initial velocity of the minimizing geodesic from p to x produced in Step 1. Hence

$$C \subseteq \exp_p(\overline{B_R(0)}), \quad \overline{B_R(0)} = \left\{v \in T_p M \mid |v|_g \leq R\right\}$$

The closed ball $\overline{B_R(0)}$ is a compact subset of $T_p M$, and $\exp_p$ is continuous and defined on all of $T_p M$ by hypothesis (iii), so $\exp_p(\overline{B_R(0)})$ is compact. Thus C is a closed subset of a compact set, hence compact. This proves (iv).

Step 3: (iv) implies (i). Let (x_k) be a d_g -Cauchy sequence. A Cauchy sequence is bounded: there is N with $d_g(x_k, x_N) \leq 1$ for all $k \geq N$, so the whole sequence lies within d_g -distance $\max_{k < N} d_g(x_k, x_N) + 1$ of x_N , a bounded set. Its closure C is then closed and bounded, hence compact by (iv). A Cauchy sequence in a compact metric space has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges to the same limit. Therefore (x_k) converges in M , proving (i).

Step 4: (i) implies (ii). Assume (M, d_g) is complete, and let $\gamma: [0, b) \rightarrow M$ be a maximal unit-speed geodesic with $b < \infty$; a contradiction will show $b = \infty$, and rescaling the velocity then handles geodesics of any speed. Take any sequence $t_k \uparrow b$. For j, k , the geodesic segment of γ between t_j and t_k has length $|t_j - t_k|$, so

$$d_g(\gamma(t_j), \gamma(t_k)) \leq |t_j - t_k|$$

As (t_k) is Cauchy in $\mathbb{R}$, the sequence $(\gamma(t_k))$ is d_g -Cauchy, hence converges to a point $x_* \in M$ by (i). This limit does not depend on the chosen sequence: any two sequences increasing to b interleave, so their images share the limit x_* , and $\gamma(t) \rightarrow x_*$ as $t \uparrow b$. Now extend γ past b . Choose a *uniformly normal* neighborhood of x_* : a neighborhood W and a radius $\eta > 0$ such that for every $x \in W$ the exponential map $\exp_x$ is a diffeomorphism on the ball $\{w \in T_x M \mid |w|_g < \eta\}$, and any two points of W within distance η are joined by a unique minimizing geodesic lying

in W . Such neighborhoods exist by the smooth dependence of geodesics on their initial data (theorem 3.1.3), applied to the exponential map of the whole tangent bundle near $(x_*, 0)$; this is a standard consequence of section 3.2 used here as a local input. For t close enough to b , $\gamma(t) \in W$ and $b - t < \eta$, so the geodesic γ restricted to $[t, b)$ is a geodesic in W of length $b - t < \eta$; being a unit-speed geodesic starting at $\gamma(t)$ with initial velocity $\dot{\gamma}(t)$, it equals $s \mapsto \exp_{\gamma(t)}((s - t)\dot{\gamma}(t))$, which is defined for s in an interval of radius η about t , in particular past b . This extends γ smoothly to a geodesic on an interval strictly containing $[0, b]$, contradicting the maximality of $b < \infty$. Hence every maximal geodesic has domain all of $\mathbb{R}$, which is (ii).

Step 5: (ii) implies (iii). If every maximal geodesic is defined on all of $\mathbb{R}$, then in particular for a fixed p every geodesic from p is defined at time 1, so $\mathcal{E}_p = T_p M$; this is (iii) for that (indeed any) p .

The five implications close the cycle (iii) $\Rightarrow$ (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), so all four conditions are equivalent. Finally, when they hold, (iii) holds at every point p , so Step 1 applies with any p as base point and produces a minimizing geodesic from p to any q . This establishes the minimizing-geodesic conclusion in full, completing the proof.

The theorem does two jobs, and it is worth separating them. As an equivalence it collapses a list of a priori distinct hypotheses into one, so that “complete” may be used unambiguously: a Riemannian manifold is complete when any one, hence all, of (i) through (iv) holds. The reduction to (iii), completeness of a single exponential map, is the sharpest of these, and it is what makes completeness checkable in examples: to know that a manifold is complete it is enough to fix one point and verify that geodesics from it run forever. The second job is the minimizing-geodesic conclusion, which is the geometric payoff. The local theory of section 3.4 guaranteed shortest paths only inside a normal ball; Hopf–Rinow removes the locality, at the cost of the completeness hypothesis, and asserts that on a complete manifold the infimum defining $d_g(p, q)$ is attained by an actual geodesic, for every pair p, q . The construction that delivers it, growing γ outward and tracking the identity $d_g(\gamma(s), q) = \rho - s$, is the geometric content: at each stage the point of a small sphere closest to q continues the geodesic without a corner, and proposition 3.4.7 is exactly what forbids the corner and forces the continuation to lie along γ itself.

Two consequences record the conclusions that are used most often downstream, and they are separated out here because later chapters will cite them by name.

Corollary 6.1.4: Minimizing Geodesics on Complete Manifolds

If (M, g) is a connected complete Riemannian manifold, then any two points $p, q \in M$ are joined by a minimizing geodesic: a geodesic γ with endpoints p and q and $L(\gamma) = d_g(p, q)$.

Proof:

Completeness is any of the equivalent conditions of theorem 6.1.3, so in particular (iii) holds at p , and the final assertion of that theorem produces the minimizing geodesic from p to q , as claimed.

Corollary 6.1.5: Compact Manifolds Are Complete

Every compact connected Riemannian manifold is geodesically complete, and hence any two of its points are joined by a minimizing geodesic.

Proof:

Let M be compact and connected. A Cauchy sequence in the metric space (M, d_g) is in particular a sequence in the compact space M , so it has a convergent subsequence, and a Cauchy sequence with a convergent subsequence converges; hence (M, d_g) is complete, condition (i) of theorem 6.1.3. By the equivalence, M is geodesically complete, and by corollary 6.1.4 any two points are joined by a minimizing geodesic, as required.

The compactness corollary is the reason the round sphere and the flat torus need no separate completeness check: both are compact, so both are complete, and every pair of points on either is joined by a shortest geodesic. On the sphere this recovers, without any new work, the fact that the distance $d_{g_r}(p, q) = r \arccos(\langle p, q \rangle / r^2)$ of example 3.4.4 is realized by a great-circle arc for every pair p, q , including the antipodal pair, where infinitely many half great circles tie for the shortest. Noncompact complete examples, such as Euclidean and hyperbolic space, require condition (iii) instead, but there the single-exponential-map criterion makes the verification immediate: one point suffices, and every geodesic from it runs forever.

Every step of the theorem used the Riemannian, positive-definite, structure of g , and the reliance is not removable. The construction of Step 1 minimized the continuous function $d_g(-, q)$ over a compact geodesic sphere, which presumes d_g is a genuine distance, and the vanishing of d_g only on the diagonal (theorem 3.4.6) rests on positive-definiteness, as recorded there. The following warning makes precise that the theorem is not simply proved using positive-definiteness but is actually false without it.

6.1.6 Warning: Hopf–Rinow fails for Lorentzian metrics The equivalence and the minimizing-geodesic conclusion of theorem 6.1.3 are special to the Riemannian case; for a pseudo-Riemannian metric of indefinite signature (definition 1.1.3), in particular a Lorentzian metric of signature $(-, +, \dots, +)$, they all fail.

The obstruction is present already in the statement. An indefinite metric assigns negative and null values of $g(v, v)$, so the length functional $L(\gamma) = \int |\dot{\gamma}|_g dt$ is not defined as a real number for curves whose velocity is timelike or null, and there is no distance function d_g of the kind built in definition 3.4.1. Conditions (i) and (iv) of the theorem, which are statements about the metric space (M, d_g) , therefore have no meaning to begin with, so the equivalence with geodesic completeness cannot even be posed in the same terms. Geodesic completeness, condition (ii), remains meaningful, since geodesics are defined by the connection alone; but it no longer forces the existence of “shortest” geodesics, because there is nothing to minimize.

The divergence is concrete. Compactness no longer implies geodesic completeness: the Clifton–Pohl torus, the quotient of $\mathbb{R}^2 \setminus \{0\}$ with the Lorentzian metric $\frac{2dx dy}{x^2 + y^2}$ by the multiplication $(x, y) \mapsto (2x, 2y)$, is a compact Lorentzian manifold that is geodesically incomplete, so the Lorentzian analogue of corollary 6.1.5 is false. The Riemannian proof of that corollary passed through metric completeness of (M, d_g) , exactly the object an indefinite metric fails to provide, and no replacement holds. Even where geodesics are complete, two points of a Lorentzian manifold causally related by a timelike curve need not be joined by a length-*maximizing* geodesic, the Lorentzian counterpart of a minimizer, without a further global hypothesis (global hyperbolicity), which has no Riemannian analogue because on a Riemannian manifold completeness alone suffices. The single hypothesis of completeness carrying so much, four equivalent formulations and a global existence statement, is thus a Riemannian phenomenon, resting on positive-definiteness at every step, and it is one of the places where the passage from the Riemannian to the Lorentzian setting is a change of substance rather than of sign.

6.2 The Cut Locus and the Injectivity Radius

The Hopf–Rinow theorem of the previous section guarantees, on a complete manifold, that any two points are joined by a minimizing geodesic. It says nothing about how long that geodesic stays minimizing if extended past its endpoint. A geodesic leaving p is shortest near p , by the local minimizing theorem of section 3.4, but the property is local: continued far enough, a geodesic can be overtaken by a competitor and cease to realize the distance from p . The round sphere already displays this. A great-circle arc from a point p is the shortest path to each point it meets, until it reaches the antipode; the moment it passes the antipode it is longer than the arc coming the other way around, and it has stopped minimizing. The antipode is the exact instant of transition, and the goal of this section is to identify and study that instant on an arbitrary manifold.

Two mechanisms can end a geodesic’s run as a minimizer, and the sphere shows both at once. The first is refocusing: the geodesic reaches a conjugate point, where the exponential map degenerates and nearby geodesics from p collide, after which minimization is impossible by the second-variation obstruction of section 5.4. The second is competition: a second minimizing geodesic of the same length arrives from p , so the point is reached two ways and neither extends as a strict minimizer. The central theorem of the section shows that at the cut point, the place where minimization fails, at least one of these two must occur. The distance from p to the nearest cut point is the injectivity radius, the exact radius out to which the exponential map is a genuine chart, and it is the scale

on which the manifold looks like its tangent space. Throughout, (M, g) is a connected, complete Riemannian manifold; completeness is used to secure minimizers, and the positive-definiteness of g is what makes distances, and hence the cut locus, well behaved. Both hypotheses fail to have their consequences in the Lorentzian setting, where Hopf–Rinow is false and the cut-locus theory takes a different form.

The starting point is to measure, for a fixed direction, exactly how long the geodesic in that direction minimizes.

Definition 6.2.1: Cut Point and Cut Time

Let (M, g) be a complete, connected Riemannian manifold, let $p \in M$, and let $v \in T_p M$ be a unit vector, $|v|_g = 1$. Consider the unit-speed geodesic $\gamma_v(t) = \exp_p(tv)$ (definition 3.2.1), defined for all $t \geq 0$ by completeness (definition 6.1.1). The *cut time* of p in the direction v is

$$t_{\text{cut}}(v) = \sup \{t > 0 \mid d_g(p, \gamma_v(t)) = t\}$$

the supremum of times up to which γ_v minimizes the distance from p . If $t_{\text{cut}}(v) < \infty$, the point $\gamma_v(t_{\text{cut}}(v))$ is the *cut point* of p along γ_v . If $t_{\text{cut}}(v) = \infty$, the geodesic γ_v minimizes from p forever and there is no cut point in the direction v .

The definition records the largest arc on which a given geodesic is a shortest path. The condition $d_g(p, \gamma_v(t)) = t$ says exactly that the arc $\gamma_v|_{[0,t]}$, of length t since γ_v has unit speed, realizes the distance from p to its endpoint, and so is minimizing. The set of such t is nonempty and contains an interval $(0, \varepsilon)$: inside a geodesic ball the radial geodesic is the unique shortest path by theorem 3.4.3, so γ_v minimizes for small t . It is also an interval starting at 0, by the following monotonicity, which is the first structural fact and is used repeatedly below.

A geodesic that minimizes up to time t_1 minimizes on every shorter initial arc. Concretely, if $d_g(p, \gamma_v(t_1)) = t_1$ then $d_g(p, \gamma_v(t)) = t$ for every $t \in [0, t_1]$. Indeed the restriction $\gamma_v|_{[0,t]}$ is a path of length t from p to $\gamma_v(t)$, so $d_g(p, \gamma_v(t)) \leq t$; if strict inequality held, a shorter path from p to $\gamma_v(t)$ followed by $\gamma_v|_{[t,t_1]}$ would join p to $\gamma_v(t_1)$ in length less than $t + (t_1 - t) = t_1$, contradicting $d_g(p, \gamma_v(t_1)) = t_1$. Hence the set $\{t > 0 \mid d_g(p, \gamma_v(t)) = t\}$ is an interval of the form $(0, t_{\text{cut}}(v)]$ or $(0, \infty)$, and γ_v minimizes on $[0, t_{\text{cut}}(v)]$ and fails to minimize strictly beyond it, as the next paragraph confirms. The cut point is therefore the last point out to which the geodesic is a shortest path.

Two facts sharpen the picture and are recorded now for use throughout. First, minimization holds up to and including the cut time: at $t = t_{\text{cut}}(v)$ one has $d_g(p, \gamma_v(t)) = t$ by continuity of d_g (theorem 3.4.6) applied to the supremum, so the geodesic arc to the cut point is still minimizing, and the cut point is the last point with this property. Second, minimization fails immediately past the cut time: for $t > t_{\text{cut}}(v)$ one has $d_g(p, \gamma_v(t)) < t$, for otherwise t would belong to the defining set and exceed its supremum. Thus the cut point is exactly the transition from minimizing to non-minimizing along γ_v .

Collecting the cut points over all directions gives a subset of the manifold that encodes the global geometry of geodesics from p .

Definition 6.2.2: Cut Locus

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$. The *cut locus* of p is the set of all cut points of p ,

$$\text{Cut}(p) = \left\{ \gamma_v(t_{\text{cut}}(v)) \mid v \in T_p M, |v|_g = 1, t_{\text{cut}}(v) < \infty \right\} \subseteq M$$

Equivalently, writing $\widetilde{\text{Cut}}(p) = \left\{ t_{\text{cut}}(v) v \mid |v|_g = 1, t_{\text{cut}}(v) < \infty \right\} \subseteq T_p M$ for the *tangent cut locus*, one has $\text{Cut}(p) = \exp_p(\widetilde{\text{Cut}}(p))$.

The cut locus is the boundary, seen from p , between the region where the geodesics from p still minimize and the region where they no longer do. Points strictly inside are reached by a unique minimizing geodesic that extends past them; points of the cut locus are reached by a minimizing geodesic that goes no farther as a minimizer; points beyond are reached only by geodesics that have already given up minimizing. The tangent cut locus $\widetilde{\text{Cut}}(p)$ is the graph, in polar form, of the function $v \mapsto t_{\text{cut}}(v)$ on the unit sphere of $T_p M$; its image under $\exp_p$ is $\text{Cut}(p)$, and the open region inside it, the set of tv with $|v| = 1$ and $t < t_{\text{cut}}(v)$, is the *segment domain*, mapped diffeomorphically to $M \setminus \text{Cut}(p)$ by the exponential map, as proposition 6.2.6 will confirm on the ball where t_{cut} is bounded below. On a compact manifold every $t_{\text{cut}}(v)$ is finite, the segment domain is a bounded star-shaped set, and M is the disjoint union of the open segment domain's image with the cut locus.

The distance from p to its cut locus is the single number that controls how large a normal ball can be, and it deserves a name.

Definition 6.2.3: Injectivity Radius

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$. The *injectivity radius* of M at p is

$$\text{inj}(p) = d_g(p, \text{Cut}(p)) = \inf \{ d_g(p, q) \mid q \in \text{Cut}(p) \}$$

with the convention $\text{inj}(p) = \infty$ when $\text{Cut}(p) = \emptyset$. The *injectivity radius* of M is

$$\text{inj}(M) = \inf_{p \in M} \text{inj}(p)$$

Equivalently, since the distance from p to a cut point $\gamma_v(t_{\text{cut}}(v))$ is $t_{\text{cut}}(v)$ (the arc to the cut point being minimizing),

$$\text{inj}(p) = \inf \left\{ t_{\text{cut}}(v) \mid v \in T_p M, |v|_g = 1 \right\}$$

The two expressions for $\text{inj}(p)$ agree because the geodesic arc from p to a cut point $\gamma_v(t_{\text{cut}}(v))$ is minimizing, so its length $t_{\text{cut}}(v)$ equals $d_g(p, \gamma_v(t_{\text{cut}}(v)))$; taking the infimum over unit v of these distances is the same as taking the infimum of $d_g(p, q)$ over $q \in \text{Cut}(p)$. The quantity $\text{inj}(p)$ is the radius of the largest metric ball about p inside which the exponential map is a diffeomorphism, so that geodesic polar coordinates and normal coordinates are valid; below that radius the manifold is a faithful copy of a ball in $T_p M$, and $\text{inj}(M)$ is the uniform such radius valid at every point. It appears as a hypothesis or a controlled quantity throughout global Riemannian geometry, and a positive lower bound on $\text{inj}(M)$ is one of the standard ways of preventing a manifold from degenerating.

A first computation on the model spaces fixes the meaning of all three definitions before the general theory. The three space forms of constant curvature realize the three possibilities: a finite cut locus that is a single point, and no cut locus at all.

Example 6.2.4: Cut locus and injectivity radius of the round sphere

Let S_r^n be the round sphere of radius r (example 1.3.4), which is compact, hence complete (corollary 6.1.5). Fix $p \in S_r^n$ and a unit vector $v \in T_p S_r^n$. The unit-speed geodesic in the direction v is the great-circle arc computed in example 3.4.4,

$$\gamma_v(t) = \cos\left(\frac{t}{r}\right)p + \sin\left(\frac{t}{r}\right)(rv)$$

traced at unit speed, closing up after arc length $2\pi r$ and passing through the antipode $-p$ at arc length πr .

The cut time is πr in every direction. By the distance formula of example 3.4.4, for any point q on this great circle at arc length $t \in [0, \pi r]$ the distance $d_{g_r}(p, q)$ equals the length t of the short arc, so γ_v minimizes on $[0, \pi r]$ and $d_{g_r}(p, \gamma_v(t)) = t$ there. For $t \in (\pi r, 2\pi r)$ the point $\gamma_v(t)$ is reached from p by the complementary arc of the same great circle, of length $2\pi r - t < t$, so $d_{g_r}(p, \gamma_v(t)) \leq 2\pi r - t < t$ and γ_v no longer minimizes. Hence

$$t_{\text{cut}}(v) = \pi r \quad \text{for every unit } v \in T_p S_r^n$$

independent of the direction, by the homogeneity and isotropy of the sphere.

The cut locus is the antipode. The cut point in the direction v is $\gamma_v(\pi r) = -p$, the antipode, for every v . As v ranges over the unit sphere the cut point is always the same point $-p$, so

$$\text{Cut}(p) = \{-p\}$$

a single point. The tangent cut locus $\widetilde{\text{Cut}}(p) = \{\pi r v \mid |v|_{g_r} = 1\}$ is the entire sphere of radius πr in $T_p S_r^n$, and $\exp_p$ collapses this whole sphere to $-p$, matching the direct formula $\exp_p(v) = (\cos(|v|/r))p + \cdots$ of example 3.2.4, which sends every v of length πr to $-p$.

The injectivity radius is πr . By definition 6.2.3,

$$\text{inj}(p) = d_{g_r}(p, \text{Cut}(p)) = d_{g_r}(p, -p) = \pi r$$

and since the sphere is homogeneous this holds at every point, so $\text{inj}(S_r^n) = \pi r$.

Both mechanisms coincide here. The cut point $-p$ is simultaneously the first conjugate point of p , at distance πr with multiplicity $n-1$ by example 5.3.4, and a point reached by infinitely many distinct minimizing geodesics, the infinitely many half great circles from p to $-p$ recorded in example 3.4.4. On the sphere the two alternatives of the cut-point characterization below occur together. For the two remaining model spaces the situation is opposite: hyperbolic space H^n has curvature -1 and Euclidean space $\mathbb{R}^n$ has curvature 0 , both satisfying $K \leq 0$, so by proposition 5.3.3 there are no conjugate points, and, as the Cartan–Hadamard theorem of section 6.4 will show, $\exp_p$ is a global diffeomorphism and every geodesic minimizes forever. Hence for H^n and $\mathbb{R}^n$ one has $t_{\text{cut}}(v) = \infty$ in every direction, $\text{Cut}(p) = \emptyset$, and $\text{inj}(p) = \infty$ at every point.

The sphere shows the two ways a geodesic can stop minimizing occurring simultaneously; on a general manifold they need not, but one of them must. This is the central theorem of the section. It identifies

the geometric event at the cut point: either the exponential map has degenerated, so the point is conjugate, or two minimizing geodesics have arrived, so the point is reached ambiguously. The proof is a limiting argument. Approach the cut point along the geodesic from just beyond it, take minimizing geodesics to those approaching points, and extract a limit; the limit is either a second minimizing geodesic distinct from γ , or, if it coincides with γ , the coincidence forces the differential of the exponential map to be singular.

Theorem 6.2.5: Characterization of the Cut Point

Let (M, g) be a complete, connected Riemannian manifold, $p \in M$, and $v \in T_p M$ a unit vector with finite cut time $t_c = t_{\text{cut}}(v) < \infty$. Let $q = \gamma_v(t_c)$ be the cut point of p along γ_v . Then at least one of the following holds:

1. q is the first conjugate point of p along γ_v ; equivalently, $d(\exp_p)_{t_c v}$ is singular.
2. There exist two distinct minimizing geodesics from p to q ; that is, there is a unit vector $w \neq v$ in $T_p M$ with $\exp_p(t_c w) = q$ and $d_g(p, q) = t_c$.

Conversely, if at time t_c either $q = \gamma_v(t_c)$ is conjugate to p along γ_v or there are two distinct minimizing geodesics of length t_c from p to q , and γ_v minimizes on $[0, t_c]$, then $t_c = t_{\text{cut}}(v)$.

Proof:

Write $\gamma = \gamma_v$ throughout and set $q = \gamma(t_c)$, so $d_g(p, q) = t_c$ because the arc to the cut point is minimizing (the discussion after definition 6.2.1).

The direct implication. Choose a strictly decreasing sequence $t_k \downarrow t_c$ with $t_k > t_c$, and set $q_k = \gamma(t_k)$. By completeness and the Hopf-Rinow theorem (theorem 6.1.3, corollary 6.1.4) there is, for each k , a minimizing geodesic from p to q_k ; write it as $\sigma_k(t) = \exp_p(t w_k)$ for a unit vector $w_k \in T_p M$, defined on $[0, \ell_k]$ with $\ell_k = d_g(p, q_k)$, so that σ_k has unit speed, $\sigma_k(0) = p$, and $\sigma_k(\ell_k) = q_k$. Because $t_k > t_c = t_{\text{cut}}(v)$, the original geodesic γ does not minimize to q_k , so

$$\ell_k = d_g(p, q_k) < t_k$$

The lengths ℓ_k are bounded: $\ell_k = d_g(p, q_k) \leq d_g(p, q) + d_g(q, q_k) = t_c + (t_k - t_c) = t_k$ by the triangle inequality, and $q_k \rightarrow q$ gives $\ell_k = d_g(p, q_k) \rightarrow d_g(p, q) = t_c$ by continuity of the distance (theorem 3.4.6). The unit vectors w_k lie on the unit sphere of $T_p M$, which is compact, so after passing to a subsequence (not relabeled) $w_k \rightarrow w$ for some unit vector $w \in T_p M$. Passing to the limit in $\exp_p(\ell_k w_k) = q_k$, using $\ell_k \rightarrow t_c$, $w_k \rightarrow w$, $q_k \rightarrow q$, and continuity of $\exp_p$ (which is defined on all of $T_p M$ by completeness),

$$\exp_p(t_c w) = q$$

Thus $\sigma(t) = \exp_p(t w)$, $t \in [0, t_c]$, is a unit-speed geodesic from p to q of length $t_c = d_g(p, q)$, so σ is a minimizing geodesic from p to q . Two cases arise according to whether σ differs from γ .

Case A: $w \neq v$. Then σ and γ are two distinct minimizing geodesics from p to q , since geodesics are determined by their initial velocity (theorem 3.1.3) and $\dot{\sigma}(0) = w \neq v = \dot{\gamma}(0)$. This is alternative (2).

Case B: $w = v$. Then $\sigma = \gamma$, and it is shown that q is conjugate to p along γ , giving alternative (1). Suppose, toward a contradiction, that q is *not* conjugate to p along γ . By theorem 5.3.2, this means $d(\exp_p)_{t_c v}$ is nonsingular, hence a linear isomorphism, so by the inverse function

theorem $\exp_p$ is a diffeomorphism from an open neighborhood $\mathcal{U}$ of $t_c v$ in $T_p M$ onto an open neighborhood $\mathcal{V}$ of q in M . For k large the vectors $t_k v$ (which converge to $t_c v$) lie in $\mathcal{U}$, and the endpoints $q_k = \gamma(t_k) = \exp_p(t_k v)$ lie in $\mathcal{V}$. The vectors $\ell_k w_k$ converge to $t_c v$ as well (as $\ell_k \rightarrow t_c$ and $w_k \rightarrow w = v$), so for k large they too lie in $\mathcal{U}$, and they satisfy $\exp_p(\ell_k w_k) = q_k = \exp_p(t_k v)$. Since $\exp_p$ is injective on $\mathcal{U}$, this forces

$$\ell_k w_k = t_k v$$

for all large k . Taking norms, $\ell_k = |\ell_k w_k| = |t_k v| = t_k$ (both w_k and v are unit), contradicting $\ell_k < t_k$. Therefore q is conjugate to p along γ . It is moreover the *first* conjugate point: no interior point $\gamma(t_0)$ with $0 < t_0 < t_c$ can be conjugate to p , because on $[0, t_c]$ the geodesic γ is minimizing, and a geodesic does not minimize past its first conjugate point (theorem 5.4.5); a conjugate point at $t_0 < t_c$ would make $\gamma|_{[0, t_c]}$ non-minimizing. Hence $q = \gamma(t_c)$ is the first conjugate point, which is alternative (1).

In either case at least one of (1), (2) holds, proving the direct implication.

The converse. Assume $\gamma = \gamma_v$ minimizes on $[0, t_c]$, so $d_g(p, q) = t_c$ and $t_{\text{cut}}(v) \geq t_c$. It remains to show $t_{\text{cut}}(v) \leq t_c$, that is, γ fails to minimize past t_c , under either hypothesis.

Suppose first there are two distinct minimizing geodesics from p to q , namely $\gamma|_{[0, t_c]}$ and a second unit-speed geodesic $\sigma|_{[0, t_c]}$ with $\dot{\sigma}(0) = w \neq v$, both of length t_c . For $\varepsilon > 0$ small, consider the point $\gamma(t_c + \varepsilon)$ on the extension of γ . The broken path formed by following σ from p to q and then the arc of γ from $q = \gamma(t_c)$ to $\gamma(t_c + \varepsilon)$ has length $t_c + \varepsilon = d_g(p, q) + \varepsilon$. This broken path has a corner at q : its incoming velocity there is $\dot{\sigma}(t_c)$ and its outgoing velocity is $\dot{\gamma}(t_c)$, and these differ, for if $\dot{\sigma}(t_c) = \dot{\gamma}(t_c)$ then σ and γ would be the same geodesic by uniqueness (theorem 3.1.3), contradicting $w \neq v$. A piecewise-smooth path with a genuine corner is not a geodesic and can be strictly shortened by rounding the corner: replacing the two segments near q by the minimizing geodesic between two nearby interior points (which is strictly shorter than going through q , since inside a normal ball about q the unique shortest path between two points off the two segments is the radial construction of theorem 3.4.3, and the broken path is not that geodesic) yields a strictly shorter path from p to $\gamma(t_c + \varepsilon)$. Hence

$$d_g(p, \gamma(t_c + \varepsilon)) < t_c + \varepsilon$$

so γ does not minimize on $[0, t_c + \varepsilon]$, whence $t_{\text{cut}}(v) \leq t_c$.

Suppose instead that $q = \gamma(t_c)$ is conjugate to p along γ . A geodesic does not minimize past its first conjugate point (theorem 5.4.5): for every $t'_c > t_c$ there is a proper variation of $\gamma|_{[0, t'_c]}$ with fixed endpoints and strictly negative second variation of energy, so $\gamma|_{[0, t'_c]}$ is not a local minimizer of energy, hence not a minimizer of length among fixed-endpoint curves; consequently $d_g(p, \gamma(t'_c)) < t'_c$. Thus γ fails to minimize on $[0, t'_c]$ for every $t'_c > t_c$, and again $t_{\text{cut}}(v) \leq t_c$.

In both cases $t_{\text{cut}}(v) = t_c$, completing the converse and the proof.

The theorem is the exact dictionary the section was built to establish. It says that a geodesic stops minimizing for one of exactly two reasons, and both are geometrically legible: refocusing of nearby geodesics (a conjugate point, where $\exp_p$ folds) or the arrival of a competing shortest path (two minimizing geodesics, where $\exp_p$ overlaps). On the sphere the two reasons coincide at the antipode, as example 6.2.4 recorded, but the theorem does not require them to; a flat cylinder, obtained by rolling up the plane, has no conjugate points at all yet a nonempty cut locus, so that there the second alternative always occurs alone. The converse half is equally usable: to locate the cut point in a

given direction, it suffices to find the first time at which either a conjugate point occurs or a second minimizing geodesic appears, and the smaller of the two candidate times is the cut time. This is the computational content behind the model-space calculations and behind the estimates of the next sections.

A symmetry of the cut relation is worth extracting from the proof, as it is used in the injectivity-radius arguments below. If q is the cut point of p along a minimizing geodesic γ , then p is the cut point of q along the reversed geodesic. Both alternatives of the theorem are symmetric in p and q : a conjugate pair is a conjugate pair for the reversed geodesic with the same multiplicity (the observation in definition 5.3.1), and two distinct minimizing geodesics from p to q are two distinct minimizing geodesics from q to p when reversed. Consequently $p \in \text{Cut}(q)$ whenever $q \in \text{Cut}(p)$, so the cut relation is symmetric. In particular $d_g(p, \text{Cut}(p))$ and $d_g(q, \text{Cut}(q))$ are linked, and inj is controlled by the shortest closed geodesic and the shortest geodesic loop, a point returned to in the compact case.

With the cut point understood, the geometric meaning of the injectivity radius becomes a theorem: it is exactly the radius out to which the exponential map is a chart.

Proposition 6.2.6: The Exponential Map is a Diffeomorphism Within the Injectivity Radius

Let (M, g) be a complete, connected Riemannian manifold and $p \in M$ with injectivity radius $\text{inj}(p) > 0$. Then $\exp_p$ restricts to a diffeomorphism from the open ball

$$B_{\text{inj}(p)}(0) = \{u \in T_p M \mid |u|_g < \text{inj}(p)\}$$

onto the open metric ball $B_g(p, \text{inj}(p)) = \{x \in M \mid d_g(p, x) < \text{inj}(p)\}$, and this ball contains no cut point of p . The radius $\text{inj}(p)$ is the largest for which this holds: $\exp_p$ is not injective on any open ball of radius exceeding $\text{inj}(p)$.

Proof:

Write $\rho = \text{inj}(p)$; the case $\rho = \infty$ is read with $B_\rho(0) = T_p M$ and $B_g(p, \rho) = M$ throughout. Set $D = \{u \in T_p M \mid |u|_g < \rho\}$, the open ball of radius ρ .

Step 1: no geodesic from p has a cut point or conjugate point before radius ρ . By the second form of definition 6.2.3, $\rho = \inf_{|v|=1} t_{\text{cut}}(v)$, so $t_{\text{cut}}(v) \geq \rho$ for every unit v . Hence for every unit v the geodesic γ_v minimizes on $[0, \rho)$, and in particular has no cut point with parameter less than ρ . Moreover no γ_v has a conjugate point at parameter $t < \rho$: a conjugate point at $t < \rho \leq t_{\text{cut}}(v)$ would, by theorem 5.4.5, make γ_v non-minimizing on $[0, t']$ for $t < t' < \rho$, contradicting $t_{\text{cut}}(v) \geq \rho$. By theorem 5.3.2, the absence of conjugate points at parameters below ρ means $d(\exp_p)_u$ is nonsingular for every $u \in D$: writing $u = tv$ with $|v| = 1$ and $t = |u| < \rho$, the differential $d(\exp_p)_{tv}$ is nonsingular because $\gamma_v(t)$ is not conjugate to p . Thus $\exp_p$ is a local diffeomorphism at every point of D .

Step 2: $\exp_p$ is injective on D . Suppose $u_1, u_2 \in D$ with $\exp_p(u_1) = \exp_p(u_2) = x$. Write $u_i = t_i v_i$ with $|v_i| = 1$ and $t_i = |u_i| < \rho$. Each arc $\gamma_{v_i}|_{[0, t_i]}$ has length $t_i < \rho \leq t_{\text{cut}}(v_i)$, so it is minimizing, giving $d_g(p, x) = t_i$; hence $t_1 = t_2 =: t < \rho$. If $v_1 \neq v_2$, then $\gamma_{v_1}|_{[0, t]}$ and $\gamma_{v_2}|_{[0, t]}$ are two distinct minimizing geodesics from p to x of common length t . By the converse in theorem 6.2.5, the existence of two distinct minimizing geodesics of length t from p to x , together

with γ_{v_1} minimizing on $[0, t]$, forces $t_{\text{cut}}(v_1) = t < \rho$, contradicting $t_{\text{cut}}(v_1) \geq \rho$. Therefore $v_1 = v_2$, and with $t_1 = t_2$ this gives $u_1 = u_2$. So $\exp_p$ is injective on D .

Step 3: $\exp_p$ maps D diffeomorphically onto $B_g(p, \rho)$. By Steps 1 and 2, $\exp_p|_D$ is an injective local diffeomorphism, hence a diffeomorphism from D onto its image $\exp_p(D)$, an open subset of M . Identify the image. If $u \in D$ then, as in Step 2, $d_g(p, \exp_p(u)) = |u| < \rho$, so $\exp_p(D) \subseteq B_g(p, \rho)$. Conversely let $x \in B_g(p, \rho)$, so $d_g(p, x) < \rho$. By corollary 6.1.4 there is a minimizing geodesic from p to x ; write it $\exp_p(u) = x$ with $u = d_g(p, x)v$, $|v| = 1$, so $|u| = d_g(p, x) < \rho$ and $u \in D$. Hence $x \in \exp_p(D)$. Therefore $\exp_p(D) = B_g(p, \rho)$, and $\exp_p: D \rightarrow B_g(p, \rho)$ is a diffeomorphism.

Step 4: the ball contains no cut point, and ρ is optimal. Every point $x = \exp_p(u) \in B_g(p, \rho)$ satisfies $d_g(p, x) = |u| < \rho \leq \inf_v t_{\text{cut}}(v)$, so x is reached before any cut time and is not a cut point; equivalently $\text{Cut}(p) \cap B_g(p, \rho) = \emptyset$, consistent with $\rho = d_g(p, \text{Cut}(p))$. For optimality, suppose $\rho < \infty$ and let $\rho' > \rho$. By definition of ρ as the infimum of the cut times there is a unit v with $t_{\text{cut}}(v) < \rho'$; set $t_c = t_{\text{cut}}(v)$ and $q = \gamma_v(t_c)$. By theorem 6.2.5, either $d(\exp_p)_{t_c v}$ is singular, so $\exp_p$ is not a local diffeomorphism at $t_c v \in B_{\rho'}(0)$, or there is a unit $w \neq v$ with $\exp_p(t_c w) = q = \exp_p(t_c v)$, and then $t_c v, t_c w$ are two distinct points of $B_{\rho'}(0)$ (both of norm $t_c < \rho'$) with the same image. In either case $\exp_p$ fails to be a diffeomorphism onto its image on the open ball $B_{\rho'}(0)$. Hence no open ball of radius exceeding ρ is carried diffeomorphically by $\exp_p$, proving optimality and completing the proof.

The proposition is the payoff of the whole section. It says that the injectivity radius is precisely the size of the largest normal ball at p : inside $B_g(p, \text{inj}(p))$ the exponential map is a chart, normal coordinates are valid, every point is reached by a unique minimizing geodesic, and the local geometry is that of a ball in the tangent space; at radius $\text{inj}(p)$ the chart breaks down, for the reason the cut-point characterization supplies, either a fold of $\exp_p$ (conjugate point) or an overlap (two minimizers). On the sphere S_r^n this recovers the concrete picture of example 6.2.4: the exponential map is a diffeomorphism on the open ball of radius πr in $T_p S_r^n$, onto the sphere minus the antipode, and no larger ball works because the sphere of radius πr collapses to the single point $-p$. For $\mathbb{R}^n$ and H^n , where $\text{inj}(p) = \infty$, the proposition reduces to the global statement that $\exp_p$ is a diffeomorphism of all of $T_p M$ onto M , which is the Cartan–Hadamard conclusion proved in section 6.4; here it is anticipated as the case $\rho = \infty$, in which no cut point and no conjugate point ever occur.

A quantitative consequence for compact manifolds is immediate and closes the section. On a compact M every cut time is finite and the injectivity radius is a positive number attained as a distance, so it can be estimated by the two obstructions of the cut-point theorem: the length of the shortest closed geodesic and the distance to the first conjugate point along any geodesic. The distance to the first conjugate point is controlled by curvature through the Rauch comparison theorem of section 6.5, and the length of the shortest closed geodesic is a global invariant; together they give the standard lower bounds for $\text{inj}(M)$. That the infimum defining $\text{inj}(M)$ is positive on a compact manifold follows because $p \mapsto \text{inj}(p)$ is continuous and positive, so it attains a positive minimum on the compact M ; the continuity is a consequence of the two mechanisms in theorem 6.2.5 varying continuously with the base point and direction. The upshot is that a compact Riemannian manifold has a well-defined positive scale $\text{inj}(M)$ below which it is uniformly modeled on Euclidean balls, and this scale is the quantitative form of the local-to-global principle that opened the chapter.

6.3 Bonnet–Myers

The second variation of energy showed that a great-circle arc on the round sphere stops minimizing once it passes the antipode: past that point the positive curvature, accumulated over a long enough geodesic, overwhelms the bending term and produces a deformation that shortens the arc. The distance to the antipode is πr on S_r^n , and the sphere has no pair of points farther apart than that. This is not a coincidence of the sphere’s symmetry. A lower bound on curvature, imposed uniformly over the manifold, caps how long a geodesic can minimize, and therefore caps the distance between any two points. The Bonnet–Myers theorem is the precise form of this principle, and it needs only a lower bound on the Ricci curvature, an average of sectional curvatures rather than a bound on each one.

The mechanism is the one already assembled. Along a minimizing geodesic the index form is nonnegative, by corollary 5.1.2; if the geodesic is too long, a family of explicit test fields makes the index form negative in aggregate, so at least one of them makes it negative, contradicting minimization. Summing over $n - 1$ orthonormal normal fields converts the curvature terms into a single Ricci curvature, which is where the hypothesis enters at exactly the right strength. The conclusion is a bound on the diameter, and from a finite diameter on a complete manifold the topology follows at once: the manifold is compact, and, applying the same bound to its universal cover, its fundamental group is finite. Throughout, (M, g) is a connected Riemannian manifold of dimension n with Levi-Civita connection ∇ , curvature endomorphism R of definition 4.1.2 in the sign convention of box 4.1.1, and Ricci curvature Ric of definition 4.4.1; D_t is the covariant derivative along a curve, and the index form I is that of definition 5.4.1. The Einstein summation convention is not needed here; sums are written explicitly.

Theorem 6.3.1: Bonnet–Myers

Let (M, g) be a complete, connected Riemannian manifold of dimension $n \geq 2$, and suppose there is a constant $k > 0$ with

$$\text{Ric} \geq (n - 1)k g,$$

meaning $\text{Ric}(X, X) \geq (n - 1)k |X|^2$ for every $X \in T_p M$ and every $p \in M$. Then the diameter of M satisfies

$$\text{diam}(M) = \sup_{p, q \in M} d_g(p, q) \leq \frac{\pi}{\sqrt{k}}$$

Consequently M is compact, and its fundamental group $\pi_1(M)$ is finite.

The proof of the diameter bound occupies the bulk of the argument; compactness and the finiteness of π_1 are drawn out from it afterward and are separated into corollary 6.3.2. The heart of the matter is the following: a minimizing geodesic cannot be longer than $\pi/\sqrt{k}$, for if it were, the second variation would detect that it can be shortened.

Proof:

The claim is the diameter bound $\text{diam}(M) \leq \pi/\sqrt{k}$; the topological consequences are deferred to corollary 6.3.2. Suppose, for contradiction, that there are points $p, q \in M$ with $d_g(p, q) = L > \pi/\sqrt{k}$. Because M is complete and connected, corollary 6.1.4 (a consequence of the Hopf–Rinow theorem theorem 6.1.3) provides a minimizing geodesic from p to q ; parametrize it by arc length as $\gamma: [0, L] \rightarrow M$, so that γ has unit speed, $\gamma(0) = p$, $\gamma(L) = q$, and $L(\gamma) = d_g(p, q) = L$. Being

minimizing, γ minimizes length, hence energy among fixed-endpoint curves of the same speed, so corollary 5.1.2 applies: for every smooth field V along γ with $V(0) = V(L) = 0$,

$$I(V, V) = \int_0^L \left(|D_t V|^2 - \langle R(V, \dot{\gamma})\dot{\gamma}, V \rangle \right) dt \geq 0 \quad (6.5)$$

The strategy is to construct fields V that violate (6.5) once $L > \pi/\sqrt{k}$.

Step 1: a parallel orthonormal normal frame. The velocity $\dot{\gamma}$ is a parallel unit field along γ , since γ is a geodesic (definition 2.4.1). Choose an orthonormal basis $\{e_1, \dots, e_{n-1}\}$ of the orthogonal complement of $\dot{\gamma}(0)$ in $T_p M$, and let E_i be the parallel field along γ with $E_i(0) = e_i$, obtained by parallel transport (definition 2.4.1). Parallel transport is a linear isometry (proposition 2.4.4) and preserves the inner product with the parallel field $\dot{\gamma}$, so for every t the vectors $E_1(t), \dots, E_{n-1}(t)$ are orthonormal and orthogonal to $\dot{\gamma}(t)$. Together with $\dot{\gamma}(t)$ they form an orthonormal basis of $T_{\gamma(t)} M$.

Step 2: the test fields and their index-form values. For each $i = 1, \dots, n-1$ define the field

$$V_i(t) = \sin\left(\frac{\pi t}{L}\right) E_i(t)$$

along γ . Each V_i is smooth and vanishes at $t = 0$ and $t = L$, since $\sin 0 = \sin \pi = 0$; thus each V_i is admissible in (6.5). Because E_i is parallel, $D_t E_i = 0$, and

$$D_t V_i = \frac{\pi}{L} \cos\left(\frac{\pi t}{L}\right) E_i, \quad |D_t V_i|^2 = \frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right),$$

using $|E_i| \equiv 1$. For the curvature term, $V_i = \sin(\pi t/L) E_i$ gives

$$\langle R(V_i, \dot{\gamma})\dot{\gamma}, V_i \rangle = \sin^2\left(\frac{\pi t}{L}\right) \langle R(E_i, \dot{\gamma})\dot{\gamma}, E_i \rangle = \sin^2\left(\frac{\pi t}{L}\right) \text{Riem}(E_i, \dot{\gamma}, \dot{\gamma}, E_i),$$

by the bilinearity of R and the definition of the $(0, 4)$ -tensor (definition 4.1.4). Substituting into the index form,

$$I(V_i, V_i) = \int_0^L \left(\frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) \text{Riem}(E_i, \dot{\gamma}, \dot{\gamma}, E_i) \right) dt$$

Step 3: summing over the frame produces the Ricci curvature. Add the $n-1$ values. The bending terms are identical, so they contribute $(n-1) \frac{\pi^2}{L^2} \cos^2(\pi t/L)$, and the curvature terms combine into a trace:

$$\sum_{i=1}^{n-1} \text{Riem}(E_i, \dot{\gamma}, \dot{\gamma}, E_i) = \sum_{i=1}^{n-1} \langle R(E_i, \dot{\gamma})\dot{\gamma}, E_i \rangle$$

This sum is exactly the Ricci curvature in the direction $\dot{\gamma}$. By definition 4.4.1, in the sign convention of box 4.1.1,

$$\text{Ric}(\dot{\gamma}, \dot{\gamma}) = \sum_{a=1}^n \langle R(f_a, \dot{\gamma})\dot{\gamma}, f_a \rangle$$

for any orthonormal basis $\{f_a\}$ of $T_{\gamma(t)} M$; take the basis $\{f_a\} = \{\dot{\gamma}, E_1, \dots, E_{n-1}\}$ of Step 1. The term with $f_a = \dot{\gamma}$ vanishes, since $\langle R(\dot{\gamma}, \dot{\gamma})\dot{\gamma}, \dot{\gamma} \rangle = \text{Riem}(\dot{\gamma}, \dot{\gamma}, \dot{\gamma}, \dot{\gamma}) = 0$ by the antisymmetry

of the curvature tensor in its first pair (proposition 4.2.1). The remaining $n - 1$ terms are precisely the sum above, so

$$\sum_{i=1}^{n-1} \text{Riem}(E_i, \dot{\gamma}, \dot{\gamma}, E_i) = \text{Ric}(\dot{\gamma}, \dot{\gamma})$$

Therefore

$$\sum_{i=1}^{n-1} I(V_i, V_i) = \int_0^L \left((n-1) \frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - \sin^2\left(\frac{\pi t}{L}\right) \text{Ric}(\dot{\gamma}, \dot{\gamma}) \right) dt$$

Step 4: the curvature hypothesis makes the sum negative. The geodesic has unit speed, $|\dot{\gamma}| = 1$, so the hypothesis $\text{Ric} \geq (n-1)k g$ gives

$$\text{Ric}(\dot{\gamma}, \dot{\gamma}) \geq (n-1)k |\dot{\gamma}|^2 = (n-1)k$$

at every point of γ . Since $\sin^2(\pi t/L) \geq 0$, this bound gives $-\sin^2(\pi t/L) \text{Ric}(\dot{\gamma}, \dot{\gamma}) \leq -(n-1)k \sin^2(\pi t/L)$, so the sum is bounded above by

$$\sum_{i=1}^{n-1} I(V_i, V_i) \leq (n-1) \int_0^L \left(\frac{\pi^2}{L^2} \cos^2\left(\frac{\pi t}{L}\right) - k \sin^2\left(\frac{\pi t}{L}\right) \right) dt$$

The two elementary integrals over a half-period are equal,

$$\int_0^L \cos^2\left(\frac{\pi t}{L}\right) dt = \int_0^L \sin^2\left(\frac{\pi t}{L}\right) dt = \frac{L}{2},$$

so

$$\sum_{i=1}^{n-1} I(V_i, V_i) \leq (n-1) \cdot \frac{L}{2} \left(\frac{\pi^2}{L^2} - k \right) = \frac{(n-1)L}{2} \left(\frac{\pi^2}{L^2} - k \right)$$

By assumption $L > \pi/\sqrt{k}$, so $L^2 > \pi^2/k$, that is $\pi^2/L^2 < k$, and the factor $\pi^2/L^2 - k$ is strictly negative. Hence

$$\sum_{i=1}^{n-1} I(V_i, V_i) < 0$$

Step 5: contradiction. A sum of $n - 1$ real numbers is negative only if at least one summand is negative, so $I(V_{i_0}, V_{i_0}) < 0$ for some index i_0 . But V_{i_0} is a smooth field along the minimizing geodesic γ vanishing at both endpoints, so (6.5) forces $I(V_{i_0}, V_{i_0}) \geq 0$. This contradiction shows that no two points can be at distance greater than $\pi/\sqrt{k}$; that is, $d_g(p, q) \leq \pi/\sqrt{k}$ for all $p, q \in M$, and therefore

$$\text{diam}(M) = \sup_{p, q \in M} d_g(p, q) \leq \frac{\pi}{\sqrt{k}},$$

as we sought to show. The topological conclusions follow in corollary 6.3.2.

The proof isolates exactly where the hypothesis is used, and shows why Ricci curvature, not sectional curvature, is the right object. A single test field V_i sees only the sectional curvature of the plane spanned by E_i and $\dot{\gamma}$; a bound on that one plane would be enough to make one $I(V_i, V_i)$ negative, but that is a bound in a single direction and cannot be assumed. Summing over the whole orthonormal

frame $E_1, \dots, E_{n-1}$ averages these planes, and the average is precisely $\text{Ric}(\dot{\gamma}, \dot{\gamma})$, so the argument needs only a lower bound on this average. This is a weaker hypothesis than a lower bound on every sectional curvature, and it is the natural one: the pigeonhole step at the end, that a negative sum has a negative summand, is what lets an averaged bound do the work of a directional one. The number $\pi/\sqrt{k}$ is the half-period of the coefficient equation $f'' + kf = 0$, the same equation that governs Jacobi fields on a space of constant curvature k (example 5.2.5), which is why the threshold coincides with the first conjugate distance $\pi r = \pi/\sqrt{k}$ on the sphere S_r^n with $r = 1/\sqrt{k}$.

With the diameter bounded, the topology of M is constrained. A complete manifold of finite diameter is compact, and a compact manifold satisfying the same Ricci bound has, after passing to its universal cover, a finite fundamental group.

Corollary 6.3.2: Compactness and Finite Fundamental Group

Let (M, g) be a complete, connected Riemannian manifold of dimension $n \geq 2$ with $\text{Ric} \geq (n-1)kg$ for a constant $k > 0$. Then:

1. M is compact.
2. The fundamental group $\pi_1(M)$ is finite.

Proof:

Both statements rest on the diameter bound $\text{diam}(M) \leq \pi/\sqrt{k}$ of theorem 6.3.1.

1. Fix $p \in M$ and set $\rho = \pi/\sqrt{k}$. Every point $q \in M$ satisfies $d_g(p, q) \leq \text{diam}(M) \leq \rho$, so M equals the closed metric ball $\overline{B}_\rho(p) = \{q \in M \mid d_g(p, q) \leq \rho\}$. This set is closed and bounded, so by the Hopf–Rinow theorem (theorem 6.1.3, the equivalence of completeness with the compactness of closed bounded sets) it is compact. Hence M is compact.
2. The manifold M is a connected smooth manifold, hence locally path-connected and semilocally simply connected, so it admits a universal covering map $\pi: \tilde{M} \rightarrow M$ with $\tilde{M}$ connected and simply connected; this is the standard existence theorem for the universal cover of a manifold (see [3]). Equip $\tilde{M}$ with the pullback metric $\tilde{g} = \pi^*g$. Since π is a local diffeomorphism, $\tilde{g}$ is a Riemannian metric and $\pi: (\tilde{M}, \tilde{g}) \rightarrow (M, g)$ is a local isometry, indeed a Riemannian covering: each point of M has an evenly covered neighborhood on which π restricts to an isometry with each sheet.

Three properties of $\tilde{M}$ are needed. First, $\tilde{M}$ is complete. A maximal geodesic $\tilde{\sigma}$ of $\tilde{M}$ projects to a geodesic $\sigma = \pi \circ \tilde{\sigma}$ of M , because π is a local isometry and local isometries carry geodesics to geodesics; conversely, since M is complete, σ extends to all of $\mathbb{R}$, and by the unique path-lifting property of the covering π its extension lifts back through $\tilde{\sigma}$'s initial data to an extension of $\tilde{\sigma}$ over all of $\mathbb{R}$. Thus every maximal geodesic of $\tilde{M}$ is defined on all of $\mathbb{R}$, so $\tilde{M}$ is geodesically complete, hence complete by theorem 6.1.3 (the equivalence of geodesic completeness with metric completeness). Second, $\tilde{M}$ satisfies the same Ricci bound. Curvature is a local isometry invariant, and π is a local isometry, so the curvature tensor of $\tilde{g}$ at a point $\tilde{x}$ is the pullback of the curvature tensor of g at $\pi(\tilde{x})$; the Ricci tensor is a contraction of the curvature tensor with the metric, so likewise $\text{Ric}_{\tilde{g}} = \pi^* \text{Ric}_g$. Because π is a local isometry, $\text{Ric}_g \geq (n-1)kg$ pulls back to $\text{Ric}_{\tilde{g}} \geq (n-1)k\tilde{g}$. Third, $\tilde{M}$ has the same dimension n .

Therefore $(\tilde{M}, \tilde{g})$ is a complete, connected Riemannian n -manifold with $\text{Ric}_{\tilde{g}} \geq (n-1)k \tilde{g}$, and theorem 6.3.1 together with part (1) applies to it: $\tilde{M}$ is compact. Now consider a fiber $\pi^{-1}(p)$ over a point $p \in M$. It is a discrete closed subset of the compact space $\tilde{M}$, hence finite. For the universal cover, the fiber $\pi^{-1}(p)$ is in bijection with the deck transformation group, which is isomorphic to $\pi_1(M, p)$; this is the standard correspondence between the fiber of the universal cover and the fundamental group (see [3]). Hence $\pi_1(M)$ is finite, completing the proof.

The two consequences are of a different character from the diameter bound but follow from it with no further geometry. Compactness is immediate: a complete manifold whose points are all within a fixed distance of a base point is a closed bounded set, which Hopf–Rinow makes compact. The finiteness of π_1 is subtler only in that it passes to the universal cover, where the same curvature bound holds because curvature is local and the covering is a local isometry; the cover is then compact for the same reason M is, and a compact cover has finite fibers, which for the universal cover count the fundamental group. The contrapositive is a source of nonexistence results: a manifold with infinite fundamental group, such as a compact hyperbolic manifold or a flat torus, admits no metric with Ricci curvature bounded below by a positive constant. The flat torus $\mathbb{R}^n/\mathbb{Z}^n$ has $\text{Ric} \equiv 0$ and infinite fundamental group $\mathbb{Z}^n$, consistent with the theorem, since 0 is not bounded below by any $(n-1)k$ with $k > 0$; and although the torus is compact, its noncompact universal cover $\mathbb{R}^n$ shows directly that no positive Ricci lower bound can hold.

It remains to see that the diameter bound cannot be improved: the sphere of the matching radius attains it. This is the exact case in which every inequality in the proof becomes an equality.

Example 6.3.3: Sharpness on the round sphere

Fix $k > 0$ and let $M = S^n(1/\sqrt{k})$ be the round sphere of radius $r = 1/\sqrt{k}$ in $\mathbb{R}^{n+1}$, with the induced metric. It is compact and complete (a compact manifold is complete, and the sphere is compact), and its curvature was computed in Chapter 4. By example 4.4.9, the sphere S_r^n of radius r has

$$\text{Ric} = \frac{n-1}{r^2} g$$

With $r = 1/\sqrt{k}$ this is $\text{Ric} = (n-1)k g$, so the hypothesis of theorem 6.3.1 holds with equality: the Ricci curvature equals its lower bound in every direction, at every point. The theorem then gives $\text{diam}(M) \leq \pi/\sqrt{k}$.

This bound is attained. The distance function on S_r^n was computed in example 3.4.4: for $p, q \in S_r^n \subset \mathbb{R}^{n+1}$, with $\langle p, q \rangle$ the ambient inner product of $\mathbb{R}^{n+1}$,

$$d_g(p, q) = r \arccos\left(\frac{\langle p, q \rangle}{r^2}\right),$$

the length of the shorter great-circle arc joining them. The largest possible value of d_g occurs when $\langle p, q \rangle = -r^2$, that is when $q = -p$ is the antipode of p ; then $\arccos(-1) = \pi$ and

$$d_g(p, -p) = r\pi = \frac{\pi}{\sqrt{k}}$$

Since no pair of points is farther apart than an antipodal pair, $\text{diam}(S^n(1/\sqrt{k})) = \pi/\sqrt{k}$, exactly the Bonnet–Myers bound. The estimate of theorem 6.3.1 is therefore sharp, and the sphere is the extremal manifold.

The sharpness is legible inside the proof of theorem 6.3.1. On S_r^n the sectional curvature is the constant $1/r^2 = k$ in every plane (example 4.3.4), so $\text{Riem}(E_i, \dot{\gamma}, \dot{\gamma}, E_i) = k$ for each parallel normal field E_i along a unit-speed great circle, and $\text{Ric}(\dot{\gamma}, \dot{\gamma}) = (n-1)k$ with no slack in the inequality of Step 4. Taking a minimizing arc of length exactly $L = \pi/\sqrt{k} = \pi r$ to the antipode, each test field $V_i = \sin(\pi t/L)E_i$ gives

$$I(V_i, V_i) = \int_0^{\pi r} \left(\frac{\pi^2}{(\pi r)^2} \cos^2\left(\frac{t}{r}\right) - k \sin^2\left(\frac{t}{r}\right) \right) dt = \int_0^{\pi r} \left(\frac{1}{r^2} \cos^2\left(\frac{t}{r}\right) - \frac{1}{r^2} \sin^2\left(\frac{t}{r}\right) \right) dt = 0,$$

using $\pi/L = 1/r$ and $k = 1/r^2$. The value is zero, not negative: at the threshold length the second variation degenerates rather than turning negative, which is the borderline signalled at the antipode in example 5.1.3. Indeed each such $V_i = \sin(t/r)E_i$ is precisely the Jacobi field of example 5.2.5 vanishing at both ends of the arc to the first conjugate point, so the antipode is simultaneously the diameter-realizing point, the first conjugate point (example 5.3.4), and the cut point of p . Any length $L > \pi r$ pushes the arc past the antipode, makes the sum of Step 3 strictly negative as in the proof, and the arc ceases to minimize. This is the same phenomenon recorded from the second-variation side in example 5.1.3, now read as the extremal case of the diameter theorem.

Positive Ricci curvature thus imposes a hard ceiling on size and forces both compactness and near-triviality of the fundamental group, with the round sphere as the extremal model. The two remaining model spaces sit on the other side of the hypothesis. Euclidean space has $\text{Ric} \equiv 0$ and hyperbolic space has $\text{Ric} = -(n-1)g$ (example 4.4.9), so neither meets a positive Ricci lower bound, and neither is compact: both are diffeomorphic to $\mathbb{R}^n$, of infinite diameter. That noncompactness is not accidental but forced by the sign of the curvature is the content of the Cartan–Hadamard theorem of the next section, which unwraps a complete simply connected manifold of nonpositive curvature onto all of $\mathbb{R}^n$.

6.4 Cartan–Hadamard

Bonnet–Myers read one sign of the curvature and returned compactness: positive Ricci curvature refocuses geodesics, bounds the diameter, and closes the manifold up. The opposite sign runs the other way. On a manifold of nonpositive sectional curvature the geodesics from a point spread at least as fast as in Euclidean space and never come back together, so the exponential map develops no singularities. The proposition of section 5.3 already recorded the infinitesimal statement, that $d(\exp_p)_v$ is nonsingular everywhere; what remains is to make it global. The Cartan–Hadamard theorem does exactly that: it upgrades the everywhere-local diffeomorphism to a genuine diffeomorphism of the whole tangent space onto the manifold, provided the manifold is complete and simply connected. The conclusion is that a complete simply connected manifold with $K \leq 0$ is diffeomorphic to $\mathbb{R}^n$ and carries a unique geodesic between any two of its points.

Three ingredients combine, and each is worth isolating. Nonpositive curvature supplies the absence of conjugate points, hence the local diffeomorphism. Completeness supplies enough room for geodesics to run forever, which is what lets a local diffeomorphism be promoted to a covering map. Simple connectivity then forces the covering to be a single-sheeted one, that is, a diffeomorphism. Removing any one of the three destroys the conclusion: the flat torus has $K \equiv 0$ and is complete but not simply connected, and it is not diffeomorphic to $\mathbb{R}^n$; an open geodesic ball in $\mathbb{R}^n$ has $K \equiv 0$ and is simply connected but incomplete, and its exponential map is defined only on a bounded set. The

argument that follows is where the completeness hypothesis, and with it the Riemannian setting through Hopf–Rinow, does its work; the Lorentzian analogue fails for the same reason Hopf–Rinow does.

Throughout, (M, g) is a connected Riemannian manifold of dimension n with Levi-Civita connection ∇ (definition 2.3.2), curvature endomorphism R (definition 4.1.2), and sectional curvature K (definition 4.3.1); γ denotes a geodesic (definition 3.1.1) and D_t the covariant derivative along it. The curvature and second-variation sign conventions of Chapters 4 and 5 are in force. The Einstein summation convention is used where indices appear.

The metric estimate underlying the theorem is a quantitative version of the convexity argument that excluded conjugate points. There it was enough that $|J|^2$ be convex; here the same convexity is pushed to a lower bound on the growth of a Jacobi field, and thereby on the differential of the exponential map.

Proposition 6.4.1: Nonpositive Curvature Makes the Exponential Map Expanding

Let (M, g) have sectional curvature $K \leq 0$. Let $\gamma: [0, \infty) \rightarrow M$ be a geodesic and J a Jacobi field along γ with $J(0) = 0$. Then

$$|J(t)| \geq t |D_t J(0)| \quad \text{for all } t \geq 0 \quad (6.6)$$

Consequently, if $p \in M$ and $v \in T_p M$ lies in the domain of $\exp_p$, then for every $w \in T_p M$ (identifying $T_v(T_p M)$ with $T_p M$ as usual)

$$|d(\exp_p)_v(w)| \geq |w| \quad (6.7)$$

In particular $d(\exp_p)_v$ is injective at every v in the domain, and $\exp_p$ is distance nondecreasing wherever it is defined.

Proof:

The bound (6.6) is proved first, then transferred to the differential of $\exp_p$ through the Jacobi-field description of theorem 5.3.2.

Step 1: the lower bound on the Jacobi field. Write $f(t) = |J(t)|^2 = \langle J, J \rangle$. Differentiating twice with metric compatibility (definition 2.2.4), and using the Jacobi equation (5.1) in the form $D_t^2 J = -R(J, \dot{\gamma})\dot{\gamma}$ exactly as in the proof of proposition 5.3.3,

$$f''(t) = 2 \langle D_t^2 J, J \rangle + 2 |D_t J|^2 = -2 \operatorname{Riem}(J, \dot{\gamma}, \dot{\gamma}, J) + 2 |D_t J|^2$$

The term $\operatorname{Riem}(J, \dot{\gamma}, \dot{\gamma}, J) = K(\Pi)(|J|^2 |\dot{\gamma}|^2 - \langle J, \dot{\gamma} \rangle^2)$ is ≤ 0 , since $K \leq 0$ and the Gram determinant is nonnegative by Cauchy–Schwarz (definition 4.3.1); this identity holds also where J and $\dot{\gamma}$ are dependent, both sides then vanishing. Hence

$$f''(t) \geq 2 |D_t J|^2 \geq 0 \quad (6.8)$$

so $f = |J|^2$ is convex on $[0, \infty)$, with $f(0) = 0$.

Set $a = |D_t J(0)|$. If $a = 0$ then $J(0) = 0$ and $D_t J(0) = 0$, so $J \equiv 0$ by uniqueness of Jacobi fields from initial data (theorem 5.2.3) and (6.6) reads $0 \geq 0$. Assume $a > 0$; then $J \not\equiv 0$, so by proposition 5.3.3 the field J has no zero on $(0, \infty)$, and the norm $g(t) = |J(t)|$ is smooth and

positive there. The claim (6.6) is that $g(t) \geq at$, and it will follow from convexity of g . Consider the ratio

$$u(t) = \frac{g(t)}{t} = \frac{|J(t)|}{t} \quad (t > 0)$$

Because $J(0) = 0$ and J is smooth with initial derivative $D_t J(0)$, one has $|J(t)| = at + o(t)$ as $t \downarrow 0$, so $u(t) \rightarrow a$ as $t \downarrow 0$. It suffices to show g is convex on $(0, \infty)$; then, as shown below, u is nondecreasing, so $u(t) \geq \lim_{s \downarrow 0} u(s) = a$, which is the claim. Where $J \neq 0$ the identity $f = g^2$ gives $f' = 2gg'$ and $f'' = 2g'^2 + 2gg''$, whence $gg'' = \frac{1}{2}f'' - g'^2$. By (6.8), $\frac{1}{2}f'' \geq |D_t J|^2$; and by Cauchy–Schwarz $g' = \langle D_t J, J \rangle / |J|$ satisfies $g'^2 \leq |D_t J|^2$. Therefore

$$gg'' = \frac{1}{2}f'' - g'^2 \geq |D_t J|^2 - |D_t J|^2 \cdot \frac{\langle D_t J, J \rangle^2}{|D_t J|^2 |J|^2} \geq 0$$

whenever $D_t J \neq 0$, and $gg'' \geq 0$ trivially where $D_t J = 0$ (there $g'' \geq 0$ from (6.8)). Since $g > 0$ on $(0, \infty)$ this gives $g'' \geq 0$, so $g = |J|$ is convex on $(0, \infty)$; and g extends continuously to $t = 0$ with $g(0) = 0$, so it is convex on $[0, \infty)$. A convex function with $g(0) = 0$ has $g(t)/t$ nondecreasing: for $0 < s < t$, convexity with $g(0) = 0$ gives $g(s) = g(\frac{s}{t} \cdot t + (1 - \frac{s}{t}) \cdot 0) \leq \frac{s}{t}g(t)$, that is $g(s)/s \leq g(t)/t$. Letting $s \downarrow 0$ yields $u(t) = g(t)/t \geq a$ for every $t > 0$, which is (6.6); at $t = 0$ the inequality is the equality $0 = 0$.

Step 2: the differential of $\exp_p$. Fix $p \in M$, let $v \in T_p M$ be in the domain of $\exp_p$, and let $w \in T_p M$. If $v = 0$ then $d(\exp_p)_0 = \text{id}$ (proposition 3.2.5) and (6.7) is an equality. Assume $v \neq 0$ and set $\gamma(t) = \exp_p(tv)$, the geodesic with $\dot{\gamma}(0) = v$, defined for t in an interval containing $[0, 1]$. By part (1) of theorem 5.3.2 the field

$$J_w(t) = d(\exp_p)_{tv}(tw)$$

is the Jacobi field along γ with $J_w(0) = 0$ and $D_t J_w(0) = w$. Applying (6.6) at $t = 1$,

$$|d(\exp_p)_v(w)| = |J_w(1)| \geq 1 \cdot |D_t J_w(0)| = |w|$$

which is (6.7). In particular $d(\exp_p)_v(w) = 0$ forces $w = 0$, so $d(\exp_p)_v$ is injective, and being a linear map between spaces of equal dimension n it is a linear isomorphism. That $\exp_p$ is distance nondecreasing is the infinitesimal statement integrated: for a smooth curve σ in the domain, the velocity of $\exp_p \circ \sigma$ is $d(\exp_p)_\sigma(\dot{\sigma})$, whose length is $\geq |\dot{\sigma}|$ by (6.7), so $L(\exp_p \circ \sigma) \geq L(\sigma)$, as claimed.

The inequality (6.7) is the metric heart of the theorem. It says that under nonpositive curvature the exponential map, viewed as a map from the flat tangent space into M , never contracts: distances measured in $T_p M$ are lower bounds for the distances of their images. The model case makes the estimate transparent. On Euclidean space $\mathbb{R}^n$, where $K \equiv 0$, the exponential map at the origin is the identity of $\mathbb{R}^n$ and (6.7) is an equality, with the Jacobi field $J_w(t) = tw$ satisfying $|J_w(t)| = t|w|$ exactly. On hyperbolic space H^n , where $K \equiv -1$, the normal Jacobi field with $J(0) = 0$ and $|D_t J(0)| = 1$ is $J(t) = \sinh(t) E(t)$ (example 5.2.5), so $|J(t)| = \sinh t \geq t$ for $t \geq 0$: the exponential map strictly expands, and the more so as t grows, which is the analytic shadow of hyperbolic space being larger than Euclidean space at every radius. The estimate degenerates for positive curvature, and indeed must: on S_r^n the coefficient $\sin(t/r)$ returns to zero at $t = \pi r$, so no lower bound of the form $t|D_t J(0)|$ can hold, and the exponential map is singular there.

With the differential of $\exp_p$ under control, the main theorem is a matter of promoting a local

diffeomorphism to a global one. The step that requires completeness is the covering-map property; the following lemma isolates it.

Lemma 6.4.2: A Distance-Nondecreasing Local Diffeomorphism from a Complete Manifold is a Covering

Let $\tilde{N}$ and N be connected Riemannian manifolds, with $\tilde{N}$ complete, and let $\Phi: \tilde{N} \rightarrow N$ be a smooth surjective local diffeomorphism such that $|d\Phi_x(u)| \geq |u|$ for every $x \in \tilde{N}$ and every $u \in T_x\tilde{N}$. Then Φ is a covering map.

Proof:

The idea is the path-lifting criterion: a local diffeomorphism Φ under which every geodesic of N lifts, from any chosen starting preimage, to a curve defined for all parameter values, is a covering map. Completeness of $\tilde{N}$ together with the expanding estimate supplies exactly this lifting. The argument is given in three parts.

1. *Φ does not decrease lengths, hence does not decrease distances.* For any piecewise-smooth curve σ in $\tilde{N}$, the hypothesis $|d\Phi(u)| \geq |u|$ gives $L(\Phi \circ \sigma) \geq L(\sigma)$ pointwise on velocities, so $L(\Phi \circ \sigma) \geq L(\sigma)$. Taking infima over curves joining two points $x, y \in \tilde{N}$, and noting $\Phi \circ \sigma$ joins $\Phi(x)$ to $\Phi(y)$,

$$d_N(\Phi(x), \Phi(y)) \leq d_{\tilde{N}}(x, y)$$

Reversing the estimate is not needed; this direction alone is used below to keep lifts within controlled distance.

2. *Geodesics of N lift for all time.* Let $c: [0, \ell] \rightarrow N$ be a geodesic (in particular a smooth curve) and let $\tilde{x}_0 \in \Phi^{-1}(c(0))$. Because Φ is a local diffeomorphism, there is a unique lift $\tilde{c}$ of c with $\tilde{c}(0) = \tilde{x}_0$ on some maximal subinterval $[0, \beta) \subset [0, \ell]$ (uniqueness and local existence of the lift are the standard consequence of Φ being a local diffeomorphism: on a neighborhood where Φ inverts, set $\tilde{c} = \Phi^{-1} \circ c$, and patch). Suppose $\beta \leq \ell$. The lift is a curve into the *complete* manifold $\tilde{N}$; its length is controlled by part (1) applied to $\Phi \circ \tilde{c} = c$: for $0 \leq s < t < \beta$,

$$d_{\tilde{N}}(\tilde{c}(s), \tilde{c}(t)) \leq L(\tilde{c}|_{[s,t]}) \leq L(c|_{[s,t]}) = \int_s^t |\dot{c}|$$

the middle inequality because the expanding estimate gives $|\dot{\tilde{c}}| = |(\Phi^{-1} \circ c)| \leq |\dot{c}|$ (from $|d\Phi(u)| \geq |u|$, the inverse contracts). Since $|\dot{c}|$ is bounded on $[0, \ell]$ (a geodesic has constant speed), the right-hand side tends to 0 as $s, t \uparrow \beta$, so $\tilde{c}(t)$ is a Cauchy net in $\tilde{N}$ as $t \uparrow \beta$. By completeness of $\tilde{N}$ (theorem 6.1.3, in the equivalence of metric completeness with geodesic completeness) the limit $\tilde{x}_\beta = \lim_{t \uparrow \beta} \tilde{c}(t)$ exists in $\tilde{N}$, and by continuity $\Phi(\tilde{x}_\beta) = c(\beta)$. Now Φ is a local diffeomorphism at $\tilde{x}_\beta$, so the lift extends past β (again by inverting Φ near $\tilde{x}_\beta$), contradicting maximality unless $\beta = \ell$. Hence $\beta = \ell$ and every geodesic of N lifts on its full parameter interval, from any prescribed starting preimage.

3. *The lifting property gives evenly covered neighborhoods.* Fix $q \in N$. Since N is a manifold, choose $\varepsilon > 0$ with $\exp_q$ a diffeomorphism of the ball $B(0, \varepsilon) \subset T_q N$ onto a geodesic ball $U = B(q, \varepsilon)$ (theorem 3.4.3 provides such a normal ball). Every point of U is joined to q

by a unique radial geodesic inside U . Let $\tilde{q} \in \Phi^{-1}(q)$ be one preimage. For $x \in U$, let c_x be the radial geodesic from q to x ; by part (2) it lifts uniquely to a curve $\tilde{c}_x$ starting at $\tilde{q}$, and define $\lambda_{\tilde{q}}(x) = \tilde{c}_x(1)$ (parametrizing c_x on $[0, 1]$). This $\lambda_{\tilde{q}}: U \rightarrow \tilde{N}$ is smooth (lifts depend smoothly on the smoothly varying geodesics), satisfies $\Phi \circ \lambda_{\tilde{q}} = \text{id}_U$, and maps q to $\tilde{q}$; so it is a smooth section of Φ over U through $\tilde{q}$, and being a right inverse of the local diffeomorphism Φ it is a diffeomorphism of U onto its image $V_{\tilde{q}} \subset \tilde{N}$, an open set. The sets $V_{\tilde{q}}$, one for each $\tilde{q} \in \Phi^{-1}(q)$, are pairwise disjoint: if $V_{\tilde{q}} \cap V_{\tilde{q}'} \neq \emptyset$, two lifts of some radial geodesic c_x would agree at their common endpoint and hence, by uniqueness of lifts (part (2)), everywhere, forcing $\tilde{q} = \tilde{q}'$. Their union is all of $\Phi^{-1}(U)$: given $\tilde{x} \in \Phi^{-1}(U)$, let $x = \Phi(\tilde{x}) \in U$ and lift the *reversed* radial geodesic c_x (from x back to q) starting at $\tilde{x}$; its endpoint is a preimage $\tilde{q}$ of q , and running the lift forward exhibits $\tilde{x} \in V_{\tilde{q}}$. Finally Φ restricts to a diffeomorphism $V_{\tilde{q}} \rightarrow U$ for each $\tilde{q}$, being the inverse of the diffeomorphism $\lambda_{\tilde{q}}$. Thus U is evenly covered. As q was arbitrary and Φ is surjective, Φ is a covering map, as we sought to show.

The lemma is the point at which completeness is spent, and it is worth naming what it buys. A local diffeomorphism need not be a covering: the exponential chart of an incomplete manifold, or the restriction of a covering to a proper open subset, is a local diffeomorphism that fails to cover. What rules out such pathologies is that geodesics downstairs can always be lifted upstairs for all time, and it is completeness of the source, delivered by Hopf–Rinow (theorem 6.1.3) in its guise as the equivalence of metric and geodesic completeness, that guarantees the lift never escapes to a boundary. This is the sole ingredient of the whole development for which the Riemannian hypothesis, rather than mere smoothness, is used, and it is the ingredient that has no Lorentzian counterpart.

Theorem 6.4.3: Cartan–Hadamard

Let (M, g) be a complete, connected, simply connected Riemannian manifold of dimension n with sectional curvature $K \leq 0$. Then for every $p \in M$ the exponential map

$$\exp_p: T_p M \longrightarrow M$$

is a diffeomorphism. In particular M is diffeomorphic to $\mathbb{R}^n$, and M has no conjugate points along any geodesic.

Proof:

Fix $p \in M$. The proof assembles three facts, in the order local diffeomorphism, then covering map, then diffeomorphism.

Step 1: $\exp_p$ is defined on all of $T_p M$ and is a local diffeomorphism. Completeness means geodesics are defined for all time (definition 6.1.1), and by Hopf–Rinow (theorem 6.1.3) this is equivalent to $\exp_p$ being defined on the whole of $T_p M$. Since $K \leq 0$, proposition 5.3.3 shows $d(\exp_p)_v$ is nonsingular at every $v \in T_p M$; equivalently, by proposition 6.4.1, it satisfies $|d(\exp_p)_v(w)| \geq |w|$ and is a linear isomorphism at each v . By the inverse function theorem $\exp_p$ is a local diffeomorphism at each point of $T_p M$.

Step 2: transporting the metric and applying the covering lemma. Equip the vector space $T_p M$ with the pullback metric $\hat{g} = \exp_p^* g$, and write $\Phi = \exp_p: (T_p M, \hat{g}) \rightarrow (M, g)$. This is a genuine Riemannian metric on $T_p M$, because $\exp_p$ is a local diffeomorphism (Step 1), so each $d\Phi_v$ is a linear isomorphism and $\hat{g}_v(u, u) = g(d\Phi_v u, d\Phi_v u) > 0$ for $u \neq 0$. By construction Φ is then a

local isometry, so $|d\Phi_v(u)|_g = |u|_{\hat{g}}$ for all u , and in particular the hypothesis $|d\Phi(u)| \geq |u|$ of lemma 6.4.2 holds with equality.

The two remaining hypotheses of the lemma are surjectivity of Φ and completeness of $(T_p M, \hat{g})$. Surjectivity: M is complete and connected, so by Hopf–Rinow (theorem 6.1.3, corollary 6.1.4) every point $q \in M$ is joined to p by a minimizing geodesic $t \mapsto \exp_p(tu)$, whence $q = \exp_p(w)$ lies in the image. Completeness: a straight line $t \mapsto tu$ through the origin of $T_p M$ is mapped by Φ to the curve $t \mapsto \exp_p(tu)$, which is a geodesic of M ; since Φ is a local isometry and carries this line to a geodesic, and a local isometry sends geodesics to geodesics and back, the line $t \mapsto tu$ is a $\hat{g}$ -geodesic. Its velocity at $t = 0$ is u (identifying $T_0(T_p M)$ with $T_p M$), so the exponential map of $(T_p M, \hat{g})$ at the origin, $\exp_0^{\hat{g}}(u) = (\text{the } \hat{g}\text{-geodesic from } 0 \text{ with velocity } u \text{ at time } 1)$, equals u and is defined for every $u \in T_p M$. Thus $\exp_0^{\hat{g}}$ is defined on all of $T_0(T_p M)$, which by criterion (iii) of Hopf–Rinow (theorem 6.1.3) makes $(T_p M, \hat{g})$ complete. All hypotheses of lemma 6.4.2 are met, so $\Phi = \exp_p$ is a covering map.

Step 3: simple connectivity forces a single sheet. The base M is simply connected. A covering map onto a simply connected, connected space, with connected total space, has exactly one sheet: the number of sheets equals the index of $\Phi_* \pi_1(T_p M)$ in $\pi_1(M)$, and $\pi_1(M) = 1$, so the covering is a homeomorphism onto M (the covering criterion and this sheet count are standard covering-space theory, [1]). The total space $T_p M$ is connected (indeed contractible), which is required for the single-sheet conclusion. Being a covering map that is bijective, $\Phi = \exp_p$ is a homeomorphism, and being in addition a local diffeomorphism it is a diffeomorphism $T_p M \rightarrow M$.

The stated consequences follow. Since $T_p M \cong \mathbb{R}^n$ as a smooth manifold and $\exp_p$ is a diffeomorphism, M is diffeomorphic to $\mathbb{R}^n$. That M has no conjugate points is proposition 5.3.3 again, now recorded as a feature of the diffeomorphism: $\exp_p$ nonsingular everywhere means no geodesic through p meets a conjugate point, and p was arbitrary. This completes the proof.

The theorem says that nonpositive curvature, imposed on a complete simply connected manifold, forces the underlying topology to be that of $\mathbb{R}^n$ and nothing else: whatever the manifold looked like, it is diffeomorphic to a vector space, and its geometry is displayed globally in a single exponential chart. This is the promised complement to Bonnet–Myers. There, positive Ricci curvature made the manifold small and compact; here, nonpositive sectional curvature makes it, topologically, as large and open as possible. The two model spaces sit at the extremes: the round sphere realizes the Bonnet–Myers conclusion, and hyperbolic space realizes the Cartan–Hadamard conclusion, with Euclidean space marking the flat boundary between them. It is instructive to see why each hypothesis is needed exactly where it is used. If M is not simply connected, the covering map of Step 2 can genuinely have several sheets, and M is a quotient $\mathbb{R}^n/\Gamma$ rather than $\mathbb{R}^n$ itself; the flat torus $\mathbb{R}^n/\mathbb{Z}^n$, with $K \equiv 0$, is the first example, complete and nonpositively curved but not simply connected and not diffeomorphic to $\mathbb{R}^n$. If M is not complete, the lifting argument of lemma 6.4.2 breaks down at the boundary; an open ball in $\mathbb{R}^n$ shows the exponential map need not even be surjective.

A word on the standing theme of the chapter. The one place completeness entered, the covering lemma, is precisely the place where the Riemannian hypothesis is indispensable. Completeness was used through Hopf–Rinow (theorem 6.1.3), whose equivalence of metric and geodesic completeness fails for Lorentzian metrics; there a geodesically complete manifold need not be metrically complete, Cauchy sequences of lifts need not converge, and geodesics need not lift for all time. The Cartan–Hadamard conclusion has no Lorentzian analogue in this form, and its collapse can be traced to exactly this failure of Hopf–Rinow, flagged when that theorem was proved. Nonpositive curvature by itself, the ingredient that gave the local diffeomorphism, transfers to the pseudo-Riemannian setting;

it is the global promotion that does not.

Two consequences record the geometric and topological content of the diffeomorphism.

Corollary 6.4.4: Unique Geodesics and Asphericity

Let M be as in theorem 6.4.3. Then any two points of M are joined by a unique geodesic, and M is contractible; in particular all higher homotopy groups vanish, so M is aspherical (a $K(\pi, 1)$ when regarded as a universal cover). Any complete connected manifold of nonpositive curvature has $\mathbb{R}^n$ as its universal cover.

Proof:

Fix $p, q \in M$. By theorem 6.4.3 the map $\exp_p: T_p M \rightarrow M$ is a diffeomorphism, so there is a unique $w \in T_p M$ with $\exp_p(w) = q$. The geodesics from p to q are exactly the curves $t \mapsto \exp_p(tw)$ with $\exp_p(w) = q$, one for each preimage of q under $\exp_p$; since there is exactly one such preimage w , there is exactly one geodesic from p to q , namely $t \mapsto \exp_p(tw)$ on $[0, 1]$. As p, q were arbitrary, any two points are joined by a unique geodesic.

Contractibility is immediate from the diffeomorphism $M \cong T_p M \cong \mathbb{R}^n$: a vector space is contractible via the straight-line homotopy $H(x, s) = (1 - s)x$ to its origin, transported through the diffeomorphism. A contractible space has all homotopy groups trivial, in particular $\pi_k(M) = 0$ for all $k \geq 1$, so M is aspherical. For a complete connected manifold M_0 of nonpositive curvature that is not assumed simply connected, its universal cover $\tilde{M}_0$ carries the pulled-back metric, is again complete (the covering is a local isometry and completeness lifts along it, by the lifting argument of lemma 6.4.2 applied to the covering projection), is connected and simply connected by construction, and has the same sectional curvature $K \leq 0$ pointwise since the covering is a local isometry. Theorem 6.4.3 applies to $\tilde{M}_0$ and gives $\tilde{M}_0 \cong \mathbb{R}^n$, as claimed.

The corollary is the statement usually attached to the phrase *Cartan–Hadamard manifold*: a complete simply connected manifold of nonpositive curvature, in which geodesics are unique globally and the space is topologically trivial. The uniqueness of geodesics is the global upgrade of the local uniqueness inside a normal ball; on a general manifold two points can be joined by many geodesics, as antipodal points on the sphere are joined by infinitely many great semicircles, but nonpositive curvature suppresses all such refocusing. That every complete nonpositively curved manifold is covered by $\mathbb{R}^n$ is the reason such manifolds are the natural home of geometric group theory: their fundamental groups act freely on a contractible space, so the manifold is a classifying space for the group.

Specializing to constant curvature closes a case left open two chapters earlier.

Corollary 6.4.5: Killing–Hopf for Nonpositive Curvature

A complete, connected, simply connected Riemannian manifold of constant sectional curvature $c \leq 0$ is isometric to Euclidean space $\mathbb{R}^n$ (if $c = 0$) or to the rescaled hyperbolic space H^n of curvature c (if $c < 0$).

Proof:

Let M have constant sectional curvature $c \leq 0$; then $K \leq 0$, so theorem 6.4.3 applies and $\exp_p: T_p M \rightarrow M$ is a diffeomorphism for each p . Fix p and let M_c denote the model space of curvature c : Euclidean space $\mathbb{R}^n$ for $c = 0$, and hyperbolic space of curvature c for $c < 0$. The model M_c is itself complete, simply connected, and of constant curvature $c \leq 0$, so $\exp_{q_0}: T_{q_0} M_c \rightarrow M_c$ is also a diffeomorphism for a chosen basepoint q_0 .

By the local classification of space forms (theorem 4.5.5), applied about p and q_0 , both metrics take, in geodesic polar coordinates on any normal ball, the common radial normal form $d\rho^2 + \psi_c(\rho)^2 g_\Sigma$, with the identical radial factor ψ_c determined by c alone ($\psi_0(\rho) = \rho$ for $c = 0$; $\psi_c(\rho) = \frac{1}{\sqrt{-c}} \sinh(\sqrt{-c}\rho)$ for $c < 0$). Choose a linear isometry $\iota: (T_p M, g_p) \rightarrow (T_{q_0} M_c, (g_{M_c})_{q_0})$ and define

$$\Psi = \exp_{q_0} \circ \iota \circ (\exp_p)^{-1}: M \longrightarrow M_c$$

This is a diffeomorphism, being a composite of the diffeomorphism $(\exp_p)^{-1}$ (from theorem 6.4.3), the linear isomorphism ι , and the diffeomorphism $\exp_{q_0}$ (from the same theorem applied to M_c). It is defined on all of M precisely because both exponential maps are global diffeomorphisms; this is exactly the input the local classification could not provide on its own. In geodesic polar coordinates Ψ is $(\rho, \theta) \mapsto (\rho, \iota\theta)$: it fixes the radial coordinate and applies the linear isometry ι on the sphere of directions. Because both metrics equal $d\rho^2 + \psi_c(\rho)^2 g_\Sigma$ with the same ψ_c , and ι carries the unit sphere $\Sigma \subset T_p M$ isometrically to the unit sphere in $T_{q_0} M_c$ (so $\iota^* g_\Sigma = g_\Sigma$), the pullback $\Psi^* g_{M_c} = g$. Hence Ψ is an isometry (definition 1.3.1) of M onto M_c , which is the assertion. For $c = 0$ this identifies M with $\mathbb{R}^n$; for $c < 0$ with rescaled hyperbolic space, matching the models of example 1.3.5.

This completes the nonpositive half of the Killing–Hopf theorem, whose statement was recorded but deferred in box 4.5.6: the global rigidity of complete simply connected space forms of nonpositive curvature is now a theorem, the completeness machinery of the present chapter supplying the global exponential-chart diffeomorphism that the local classification of theorem 4.5.5 lacked. The positive-curvature case $c > 0$ is different in character: there conjugate points appear, $\exp_p$ is no longer a diffeomorphism, and the sphere is compact, so the argument by a global normal chart cannot run. That case is handled separately, and its proof does not follow from Cartan–Hadamard; the round sphere S^n , with cut locus a single antipodal point and injectivity radius π (example 5.3.4 and the cut-locus discussion of section 6.2), is the model whose global geometry the positive Killing–Hopf statement pins down.

6.5 The Rauch Comparison Theorem

The three chapters behind this one have measured how geodesics spread through a single quantity: the norm of a Jacobi field. On a space of constant curvature that norm is a trigonometric or hyperbolic function whose behavior is transparent, and the trichotomy of the model spaces is read off from it directly, refocusing on the sphere, linear growth in Euclidean space, exponential growth in hyperbolic space. On a general manifold the Jacobi equation $D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$ still governs the spreading, but its curvature coefficient varies from point to point and direction to direction, and the norm $|J|$ is no longer a function one can write down. What remains available is a comparison. A one-sided bound on the sectional curvature does not determine $|J|$, but it does trap it between the model values, and the direction of the trapping is dictated by the direction of the curvature bound.

The Rauch comparison theorem is the exact statement of this trapping, and it is the engine behind the global results of the chapter. Its content is a monotonicity: where the curvature is smaller, Jacobi fields grow faster, so geodesics spread more and refocus later; where the curvature is larger, they grow more slowly and refocus sooner. Cast against the constant-curvature models it recovers, in a single argument, that positive curvature forces a conjugate point at a computable distance (the mechanism behind the diameter bound of Bonnet–Myers) and that nonpositive curvature forbids conjugate points entirely (the mechanism behind Cartan–Hadamard). The proof is a comparison of index forms, run infinitesimally: the logarithmic derivative of $|J|$ is expressed as a ratio built from the index form on a shrinking interval, and the index lemma of section 5.4 makes that ratio monotone in the curvature. Throughout, M and $\tilde{M}$ are Riemannian manifolds, $\gamma: [0, b] \rightarrow M$ and $\tilde{\gamma}: [0, b] \rightarrow \tilde{M}$ are unit-speed geodesics with covariant derivatives D_t and $\tilde{D}_t$ and curvature endomorphisms R and $\tilde{R}$, and the curvature sign convention of box 4.1.1 is in force. That both manifolds are Riemannian, and not of mixed signature, is used at the one point where positive-definiteness is needed, and is flagged there.

The comparison rests on rewriting the index form of a Jacobi field vanishing at the initial point as a boundary term. This is the same integration by parts that drove the index lemma, specialized to a Jacobi field and to a variable upper endpoint.

Lemma 6.5.1: The index form of a Jacobi field as a boundary term

Let $\gamma: [0, b] \rightarrow M$ be a unit-speed geodesic and J a Jacobi field along γ with $J(0) = 0$. For $t \in (0, b]$ define the index form of J over the subinterval $[0, t]$ by

$$I_t(J, J) = \int_0^t \left(|D_s J|^2 - \langle R(J, \dot{\gamma})\dot{\gamma}, J \rangle \right) ds$$

Then

$$I_t(J, J) = \langle D_t J(t), J(t) \rangle$$

Consequently, wherever $J(t) \neq 0$,

$$\frac{d}{dt} \log |J(t)| = \frac{\langle D_t J(t), J(t) \rangle}{|J(t)|^2} = \frac{I_t(J, J)}{|J(t)|^2}$$

Proof:

On $[0, t]$ the field J is smooth, so metric compatibility of ∇ (definition 2.2.4) gives $\frac{d}{ds} \langle D_s J, J \rangle = \langle D_s^2 J, J \rangle + \langle D_s J, D_s J \rangle$, that is $\langle D_s J, D_s J \rangle = \frac{d}{ds} \langle D_s J, J \rangle - \langle D_s^2 J, J \rangle$. Integrating over $[0, t]$,

$$\int_0^t |D_s J|^2 ds = \left[\langle D_s J, J \rangle \right]_0^t - \int_0^t \langle D_s^2 J, J \rangle ds = \langle D_t J(t), J(t) \rangle - \int_0^t \langle D_s^2 J, J \rangle ds$$

the boundary contribution at $s = 0$ vanishing because $J(0) = 0$. The Jacobi equation (5.1) gives $D_s^2 J = -R(J, \dot{\gamma})\dot{\gamma}$, so $-\langle D_s^2 J, J \rangle = \langle R(J, \dot{\gamma})\dot{\gamma}, J \rangle$ and

$$\int_0^t |D_s J|^2 ds = \langle D_t J(t), J(t) \rangle + \int_0^t \langle R(J, \dot{\gamma})\dot{\gamma}, J \rangle ds$$

Rearranging is exactly $I_t(J, J) = \langle D_t J(t), J(t) \rangle$. For the last statement, metric compatibility

gives $\frac{d}{dt} |J|^2 = 2 \langle D_t J, J \rangle$, so $\frac{d}{dt} \log |J| = \frac{1}{2} \frac{d}{dt} \log |J|^2 = \frac{\langle D_t J, J \rangle}{|J|^2}$, and substituting $\langle D_t J, J \rangle = I_t(J, J)$ yields the claim, as required.

The identity turns a statement about the growth rate of $|J|$ into a statement about the index form, and the index form is where the curvature bound bites. The index lemma of section 5.4 says that on a geodesic without interior conjugate points the Jacobi field minimizes the index form among all fields with the same endpoint values. Comparing the two manifolds through this minimizing property, with a transplant of one field onto the other manifold via parallel frames, is the substance of the theorem. The transplant is set up now.

Theorem 6.5.2: The Rauch Comparison Theorem

Let $\gamma: [0, b] \rightarrow M$ and $\tilde{\gamma}: [0, b] \rightarrow \tilde{M}$ be unit-speed geodesics in Riemannian manifolds of the same or different dimension, and let J and $\tilde{J}$ be Jacobi fields along γ and $\tilde{\gamma}$ respectively, normal to the geodesics, with

$$J(0) = 0, \quad \tilde{J}(0) = 0, \quad |D_t J(0)| = |\tilde{D}_t \tilde{J}(0)|$$

Suppose that $\tilde{\gamma}$ has no point conjugate to $\tilde{\gamma}(0)$ on $(0, b]$, and that the sectional curvatures satisfy

$$K_M(\Pi) \leq K_{\tilde{M}}(\tilde{\Pi})$$

for every $t \in [0, b]$, every 2-plane $\Pi \subseteq T_{\gamma(t)} M$ containing $\dot{\gamma}(t)$, and every 2-plane $\tilde{\Pi} \subseteq T_{\tilde{\gamma}(t)} \tilde{M}$ containing $\dot{\tilde{\gamma}}(t)$. Then

$$|J(t)| \geq |\tilde{J}(t)| \quad \text{for all } t \in [0, b]$$

The hypotheses require a word before the proof. The curvature comparison is between *all* planes tangent to γ on the M side and *all* planes tangent to $\tilde{\gamma}$ on the $\tilde{M}$ side, so the statement is that the largest sectional curvature encountered along γ is no larger than the smallest encountered along $\tilde{\gamma}$; in the applications one manifold is a constant-curvature model and the bound reduces to a single number. The assumption $|D_t J(0)| = |\tilde{D}_t \tilde{J}(0)|$ normalizes the two fields to leave the initial point at the same infinitesimal rate, which is what makes their subsequent growth comparable; without it the inequality could be scaled either way. The no-conjugate-point hypothesis is imposed on the *comparison* manifold $\tilde{M}$, the one whose Jacobi field is the smaller, and it is exactly what the index lemma needs. The conclusion says that the field on the less curved manifold is at least as long throughout: less curvature, more spreading.

Proof:

If $D_t J(0) = 0$ then, since $J(0) = 0$ as well, $J \equiv 0$ by the uniqueness of Jacobi fields from initial data (theorem 5.2.3); the hypothesis then forces $\tilde{D}_t \tilde{J}(0) = 0$ and $\tilde{J} \equiv 0$, and the inequality holds trivially. Assume henceforth $\lambda := |D_t J(0)| = |\tilde{D}_t \tilde{J}(0)| > 0$.

The proof proceeds in four steps: first the behavior of the two norms near $t = 0$, then the reduction of the inequality to a monotonicity of the ratio, then the index-form comparison that drives the monotonicity, and finally the assembly.

Step 1: initial behavior and non-vanishing of $\tilde{J}$. Because $\tilde{\gamma}$ has no conjugate point on $(0, b]$ and $\tilde{J}$ is a nonzero Jacobi field with $\tilde{J}(0) = 0$, the field $\tilde{J}$ does not vanish on $(0, b]$: a zero at $t_0 \in (0, b]$ would make $\tilde{J}$ a nonzero Jacobi field vanishing at 0 and t_0 , so $\tilde{\gamma}(t_0)$ would be conjugate to $\tilde{\gamma}(0)$ (definition 5.3.1), contrary to hypothesis. Thus $|\tilde{J}(t)| > 0$ for $t \in (0, b]$, and the ratio

$|J(t)| / |\tilde{J}(t)|$ is defined there wherever the denominator is positive.

Near $t = 0$ both fields grow linearly. Expanding a Jacobi field with $J(0) = 0$ in a Taylor expansion, using $D_t^2 J(0) = -R(J(0), \dot{\gamma})\dot{\gamma} = 0$ from the Jacobi equation and $J(0) = 0$, gives $J(t) = t P_0 D_t J(0) + O(t^3)$, where P_0 denotes parallel transport from $\gamma(0)$; hence $|J(t)| = \lambda t + O(t^3)$, and likewise $|\tilde{J}(t)| = \lambda t + O(t^3)$. Therefore

$$\lim_{t \rightarrow 0^+} \frac{|J(t)|}{|\tilde{J}(t)|} = 1 \quad (6.9)$$

and both $|J|$ and $|\tilde{J}|$ are positive on some interval $(0, \varepsilon)$.

Step 2: reduction to a pointwise index inequality. Let

$$c^* = \sup \{t \in (0, b] \mid |J(s)| > 0 \text{ for all } s \in (0, t]\}$$

which is positive because $|J| > 0$ on $(0, \varepsilon)$ by Step 1. On the open interval $(0, c^*)$ both $|J|$ and $|\tilde{J}|$ are positive, so the ratio

$$\rho(t) = \frac{|J(t)|^2}{|\tilde{J}(t)|^2}$$

is smooth and positive there, with $\rho(0^+) = 1$ by (6.9). Its logarithmic derivative is

$$\frac{1}{2} \frac{\rho'(t)}{\rho(t)} = \frac{d}{dt} \log |J(t)| - \frac{d}{dt} \log |\tilde{J}(t)| \quad (6.10)$$

and by lemma 6.5.1, applied on each manifold, the two logarithmic derivatives are the index-form ratios

$$\frac{d}{dt} \log |J(t)| = \frac{I_t(J, J)}{|J(t)|^2}, \quad \frac{d}{dt} \log |\tilde{J}(t)| = \frac{\tilde{I}_t(\tilde{J}, \tilde{J})}{|\tilde{J}(t)|^2} \quad (6.11)$$

where I_t and $\tilde{I}_t$ are the index forms over $[0, t]$ on M and $\tilde{M}$. Combining (6.10) and (6.11), the sign of $\rho'(t)$ is the sign of

$$\frac{I_t(J, J)}{|J(t)|^2} - \frac{\tilde{I}_t(\tilde{J}, \tilde{J})}{|\tilde{J}(t)|^2}$$

so ρ is nondecreasing on $(0, c^*)$ once it is shown that, for every fixed $c \in (0, c^*)$,

$$\frac{I_c(J, J)}{|J(c)|^2} \geq \frac{\tilde{I}_c(\tilde{J}, \tilde{J})}{|\tilde{J}(c)|^2} \quad (6.12)$$

Step 3 proves (6.12) at each such c .

Step 3: the transplant and the index-form comparison. Fix $c \in (0, c^*)$. Rescale both fields to unit length at $t = c$: put $J^* = J/|J(c)|$ and $\tilde{J}^* = \tilde{J}/|\tilde{J}(c)|$, which are Jacobi fields (the Jacobi equation is linear) with $J^*(0) = \tilde{J}^*(0) = 0$ and $|J^*(c)| = |\tilde{J}^*(c)| = 1$. Because I_c and $\tilde{I}_c$ are quadratic, $I_c(J^*, J^*) = I_c(J, J)/|J(c)|^2$ and $\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) = \tilde{I}_c(\tilde{J}, \tilde{J})/|\tilde{J}(c)|^2$, so (6.12) is the normalized inequality

$$I_c(J^*, J^*) \geq \tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \quad (6.13)$$

Transplant J^* from γ onto $\tilde{\gamma}$ by matching parallel frames. Both geodesics are unit speed, so $\dot{\gamma}$ and $\dot{\tilde{\gamma}}$ are parallel unit fields and their normal bundles $\dot{\gamma}^\perp, \dot{\tilde{\gamma}}^\perp$ over $[0, c]$ carry parallel orthonormal

frames obtained by parallel-transporting orthonormal bases of the normal spaces at $t = 0$. Let $n = \dim M$ and $\tilde{n} = \dim \tilde{M}$, and choose a parallel orthonormal frame $(E_2, \dots, E_n)$ of $\dot{\gamma}^\perp$ along γ . Write the normal field J^* in this frame,

$$J^*(t) = \sum_{a=2}^n \varphi^a(t) E_a(t)$$

with scalar coefficients φ^a , smooth on $[0, c]$ and vanishing at $t = 0$. Fix a linear isometry $L: \dot{\gamma}(0)^\perp \rightarrow \dot{\tilde{\gamma}}(0)^\perp$ into the target normal space, which exists because $\dim \dot{\gamma}(0)^\perp = n - 1 \leq \tilde{n} - 1 = \dim \dot{\tilde{\gamma}}(0)^\perp$ and a linear map preserving inner products can be defined on an orthonormal basis; in the equidimensional case $n = \tilde{n}$ relevant to every application below, L is a linear isometry of the two normal spaces. Let (LE_a) denote the parallel field along $\tilde{\gamma}$ with initial value $L(E_a(0))$, and define the transplanted field along $\tilde{\gamma}$ by

$$\tilde{V}(t) = \sum_{a=2}^n \varphi^a(t) (LE_a)(t)$$

Since L is a linear isometry and parallel transport preserves inner products on both manifolds, at every t the vectors $\{(LE_a)(t)\}$ are orthonormal in $\dot{\tilde{\gamma}}(t)^\perp$ with the same Gram relations as $\{E_a(t)\}$; hence the transplant is norm-preserving pointwise,

$$|\tilde{V}(t)|^2 = \sum_a (\varphi^a(t))^2 = |J^*(t)|^2, \quad |\tilde{D}_t \tilde{V}(t)|^2 = \sum_a (\dot{\varphi}^a(t))^2 = |D_t J^*(t)|^2 \quad (6.14)$$

the derivative identity holding because both frames are parallel, so covariant differentiation acts only on the coefficients φ^a . Thus $\tilde{V}$ is a smooth normal field along $\tilde{\gamma}$ with $\tilde{V}(0) = 0$ and $|\tilde{V}(c)| = |J^*(c)| = 1$. Choose L so that in addition $\tilde{V}(c) = \tilde{J}^*(c)$; this is arranged by composing L with a rotation of the target normal space carrying the unit vector $\tilde{V}(c)/|\tilde{V}(c)|$ to $\tilde{J}^*(c)$, a rotation that preserves all the inner-product relations used above.

The comparison now runs in two links.

Link (a): the index lemma on $\tilde{M}$. The comparison manifold $\tilde{\gamma}$ has no point conjugate to $\tilde{\gamma}(0)$ on $(0, c]$, since $c < c^* \leq b$ and no conjugate point exists on $(0, b]$ by hypothesis. The field $\tilde{J}^*$ is the Jacobi field along $\tilde{\gamma}$ with $\tilde{J}^*(0) = 0$ and endpoint value $\tilde{J}^*(c) = \tilde{V}(c)$, and $\tilde{V}$ is a normal field along $\tilde{\gamma}$ with the same endpoint data $\tilde{V}(0) = 0$, $\tilde{V}(c) = \tilde{J}^*(c)$. The index lemma (theorem 5.4.4) applies verbatim on $\tilde{M}$ and gives

$$\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \leq \tilde{I}_c(\tilde{V}, \tilde{V}) \quad (6.15)$$

with equality iff $\tilde{V} = \tilde{J}^*$.

Link (b): the curvature term is larger on $\tilde{M}$. Compute the two index forms in their frames, abbreviating $\Pi_t = \text{span}(J^*(t), \dot{\gamma}(t))$ and $\tilde{\Pi}_t = \text{span}(\tilde{V}(t), \dot{\tilde{\gamma}}(t))$,

$$I_c(J^*, J^*) = \int_0^c \left(|D_t J^*|^2 - \langle R(J^*, \dot{\gamma}) \dot{\gamma}, J^* \rangle \right) dt, \quad \tilde{I}_c(\tilde{V}, \tilde{V}) = \int_0^c \left(|\tilde{D}_t \tilde{V}|^2 - \langle \tilde{R}(\tilde{V}, \dot{\tilde{\gamma}}) \dot{\tilde{\gamma}}, \tilde{V} \rangle \right) dt$$

The derivative-energy integrands agree pointwise by (6.14): $|\tilde{D}_t \tilde{V}|^2 = |D_t J^*|^2$. For the curvature integrands, the definition of sectional curvature (definition 4.3.1) gives, where J^* and $\dot{\gamma}$ are independent,

$$\langle R(J^*, \dot{\gamma}) \dot{\gamma}, J^* \rangle = K_M(\Pi_t) (|J^*|^2 - \langle J^*, \dot{\gamma} \rangle^2) = K_M(\Pi_t) |J^*|^2$$

since $J^* \perp \dot{\gamma}$ and $|\dot{\gamma}| = 1$; where J^* and $\dot{\gamma}$ are dependent the Gram determinant $|J^*|^2 - \langle J^*, \dot{\gamma} \rangle^2$ vanishes and so does the pairing, so the identity $\langle R(J^*, \dot{\gamma})\dot{\gamma}, J^* \rangle = K_M(\Pi_t) |J^*|^2$ still holds with $K_M(\Pi_t)$ read as any value. The identical computation on $\tilde{M}$ gives $\langle \tilde{R}(\tilde{V}, \dot{\gamma})\dot{\gamma}, \tilde{V} \rangle = K_{\tilde{M}}(\tilde{\Pi}_t) |\tilde{V}|^2$. By (6.14) the two lengths agree, $|\tilde{V}(t)|^2 = |J^*(t)|^2$, and by the curvature hypothesis $K_M(\Pi_t) \leq K_{\tilde{M}}(\tilde{\Pi}_t)$ for the two tangent-plane families. Therefore, pointwise in t ,

$$\langle R(J^*, \dot{\gamma})\dot{\gamma}, J^* \rangle = K_M(\Pi_t) |J^*|^2 \leq K_{\tilde{M}}(\tilde{\Pi}_t) |\tilde{V}|^2 = \langle \tilde{R}(\tilde{V}, \dot{\gamma})\dot{\gamma}, \tilde{V} \rangle$$

Here the Riemannian hypothesis enters: $|J^*|^2 = |\tilde{V}|^2 \geq 0$ is a genuine nonnegative length, so multiplying the curvature inequality $K_M \leq K_{\tilde{M}}$ by it preserves the direction. In a mixed-signature setting a normal field can have $\langle J^*, J^* \rangle$ of either sign, this step fails, and indeed Hopf–Rinow and the whole comparison theory break down; this is the analytic reason the theorem is confined to Riemannian metrics. Subtracting the larger curvature term on $\tilde{M}$ leaves the smaller integrand, so

$$\tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*) \quad (6.16)$$

Chaining (6.15) and (6.16),

$$\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \leq \tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*)$$

which is the normalized inequality (6.13), hence (6.12).

Step 4: integration and the boundary case. By Step 3 the inequality (6.12) holds for every $c \in (0, c^*)$, so by Step 2 the ratio ρ is nondecreasing on $(0, c^*)$. Since $\rho(0^+) = 1$, this gives $\rho(t) \geq 1$, that is

$$|J(t)| \geq |\tilde{J}(t)| \quad \text{for } t \in (0, c^*)$$

It remains to show $c^* = b$, so that the inequality holds on all of $(0, b]$. Suppose $c^* \leq b$. On $(0, c^*)$ one has $|J(t)| \geq |\tilde{J}(t)|$, and $|\tilde{J}|$ is continuous and positive on $(0, b] \supseteq (0, c^*)$ by Step 1; letting $t \uparrow c^*$ and using continuity of $|J|$ gives $|J(c^*)| \geq |\tilde{J}(c^*)| > 0$. Thus $|J|$ is positive at c^* and, being continuous, positive on a neighborhood of c^* , contradicting the definition of c^* as the supremum of parameters up to which $|J|$ stays positive unless $c^* = b$. Hence $c^* = b$, and the inequality $|J(t)| \geq |\tilde{J}(t)|$ holds for all $t \in (0, b]$; at $t = 0$ both sides vanish. This is the assertion, completing the proof.

The proof deserves a plain reading, because its structure is the template for every comparison argument to come. The quantity that carries the geometry is the logarithmic derivative of $|J|$, which lemma 6.5.1 identifies with the index form of the Jacobi field over a shrinking interval, divided by $|J|^2$. Comparing that ratio across the two manifolds is done by transplanting the field J^* from the less-curved manifold M onto the comparison geodesic $\tilde{\gamma}$, where two facts then push the index form in the same direction. The index lemma says the Jacobi field $\tilde{J}^*$ is the most economical field on $\tilde{M}$ with its endpoint value, so the transplant $\tilde{V}$ has at least as large an index form; and the curvature bound says that, term by term, the larger curvature of $\tilde{M}$ subtracts more, making the index form of the same field smaller on $\tilde{M}$ than on M . Reading the two facts along the chain $\tilde{I}_c(\tilde{J}^*, \tilde{J}^*) \leq \tilde{I}_c(\tilde{V}, \tilde{V}) \leq I_c(J^*, J^*)$ produces the normalized inequality that is the differential form of the conclusion. Integrating the resulting monotonicity from the common linear start at $t = 0$ delivers $|J| \geq |\tilde{J}|$. The transplant is the only nontrivial device, and it works precisely because parallel frames along a unit-speed geodesic carry the metric structure faithfully, so the derivative-energy term is transported unchanged and only the curvature term is allowed to differ.

The first use of the theorem is to convert one-sided curvature bounds into conjugate-point conclusions, by comparing against the constant-curvature models whose Jacobi fields are the explicit functions of example 5.2.5.

Corollary 6.5.3: Curvature bounds and conjugate points

Let (M, g) be a Riemannian manifold and $\gamma: [0, \infty) \rightarrow M$ a unit-speed geodesic issuing from $p = \gamma(0)$.

1. If the sectional curvature satisfies $K_M \leq \kappa$ along γ for a constant κ , then γ has no point conjugate to p before parameter $\pi/\sqrt{\kappa}$ when $\kappa > 0$, and no conjugate point at all when $\kappa \leq 0$. In particular $K_M \leq 0$ forces no conjugate points, recovering proposition 5.3.3.
2. If the sectional curvature satisfies $K_M \geq \kappa$ along γ for a constant $\kappa > 0$, and M has dimension at least two, then γ has a point conjugate to p at some parameter $t \leq \pi/\sqrt{\kappa}$.

Proof:

Let M_κ denote the simply connected model space of constant curvature κ , and $\tilde{\gamma}$ a unit-speed geodesic in it. Fix a normal Jacobi field J along γ with $J(0) = 0$ and $|D_t J(0)| = 1$, and let $\tilde{J}$ be the normal Jacobi field along $\tilde{\gamma}$ with $\tilde{J}(0) = 0$ and $|\tilde{D}_t \tilde{J}(0)| = 1$; by example 5.2.5, $|\tilde{J}(t)| = s_\kappa(t)$, the solution of $s'' + \kappa s = 0$ with $s(0) = 0$, $s'(0) = 1$, that is $s_\kappa(t) = \sin(\sqrt{\kappa}t)/\sqrt{\kappa}$ for $\kappa > 0$, t for $\kappa = 0$, and $\sinh(\sqrt{-\kappa}t)/\sqrt{-\kappa}$ for $\kappa < 0$.

1. Suppose $K_M \leq \kappa$ along γ . Fix any $b > 0$ with $b < \pi/\sqrt{\kappa}$ when $\kappa > 0$, and any $b > 0$ when $\kappa \leq 0$; then on $(0, b]$ the model function s_κ is positive, so $\tilde{\gamma}$ has no conjugate point on $(0, b]$. The Rauch theorem theorem 6.5.2, with the roles $M \rightarrow M$ and $\tilde{M} \rightarrow M_\kappa$ (the curvature bound $K_M \leq \kappa = K_{M_\kappa}$ is the required $K_M \leq K_{\tilde{M}}$), gives $|J(t)| \geq |\tilde{J}(t)| = s_\kappa(t) > 0$ on $(0, b]$. Hence J does not vanish on $(0, b]$; as b is arbitrary below $\pi/\sqrt{\kappa}$ (or arbitrary when $\kappa \leq 0$), J does not vanish on $(0, \pi/\sqrt{\kappa})$ (respectively on $(0, \infty)$). The same holds for every normal Jacobi field vanishing at 0, by taking it as J , so no such field vanishes before $\pi/\sqrt{\kappa}$, and γ has no conjugate point of p there. When $\kappa \leq 0$ this is the absence of conjugate points of proposition 5.3.3, now obtained as a special case of Rauch.
2. Suppose $K_M \geq \kappa > 0$ along γ , and $n = \dim M \geq 2$. The comparison runs with the roles reversed: the less-curved manifold is now the model M_κ , and the more-curved comparison manifold is M , whose geodesic γ must be shown conjugate-point-free before Rauch can be applied to it. Argue by contradiction: suppose γ has no conjugate point of p on $(0, \pi/\sqrt{\kappa}]$. Then theorem 6.5.2 applies with base geodesic $\tilde{\gamma} \subset M_\kappa$ and comparison geodesic $\gamma \subset M$, the curvature hypothesis $\kappa \leq K_M$ being the required $K_{M_\kappa} \leq K_M$ and the comparison manifold M being conjugate-point-free on $(0, \pi/\sqrt{\kappa}]$ by assumption; taking the model field $\tilde{J}$ along $\tilde{\gamma}$ (norm s_κ) and the field J along γ with the matched initial derivative $|D_t J(0)| = 1 = |\tilde{D}_t \tilde{J}(0)|$, its conclusion is that the model field is the longer, $s_\kappa(t) = |\tilde{J}(t)| \geq |J(t)|$ on $(0, \pi/\sqrt{\kappa}]$. But $s_\kappa(\pi/\sqrt{\kappa}) = \sin(\pi)/\sqrt{\kappa} = 0$, so $|J(\pi/\sqrt{\kappa})| \leq 0$, forcing $J(\pi/\sqrt{\kappa}) = 0$. Since $n \geq 2$ there is a nonzero normal Jacobi field J with $J(0) = 0$, and it vanishes at $\pi/\sqrt{\kappa}$, so $\gamma(\pi/\sqrt{\kappa})$ is conjugate to p , contradicting the assumption. Hence γ has a conjugate point at some $t \leq \pi/\sqrt{\kappa}$, as claimed.

Both parts follow, completing the proof.

The two halves of the corollary are the quantitative skeleton of the two global theorems of the chapter. Part (2), that curvature bounded below by $\kappa > 0$ forces a conjugate point by parameter $\pi/\sqrt{\kappa}$, is the geodesic-focusing mechanism behind Bonnet–Myers: past a conjugate point a geodesic no longer minimizes (theorem 5.4.5), so no minimizing geodesic can be longer than $\pi/\sqrt{\kappa}$, and the diameter is bounded. Bonnet–Myers in fact needs only a lower bound on the Ricci curvature rather than on all sectional curvatures, so it is proved by a direct second-variation computation rather than by this corollary, but the scale $\pi/\sqrt{\kappa}$ is the same and its origin is the model conjugate distance seen here. Part (1), that curvature bounded above by 0 forbids conjugate points, is the local input to Cartan–Hadamard, giving a second proof of proposition 5.3.3 that exhibits it as one end of a continuum: as the upper bound κ decreases through 0, the first possible conjugate distance $\pi/\sqrt{\kappa}$ increases to infinity and then disappears.

Against a constant-curvature model the theorem sharpens to a two-sided estimate that pins the growth of any Jacobi field between the two model solutions.

Example 6.5.4: Two-sided comparison against the constant-curvature models

Let (M, g) be a Riemannian manifold whose sectional curvature along a unit-speed geodesic γ is pinched between two constants,

$$\delta \leq K_M \leq \Delta$$

for every tangent 2-plane containing $\dot{\gamma}$. Let J be a normal Jacobi field along γ with $J(0) = 0$ and $|D_t J(0)| = 1$. Write s_κ for the model solution of $s'' + \kappa s = 0$ with $s(0) = 0$, $s'(0) = 1$, namely

$$s_\kappa(t) = \begin{cases} \frac{\sin(\sqrt{\kappa} t)}{\sqrt{\kappa}} & \kappa > 0 \\ t & \kappa = 0 \\ \frac{\sinh(\sqrt{-\kappa} t)}{\sqrt{-\kappa}} & \kappa < 0 \end{cases}$$

as in example 5.2.5. Then, on the interval $[0, b]$ up to the first zero of s_Δ (which is $\pi/\sqrt{\Delta}$ if $\Delta > 0$ and $+\infty$ otherwise),

$$s_\Delta(t) \leq |J(t)| \leq s_\delta(t) \quad (6.17)$$

The two inequalities are two applications of theorem 6.5.2 with opposite role assignments.

For the lower bound $s_\Delta(t) \leq |J(t)|$, compare M against the model M_Δ of constant curvature Δ . The hypothesis $K_M \leq \Delta = K_{M_\Delta}$ is the required $K_M \leq K_{\tilde{M}}$ with $\tilde{M} = M_\Delta$; the model geodesic $\tilde{\gamma}$ has no conjugate point on $(0, b]$ because $s_\Delta > 0$ there; and the model Jacobi field $\tilde{J}$ with matched initial data has $|\tilde{J}| = s_\Delta$ by example 5.2.5. Rauch gives $|J(t)| \geq |\tilde{J}(t)| = s_\Delta(t)$.

For the upper bound $|J(t)| \leq s_\delta(t)$, compare the model M_δ of constant curvature δ against M . Now the base manifold is M_δ and the comparison manifold is M : the hypothesis $\delta \leq K_M$ is $K_{M_\delta} \leq K_M$, the required inequality with the model in the role of the less-curved manifold; the comparison manifold M has no conjugate point on $(0, b]$ because $b \leq \pi/\sqrt{\Delta}$ and part (1) of corollary 6.5.3 (using $K_M \leq \Delta$) excludes conjugate points before $\pi/\sqrt{\Delta}$; and the model field on M_δ with matched initial data has norm s_δ . Rauch gives $s_\delta(t) \geq |J(t)|$, which is the upper bound.

The two model space forms make (6.17) concrete. On the round sphere S_r^n , where $K \equiv 1/r^2$, both bounds coincide and force $|J(t)| = r \sin(t/r)$, in agreement with example 5.2.5; the pinching is trivial because the space is already a model. For a genuine pinch, take a metric with $1/4 \leq K \leq 1$ on the unit-normalized scale, the setting of the sphere theorem. Then (6.17) reads $\sin t \leq |J(t)| \leq$

$2 \sin(t/2)$ on $[0, \pi]$, so a Jacobi field vanishing at 0 with unit initial derivative cannot vanish again before $t = \pi$ (the lower bound $\sin t$ is positive on $(0, \pi)$) and must vanish by $t = 2\pi$ (the upper bound $2 \sin(t/2)$ vanishes at 2π); the first conjugate point lies in $[\pi, 2\pi]$, trapping the injectivity-radius scale between the two models. On hyperbolic space H^n , where $K \equiv -1$, both bounds again coincide and give $|J(t)| = \sinh t$, and for a manifold with $-4 \leq K \leq -1$ the estimate becomes $\sinh t \leq |J(t)| \leq \frac{1}{2} \sinh(2t)$, both sides growing without bound, so no conjugate point ever appears, consistent with corollary 6.5.3(1).

The comparison theorem and its corollary complete the pattern promised at the start of the chapter. Curvature, an infinitesimal quantity attached to each 2-plane, controls the global spreading of geodesics through the single monotone comparison of Rauch, and the two model-space bounds bracket every intermediate behavior. Where the curvature is large the geodesics refocus early, capping the diameter and the injectivity radius; where it is small they spread freely, and in nonpositive curvature they never refocus at all. The conjugate-point statements of section 5.3, proved there by direct constant-curvature computation and by the convexity of $|J|^2$, reappear here as the two endpoints of a single inequality, and the same machinery, applied with a Ricci bound in place of a sectional one, drives the diameter and topology theorems that the models exemplify: the sphere compact and refocusing, hyperbolic space open and expanding, Euclidean space the flat boundary between them.

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